4A Neutrophils I

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INTRODUCTION

Polymorphonuclear leukocytes or granulocytes are hematopoietically derived phagocytes characterized by multilobed nuclei and the presence of multiple, distinct granules within their cytoplasm. Three different polymorphonuclear leukocytes are distinguished according to their granular staining properties: neutrophils (polymorphonuclear neutrophils or PMNs), basophils, and eosinophils. Neutrophil granules stain preferentially with neutral dyes, whereas basophil granules stain with basic dyes, and eosinophilic granules stain with acidic colorants such as eosin. These three types of leukocytes differ not only in their tinctorial properties, but also in their functions and roles during the inflammatory process. They constitute key effector cells in innate immunity, and the frontline of host defense in response to foreign antigens and microorganisms. In this chapter, we will focus on the biology and role of neutrophils. The other polymorphonuclear leukocytes are discussed elsewhere.

NEUTROPHIL HOMEOSTASIS: MYELOPOIESIS AND DESTRUCTION

Neutrophil myelopoiesis is a closely regulated process that begins with the differentiation of pluripotent stem cells into primitive myeloid progenitors, which in turn differentiate into specific myeloid precursors. Contact with specific adhesion molecules, hematopoietic growth factors, and cytokines promotes the progression of myeloblasts along unique pathways to mature as neutrophils, eosinophils, and basophils, as well as monocytes. The sequence that leads to neutrophil formation begins with the neutrophilic promyelocyte and progresses through several maturation steps (neutrophilic myelocyte, metamyelocyte, band cell, and mature neutrophil) (Figure 4A.1). This process of neutrophil development takes approximately 14 days; cell division is terminated at the metamyelocyte stage, while granule development continues. Eventually, terminally differentiated neutrophils are released from the bone marrow. At that point they are unable to divide, and their synthetic machinery has become partially inactivated. During steady-state granulopoiesis, roughly 10^{11} neutrophils are released into the bloodstream daily. Neutrophil half-life in blood is 6–8 hours and, under normal circumstances, neutrophils constitute roughly 60% of all leukocytes in human peripheral blood [1].

The coordinated expression of a number of myeloid transcription factors (including PU.1, CCAAT enhancer binding proteins α and ε [C/EBPα and C/EBPε], and GFI-1) is necessary for the regulation of neutrophil development [2]. Among the extracellular factors that direct pluripotent stem cells to differentiate into neutrophils, granulocyte colony-stimulating factor (G-CSF) plays an essential role [3]. G-CSF has been shown to induce myeloid differentiation [4], stimulate proliferation of granulocytic precursors, and provoke neutrophil release from the bone marrow [5]. The biological effects of G-CSF are mediated via its receptor (G-CSFR or CD114), a member of the hematopoietic cytokine receptor family [6]. Other hematopoietic cytokines contributing to neutrophil development in vivo include granulocyte-macrophage colony-stimulating factor (GM-CSF), interleukin-6 (IL-6), and IL-3 [7–9].

In the absence of inflammatory stimuli, neutrophil populations are maintained within a relatively narrow range. Although incompletely understood, the mechanisms controlling neutrophil homeostasis regulate both neutrophil production and clearance. Recent observations implicate the SDF-1/CXCR4 signaling system in the process of neutrophil clearance. SDF-1 (CXCL12) is a CXC chemokine that is secreted from bone marrow and attracts neutrophils by engaging the CXCR4 receptor. Senescent neutrophils upregulate CXCR4 and acquire the ability to migrate

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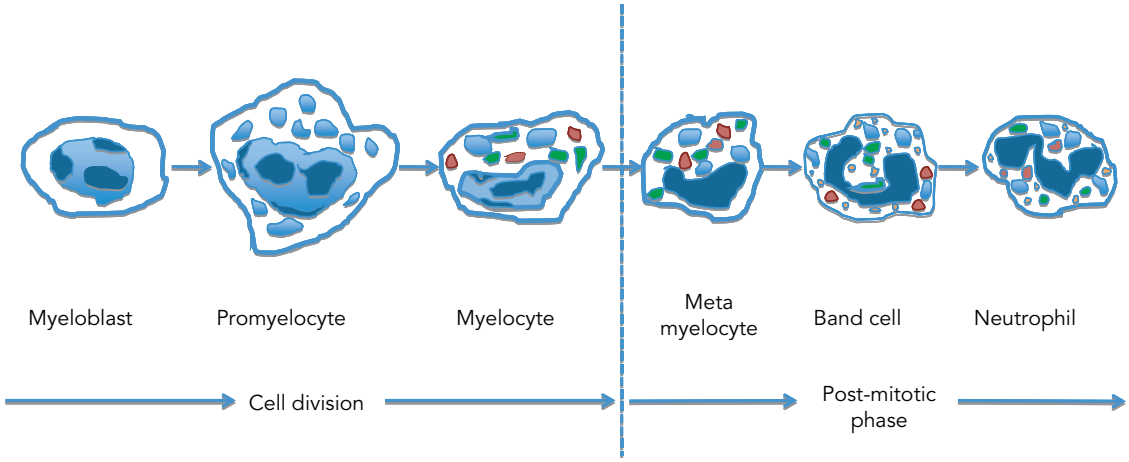


Figure 4A.1. Neutrophil development.

toward SDF-1, leading to homing to the bone marrow and clearance of aging cells from the blood [10,11]. In the bone marrow, as well as in the spleen and liver, damaged and aging neutrophils are cleared by tissue macrophages. Reduction of neutrophil populations is also effected through induction of apoptosis. For example, aged neutrophils undergo programmed cell death when exposed to tumor necrosis factor alpha (TNFα) or Fas ligand (CD95) [12,13].

A feedback loop downregulating neutrophil production has recently been identified. Following phagocytosis of apoptotic neutrophils by tissue macrophages, the latter shut down their secretion of IL-23. In consequence, IL-17 production by TH17 cells is reduced. Lack of IL-17 then results in decreased levels of G-CSF and, consequently, reduced neutrophil development and release. In the setting of increased neutrophil populations (and hence increased populations of apoptotic neutrophils), this system is poised to maintain neutrophil populations within a constant range [14].

NEUTROPHIL ANATOMY: MORPHOLOGY AND GRANULES

The development and formation of neutrophil granules occurs in a sequential process during myeloid cell differentiation (Figure 4A.1). Granule formation begins in the early promyelocyte stage, a period during which the majority of nascent granules are rich in myeloperoxidase (MPO), an enzyme that catalyzes the formation of hypochlorous acid. These *primary* granules are also referred to as *azurophilic*, owing to their affinity for the basic dye azure A [15]. They are functionally similar, although not identical, to lysosomes of other cells. Azurophilic granule maturation is largely complete at the myelocyte stage of neutrophil development, at which point peroxidase-negative granules begin to form. On the basis of their time of appearance and content, the latter are subdivided into secondary (or specific) and tertiary (or gelatinase) granules. Specific granules develop in myelocytes and metamyelocytes and are rich in lactoferrin, whereas gelatinase granules form later (neutrophil band stage) and lack lactoferrin [16]. A fourth category of granules, secretory vesicles, are smaller than the others and appear during the late stage of nuclear neutrophil segmentation [17].

In addition to MPO, azurophilic granules contain a variety of acidic hydrolases and serine proteinases (e.g., elastase, proteinase-3, and cathepsins), as well as enzymes directed at nucleic acids and sugars (Table 4A.1). Like lysosomes, azurophilic granules contain granulophysin (CD63) in their membrane [18]. In contrast to lysosomes, however, primary granules are deficient in lysosome-associated membrane proteins 1 and 2 (LAMP-1 and LAMP-2). Similarly, the mannose-6-phosphate receptor system – essential for lysosomal enzyme targeting [19] – is not utilized by neutrophil azurophilic granules [20]. Thus, some authors consider that azurophilic granules are best described as regulated secretory granules, rather than as specialized lysosomes [21].

In addition to soluble vesicular contents, specific granules possess an extensive array of membrane-associated proteins including cytochromes, signaling molecules, and receptors (Table 4A.1). Specific granules thus form a storage depot of proteins that can be directed to the surfaces of phagocytic vacuoles as well as the plasma membrane. One particularly important family of soluble proteinases found in neutrophil-specific granules are the matrix metalloproteinases (MMPs), including neutrophil collagenase-2 (MMP-8), gelatinase-B (MMP-9), and leukolysin (MMP-25). Many of these are stored as inactive proenzymes. They undergo proteolytic activation following granule

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TABLE 4A.1. Content of neutrophil granules

Azurophil granules	Specific granules	Gelatinase granules	Secretory vesicles
CD63 (granulophysin)	fMLP receptor	fMLP receptor	fMLP receptor
CD68	CD11b/CD18	CD11b/CD18	CD11b/CD18
Stomatin	CD66	Cytochrome b_{558}	Cytochrome b_{558}
Presenilin 1	CD67	SCAMP	Leukolysin (MMP-25)
	CD15	SNAP-23, -25	CD10
	Cytochrome b_{558}	Leukolysin (MMP-25)	CD13
	TNF receptor		CD14
	Fibronectin receptor		CD16
	Laminin receptor		CD45
	Thrombospondin receptor		CR1
	Vitronectin receptor		C1q receptor
	SCAMP		DAF
	SNAP-23, -25		
	Leukolysin (MMP-25)		
	NB1 antigen		
Myeloperoxidase	Lysozyme	Gelatinase (high concentrations)	Plasma proteins
Acid mucopolysaccharide	Lactoferrin	Lysozyme	
α-1 Antitrypsin	Gelatinase (MMP-9)	Acetyltransferase	
α-Mannosidase	Collagenase (MMP-8)	β2-Microglobulin	
β-Glycerophosphatases	β2-Microglobulin		
β-Glucuronidase	Histaminase		
Cathepsins	Heparanase		
Defensins	Sialidase		
Elastase	hCAP-18		
Lysozyme			
Proteinase-3			
BPI			

fusion and interaction with azurophilic granule contents, and are capable of disrupting major structural components of bacteria and/or the extracellular membrane. Neutrophil MMPs appear to be crucial not only for bacterial killing, but also for neutrophil extravasation and migration [22].

The remaining two classes of vesicles are less well studied. Gelatinase granules resemble specific granules in size and density, and share some proteins in common with them. Gelatinase granules differ from specific granules, however, in that they contain high concentrations of gelatinase, a latent enzyme with the capacity for tissue destruction. Secretory vesicles are smaller and lighter than the other classes. They lack proteolytic enzymes but are rich in membrane-associated proteins, including receptors otherwise identified with the plasma membrane. Moreover, they are preferentially directed not to phagosomes, but to the plasma membrane. Thus, secretory vesicles appear to serve mainly as a reservoir of neutrophil plasma membrane proteins (Table 4A.1).

NEUTROPHIL PHYSIOLOGY: ACTIVATION AND FUNCTION

In order to carry out their role in acute inflammation, bloodstream neutrophils must first sense chemical mediators (chemoattractants) via cell membrane receptors. Next, neutrophils must attach to the activated endothelium through several interactions involving adhesion molecules and their receptors (rolling and adhesion). After passing through postcapillary

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venules (diapedesis), neutrophils migrate toward sites of inflammation where they recognize their target, engulf it (phagocytosis), and finally destroy it. Activated neutrophils also play an important role in further upregulating the inflammatory process. Eventually, neutrophils participating in acute inflammation must be cleared from the inflammatory site as the inflammation resolves.

Stimuli and Receptors

Chemoattractants playing a role in neutrophil activation include lipid mediators (e.g., leukotriene B_4 [LTB_4] and platelet-activating factor), as well as proteins and peptides such as bacterial formylated peptides and the complement split product C5a. One important group of neutrophil chemoattractants are the CXC chemokines, characterized by the presence of paired cysteines ("C"), separated by any other amino acid ("X") at the C-terminus. CXC chemokines that appear to be most critical for neutrophil recruitment include IL-8 (CXCL8), MIP-2 (CXCL2), and KC (CXCL1). At sites of inflammation, chemoattractants are either produced by inflammatory cells (e.g., LTB_4 or IL-8), liberated from previously synthesized proteins (e.g., C5a), or derived from bacteria themselves (e.g., fMLP). Although all neutrophil chemoattractants can stimulate multiple aspects of neutrophil activation, they differ in both their relative potencies and the kinetics of their effects [23]. For example, neutrophil mobilization from the bone marrow by G-CSF takes 4–6 hours, whereas IL-8-driven neutrophil mobilization into peripheral tissues takes only minutes [24,25].

Neutrophil responses to chemoattractants and other stimuli depend on the presence of specific surface receptors. On the basis of microarray analysis, receptors for various inflammatory stimuli are significantly expressed only in terminally differentiated neutrophils [26]. Important examples include receptors for interferon (IFN) α and γ; IL receptors IL-1R, IL4R, IL-6R, IL-10R, and IL-17R; tumor necrosis factor (TNF) receptors 1 and 2; and CXC and CC chemokine receptors such as IL-8R-α and β, CXCR-4 and CCR-1, 2, and 3. Whereas growth factors and cytokines signal through various classes of protein tyrosine kinase–based receptors, chemoattractants signal mostly via G-protein–coupled receptors (GPCRs).

Once a GPCR is activated by its ligand, the receptor undergoes a conformational change, causing activation of an associated heterotrimeric GTP-binding protein (G protein). In neutrophils, the predominant G proteins are of the G_i family [27]. All G proteins function as "molecular switches," alternating between an inactive guanosine diphosphate (GDP)- and active guanosine triphosphate (GTP)-bound state, ultimately going on to regulate downstream cell processes.

Intracellular Signaling and Second Messengers

Multiple intracellular systems propagate signals from the surface of neutrophils to their interiors. One particularly important system is constituted by low molecular weight (20–25 kDa) GTP-binding proteins (LMW-GBPs), also known as Ras-related or Ras superfamily proteins. In contrast to heterotrimeric G proteins, these proteins are monomeric. However, they share with G proteins the ability to bind and hydrolyze GTP and serve as molecular switches. Four LMW-GBP subfamilies play important roles in neutrophils: the Ras subfamily itself, whose members regulate cell growth and division; the Rho subfamily, functioning in cytoskeletal rearrangements and superoxide generation [28]; and the Rab and Arf subfamilies, critical for vesicular and endomembrane trafficking [29].

Second Messengers

Neutrophils also employ second messenger systems, in which small molecules generated at the plasma membrane diffuse through the cell to affect targets. Among the best-studied of the systems in neutrophils is phospholipase C (PLC). At the cell membrane, activated PLC cleaves phosphoinositol bisphosphate (PIP_2) into diacyl glycerol (DAG) and inositol trisphosphate (IP_3), which in turn increases calcium influx to induce protein kinase C activation. A second critical phospholipase in neutrophils is $cPLA_2$, which hydrolyzes the sn-2 position of membrane glycerophospholipids to liberate arachidonic acid (AA), a precursor of eicosanoids including prostaglandins (PGs) and leukotrienes (LTs) [30,31].

Nonlipid second messengers regulating neutrophil activation include cAMP, whose intracellular concentrations can rise rapidly upon neutrophil stimulation. It is thought that cAMP provides a mostly negative regulatory (off) signal, as direct exposure to membrane-permeable analogs of cAMP inhibits many neutrophil responses, probably via activation of protein kinase A [32,33]. cGMP can also be generated by neutrophils, and appears to have a modest enhancing effect on some neutrophil responses.

Low levels of nitric oxide (NO) are also expressed in neutrophils following stimulation [34]. In addition to phagocytic and tumoricidal activity, NO in neutrophil appears to play an important role in signal transduction. Exogenously added NO exerts a variety of effects on neutrophils, including inhibition of the NADPH oxidase, actin polymerization, and chemotaxis [35]. Excessive NO production is seen during the course of a variety of rheumatic and inflammatory diseases, attesting to the possible importance of its effects on neutrophils [36]. See Chapters 14 through 16 for further discussion on second messengers and NO functions.

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Kinases and Kinase Cascades

Intracellular kinases and kinase cascades also participate in neutrophil regulation. Protein kinase C (PKC) is a family composed of three subsets (PKC-α, β, and γ) with overlapping but distinct functions. PKCs are activated in response to several chemoattractants, and studies support a role for PKC in neutrophil survival and function. Activation of specific isotypes of PKC is known to be involved in membrane alteration and motility, oxidative phosphorylation, and apoptosis modulation of neutrophils. Exogenous activators of PKCs, including phorbol myristate acetate (PMA) and palmitoyl-2-linoleoylglycerol hydroperoxide (PLG-OOH), stimulate neutrophil responses such as adhesion and O_2-generation [37,38]. Moreover, inhibitors of PKC (including chelerythrine chloride and staurosporine) block stimulation of neutrophil functions [39].

Mitogen-activated protein kinases (MAPK) are a family of serine/threonine-specific protein kinases that respond to extracellular stimuli (mitogens) and regulate cellular activities including gene expression, mitosis, differentiation, and cell survival/apoptosis. MAPKs include the Erk, p38, and Jun kinase (Jnk) subfamilies. In neutrophils, chemoattractants, colony stimulating factors, and other stimuli such as LPS are capable of activating all three classes of MAPK [40–42]. Current evidence support a role for Erk in neutrophil activation [43,44], including adhesion and phagocytosis processes [32,45].

Phosphatidylinositol 3-kinases (PI 3-Ks) phosphorylate lipid (phosphotidylinositol) targets. The main active product of class I PI 3-Ks appears to be phosphatidylinositol trisphosphate (PIP_3). Chemoattractants such as *N*-formyl-methionyl-leucyl-phenylalanine (FMLP) rapidly activate PI 3-K in neutrophils, regulating O_2^- generation, adhesion, chemotaxis [46], and degranulation [47]. A role for PI 3-K in the regulation of neutrophil survival and apoptosis has also been established [48].

THE CONSEQUENCES OF NEUTROPHIL ACTIVATION: ADHESION, MIGRATION, AND BACTERIAL KILLING

Neutrophil Adhesion and Migration

The ability of bloodstream neutrophils to adhere to endothelium is an essential step required to concentrate them and direct them to sites of inflammation. Membranes on both neutrophils and endothelial cells have the capacity to express multiple families of interacting adhesion molecules. These include selectins, integrins, and intercellular adhesion molecules (ICAMs), as well as sialylated glycoproteins. Under noninflammatory conditions, neutrophils and other leukocytes travel primarily through the center of the blood vessel lumen, where flow is fastest. In response to proinflammatory signals, both the neutrophils and the blood vessels undergo a series of changes. As a consequence of vascular dilatation blood flow slows, facilitating leukocyte interactions with endothelial cells. A process of rolling adhesion ensues, in which P- and L-selectins on neutrophils, and E-selectins on endothelia, interact with sialyl-Lewis^x moieties (s-Le^x) on their respective partners. These interactions are reversible and transient, and prepare the cells for a tighter binding step to follow. In response to CXCL8 (released by tissue macrophages and transported actively through the venular endothelial cells [49]), complement or formyl peptide fragments generated at the inflammatory site, integrins (predominantly LFA-1 and CR3 [Mac-1]) already expressed on neutrophil surfaces are stimulated to undergo conformational changes that render them adhesive for their cognate receptors. Concurrently, macrophage-derived TNFα and/or IL-1β induce endothelial cell expression of the integrins' cognate ligands, ICAM-1 and ICAM-2. The result is tight binding, and arrest of neutrophil motion [50].

Movement of neutrophils out of the circulatory system, or diapedesis, requires first the interaction between molecules of CD31 (expressed by PMNs and also by the intercellular junctions of endothelial cells) allowing the cell to pass through the endothelium [51]. The secretion of a broad range of MMPs degrades the basement membrane and permits the neutrophil passage through the acellular matrix. By mechanisms not fully established, the basement membrane appears to reseal behind the exited neutrophil. Once in the interstitial compartment, neutrophils migrate along the chemotactic gradient toward the site of injury or infection [52]. The mechanism of chemotaxis involves, in part, directed localization of chemoattractant receptors to the leading edge of the neutrophil ("headlight phenomenon") and cytoskeletal rearrangements to permit unidirectional motion.

Phagocytosis and Bacterial Killing

Once recruited to the inflammatory site, neutrophils directly recognize, phagocytose, and destroy foreign pathogens. Phagocytosis – a specialized form of endocytosis – is the cellular process of engulfing particles by the cell membrane to form an internal phagosome. Initiation of phagocytosis can be achieved by neutrophil recognition of pathogen-associated molecular patterns, or PAMPs, small repetitive molecular motifs that are found on bacteria and/or viruses but not on mammalian cells. PAMPs are recognized by Toll-like receptors (TLRs) [53,54] and other pattern recognition receptors (PRRs). Indeed, it has been demonstrated

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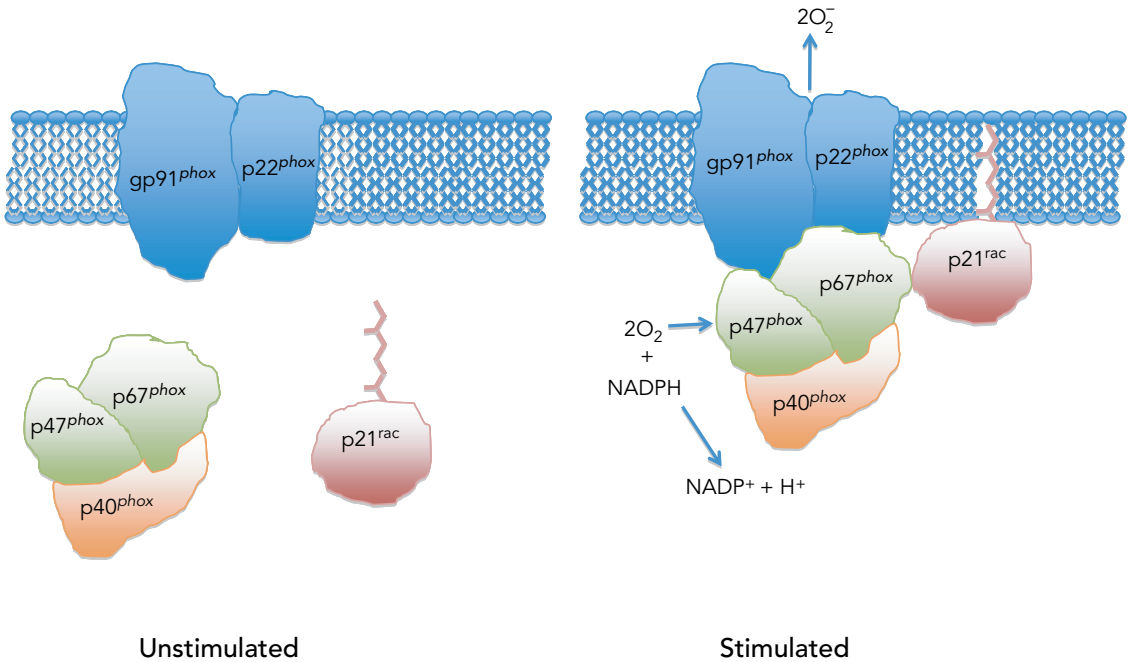


Figure 4A.2. Assembly and activation of the phagocyte oxidase system, resulting in the production of superoxide anions.

that human neutrophils are able to express all TLRs except for TLR3. Upon TLR stimulation, neutrophil shape is altered and phagocytosis is increased [55].

Neutrophil phagocytosis is greatly facilitated by opsonization, or the coating of bacterial or other targets with immunoglobulins and complement C3b fragments. Antibody-coated pathogens are recognized by both complement and immunoglobulin Fc receptors on the neutrophil surface [56,57]. Neutrophils possess receptors to the IgG isotypes IgG_1 and IgG_3 (FcγRI [CD64], FcγRII [CD32], and FcγRIII [CD16]). Opsonization of bacteria by immunoglobulins plays a particularly important role in the response of neutrophils to bacteria with polysaccharide capsules, since their capsules help them evade direct phagocytosis [58].

Neutrophil mechanisms of pathogen destruction are multiple, and involve granule fusion, toxic oxygen radical production, activation of latent proteolytic enzymes, and the activity of antibacterial proteins. The phagosome undergoes fusion with neutrophil granules to form a phagolysosome, the protected space in which pathogen degradation occurs.

Oxygen-dependent degradation of pathogens requires a NADPH oxidase system and a *respiratory burst*: a rapid production and release of reactive oxygen species (ROS) such as NO, superoxide anion, and hydrogen peroxide, all of which are highly toxic to bacteria. The NADPH oxidase enzyme is a membrane-bound multimeric complex that assembles from membrane and cytoplasmic proteins upon cell stimulation (Figure 4A.2). The NADPH oxidase system converts oxygen molecules (O_2) into superoxide anions (O_2^-). A second enzyme, superoxide dismutase (SOD), then converts O_2^- into hydrogen (H_2O_2), which can kill microorganisms. H_2O_2 can be further converted by myeloperoxidase into hypochlorous acid (HOCl, chlorine bleach), which also has potent antibacterial activity. Whereas the membrane components of the NADPH oxidase are localized to specific granules, myeloperoxidase is localized to primary granules; thus, the production of HOCl cannot occur until both granule classes fuse into the phagolysosome.

Oxygen-independent degradation also depends on the fusion of granules into the phagolysosome. As detailed at the beginning of this chapter, primary, specific, and gelatinase granules contain proteolytic enzymes such as lysozyme and metalloproteinases, as well as defensins and cationic proteins (e.g., bactericidal permeability inducing protein, BPI) with intrinsic antimicrobial properties required for the destruction of invading microorganisms [59].

Most recently, a unique, nonphagocytic mechanism of bacterial killing has been identified in neutrophils. Neutrophil extracellular traps (NETs) are extracellular neutrophil structures composed of chromatin and granule proteins that bind and kill microorganisms [60]. In settings of extreme stimulation, and in the presence of stress from ROS, neutrophils may undergo a novel form of cell death characterized by cell membrane breakdown and the release of NETs. These NETs can bind and neutralize extracellular

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pathogens. Because this process requires the sacrifice of the neutrophil, the term "beneficial suicide" has been proposed [61].

NEUTROPHIL PROPAGATION OF INFLAMMATION: PRODUCTION OF INFLAMMATORY MEDIATORS

In addition to engulfing and destroying foreign invaders, activated neutrophils enhance the inflammatory response by producing mediators that promote additional inflammation and facilitate host defense.

Arachidonic Acid Metabolites

Upon stimulation, neutrophils release AA from membrane phospholipids. While AA can act as a neutrophil chemoattractant [45,62], its main role is as a precursor for the generation of other, more potent lipid compounds, most prominently the prostaglandins and leukotrienes.

Prostaglandins are generated by the action of cyclooxygenases (COXs, endoperoxide synthases) on AA. COXs are heterobifunctional enzymes that first reduce AA to PGG_2, then convert PGG_2 into PGH_2. PGH_2 serves as a substrate for generating multiple prostanoids. Among the latter, the most directly relevant to inflammation is PGE_2. Proinflammatory effects of PGE_2 include vasodilation, increased vascular permeability, and pain. COX activity in resting neutrophils is negligible, but activated neutrophils upregulate COX-2 to produce PGE_2.

Neutrophils generate LTs through the action of 5-lipoxygenase (5-LO), like COX a heterobifunctional enzyme. 5-LO first converts AA to 5-hydroxyeicosatetraenoic acid (5-HETE), then converts 5-HETE into LTA_4. Both 5-HETE [45] and LTA_4 possess intrinsic biological activity, but mainly serve as substrates for additional reactions. Neutrophils convert LTA_4 into LTB_4 via the action of LTA_4 hydrolase; LTB_4 is itself a potent chemoattractant and activator of neutrophils.

Interestingly, some neutrophil lipid products have anti-inflammatory effects. Lipoxins are generated by the sequential actions of neutrophil 5-lipoxygenase and a related enzyme (either 12- or 15-lipoxygenase) in another cell type, either platelets or endothelial cells [63,64]. The requirement for two cells, passing substrates between them, suggests that lipoxins may not be produced until an inflammatory response is well underway, when they may serve to promote the resolution of inflammation [65]. Some prostaglandins – notably 15d-PGJ_2 and possibly $PGF_2\alpha$ – also characteristically have anti-inflammatory effects [66]. Even PGE_2 may, in some cases, have anti-inflammatory effects, underlining the importance of context on the determination of inflammatory outcomes.

Cytokine Production

Although the neutrophil protein synthetic apparatus is partly disabled, neutrophils remain capable of limited protein synthesis, including the production of cytokines. Cytokines released by neutrophils include MIP-1-α and β (CCL3 and CCL4), transforming growth factor-β (TGF-β), as well as IL-8 (CXC8), growth-related oncogene-α (Gro-α/CXCL1), and the granulocyte chemotactic protein-2 (CXCL6) [67]. IL-1α and β, TNFα, and IL-6 may also be produced by neutrophils, although this remains a matter of discrepancy [68]. The production of each chemokine requires a relatively selective combination of stimulants [69]. For example, in the presence of LPS and TNFα, only IL-8, Gro-α, and MIP-1-α are produced, while LPS and IFNγ additionally induce the synthesis of CXCL 9 and 10. Neutrophils under conditions of inflammation may upregulate the synthesis not only of the cytokines, but also of their cognate receptors, leading to autocrine-stimulation loops that regulate cell trafficking and immune responses [70].

B-lymphocyte stimulator (BLyS), a member of the TNF superfamily of ligands, and TNF-related apoptosis-inducing ligand (TRAIL) have been reported to be secreted by G-CSF and IL-8–timulated neutrophils, respectively [71,72]. Whereas BlyS supports the survival and maturation of B cells, TRAIL acts to stimulate the antitumor and antiviral actions of T cells. Through these and other products, neutrophil activation may not only provide direct attack on pathogens, but also support the transition from innate to adaptive immunity.

Just as neutrophils have the capacity to make anti-inflammatory lipids, they may also produce anti-inflammatory proteins. TGF-β has a dual role. On the one hand, it is a potent chemoattractant for human neutrophils [73], and enhances neutrophil cytokine production, including that of TGF-β itself [74]. On the other hand, TGF-β also has potent antiinflammatory effects [75] that may contribute to the resolution of the inflammatory state. Activated neutrophils also produce IL-1 receptor antagonist (IL-1Ra), an endogenous inhibitor of IL-1β signaling [76]. The value of recombinant IL-1Ra (anakinra) in the treatment of autoinflammatory conditions such as Still's disease [77] highlights the biological importance of endogenous IL-1Ra.

REFERENCES

1. Bainton, D.F., Ullyot, J.L., and Farquhar, M.G. 1971. The development of neutrophilic polymorphonuclear leukocytes in human bone marrow. *J Exp Med* **134**(4):907–934.

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2. Rosmarin, A.G., Yang, Z., and Resendes, K.K. 2005. Transcriptional regulation in myelopoiesis: hematopoietic fate choice, myeloid differentiation, and leukemogenesis. *Exp Hematol* **33**(2):131–143.
3. Gallin, J.I., and Snyderman, R. 1999. *Inflammation: Basic Principles and Clinical Correlates*. Philadelphia, PA: Lippincott Williams & Wilkins, 3rd ed., p. xxiii, 1335.
4. Richards, M.K., Liu, F., Iwasaki, H., Akashi, K., and Link, D.C. 2003. Pivotal role of granulocyte colony-stimulating factor in the development of progenitors in the common myeloid pathway. *Blood* **102**(10):3562–3568.
5. Lord, B.I., Bronchud, M.H., Owens, S., et al. 1989. The kinetics of human granulopoiesis following treatment with granulocyte colony-stimulating factor in vivo. *Proc Natl Acad Sci USA* **86**(23):9499–9503.
6. Fukunaga, R., Seto, Y., Mizushima, S., and Nagata, S. 1990. Three different mRNAs encoding human granulocyte colony-stimulating factor receptor. *Proc Natl Acad Sci USA* **87**(22):8702–8706.
7. Pojda, Z., and Tsuboi, A. 1990. In vivo effects of human recombinant interleukin 6 on hemopoietic stem and progenitor cells and circulating blood cells in normal mice. *Exp Hematol* **18**(9):1034–1037.
8. Metcalf, D., Begley, C.G., Johnson, G.R., Nicola, N.A., Lopez, A.F., and Williamson, D.J. 1987. Quantitative responsiveness of murine hemopoietic populations in vitro and in vivo to recombinant multi-CSF (IL-3). *Exp Hematol* **15**(3):288–295.
9. Metcalf, D., Begley, C.G., Johnson, G.R., Nicola, N.A., Lopez, A.F., and Williamson, D.J. 1986. Effects of purified bacterially synthesized murine multi-CSF (IL-3) on hematopoiesis in normal adult mice. *Blood* **68**(1):46–57.
10. Nagase, H., Miyamasu, M., Yamaguchi, M., et al. 2002. Cytokine-mediated regulation of CXCR4 expression in human neutrophils. *J Leukoc Biol* **71**(4):711–717.
11. Martin, C., Burdon, P.C.E., Bridger, G., Gutierrez-Ramos, J-C., Williams, T.J., and Rankin, S.M. 2003. Chemokines acting via CXCR2 and CXCR4 control the release of neutrophils from the bone marrow and their return following senescence. *Immunity* **19**(4):583–593.
12. Murray, J., Barbara, J.A., Dunkley, S.A., et al. 1997. Regulation of neutrophil apoptosis by tumor necrosis factor-alpha: requirement for TNFR55 and TNFR75 for induction of apoptosis in vitro. *Blood* **90**(7):2772–2783.
13. Tortorella, C., Piazzolla, G., Spaccavento, F., Pece, S., Jirillo, E., and Antonaci, S. 1998. Spontaneous and Fas-induced apoptotic cell death in aged neutrophils. *J Clin Immunol* **18**(5):321–329.
14. Stark, M.A., Huo, Y., Burcin, T.L., Morris, M.A., Olson, T.S., and Ley, K. 2005. Phagocytosis of apoptotic neutrophils regulates granulopoiesis via IL-23 and IL-17. *Immunity* **22**(3):285–294.
15. Borregaard, N., Lollike, K., Kjeldsen, L., et al. 1993. Human neutrophil granules and secretory vesicles. *Eur J Haematol* **51**(4):187–198.
16. Borregaard, N., Sehested, M., Nielsen, B.S., Sengeløv, H., and Kjeldsen, L. 1995. Biosynthesis of granule proteins in normal human bone marrow cells. Gelatinase is a marker of terminal neutrophil differentiation. *Blood* **85**(3):812–817.
17. Cowland, J.B., and Borregaard, N. 1999. The individual regulation of granule protein mRNA levels during neutrophil maturation explains the heterogeneity of neutrophil granules. *J Leukoc Biol* **66**(6):989–995.
18. Cham, B.P., Gerrard, J.M., and Bainton, D.F. 1994. Granulophysin is located in the membrane of azurophilic granules in human neutrophils and mobilizes to the plasma membrane following cell stimulation. *Am J Pathol* **144**(6):1369–1380.
19. Dahms, N.M., Lobel, P., and Kornfeld, S. 1989. Mannose 6-phosphate receptors and lysosomal enzyme targeting. *J Biol Chem* **264**(21):12115–12118.
20. Nauseef, W.M., McCormick, S., and Yi, H. 1992. Roles of heme insertion and the mannose-6-phosphate receptor in processing of the human myeloid lysosomal enzyme, myeloperoxidase. *Blood* **80**(10):2622–2633.
21. Cieutat, A.M., Lobel, P., August, J.T., et al. 1998. Azurophilic granules of human neutrophilic leukocytes are deficient in lysosome-associated membrane proteins but retain the mannose 6-phosphate recognition marker. *Blood* **91**(3):1044–1058.
22. Owen, C.A., and Campbell, E.J. 1999. The cell biology of leukocyte-mediated proteolysis. *J Leukoc Biol* **65**(2):137–150.
23. Kobayashi, Y. 2006. Neutrophil infiltration and chemokines. *Crit Rev Immunol* **26**(4):307–316.
24. Cohen, A.M., Zsebo, K.M., Inoue, H., et al. 1987. In vivo stimulation of granulopoiesis by recombinant human granulocyte colony-stimulating factor. *Proc Natl Acad Sci USA* **84**(8):2484–2488.
25. Hechtman, D.H., Cybulsky, M.I., Fuchs, H.J., Baker, J.B., and Gimbrone, M.A., Jr. 1991. Intravascular IL-8 inhibitor of polymorphonuclear leukocyte accumulation at sites of acute inflammation. *J Immunol* **147**(3):883–892.
26. Theilgaard-Monch, K., Porse, B.T., and Borregaard, N. 2006. Systems biology of neutrophil differentiation and immune response. *Curr Opin Immunol* **18**(1):54–60.
27. Bokoch, G.M. 1995. Chemoattractant signaling and leukocyte activation. *Blood* **86**(5):1649–1660.
28. Hall, A. 2005. Rho GTPases and the control of cell behaviour. *Biochem Soc Trans* **33**(Pt 5):891–895.
29. Kawasaki, M., Nakayama, K., and Wakatsuki, S. 2005. Membrane recruitment of effector proteins by Arf and Rab GTPases. *Curr Opin Struct Biol* **15**(6):681–689.
30. Morgan, C.P., Sengelov, H., Whatmore, J., Borregaard, N., and Cockcroft, S. 1997. ADP-ribosylation-factor-regulated phospholipase D activity localizes to secretory vesicles and mobilizes to the plasma membrane following N-formylmethionyl-leucyl-phenylalanine stimulation of human neutrophils. *Biochem J* **325**(Pt 3):581–585.
31. Murakami, M., and Kudo, I. 2002. Phospholipase A2. *J Biochem* **131**(3):285–292.
32. Pillinger, M.H., Feoktistov, A.S., Capodici, C., et al. 1996. Mitogen-activated protein kinase in neutrophils and enucleate neutrophil cytoplasts: evidence for regulation of cell-cell adhesion. *J Biol Chem* **271**(20):12049–12056.
33. Smolen, J.E., Stoehr, S.J., and Kuczynski, B. 1991. Cyclic AMP inhibits secretion from electroporated human neutrophils. *J Leukoc Biol* **49**(2):172–179.
34. Amin, A.R., Attur, M., Vyas, P., et al. 1995. Expression of nitric oxide synthase in human peripheral blood mononuclear cells and neutrophils. *J Inflamm* **47**(4): 190–205.
35. Scher, J.U., Pillinger, M.H., and Abramson, S.B. 2007. Nitric oxide synthases and osteoarthritis. *Curr Rheumatol Rep* **9**(1):9–15.
36. Clancy, R.M., Amin, A.R., and Abramson, S.B. 1998. The role of nitric oxide in inflammation and immunity. *Arthritis Rheum* **41**(7):1141–1151.
37. Nauseef, W.M., Volpp, B.D., McCormick, S., Leidal, K.G., and Clark, R.A. 1991. Assembly of the neutrophil respiratory burst oxidase. Protein kinase C promotes cytoskeletal and membrane association of cytosolic oxidase components. *J Biol Chem* **266**(9):5911–5917.

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Fundamentals of Inflammation, edited by Charles N. Serhan, et al., Cambridge University Press, 2010. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/yale-ebooks/detail.action?docID=802921.
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38. Kambayashi, Y., Takekoshi, S., Tanino, Y., et al. 2007. Various molecular species of diacylglycerol hydroperoxide activate human neutrophils via PKC activation. *J Clin Biochem Nutr* **41**(1):68–75.
39. Yamamoto, Y., Kambayashi, Y., Ito, T., Watanabe, K., and Nakano, M. 1997. 1,2-Diacylglycerol hydroperoxides induce the generation and release of superoxide anion from human polymorphonuclear leukocytes. *FEBS Lett* **412**(3):461–464.
40. Suzuki, K., Hino, M., Hato, F., Tatsumi, N., and Kitagawa, S. 1999. Cytokine-specific activation of distinct mitogen-activated protein kinase subtype cascades in human neutrophils stimulated by granulocyte colony-stimulating factor, granulocyte-macrophage colony-stimulating factor, and tumor necrosis factor-alpha. *Blood* **93**(1):341–349.
41. Nick, J.A., Avdi, N.J., Young, S.K., et al. 1999. Selective activation and functional significance of p38alpha mitogen-activated protein kinase in lipopolysaccharide-stimulated neutrophils. *J Clin Invest* **103**(6):851–858.
42. Avdi, N.J., Nick, J.A., Whitlock, B.B., et al. 2001. Tumor necrosis factor-alpha activation of the c-Jun N-terminal kinase pathway in human neutrophils. Integrin involvement in a pathway leading from cytoplasmic tyrosine kinases apoptosis. *J Biol Chem* **276**(3):2189–2199.
43. Chang, L.C., and Wang, J.P. 1999. Examination of the signal transduction pathways leading to activation of extracellular signal-regulated kinase by formyl-methionyl-leucyl-phenylalanine in rat neutrophils. *FEBS Lett* **454**(1–2):165–168.
44. Chen, L.W., Lin, M.W., and Hsu, C.M. 2005. Different pathways leading to activation of extracellular signal-regulated kinase and p38 MAP kinase by formyl-methionyl-leucyl-phenylalanine or platelet activating factor in human neutrophils. *J Biomed Sci* **12**(2):311–319.
45. Capodici, C., Pillinger, M.H., Han, G., Philips, M.R., and Weissmann, G. 1998. Integrin-dependent homotypic adhesion of neutrophils. Arachidonic acid activates Raf-1/Mek/Erk via a 5-lipoxygenase-dependent pathway. *J Clin Invest* **102**(1):165–175.
46. Thelen, M., Uguccioni, M., and Bosiger, J. 1995. PI 3-kinase-dependent and independent chemotaxis of human neutrophil leukocytes. *Biochem Biophys Res Commun* **217**(3):1255–1262.
47. Capodici, C., Hanft, S., Feoktistov, M., and Pillinger, M.H. 1998. Phosphatidylinositol 3-kinase mediates chemoattractant-stimulated, CD11b/CD18-dependent cell-cell adhesion of human neutrophils: evidence for an ERK-independent pathway. *J Immunol* **160**(4):1901–1909.
48. Kilpatrick, L.E., Lee, J.Y., Haines, K.M., Campbell, D.E., Sullivan, K.E., and Korchak, H.M. 2002. A role for PKC-delta and PI 3-kinase in TNF-alpha-mediated antiapoptotic signaling in the human neutrophil. *Am J Physiol Cell Physiol* **283**(1):C48–C57.
49. Middleton, J., Neil, S., Wintle, J., et al. 1997. Transcytosis and surface presentation of IL-8 by venular endothelial cells. *Cell* **91**(3):385–395.
50. Diacovo, T.G., Roth, S.J., Buccola, J.M., Bainton, D.F., and Springer, T.A. 1996. Neutrophil rolling, arrest, and transmigration across activated, surface-adherent platelets via sequential action of P-selectin and the beta 2-integrin CD11b/CD18. *Blood* **88**(1):146–157.
51. Muller, W.A., Weigl, S.A., Deng, X., and Phillips, D.M. 1993. PECAM-1 is required for transendothelial migration of leukocytes. *J Exp Med* **178**(2):449–460.
52. Baggiolini, M., and Clark-Lewis, I. 1992. Interleukin-8, a chemotactic and inflammatory cytokine. *FEBS Lett* **307**(1):97–101.
53. Underhill, D.M., and Ozinsky, A. 2002. Toll-like receptors: key mediators of microbe detection. *Curr Opin Immunol* **14**(1):103–110.
54. Ozinsky, A., Underhill, D.M., Fontenot, J.D., et al. 2000. The repertoire for pattern recognition of pathogens by the innate immune system is defined by cooperation between toll-like receptors. *Proc Natl Acad Sci USA* **97**(25):13766–13771.
55. Hayashi, F., Means, T.K., and Luster, A.D. 2003. Toll-like receptors stimulate human neutrophil function. *Blood* **102**(7):2660–2669.
56. Indik, Z.K., Park, J.G., Hunter, S., and Schreiber, A.D. 1995. The molecular dissection of Fc gamma receptor mediated phagocytosis. *Blood* **86**(12):4389–4399.
57. Ravetch, J.V., and Kinet, J.P. 1991. Fc receptors. *Annu Rev Immunol* **9**:457–492.
58. Karakawa, W.W., Sutton, A., Schneerson, R., Karpas, A., and Vann, W.F. 1988. Capsular antibodies induce type-specific phagocytosis of capsulated *Staphylococcus aureus* by human polymorphonuclear leukocytes. *Infect Immun* **56**(5):1090–1095.
59. Elsbach, P., and Weiss, J. 1998. Role of the bactericidal/permeability-increasing protein in host defence. *Curr Opin Immunol* **10**(1):45–49.
60. Brinkmann, V., Reichard, U., Goosmann, C., et al. 2004. Neutrophil extracellular traps kill bacteria. *Science* **303**(5663):1532–1535.
61. Brinkmann, V., and Zychlinsky, A. 2007. Beneficial suicide: why neutrophils die to make NETs. *Nat Rev Microbiol* **5**(8):577–582.
62. Abramson, S.B., Leszczynska-Piziak, J., and Weissmann, G. 1991. Arachidonic acid as a second messenger. Interactions with a GTP-binding protein of human neutrophils. *J Immunol* **147**(1):231–236.
63. Samuelsson, B., Dahlen, S.E., Lindgren, J.A., Rouzer, C.A., and Serhan, C.N. 1987. Leukotrienes and lipoxins: structures, biosynthesis, and biological effects. *Science* **237**(4819):1171–1176.
64. Chiang, N., Arita, M., and Serhan, C.N. 2005. Anti-inflammatory circuitry: lipoxin, aspirin-triggered lipoxins and their receptor ALX. *Prost Leukot Essent Fatty Acids* **73**(3–4):163–177.
65. Serhan, C.N., Hong, S., Gronert, K., et al. 2002. Resolvins: a family of bioactive products of omega-3 fatty acid transformation circuits initiated by aspirin treatment that counter proinflammation signals. *J Exp Med* **196**(8):1025–1037.
66. Scher, J.U., and Pillinger, M.H. 2005. 15d-PGJ2: the anti-inflammatory prostaglandin? *Clin Immunol* **114**(2):100–109.
67. Kasama, T., Miwa, Y., Isozaki, T., Odai, T., Adachi, M., and Kunkel, S.L. 2005. Neutrophil-derived cytokines: potential therapeutic targets in inflammation. *Curr Drug Targets Inflamm Allergy* **4**(3):273–279.
68. Schroder, A.K., von der Ohe, M., Kolling, U., et al. 2006. Polymorphonuclear leucocytes selectively produce anti-inflammatory interleukin-1 receptor antagonist and chemokines, but fail to produce pro-inflammatory mediators. *Immunology* **119**(3):317–327.
69. Scapini, P., Lapinet-Vera, J.A., Gasperini, S., Calzetti, F., Bazzoni, F., and Cassatella, M.A. 2000. The neutrophil as a cellular source of chemokines. *Immunol Rev* **177**:195–203.
70. Theilgaard-Monch, K., Jacobsen, L.C., Borup, R., et al. 2005. The transcriptional program of terminal granulocytic differentiation. *Blood* **105**(4):1785–1796.
71. Scapini, P., Carletto, A., Nardelli, B., et al. 2005. Proinflammatory mediators elicit secretion of the

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Fundamentals of Inflammation, edited by Charles N. Serhan, et al., Cambridge University Press, 2010. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/yale-ebooks/detail.action?docID=802921.
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intracellular B-lymphocyte stimulator pool (BLyS) that is stored in activated neutrophils: implications for inflammatory diseases. *Blood* **105**(2):830–837.
72. Cassatella, M.A., Huber, V., Calzetti, F., et al. 2006. Interferon-activated neutrophils store a TNF-related apoptosis-inducing ligand (TRAIL/Apo-2 ligand) intracellular pool that is readily mobilizable following exposure to proinflammatory mediators. *J Leukoc Biol* **79**(1):123–132.
73. Reibman, J., Meixler, S., Lee, T.C., et al. 1991. Transforming growth factor beta 1, a potent chemoattractant for human neutrophils, bypasses classic signal-transduction pathways. *Proc Natl Acad Sci USA* **88**(15):6805–6809.
74. Fava, R.A., Olsen, N.J., Postlethwaite, A.E., et al. 1991. Transforming growth factor beta 1 (TGF-beta 1) induced neutrophil recruitment to synovial tissues: implications for TGF-beta-driven synovial inflammation and hyperplasia. *J Exp Med* **173**(5):1121–1132.
75. Huynh, M.L., Fadok, V.A., and Henson, P.M. 2002. Phosphatidylserine-dependent ingestion of apoptotic cells promotes TGF-beta1 secretion and the resolution of inflammation. *J Clin Invest* **109**(1):41–50.
76. McColl, S.R., Paquin, R., Menard, C., and Beaulieu, A.D. 1992. Human neutrophils produce high levels of the interleukin 1 receptor antagonist in response to granulocyte/macrophage colony-stimulating factor and tumor necrosis factor alpha. *J Exp Med* **176**(2):593–598.
77. Fitzgerald, A.A., LeClercq, S.A., and Yan, A. 2005. Rapid responses to anakinra in patients with refractory adult-onset Still's disease. *Arthritis Rheum* **52**(6):1794–1803.

SUGGESTED READINGS

Burg, N.D., and Pillinger, M.H. 2001. The neutrophil: function and regulation in innate and humoral immunity. *Clin Immunol* **99**(1):7–17.

Kobayashi, Y. 2006. Neutrophil infiltration and chemokines. *Crit Rev Immunol* **26**(4):307–316.

Theilgaard-Monch, K., Porse, B.T., and Borregaard, N. 2006. Systems biology of neutrophil differentiation and immune response. *Curr Opin Immunol* **18**(1):54–60.

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4B Neutrophils II

Marco A. Cassatella

INTRODUCTION

Polymorphonuclear neutrophil leukocytes (PMN) are "professional" phagocytic cells of the innate immune system that act as the first line of defense against invading pathogens, principally bacteria and fungi but also viruses. Because of their powerful microbicidal equipment, they have a major role in inflammatory responses other than anti-infectious defenses. In fact, after agonist challenge, neutrophils have the capacity to generate reactive oxygen species (ROS) and release lytic enzymes with potent antimicrobial activity, which are all essential for pathogen killing. Conversely, PMN-derived ROS and proteases may also damage the surrounding tissues if released in an uncontrolled manner, as observed in inflammatory diseases dominated by neutrophils. Nevertheless, it is now clear that the role of neutrophils goes far beyond phagocytosis and pathogen killing, as uncovered in the past two decades (Figure 4B.1). For instance, it has been documented that neutrophils have the capacity to migrate toward the lymph nodes and to express major histocompatibility complex class II (MHC II) molecules, once appropriately activated. An additional and fascinating aspect that has gradually come to light is the ability of neutrophils to newly express a number of genes, whose products lie at the core of inflammatory and immune responses. Not only neutrophils synthesize numerous proteins involved in their effector functions, including some complement components and Fc receptors, but they also produce a variety of cytokines and chemokines. Since the latter molecules exert a broad spectrum of biological activities, it is reasonable to assume that neutrophils, by producing and releasing cytokines, may significantly contribute to the regulation of many different processes and, in turn, act as key regulators of cross talks among immune, endothelial, stromal, and parenchymal cells, as well as important players in autoimmune diseases and antitumoral responses. Finally, the application to neutrophils of the new paradigm of apoptosis, defined as the process of regulated cell death that prevents release of their cytotoxic and proteolytic contents, is now used to explain how some processes lead to resolution of inflammation or infection without tissue damage, simply by allowing neutrophil apoptosis to occur.

NEUTROPHIL DIFFERENTIATION, MATURATION, AND OTHER GENERALITIES

Polymorphonuclear neutrophils are the most abundant circulating leukocyte type in humans, normally present at 2.5–7.5 $\times$ 10^9 cells/L in the blood. PMN are members of the granulocyte family of leukocytes, which also comprises eosinophils and basophils and that were thus named because they are not well stained either by eosin, a red acidic stain, or by methylene blue, a basic or alkaline stain. All three types of granulocytes contain heterogeneous cytoplasmic granules (500–1,500/neutrophil), which are storage pools for mostly cell-specific intracellular enzymes, cationic protein, receptors, and discrete proteins. A morphological peculiarity of the neutrophil concerns their nucleus, which is polymorphous and usually consists of three to five sausage-shaped masses of chromatin connected by fine threads.

Mature neutrophils are terminally differentiated, nondividing cells, with a rich supply of cytoplasmic glycogen. They develop and maturate in the bone marrow from pluripotent CD34+ stem cells (over a 7–14 day period, at a rate of approximately 5–10 $\times$ 10^{10} cells on a daily basis), under the regulatory effects of several colony-stimulating factors (CSFs), including interleukin-3 (IL-3), granulocyte-macrophage CSF (GM-CSF) and granulocyte CSF (G-CSF). Transcriptional profiling studies have suggested that, whereas macrophages represent the default myeloid cells that mature,

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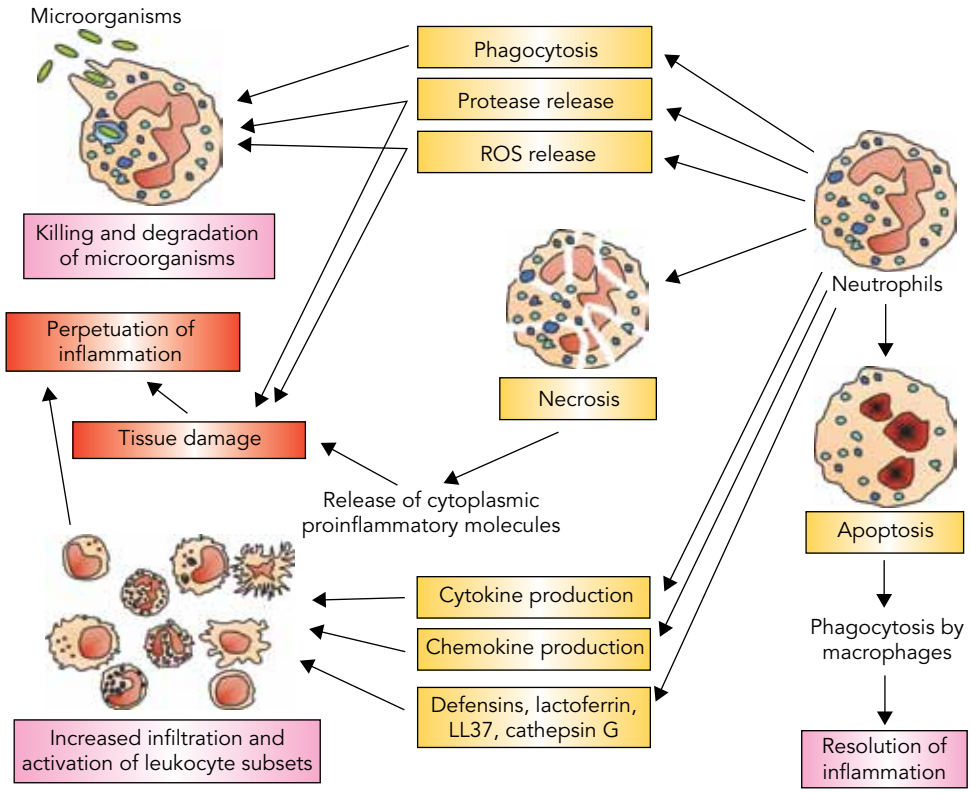


Figure 4B.1. Potential effector functions of neutrophils at sites of infection and inflammation.

granulocytes arise through the selective expression of a subset of transcription factors (Egr1, HoxB7, STAT3) during differentiation. Colony-stimulating factors exert their growth and differentiation effects on progressively more committed stages of cell maturation, starting from the myeloblast, the promyelocyte, the myelocyte, the metamyelocyte, the nonsegmented granulocyte (band neutrophils), and, finally, the mature neutrophil with segmented nuclei. Targeted genetic disruption experiments in mice have clearly shown that G-CSF and its receptor are essential for the maintenance of normal levels of blood neutrophils. Importantly, both GM-CSF and G-CSF can also positively influence the survival and functions of mature neutrophils.

Within the circulation, PMNs are 10–20 nm in diameter and exist in two pools in a dynamic equilibrium: a circulating pool and a "marginated" pool, the latter believed to be sequestered within the microvasculature of many organs. Cell-labeling experiments have shown that the lifespan of neutrophils in the circulation is short, with a half-life of approximately 7–12 hours. In the resting uninfected host, the production and elimination of neutrophils are balanced, resulting in fairly constant concentration in peripheral blood. Neutrophil turnover must therefore be under strict control, as demonstrated by the spontaneous apoptosis that neutrophils undergo before their removal by macrophage in the lung, spleen, and liver. During pathological conditions (for instance, during a bacterial infection), the number of circulating neutrophils dramatically increases (even up to 10-fold, a condition called neutrophilia), as a result of either an accelerated release of neutrophils from the bone marrow combined with a stimulated maturation of immature neutrophils by CSFs, or a demargination from the lungs or the spleen. Under these conditions, the generation of specific chemotactic agents triggers the migration of neutrophils to the site of infection where their phagocytic activities and defensive functions are exerted. In the absence of an inflammatory stimulus, one mechanism shown to control the release and reuptake of neutrophils in the mouse bone marrow involves the interaction of stromal cell–derived factor-1α/CXCL12 and its cognate receptor on the neutrophil, CXCR4. Expression of CXCR4 increases, in fact, as neutrophils age, and consequently is thought to result in cells homing to murine bone marrow. The importance of CXCR4 is illustrated by the findings in patients with the myelokathexis syndrome, also referred to as WHIM (warts, hypogammaglobulinemia, infections, and myelokathexis) syndrome, which is now specifically attributable to a defect in CXCR4 expression. Patients who have this disorder have severe leukopenia

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and neutropenia, with accumulation of neutrophils in the bone marrow and alterations in the trafficking of lymphocytes and hematopoietic progenitor cells as well.

NEUTROPHIL MICROBICIDAL MECHANISMS

Neutrophils have been described as mobile arsenals that seek and destroy a variety of targets, mainly following phagocytosis. Two primary processes are utilized by neutrophils to eliminate invading pathogens. One of them involves the generation of ROS, which include O_2^- (superoxide anion), H_2O_2 (hydrogen peroxide), singlet oxygen, and other products derived from the metabolism of H_2O_2. The other process involves the release of their intracellular granules, which contain lytic enzymes and antimicrobial polypeptides. The former mechanism is referred to as the "respiratory burst," which is defined as an increase in the oxidative metabolism following the uptake of particles, or in response to soluble inflammatory stimuli, leading to the generation of O_2^- through the activation of an enzymatic system that is unique to phagocytic cells (neutrophils, eosinophils, monocytes, and macrophages), the NADPH oxidase. This latter enzyme consists of a multiprotein complex which is dissociated and thus dormant in unstimulated PMN, formed by a flavocytochrome-b_{558}, which is an heterodimer of $gp91^{phox}$ and $p22^{phox}$ (*phox* standing for phagocyte oxidase) chains, four cytoplasmic components, namely $p40^{phox}$, $p47^{phox}$, $p67^{phox}$, and the small GTP-binding regulatory protein, Rac1/2. Upon cell stimulation, the cytosolic components become phosphorylated and assemble together with the cytochrome and Rac1/2 on the plasma membrane, thus forming the active enzyme. NADPH oxidase then produces the superoxide free radical by catalyzing the transfer of electrons from NADPH to molecular oxygen. Once discharged into the phagosome, O_2^- is converted to hydrogen peroxide (by superoxide dismutase), which, in the presence of neutrophil myeloperoxidase and the abundant Cl^- ions taken up from extracellular fluids, is metabolized into hypochlorous acid (HOCl). The latter, in addition to numerous other microbicidal oxidants, synergizes with granule proteins to kill microbes in the neutrophil phagosome. While O_2^- and H_2O_2 are moderately bactericidal, the production of HOCl is one of the neutrophil's major weapons against microbes. Furthermore, O_2^- can act as a substrate for the reactions with other cellular radicals, for example, O_2^- reacts with nitric oxide to form the very reactive peroxynitrite ($OONO^-$) molecule, a potent cytotoxic oxidant. In the phagosome, the high concentrations of ROS are very toxic, and, ultimately, contribute to the killing of the phagocytosed microbe. The same oxidants can however damage "innocent bystanders" and provoke the tissue destruction. Interestingly, neutrophils contain large reserves of endogenous antioxidants, including glutathione and ascorbate, which serve to protect them from the oxidative stress.

The critical role of NADPH oxidase and its products in host defense is best illustrated by chronic granulomatous disease (CGD), which is a rare condition with an incidence of about 1/250,000 individuals. CGD is the most common clinically significant inherited disorder of neutrophil function, in which mutations in any of the NADPH oxidase complex subunits (except $p40^{phox}$) can lead to a severe immunodeficiency characterized by defective killing of phagocytosed pathogens by phagocytic leukocytes because of their inability to generate ROS. This condition leads to recurrent life-threatening infections that are difficult to treat and which can lead to early death. These infections typically involve microorganisms for which oxidant-mediated killing is particularly critical for effective host defense, including *Staphylococcus aureus*, *Aspergillus* species, *Nocardia*, and a variety of Gram-negative enteric bacilli. Additional and distinctive hallmarks of CGD patients are aberrant inflammatory responses and the formation of inflammatory tissue granulomas, which can obstruct the gastric outlet or urinary tract or cause granulomatous colitis and the symptoms of inflammatory bowel disease. Such chronic inflammatory complications are thought to reflect an important role for superoxide in downregulating the inflammatory response, even though a definitive confirm is required.

The second mechanism used by neutrophils for the elimination of pathogens relies on the release and activity into the phagosome of potent degradative enzymes contained in their granules (Figure 4B.2). Granule proteins are cytotoxic and immune regulatory molecules synthesized at defined stages of neutrophil maturation in the bone marrow, with no further de novo expression occurring once the PMN has entered the bloodstream. Traditionally, neutrophil granules are subdivided into peroxidase-positive granules (based on the presence of myeloperoxidase), also called primary or azurophil granules, and peroxidase-negative granules that include the specific (secondary) granules, the gelatinase (tertiary) granules, and the secretory vesicles. Distinctions between granule classes can be made upon analysis of marker proteins, even if their strict compartmentalization is not a dogma (Figure 4B.2). For instance, proteinase 3, a serine protease described in azurophil granules is also localized in the membrane of secretory vesicles, the most mobilizable compartment of neutrophils, and in the plasma membrane as well.

The *azurophilic* or primary granules, so named because they are the first to appear during hemopoiesis in the promyelocyte stage, constitute the densest

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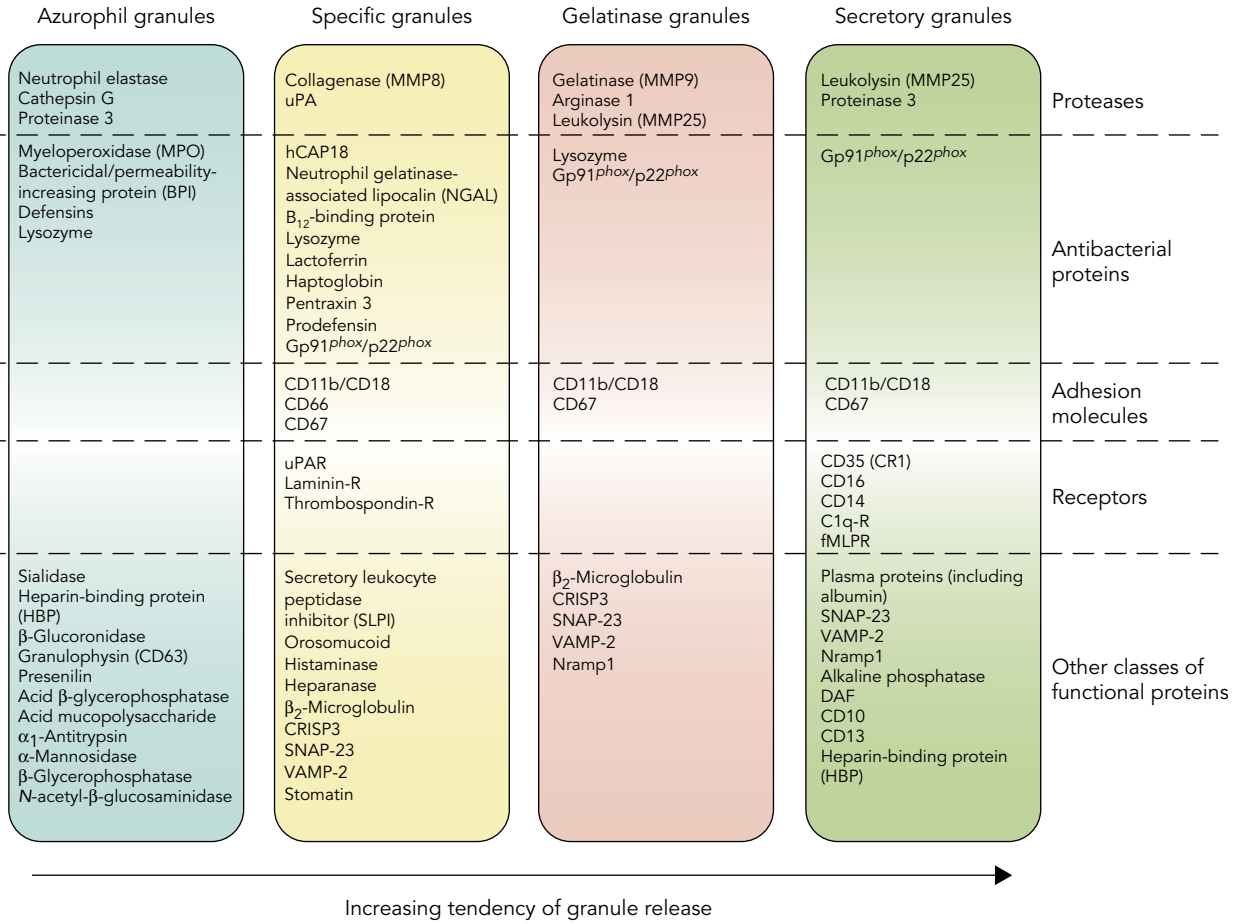


Figure 4B.2. Main constituents of neutrophil granules.

granule populations and undergo limited exocytosis in response to stimulation. Azurophil granules are packaged with acidic hydrolases and antimicrobial proteins and they are believed to contribute primarily to the killing and degradation of engulfed microorganisms that take place in the phagolysosome. They contain myeloperoxidase (MPO, which is responsible for the characteristic greenish color of pus and is often used as specific marker for *azurophilic* granules), hydrolases, lysozyme, matrix metalloproteinases, and three structurally related serprocidins (serine proteases with microbicidal activity): proteinase-3, cathepsin G, and elastase, which display proteolytic activity against a variety of extracellular matrix components, such as elastin, fibronectin, laminin, type IV collagen, and vitronectin. Azurophil granules also contain antimicrobial molecules such as bactericidal/permeability-increasing protein/BPI (which is important for killing Gram-negative bacteria) and α-defensins, a family of small cysteine-rich antibiotic peptides with broad antimicrobial activity against bacteria, fungi, and certain enveloped viruses.

The second major type of granule is termed specific or secondary. Specific granules are formed in the myelocytic stage, are less dense and smaller than the azurophil ones, and stain heavily for glycoproteins. They are known to contain relatively unique constituents, including collagenase, haptoglobin, and vitamin B_{12}-binding protein. Secondary granules mainly participate in the antimicrobial activities of the PMN by mobilization of their arsenal of antimicrobial substances, for example, lactoferrin, NGAL (neutrophil gelatinase-associated lipocalin), lysozyme, hCAP18 (cathelicidin human cationic antimicrobial protein of 18 kDa, the proform of LL-37), and pentraxin-3, either to the phagosome or the exterior of the cell, their mobilization being very important in modulating inflammation. In general, the mechanisms of the inhibitory effect of antimicrobial proteins include depriving ions essential for microbial survival, degrading structural components of microorganisms (e.g., peptidoglycan), and disrupting the integrity of target cell membrane by punching pores in the membrane or by perturbing membrane integrity.

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Tertiary granules (whose primary constituent is gelatinase, a metallo-enzyme) are produced at the metamyelocyte stage during differentiation and more easily exocytosed by mature neutrophils than secondary granules. These characteristics reflect the fact that tertiary granules are important primarily as a reservoir of matrix-degrading enzymes and membrane receptors needed during PMN extravasation and diapedesis. Thus, whereas the larger specific granules are rich in antibiotic substances, the smaller gelatinase granules are not. Finally, the secretory vesicles (the fourth type of granules) are created by endocytosis during terminal neutrophil maturation in the bone marrow and are the most readily mobilizable. The propensity for exocytosis is reflected in the density of vesicle-associated membrane protein (VAMP-2), a fusogenic protein associated with the granule membrane. Secretory vesicles contain plasma proteins like albumin and thus do not release toxic substances when they fuse with the plasma membrane and empty their content. Their importance lies in their membrane, which is the main store of receptors (including β2-integrins, the complement receptor 1 [CR1], receptors for formylated bacterial peptides (formylmethionyl-leucyl-phenylalanine [fMLP]-receptors), CD14, the FcγIII receptor CD16, and the metalloprotease leukolysin), which, when translocated to the cell surface, allow the neutrophil to interact with other ligands and to respond to soluble mediators. Thereby, the relatively inactive, on the endothelium rolling PMN, is transformed into a cell that is capable of interacting with the endothelium, monocytes, and dendritic cells (DC) and to better receive inflammatory signals from the environment. An additional protein described to be stored in secretory vesicles is heparin-binding protein (HBP, also known as CAP37 and azurocidin), whose release at the initial stage of PMN extravasation is thought to be of essential importance in the PMN-induced increase in vascular permeability. Finally, peroxidase-negative granules also contain three metalloproteases with great physiological and pathophysiological significance, namely collagenase, matrix metalloproteinase-8 (MMP-8), gelatinase (MMP-9), and leukolysin (MMP-25) (Figure 4B.1).

It is now established that PMN granules are not just emptied like bags, but are rather released upon specific signals that the PMN senses in the local environment. The mechanisms underlying the secretion of the four morphologically distinct populations of granules may be under separate control, as reflected by the order of exocytosis observed after ionophore-induced progressive elevation of cytosolic calcium: secretory vesicles, gelatinase granules, specific granules, and, lastly, azurophilic granules. It is thus plausible that during its journey from the bloodstream to the inflammatory site, the PMN releases its granules in a hierarchical order because not all stored proteins are needed at the same time. Primary neutrophil defects causing disorders of degranulation are very rare. Two such genetic disorders affecting neutrophil granules have been described, the Chediak–Higashi syndrome and the neutrophil-specific granule deficiency, both of which are associated with increased frequency of bacterial infections. Another common disorder of granules is the MPO deficiency, the most common inherited disorder of phagocytes, in which deficiency in MPO prevents the formation of hypochlorous acid from chloride and hydrogen peroxide. There is a remarkable lack of clinical symptoms in most individuals with MPO deficiency, despite in vitro defects in the neutrophil ability to kill *Candida albicans* and *Aspergillus fumigatus* hydra.

NEUTROPHIL RECEPTORS

Since their interactions with the external milieu are crucial for innate responses, neutrophils have developed a recognition apparatus that is able to specifically bind a wide range of extracellular ligands. Following ligation of one or more types of surface receptors, neutrophils generate a number of activation steps, via the generation of a cascade of intracellular "second messengers." These steps are biochemical events which mediate the transmission of biological information between membrane receptors and various intracellular components involved in specific effector functions. Consequently, neutrophil receptors are able to regulate a wide range of functions, including adhesion, migration, phagocytosis, survival, cell activation, gene expression induction, proinflammatory mediator release, and target cell killing. The transduction machinery is mainly, but not exclusively, located in the plasma membrane and is composed of a series of enzymes (i.e., kinases, phosphatases, adenylate cyclase, phospholipases, and other enzymes involved in lipid metabolism) or regulatory proteins (G-protein subunits, channel proteins, anchoring and adaptor proteins) which, in turn, generate, or whose activities are regulated by, several other second messengers (calcium ions, inositol phosphates, diacyglycerol, phosphatidate, cAMP, and so forth). A nonexhaustive list of neutrophil receptors includes (i) receptors for proinflammatory mediators (i.e., the anaphylotoxin complement component C5a, leukotriene B_4 [LTB_4], platelet-activating factor [PAF], substance P, and bacterial formylated peptides typified by fMLP); (ii) receptors for cytokines such as interferon-γ (IFNγ), IL-1, IL-4, IL-6, IL-10, IL-13, IL-15, IL-18, tumor necrosis factor-α (TNFα), G-CSF, GM-CSF, and many others; (iii) receptors for chemokines, including those for IL-8/CXCL8 and GROα/CXCL1, called CXCR1 and CXCR2; (iv) receptors/adhesion molecules for the endothelium (see later); (v) receptors for tissue matrix proteins and for lectins; and (vi) opsonin receptors,

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such as those for the Fc portion of γ-immunoglobulins (FcγRs) and those for the major cleavage fragments of the complement system (CRs). Neutrophils constitutively express the low-affinity FcγRs (FcγRIIA/CD32A and FcγRIIIA/CD16A), and, when exposed to IFNγ or G-CSF, the high-affinity FcγR (FcγRI/CD64) as well. CR expressed by neutrophils are CR1 (also known as CD35), which binds to complement components C1q, C4b, C3b, and Mannan-binding lectin (MBL); CR3 ($\alpha_M\beta_2$ integrin, CD11b/CD18, or MAC-1), which binds to iC3b, ICAM-1, and some microbes; CR4 ($\alpha_X\beta_2$ integrin, CD11c/CD18), which binds to iC3b. By expressing these latter receptors, neutrophils are able to recognize and bind, in a cooperative manner, IgG-opsonized particles and/or complement-opsonized microbes, and then activate their phagocytosis.

Neutrophils also express a variety of pattern recognition receptors (PRRs). The latter represent an emerging class of sensors that function by recognizing the so-called pathogen-associated molecular patterns (PAMPs). The latter ligands constitute a limited set of conserved molecular patterns that are unique to microbes and that induce signaling cascades that ultimately activate a plethora of effector mechanisms by immune and nonimmune cells, which all serve to clear the invading pathogens. PRRs are involved in the initiation of the inflammatory response upon infection, activating multiple pathways of host defense, cytokine production, cell survival, and ROS production, without, by themselves, triggering phagocytosis. Examples among PRRs include the Toll-like receptors (TLRs) and nucleotide-binding oligomerization domain (NOD) proteins, two families of innate immune recognition receptors which are required for the detection of a broad range of microbial products including, for instance, lipopolysaccharide (LPS, recognized by TLR4), peptidoglycan and bacterial lipopeptides (recognized by TLR2), foreign DNA (recognized by TLR9), viral RNA (recognized by TLR3, 7, and 8), and muramyldipeptide (recognized by NOD1). Neutrophils express all TLRs (with the exception of TLR3), whose activation has been shown to influence many functional responses. For instance, engagement of TLR can (a) modulate neutrophil expression of adhesion molecules; (b) regulate neutrophil recruitment directly through effects on chemokine receptor expression and indirectly through effects on CXCL8 generation by neutrophils themselves; (c) "prime" (see later) neutrophils for enhanced ROS production; (d) prolong neutrophil survival both directly and indirectly (via monocyte activation). Crucial in the regulation of many neutrophil effector functions is TLR4. Being the specific receptor for LPS, great efforts have been made to characterize the molecular mechanisms whereby TLR4 regulates the expression of target genes in myeloid cells. From studies mainly performed in macrophages or DCs of gene-targeted mice, it has emerged that LPS triggers two classes of genes via TLR4, whose regulatory signals are transduced by discrete intracellular adaptor molecules with TIR (Toll-IL-1 receptor) domains that bind to the same TLR4. One class is defined as "MyD88 (myeloid differentiation factor-88)-dependent," because it is not induced in MyD88$^{-/-}$ mice, and that mostly includes proinflammatory mediators such as TNFα, IL-1, IL-12p40, and CXCL8. In this pathway, MyD88 and TIRAP/MAL (TIR domain containing adapter protein/myeloid differentiation factor-88 adapter-like protein) mediate a rapid and early activation of the transcription factor NF-κB, whose nuclear translocation is essential for transcriptional induction of the above-mentioned proinflammatory genes. The other class is defined as "MyD88-independent" because it relies on a more delayed activation of both NF-κB and IRF (IFN-regulatory factor)-3 transcription factors in MyD88$^{-/-}$ mice, which ultimately lead to the expression of IFNβ and subsequently to an IFNβ-dependent STAT1 activation. Genes that belong to this last class are thus regulated by endogenous IFNβ and activated in a more delayed manner compared to those of the "MyD88-dependent" pathway. They include, for instance, a number of antiviral genes, Th1-activating chemokines such as MIG (monokine induced by IFNγ)/CXCL9, IP-10 (IFN-inducible protein-10)/CXCL10 and I-TAC (IFN-inducible T-cell alpha chemoattractant)/CXCL11, anticancer molecules such as TRAIL (TNF-related apoptosis-inducing ligand), and iNOS (inducible NO synthase). Along the "MyD88-independent" pathway, both the TRIF (TIR domain-containing adapter inducing IFNβ) and TRAM (TIR domain-containing adapter-inducing IFNβ-related adapter molecule) adaptor proteins transduce the activation of redundant protein kinases, TBK1 (TRAF family-associated NF-κB binding kinase [TANK]-binding kinase-1) and IKK (IκB kinase, IKK)-ε, which phosphorylate IRF3 on Ser/Thr residues. As a result, IRF3 dimerizes, translocates to the nucleus, associates with other coactivators (for instance, CBP/p300), and ultimately contributes to activate IFNβ gene transcription.

Recent studies, aimed at clarifying the molecular mechanisms whereby the LPS–TLR4 complex transduces its signals in human neutrophils, have uncovered that LPS does not mobilize the so-called MyD88-independent pathway for its inability to activate TBK1 and, in turn, IRF3 phosphorylation (Figure 4B.3). Consequently, neither IFNβ nor all the genes classically grouped as "IFNβ-dependent" is inducible in LPS-treated neutrophils. Curiously, lack of TBK1 activation by TLR4 ligands occurs despite the constitutive expression of all the intracellular signaling and regulatory components involved along

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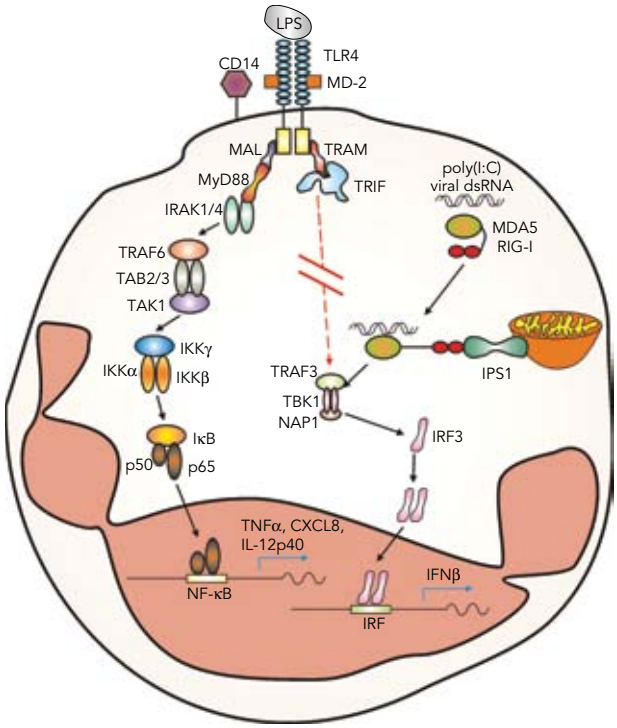


Figure 4B.3. TLR4- and helicase-mediated signal transduction pathways operating in human neutrophils.

the MyD88-independent cascade, including TRAM, TRIF, NAP1, TRAF3, TBK1, IKKε, and IRF3 in neutrophils. In view of the fact that the other known TLR-dependent transduction pathway ultimately signaling through TBK1 and IRF3, namely the "TLR3/TRIF-dependent pathway," is also nonfunctional in neutrophils (for their lack of TLR3 expression) it is intriguing that human neutrophils should constitutively express TBK1 and IRF3. This puzzle has been recently clarified, as revealed by capacity of human neutrophils to trigger an immunoregulatory and antiviral gene expression program, in response to intracellularly administered polyinosinic:polycytidylic acids [poly(I:C)], specifically via TBK1 and IRF3 activation (Figure 4B.3). Poly(I:C) is a synthetic mimetic of viral double-stranded RNA (dsRNA), which is known to interact either with endosomal TLR3 or with cytoplasmic RNA helicases (involved in viral RNA recognition) such as MDA5 (melanoma differentiation-associated gene 5) and RIG-I (retinoic acid–inducible gene I). Accordingly, human neutrophils constitutively express significant levels of both MDA5 and RIG-I. The biological relevance of these findings were strengthened by the observations that murine neutrophils infected with picornaviruses such as encephalomyocarditis virus (EMCV), whose dsRNA recognition specifically requires MDA5 (Figure 4B.3), were found to produce large amounts of either IFNβ or TNFα. Taken together, these data have revealed that neutrophils are able to activate antiviral responses via helicase recognition and thus contribute to local or systemic IFNβ production.

THE ROLE OF NEUTROPHILS IN ACUTE INFLAMMATION

When pathogenic agents enter into (or localize within) a tissue, neutrophils are rapidly recruited to the infected site by a sequence of events that begins with the elaboration of mediators able to promote their migration from the intravascular compartment. Such inflammatory mediators include not only cytokines such as TNFα, IL-1β, or IL-17A (this latter being one of the most abundant product of the Th17 subset of proinflammatory T lymphocytes), which orchestrate neutrophil recruitment through their capacity to trigger the production of chemotactic factors, but also various products derived from numerous sources, including activated plasma components and tissue macrophages or endothelial cells. Classical chemotactic factors include C5a, LTB_4, fMLP, and chemokines of the CXC family (CXCL8/IL-8, CXCL1/GROα, CXCL5/epithelial cell neutrophil-activating protein-78, etc.), the latter molecules representing the main and more specific chemoattractant involved in neutrophil recruitment, according to many in vivo (gene-targeted) models.

When neutrophils need to be recruited to an inflammatory site, they adhere to endothelial cells, transmigrate by diapedesis (the process of squeezing between endothelial lining cells), and crawl toward a corresponding chemotactic gradient (a process called chemotaxis) to the injury site. The detection of chemotactic stimuli by neutrophils leads, in fact, to the activation of signaling pathways, cytoskeletal rearrangement, and changes in cell surface molecules that coordinate to facilitate migration. Recently, chemotaxis has been shown to be dependent on phosphoinositide 3-kinase (PI3K)-γ, which belongs to a family of class I PI3Ks. Adhesion of blood neutrophils (and of other leukocytes as well) to the endothelium during inflammation is a multistep and highly complex phenomenon, which requires specific leukocyte–endothelial interactions involving different families of adhesion molecules, which may vary according to the precise stimulatory conditions. Adhesion molecules include members of the selectin family and their cognate carbohydrate and glycoprotein ligands, which mediate leukocyte deceleration along the vessel wall (a process called "rolling"), as well as the integrins and their cognate immunoglobulin superfamily ligands, which mediate high-affinity adhesion of leukocytes to venules. Neutrophil integrins consist of a family of at least three different glycoproteins, each one composed of an identical β-subunit (β_2, known as CD18) noncovalently linked to different α-subunits (known as CD11a/CD18, lymphocyte

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function antigen [LFA-1], CD11b/CD18 [Mac-1], CD11c/CD18 [p150,95], and CD11d/CD18). Upon encountering chemoattractants, probably displayed bound to glycosaminoglycans on the vessel wall, integrins are converted from an inactive to an active conformation. Activated integrins then interact with counter-receptors (e.g., ICAM-1 [intercellular adhesion molecule-1] and ICAM-2) on the surface of the endothelium, to lead to the strong adhesion and arrest of the leukocyte. The neutrophils then flatten and migrate between and right through the endothelial cells of postcapillary venules to the surrounding tissue, following the gradient of specific chemotactic factor(s). The former process involves homophilic interaction of CD31 and JAM-A on neutrophils and endothelial cells where CD31 and JAM-A act sequentially to mediate neutrophil migration through the venular walls. The process of transmigration under the influence of adhesion molecules or chemotactic factors, as well as the exposure to other stimuli in local tissues, often result in a marked release by neutrophils not only of preformed proinflammatory mediators, such as those contained in the granules, but also in the rapid production of ROS or the activation of cytokine gene expression.

The fundamental importance of adhesive glycoproteins for the neutrophil functions in vivo is illustrated by those individuals, genetically lacking the leukocyte adhesive proteins, who display an abnormally high susceptibility to bacterial infections. Several types of leukocyte adhesion deficiency (LAD) disease have been identified. LAD-I is an autosomal recessive disorder in which mutations of the β2 integrin family typically eliminate expression of all three integrin complexes. Because of the multiple defects in adhesion-related functions, LAD-I patients develop recurrent bacterial and fungal infections, typically with *S. aureus* or Gram-negative enteric microbes. Characteristic clinical features include frequent skin and periodontal infections, delayed separation of the umbilical cord and omphalitis, and deep tissue abscesses. Neutrophilia with paucity of neutrophils at inflamed or infected sites is also characteristic. LAD-II, caused by mutations in the membrane transporter for fucose, is associated with loss of expression of fucosylated glycans on the cell surface. Fucosylated proteins such as sialyl-Lewis X (CD15s) are ligands for endothelial selectins and are important for the early phases of leukocyte adhesion to endothelial cells. Patients with LAD-II also have leukocytosis and form pus poorly, although infections tend to be less severe in LAD-II patients compared to LAD-I patients. Finally, LAD-III has been described in four patients and appears to be an autosomal recessive disorder. LAD-III is characterized by defects in cell signaling that interfere with activation of multiple classes of integrins downstream of G-protein–coupled receptors. The phenotype of this disorder is similar to LAD-I, but it can also be associated with defects in integrin-mediated platelet aggregation and excessive bleeding.

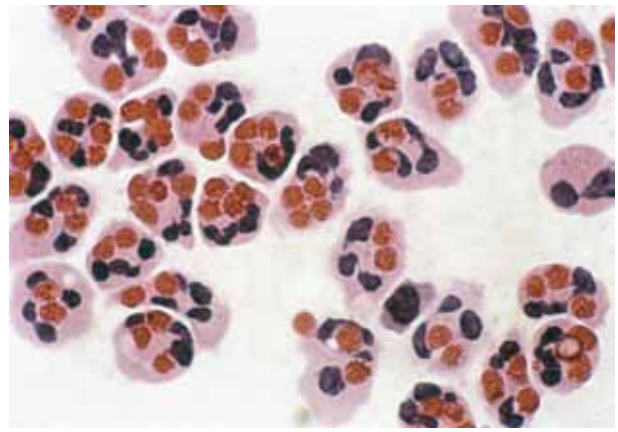


Figure 4B.4. Smear of neutrophils that have phagocytosed opsonized red cells.

After the process of emigration, during which neutrophils undergo shape changes and mobilize their granules to the cell surface, cells arrive to infection foci, where they adhere to extracellular matrix components such as laminin and fibronectin, and begin to react with the etiopathogenetic agent. When neutrophils are exposed to inflammatory mediators such as cytokines (IFNγ, GM-CSF, G-CSF) or LPS, they become "primed" or hyperresponsive to subsequent activating stimuli, in this way acquiring the capacity to exhibit maximal degranulation and respiratory burst responses and enhanced cytokine and lipid mediator release. Accordingly, agents that "prime" neutrophils usually also extend their lifespan by inhibiting apoptosis. In addition, activated neutrophils at inflammatory sites also synthesize large quantities of important inflammatory lipid mediators, including LTB_4 and PAF, which together facilitate the extravasation and recruitment of additional waves of granulocytes to inflamed tissues.

Although neutrophil granulocytes do not apparently show any particular specificity for antigens, they nevertheless play an important role in host protection against microorganisms, by eliminating them through phagocytosis (the process of eating particles). In fact, many cells in our body are capable of phagocytosis, but only neutrophils and macrophages do it to an extent sufficient to consider them as "professional phagocytes" (up to 10–12 particles [e.g., bacteria] can be engulfed by a single neutrophil [see Figure 4B.4]). Phagocytosis is triggered upon the binding of opsonized microorganisms through, for instance, FcγRs and CRs, or through nonspecific glycosylated receptors that recognize certain lectins on target microorganisms. During

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the phagocytic process, the foreign particle is internalized, initially through membrane recruitment to the site of particle contact, and then via membrane extensions outward to surround the particle. Subsequently, particle engulfment within cytoplasmic phagosomes occurs, which eventually fuse with granules, thereby forming phagolysosomes. This, in turn, results in the subsequent release of proteolytic enzymes and other bactericidal components into the phagolysosome. At the same time, NADPH oxidase assembles on the phagosome membrane after phagocytosis and starts to generate ROS into the phagolysosome to kill bacteria by oxidizing microbial proteins and lipids. The activity of NADPH oxidase also leads to the acidification of the phagosome, which enhances the effectiveness of pH-sensitive antimicrobial compounds. In addition, activation of gene transcription and cytokine production occur during phagocytosis, representing another critical feature of neutrophils for the development of an effective immune response. For instance, recruited neutrophils that encounter a pathogen respond by producing further chemokines, in particular CXCL8, to amplify the innate immune response.

Beside oxygen-dependent and independent killing mechanisms, neutrophils also use powerful structures to capture and kill microbes in the extracellular space, the so-called NETs (neutrophil extracellular traps), thus fulfilling their antimicrobial function even beyond their life span. NETs consist of threads of nuclear chromatin, with a diameter of approximately 15 nm, which are decorated with antimicrobial peptides and enzymes (e.g., BPI, neutrophil elastase, and cathepsin G), but lack membranes and cytosolic markers. Typically during NET formation, the nuclear envelope and the granule membranes gradually dissolve, two phenomena that allow mixing of the cytoplasmic components. Pathogens trapped by NETs are killed by the high local concentration of antimicrobial peptides and enzymes, similar to the situation in the phagolysosome. NETs entrap not only various Gram-positive and Gram-negative bacteria, such as *S. aureus*, *Salmonella typhimurium*, *Streptococcus pneumoniae*, and group A streptococci, but also pathogenic fungi, such as *C. albicans*. Current data suggest that the formation of NETs involves a novel cell death mechanism, which is neither apoptotic nor necrotic, and that does not seem to be dependent on the activation of caspases. NADPH oxidase–dependent ROS are instead formed, and subsequent signaling ultimately leads to the disintegration of nuclear and cellular membranes and to the release of DNA and effector molecules that mix to form extracellular traps. Accordingly, NETs are now thought to be important in CGD, as neutrophils from these individuals do not form NETs. This raises the possibility that the absence of NETs might be another major explanation for the pathology seen in CGD.

While all of the above-mentioned events are necessary and essential for neutrophil-mediated microbicidal activity and digestion of engulfed particles, the same effector mechanisms can also be involved in other effects of PMN, such as amplification of the inflammatory process, preparation of the tissue-healing processes, tumoricidal activity and cytotoxicity, and tissue matrix injury if ROS and proteases are massively released or leaked into the external milieu.

NEUTROPHIL-DERIVED CYTOKINES

Cytokines orchestrate the complex networks mediating the myriads of cellular interactions that ultimately regulate effector cell functions. T cells, DC and natural killer (NK) cells, monocytes, and macrophages are considered to be major producers of cytokines during natural and adaptive immunity. However, numerous in vitro and in vivo studies have made it clear that PMNs also have the ability to synthesize and release immunoregulatory cytokines. Such findings, while challenging the traditional concept of neutrophils viewed as "terminally differentiated cells, devoid of transcriptional and protein synthesis activity," have made it clear that the contribution of neutrophils to host defence and natural immunity extends well beyond their traditional role as professional phagocytes. It is important to mention that, at least in vitro, neutrophils usually (but not always) make fewer molecules of a given cytokine than do monocytes/macrophages or lymphocytes on a per-cell basis. In vivo, however, neutrophils constitute the majority of infiltrating cells in inflamed tissues and often outnumber mononuclear leukocytes by one to two orders of magnitude. Thus, the fact that neutrophils clearly predominate over other cell types under various in vivo conditions suggests that, under those circumstances, the contribution of neutrophil-derived cytokines can be of foremost importance. Interestingly, at the molecular level, studies addressing cytokine release by neutrophils have revealed that the induction of cytokine production in neutrophils is usually preceded by an increased accumulation of the related mRNA transcripts. Similarly to other cell types, cytokine expression in neutrophils can be regulated at the transcriptional, posttranscriptional, translational, and posttranslational levels. Elegant studies have also shown that the inflammatory cytokines produced by neutrophils are generally encoded by immediate-early response genes, which in turn depend on the activation of transcription factors such as those belonging to the NFκB, STAT (signal transducers and activators of transcription), ETS, CCAAT/enhancer-binding proteins (C/EBP), and AP-1 families of transcription factors. Moreover, a wide range

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TABLE 4B.1. Cytokines expressed in resting or activated human neutrophils

C-X-C chemokines	***Proinflammatory cytokines***	***Colony-stimulating factors***
IL-8/CXCL8	**TNFα**	G-CSF
GROα, GROβ, GROγ	**IL-1α, IL-1β**	M-CSF(?)
ENA-78/CXCL5	IL-16(?), **IL-17**(?)	GM-CSF(?)
CINC-1*, CINC-2α	IL-18	IL-3(?)
CINC-3/MIP-2/CXCL1	MIF	SCF*(?)
PF4/CXCL4	**IL-6**(?), IL-7, IL-9	
GCP-2/CXCL6		***Angiogenic and fibrogenic factors***
IP-10/CXCL10	***Anti-inflammatory cytokines***	**VEGF**
MIG/CXCL9	**IL-1ra**	HB-EGF
I-TAC/CXCL11	**$TGF\beta_1$, $TGF\beta_2$**	**FGF-2**
	IL-4(?), **IL-10**(?)	**TGFα**
C-C chemokines		**HGF**
MCP-1/CCL2	***Immunoregulatory cytokines***	
MIP-1α, CCL3	IFNβ*, **IFNβ**, **IFNγ**(?)	***TNF superfamily members***
MIP-1β, CCL4	**IL-12**	**FasL**
MIP-3α, CCL20	IL-23(?)	CD30L
MIP-3β, CCL19		TRAIL
PARC/CCL18	***Other cytokines***	**LIGHT***
MDC/CCL22*	**Oncostatin M**	**Lymphotoxin-β***
	GDF (?)	APRIL, **BAFF/BLyS**
	NGF*, BDNF*, NT4*	RANKL

Cytokines in bold refer to neutrophil studies in animal models confirming human findings. *: mRNA only; (?): it requires definitive corroboration.

of stimuli that are able to induce cytokine synthesis in PMN has been already identified. It is clear, for instance, that cytokines themselves, chemotactic factors (fMLP, LTB_4, PAF, C5a, and CXCL8), phagocytic particles, microorganisms (such as fungi, viruses, and bacteria), and PRR ligands can all induce the synthesis and release of cytokines by neutrophils. Not only do the magnitudes and kinetics of cytokine release vary substantially depending on the stimulus used, but also the pattern of production is influenced to a great extent by the stimulus used, each one inducing a characteristic signature. Thus, since neutrophils might usually represent the first cell type encountering, and interacting with, the etiologic agent in an inflammatory context, a stimulus-specific response of neutrophils in terms of cytokine production might help in predicting the evolution of certain types of inflammatory and immune reactions.

Table 4B.1 lists all the cytokines that, to date, have been shown to be released by neutrophils in vitro, either constitutively or following appropriate stimulation, or in vivo. Numerous in vivo observations, in fact, not only have confirmed and reproduced the in vitro findings, but often have clarified their biological meanings and implications. As outlined in Table 4B.1, neutrophils can produce proinflammatory, anti-inflammatory, immunoregulatory, angiogenic and fibrogenic cytokines, chemokines, and ligands belonging to the TNF superfamily. At first sight, an analysis of the table immediately suggests that the ability of neutrophils to produce such a variety of cytokines enables them to significantly influence many processes, not only the multiple aspects of the inflammatory and immune responses, but also hematopoiesis, wound healing, and antiviral defense. For instance, other than typical proinflammatory cytokines such as TNFα and IL-1β, being expected to be produced by inflammatory cells, neutrophils have been shown to also secrete BLyS/BAFF (B-lymphocyte stimulator/B-cell activating factor) and APRIL (a Proliferation-Inducing Ligand). Since BAFF and APRIL are well known to be essential for B-lymphocyte homeostasis and related pathologies, it is plausible to assume a role of neutrophils not only in sustaining B and plasma cell antibody production and survival, but also in promoting B-cell-dependent autoimmune diseases and tumors, as already elegantly demonstrated in the case of B-cell lymphoma. Along the same line, the recent findings that neutrophils exposed to type I and II IFN express and produce another TNF superfamily member selectively involved in tumor cell killing and autoimmunity, namely TRAIL, have opened an additional perspective to exploit neutrophils for novel roles in anticancer responses, whose importance has been recently highlighted in the BCG immunotherapy for bladder cancer. Experimental studies of tumor cure and prevention have, in fact, suggested that, at least in some models, engagement of neutrophil functions can be crucial for the establishment of an

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effective antitumoral immune response and immune memory reaction. Cytotoxic mediators of tumor and endothelial cell killing produced by neutrophils include TNFα, defensins, proteases (such as elastase and cathepsin G), ROS, nitric oxide, and angiostatic chemokines (CXCL9/MIG, CXCL10/IP-10, and CXCL11/I-TAC). In addition, it has been observed that neutrophils may indirectly generate massive amounts of bioactive, angiostatin-like fragment, inhibiting basic FGF (fibroblast growth factor) plus VEGF (vascular endothelial growth factor)-induced endothelial cell proliferation. However, as described for macrophages, neutrophils can also favor malignant growth and progression, in relation to the type of tumor environment in which they reside: they may do so, for instance, via the production of proangiogenic cytokines such as VEGF and CXCL8, or acting as a distinct myeloid-derived suppressor cell subpopulation releasing, at least in mice, arginase 1–mediated urea and ornithine.

NEUTROPHIL CROSS TALK WITH OTHER CELL TYPES: ROLE OF NEUTROPHIL-DERIVED CHEMOKINES

Amongst the cytokines produced by neutrophils, chemokines are particularly relevant because of their ability to selectively recruit discrete cell populations into sites of injury and thereby effectively regulate leukocyte trafficking. In addition, chemokines play fundamental roles in coordinating the immune system responses, in regulating B- and T-cell development and in modulating angiogenesis. As displayed in Figure 4B.5, activated neutrophils may potentially produce several chemokines, including IL-8/CXCL8, GROα/CXCL1, monocyte chemotactic protein-1 (MCP-1/CCL2), macrophage inflammatory protein-1α (MIP-1α/CCL3), and MIP-1β/CCL4 and MIG/CXCL9, IP-10/CXCL10, and I-TAC/CXCL11. As examples, LPS or TNFα-treated neutrophils produce CXCL8, CXCL1, CCL3, and CCL4; fMLP-treated neutrophils produce CXCL8 only; and IFNγ plus LPS- or IFNγ plus TNFα-treated neutrophils specifically produce, among other chemokines, CXCL9, CXCL10, and CXCL11. Because the chemokines produced by neutrophils are primarily chemotactic for neutrophils, monocytes, DCs, NK, and T-helper type 1 and type 17 cells, a potential role for neutrophils in orchestrating the sequential recruitment to, and activation of, distinct leukocyte types in the inflamed tissue is plausible. Broadly speaking, a general scenario that has already been validated in various experimental models is that, after migration to a focus of infection, neutrophils are stimulated by the etiological agent to initially produce the chemokines that are chemoattractive for neutrophils themselves, including CXCL8. This would

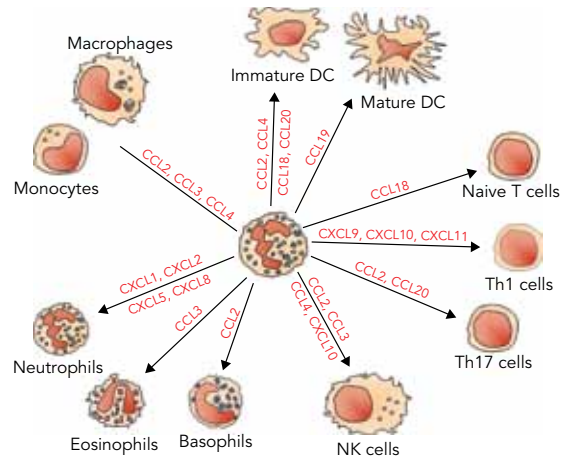


Figure 4B.5. Chemokines produced by neutrophils and their target cells.

activate a positive-feedback loop to induce accumulation of a larger number of neutrophils. Then, specific CC-chemokines would be produced by neutrophils to serve an instrumental role in recruiting the required leukocyte types necessary for the subsequent phases of infection.

Remarkably, neutrophils have also been shown to produce biologically active MIP-3α/CCL20 and MIP-3β/CCL19, two structurally related CC-chemokines that have been suggested to play a fundamental role in trafficking of, respectively, immature and mature DC to mucosal surfaces and lymphoid organs. DCs are professional antigen-presenting cells that are pivotal in the induction of T-cell responses to combat infection. Immature DCs are scattered throughout the peripheral tissues, where they sample antigens and process these into peptides that are loaded onto MHC II molecules for presentation to T cells. Such a process occurs during maturation of DC, which is paralleled by upregulation of MHC II molecules, costimulatory molecules and proinflammatory cytokines that enable DC to stimulate pathogen-specific T cells efficiently. DC can produce IL-12 in response to intracellular bacteria and instruct Th1 polarization to produce IFNγ and trigger effective cellular immune responses. By contrast, DC instruct Th2 polarization in response to extracellular parasites, to produce IL-4, IL-5, and IL-13 and to induce appropriate anti-parasite immunity. Thus, DC tailor immune responses to the type of pathogen for rapid and complete clearance of infection. It has been postulated that neutrophil-derived MIP-3α/CCL20 and MIP-3β/CCL19 might contribute to the recruitment of DC, at various stages of maturation, to sites of inflammation and disease for cross talks with immunocompetent T lymphocytes and regulation of the immune response. Likewise, several antimicrobial compounds released

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by neutrophils, for instance, α-defensins, lactoferrin, LL-37, and cathepsin G (Figure 4B.1) have been found to act as chemoattractants for immature DC and T cells. In addition, neutrophils proteolytically activate prochemerin to generate chemerin, one of the few chemokines that attracts both immature DC and plasmacytoid DC. On the other hand, immature DC are also known to produce CXCL8/IL-8 early upon stimulation, thereby attracting more neutrophils for a more proximal colocalization. This ensures, for instance, that neutrophils and DC are present at the same place at the same time during pathogenic challenge to enable cross talk between these cells. Current evidence now indicates that neutrophils can indirectly play a role in adaptive immunity, not only by recruitment of immune cells, such as DC or specific T lymphocyte subsets, for instance Th17 or Th1, but also by instructing directly DC and inducing adaptive immune responses. Upon coculture of immature DC with activated neutrophils, in fact, DC maturate, as demonstrated by upregulation of HLA-DR, the maturation marker CD83, the co-stimulatory molecules CD86 and CD40, and IL-12-production, and, in turn, trigger T-cell proliferation and strong Th1 cell responses. Such neutrophil-induced maturation of DC is in part mediated by neutrophil-derived TNFα, as well as by cellular contacts regulated by a defined set of receptors expressed by both neutrophils and DC. This is also reflected in vivo, for instance during Crohn's disease, which is characterized by inflammation of the colon mucosa and Th1 responses, in which DC and neutrophils can be found in close proximity.

Recent reports suggest that neutrophils might travel to the lymph nodes during infections and express MHC II and co-stimulatory molecules, in the latter case when appropriately activated by IFNγ, TNFα, and GM-CSF. However, whether neutrophils transmit signals to naive T cells remains still puzzling. It is possible that within the lymph nodes neutrophils undergo apoptosis and are taken up by DC. As a consequence, DC can present PMN-derived antigens to T cells. Other evidence indicates that neutrophils have also a role in directing T-cell polarization, for instance through their capacity to produce the Th1-inducing cytokine, IL-12. The latter has been clearly demonstrated in mice, in which strong Th1-dependent T-cell responses that result in pathogen clearance are elicited upon infection with *C. albicans*, *Helicobacter pylori*, or *Legionella pneumophila*: strikingly, depletion of neutrophils reverses the Th1 responses into a predominant Th2-response, therefore making the mice susceptible to infection. Another mechanism whereby neutrophils may interact with Th1 cells is through the production of chemokines such as CCL3/CCL4, CXCL9, CXCL10, and CXCL11, which act on CCR5 and CXCR3 of Th1 cells but not on the chemokine receptors expressed by Th2 cells. Indeed, secretion of these chemokines has been shown to enable neutrophils to augment Th1 cell responses by preferential attraction of Th1 cells to the sites of infection. Therefore, it is tempting to speculate that the production of cytokines and chemokines by neutrophils not only influences the development and control of Th1 responses, but may also be involved in the cross talk with Th17 cells. Finally, it has also been proposed that mature postmitotic neutrophils can also "transdifferentiate" into much-longer-lived cells with macrophage- or DC-like characteristics, which might constitute a further way for neutrophils to act as regulatory cells of the adaptive immune response.

NEUTROPHIL APOPTOSIS

As previously mentioned, neutrophils are short-lived cells. In culture, neutrophils undergo spontaneous death, with a half-life of around 12–16 hours. It is believed that this behavior reflects a phenomenon that has adapted to ensure a strict control on neutrophil turnover and maintain the number of neutrophils relatively stable in healthy individuals. Deregulation of the neutrophil death rate may in fact cause unwanted and exaggerated inflammation, autoimmunity, or cancer, instead of providing a perfect balance between their immune functions and their safe clearance. An accelerated neutrophil death may lead to a decrease of neutrophil counts (neutropenia), which augments the chance of contracting bacterial or fungal infections and impairing the resolution of such infections, while a delayed neutrophil death elevates neutrophil counts (neutrophilia). The importance of appropriately controlling the turnover of neutrophils is clearly manifest at inflammatory foci, where it contributes to limit neutrophil activities and minimize the risk of externalization of excessive amounts of cytotoxic molecules (i.e., from the mobilization of neutrophil cytoplasmic granules). Conversely, the removal of apoptotic neutrophils is thought to be another important step in preventing the release of granule contents and cytoplasmic danger signals into the extracellular fluid, thereby halting further injury during resolution of inflammation. One can, in fact, imagine that suppression or delay of the basal apoptotic rate of PMN by an agent would be deleterious for the host during inflammation, because PMN might remain activated and thus would perpetuate tissue damage by releasing their toxic products. A paradigmatic example for complete resolution of inflammation, without relevant subsequent lung injury, is pneumococcal pneumonia, which is characterized by an extensive neutrophil accumulation that is followed by neutrophil apoptosis and timely removal of apoptosing cells by lung macrophages.

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Neutrophil spontaneous death in culture shares many features of classical apoptosis, such as cell body shrinkage, cellular crenation, exteriorization of phosphatidylserine (PS) from the inner to the outer leaflet of the plasma membrane, exposition of "eat-me" signals for phagocytosis by scavengers, vacuolated cytoplasm, mitochondria depolarization, nuclear condensation, and internucleosomal DNA fragmentation. Neutrophil apoptosis proceeds, in vitro, through an execution phase during which cell dismantling is initiated, with or without fragmentation into apoptotic bodies but with maintenance of a near-intact cytoplasmic membrane. This cascade is ultimately followed by a transition to a necrotic cell elimination traditionally called "secondary necrosis." Secondary necrosis involves an activation of self-hydrolytic enzymes and swelling of the cell or of the apoptotic bodies, as well as a generalized and irreparable damage to the cytoplasmic membrane culminating with cell disruption. For instance, elimination of alveolar macrophages in a model of pneumococcal pneumonia was shown to affect the clearance of accumulated neutrophils and to result in extensive apoptotic secondary necrosis of neutrophils followed by an exaggerated lung inflammation and increased lethality.

Neutrophil apoptosis is also associated with the loss of receptor expression (for instance, CD16) and greatly reduced responsiveness to external stimuli, as revealed by impaired chemotaxis, respiratory burst, degranulation, phagocytosis, and cytokine synthesis. It is thus plausible that, within an inflammatory context, neutrophils become functionally isolated from their environment. Another form whereby neutrophils can die, for instance following a dramatic insult by an exterior aggression damaging enough to produce irreversible alterations, is primary necrosis. In this mode of cell death, there is an immediate, extensive, and irreparable damage to the cytoplasmic membrane associated with loss of plasma membrane integrity, so that release of harmful neutrophil contents is not prevented. Interestingly, some of the features of classical apoptosis, including those observed in constitutive neutrophil death, are also shared by autophagic cell death, another recently characterized physiological cell death process. Autophagy is a caspase-independent, cell death program, featured by the degradation of cellular components in autophagic vacuoles within the dying cell and little chromatin condensation. The autophagosomes fuse with endosomes or lysosomes to form amphisomes or autolysosomes, respectively. Lysosomal proteases ultimately degrade the luminal content of the autophagic vacuole of the autolysosome. In vivo, it has been demonstrated that cells dying through autophagy are phagocytosed by surrounding cells. In primary human neutrophils, this form of death has been shown to be induced by Siglec-9 (sialic acid binding immunoglobulin-like lectin-9) ligation, with the concurrent stimulation with certain proinflammatory cytokines. While there is almost nothing known about the role and the mechanisms triggering autophagy in neutrophils, one critical question is whether the autophagic-like neutrophil death shares the anti-inflammatory properties of apoptosis or whether it induces additional inflammation.

Whatever the case is, the key role of apoptosis in resolution of neutrophilic inflammation has gained increasing attention over the past few years and is appreciated as crucial to allow safe clearance of potentially dangerous neutrophils from tissues. In view of its relevance, studying and discovering molecules that modulate the PMN apoptotic rate is of great importance in biology and medicine. Many of the molecular events involved in the apoptosis pathway have been already identified, with their exquisite regulation becoming better appreciated. For instance, evidence now suggests that neutrophil death is mediated by a complex network of intracellular death/survival signaling pathways, which can be modulated by a variety of extracellular stimuli. Signals such as adhesion, transmigration, hypoxia, prosurvival factors, and cytokines (LPS, G-CSF, GM-CSF, and IFNγ), can all delay neutrophil apoptosis and extend their lifespan, thereby enhancing the ability of neutrophils to function during inflammatory challenge. Even glucocorticoids have been shown to delay neutrophil apoptosis in vitro, a finding which might have potential clinical implications. For instance, while in severe asthma, eosinophils and neutrophils are often found together, neutrophils may gradually replace eosinophils in proportion to the severity and/or duration of the disease, perhaps reflecting the ability of corticosteroids to induce eosinophil apoptosis while inhibiting this process in neutrophils. By contrast, death signals (for instance, "death" receptor engagement, cytokine depletion, or ingestion of bacterial pathogens such as *Escherichia coli* or *S. aureus*) intervene to trigger apoptosis once neutrophil function is complete, and prepare their phagocytic removal. At the molecular level, neutrophil apoptosis is a highly and exquisitely regulated process involving many molecules and proteolytic cascades such as FasL, TNFα, TRAIL, ROS, caspases, IAP (inhibitor of apoptosis protein), survivin, and Bcl-2 family members such as Mcl-1 and A1. Both intrinsic (via mitochondria) and extrinsic (via death receptor signaling) apoptotic pathways have been described in neutrophils. For instance, neutrophils often undergo apoptosis after phagocytosis, in a ROS-dependent fashion and following the cleavage and activation of caspases 8 and 3. A novel, recently identified but unexpected player

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in the regulation of PMN apoptosis is CDK (cyclin-dependent protein kinase), which has been shown to induce PMN apoptosis and overrides the effect of potent survival mediators, such as LPS and GM-CSF, by a mechanism involving caspase activation and reduction of Mcl-1. The possible use of synthetic CDK inhibitors in the nonproliferating neutrophils is exciting and opens new opportunities of therapeutic strategies, for instance in models of pulmonary inflammatory diseases.

Although it is of great importance to discover and explain the mode of action of apoptotic neutrophil modulators, it is also of major interest to fully understand the mechanisms involved in the recognition of apoptotic PMNs. First of all, apoptotic cells release lysophosphatidylcholine to attract monocytes and macrophages. To allow macrophages to ingest apoptotic PMN, and differentiate them from viable cells, neutrophil membrane is altered in the way that they express specific molecules and receptors. Recognition and engulfment is, for instance, facilitated by the loss of phospholipids asymmetry and phosphatidylserine (PS) exposure ("eat-me signal") to the outer leaflet of the apoptotic cell surface. However, this phenomenon does not seem to be sufficient for the recognition of apoptotic cells. Other proteins, such as calreticulin or annexin-1, the latter being externalized on apoptotic cell membrane colocalized with PS, might participate in the removal of neutrophils by macrophages. Several other receptors have been reported to play a role in the ingestion of apoptotic cells by macrophages: some of them involved in tethering to the phagocyte and others in signaling pathways leading to actin cytoskeleton rearrangement and engulfment during membrane extension and fusion. These receptors on phagocytes include the controversial PSR, the vitronectin receptor, scavenger receptors like CD36 and CD14, lectins, Mer, integrins ($\alpha_v\beta_5$ and $\alpha_v\beta_3$), CR2 (complement receptor-2), CR3, β2-GPI receptor, and CD91. They can interact directly with apoptotic cells or indirectly through bridging proteins, like C1q, β_2-GPI, Gas6, and MFG-E8. In addition, components of the innate immune system play a role in regulating apoptotic cell clearance since the opsonization of apoptotic cells increases their removal by macrophages. Interestingly, although the presence of such "eat me" signals has been largely documented, the presence of "don't eat me" signals on the surface of viable nonapoptotic cells has also been recognized. In this respect, CD47 or CD31 have been identified as "don't eat me" molecule and the proposed model is that loss or inactivation of CD47 effects accompanies apoptosis and allows the apoptotic cells to be recognized and cleared. On the other hand, it appears that this phenomenon of discrimination between apoptotic and viable cells involves redistribution of eat-me signals into patches (to increase their avidity toward receptors) as well. Engulfment of apoptotic cells by macrophages (also called "efferocytosis") is influenced by cytokines such as IL-1 and TNFα, glucocorticoid hormones, prostaglandins, and also interaction of surface adhesion molecules with neighboring cells and extracellular matrix components. Lipoxins also stimulate macrophages to phagocytose apoptotic PMN. Expression of 15-lipoxygenase on nonphlogistic macrophages, which had phagocytosed apoptotic PMN, induces lipoxin A4 release, thus resulting in a decreased PMN recruitment and in an increase in apoptotic PMN removal. After apoptotic cell ingestion, macrophages phenotypes switch to induce tissue repair or emigrate in the lymphatic system, suggesting that phagocytosis of apoptotic cells is involved in the negative regulation of macrophage activation and that PMN apoptosis could be considered as endogenous "active" anti-inflammatory process. Furthermore, phagocytosis of apoptotic neutrophils by human macrophages in vitro suppresses the release of macrophage-derived proinflammatory mediators such as TNFα, IL-1β, or GM-CSF and it induces the release of anti-inflammatory mediators, like TGF-β1, IL-10, PGE_2, leukocyte protease inhibitor, and reparative growth factor. This process thus confers an anti-inflammatory phenotype to macrophages that helps to lead to resolution of inflammation, but that could predispose to development of autoimmunity if not efficient.

NEUTROPHILS IN HUMAN DISEASES

Previous sections have already summarized some of the most common inherited disorders concerning the functions of neutrophils that impair critical responses for host defense. Typically, patients with defects in neutrophil function present, in infancy or childhood, with recurrent and/or difficult to treat bacterial infections. The microorganisms causing these infections are often unusual or opportunistic pathogens and typically infect the skin, mucosa, gums, lung, or draining lymph nodes, or cause deep tissue abscesses. However, many of these disorders have characteristic clinical and microbiological features that are related to the specific nature of the defect in neutrophil function.

Other than in inherited impairments of neutrophil functions, dramatic clinical consequences may be observed also during acquired neutropenia. For instance, susceptibility to infectious diseases increases sharply when neutrophil counts fall below 1,000 cells/μL. When the absolute neutrophil count

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falls to < 500 cells/μL, control of endogenous microbial flora (e.g., mouth, gut) is critically impaired; when it is < 200/μL, the inflammatory process is absent. Neutropenia can be due to depressed production (i.e., hereditary neutropenias), increased peripheral destruction, or excessive peripheral pooling. The most common neutropenias are iatrogenic, resulting from the use of cytotoxic or immunosuppressive therapies for malignancy or control of autoimmune disorders. Interestingly, acute neutropenia, such as that caused by cancer chemotherapy, is more likely to be associated with increased risk of infection than neutropenia of long duration. Recombinant human G-CSF usually reverses this form of neutropenia. A neutrophilia occurs instead when the number of circulating neutrophils dramatically increase (even up to 10-fold), for instance from increased neutrophil production, increased marrow release, or defective margination. The most important acute cause of neutrophilia is infection, which is caused by both increased production and increased marrow release of neutrophils. Increased marrow release and mobilization of the marginated leukocyte pool are also induced by glucocorticoids. Persistent neutrophilia with cell counts of 30,000–50,000/μL is called a leukemoid reaction, a term often used to distinguish this degree of neutrophilia from leukemia. In a leukemoid reaction, the circulating neutrophils are usually mature and not clonally derived.

Apart from the disorders associated with genetic dysfunctions or quantitative alterations of neutrophils, there are certain pathological situations in which neutrophils themselves become the predominant contributors to tissue injury, especially when the mechanisms supposed to control and inactivate their hypothetical beneficial and protective effector functions are deregulated. For instance, the latter may occur when the acute insult cannot be resolved, when the shut off of the neutrophil influx (that may involve inactivation of proinflammatory mediators and/or temporal change in the pattern of chemokines produced) is impaired, or when the safe clearance of dying neutrophils from the inflammatory site is altered. An uncontrolled production and release of oxidants and proteases by neutrophils, acting either individually or in concert, appear to be responsible for much (but not all) of the tissue injury observed in such cases. Examples of such pathologies, tentatively classified according to the major neutrophil-activating event (some of them covered in other chapters), are (i) diseases caused by ischemia-reperfusion injury (i.e., myocardial infarction, transplantation); (ii) bacterial infections (adult respiratory distress syndrome, endotoxic shock, osteomyelitis, endocarditis, acute and chronic pyelonephritis); (iii) cytokine-mediated diseases (rheumatoid arthritis, inflammatory bowel diseases); (iv) diseases caused by crystal deposition (gout, articular chondrocalcinosis); (v) immune complex–mediated diseases (vasculitis, lupus, Goodpasture's syndrome); (vi) antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (Wegener's granulomatosis, pauci-immune necrotizing crescentic glomerulonephritis); (vii) airway diseases in which considerable observational and experimental data support an association between neutrophils and the severity and progression (chronic obstructive pulmonary disease, bronchiectasis, bronchiolitis, cystic fibrosis, and even certain forms of asthma are characterized by neutrophil infiltration of the airway wall). One of the key challenges in neutrophil-dominated conditions is how to manipulate neutrophil function to abolish their destructive potential in a way that does not compromise their antibacterial and antifungal capacity. This has been difficult to achieve, as successful treatments in animal models have frequently proven ineffective or limited by side effects when used in human inflammatory diseases.

CONCLUSIONS

Contrary to what is traditionally thought, the neutrophil is now recognized as a highly versatile and sophisticated cell with significant synthetic capacity and an important role in linking the innate and adaptive arms of the immune response. Even though the ability of neutrophils to transcribe many genes is no longer a matter of debate, the research conducted in the recent years has brought forward exciting discoveries that have greatly broadened our knowledge on the functional role of this cell type and uncovered novel links involving neutrophils to unsuspected physiopathologic processes. For instance, evidence on their capacity to change phenotype under specific circumstances; or on their active involvement in the resolution of inflammation (other than in its regulation); or also on their unquestionable regulatory role in angiogenesis and tumor fate; or else on their response to, and release of, a wide variety of cytokines and chemotactic molecules have made it clear that the obsolete concept of the neutrophil as a "terminally differentiated, synthetically inert cell" found in most biomedical textbooks is clearly no longer tenable. Time is now mature for pathologists, cell biologists, physicians, and, why not, the same immunologists to definitively change such an outdated view and start thinking the neutrophil biology in a more modern way.

SUGGESTED READINGS

Borregaard, N., Sørensen, O.E., and Theilgaard-Mönch, K. 2007. Neutrophil granules: a library of innate immunity proteins. *Trends Immunol* **28**(8)**:**340–345.

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Fundamentals of Inflammation, edited by Charles N. Serhan, et al., Cambridge University Press, 2010. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/yale-ebooks/detail.action?docID=802921.
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Cassatella, M.A., ed. 2003. The neutrophil: an emerging regulator of inflammatory and immune response. *Chem Immunol Allergy* **83:**1–225.

Luo, H.R., and Loison, F. 2008. Constitutive neutrophil apoptosis: mechanisms and regulation. *Am J Hematol* **83**(4)**:**288–295.

Nathan, C. 2006. Neutrophils and immunity: challenges and opportunities. *Nat Rev Immunol* **6**(3)**:**173–182.

van Gisbergen, K.P., Geijtenbeek, T.B., and van Kooyk, Y. 2005. Close encounters of neutrophils and DCs. *Trends Immunol* **26**(12)**:**626–631.

Witko-Sarsat, V., Rieu, P., Descamps-Latscha, B., Lesavre, P., and Halbwachs-Mecarelli, L. 2000. Neutrophils: molecules, functions and pathophysiological aspects. *Lab Invest* **80**(5)**:**617–653.

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5 Mast Cells as Sentinels of Inflammation

Joshua A. Boyce

INTRODUCTION

Mast cells are tissue-dwelling hematopoietic effector cells that are endowed with a range of potent inflammatory effector molecules. They are implicated in both allergic and nonallergic diseases, as well as in innate and adaptive immunity to infectious agents based on animal studies. Mast cells constitutively reside in a perivascular distribution in connective tissues and at mucosal surfaces (Figure 5.1). They are especially abundant in tissues that form interfaces with the external environment (skin, conjunctivae, intestinal, and airway mucosa), suggesting a strategic placement so as to function in a first line of host defense. Mast cells are best known for their role as effectors of classic type 1 hypersensitivity (allergic) reactions. In such reactions, mast cell activation is initiated by the binding of multivalent allergen to membrane-bound IgE that is coupled with the tetrameric high-affinity Fc receptor for IgE (FcεRI) on mast cells. IgE-dependent activation of mast cells results in their release of preformed inflammatory mediators that are stored in their secretory granules, including histamine, neutral proteases, preformed cytokines, and proteoglycans (Figure 5.2). In addition, activated mast cells secrete newly synthesized lipid mediators that are the products of endogenous arachidonic acid metabolism, such as prostaglandin (PG)D_2, leukotriene (LT)B_4, and LTC_4, the parent molecule of the cysteinyl leukotrienes (cys-LTs). Finally, activated mast cells synthesize and secrete a host of proinflammatory cytokines over a period of hours. Thus, the net result of IgE-dependent tissue mast cell activation includes the rapid development of plasma extravasation, tissue edema, bronchoconstriction, leukocyte recruitment, and persistent inflammation. These events contribute to the pathogenesis of anaphylaxis, urticaria, angioedema, and acute exacerbations of asthma, each of which are diseases in which mast cells play a role.

In the past decade, experimental models have linked mast cells and their products in a broad range of inflammatory diseases, as well as host defense. Moreover, numerous activating stimuli for mast cells have been identified that do not require signaling through FcεRI. These findings suggest several contexts in which mast cells contribute to protective immunity, and disease states in which IgE-independent mechanisms of mast cell activation are functionally relevant. It therefore seems likely that mast cell–targeted treatment strategies have potential efficacy in the context of nonallergic inflammatory diseases, as well as in their traditionally accepted place in allergy.

CHARACTERISTICS OF MAST CELLS

Mature mast cells can be found in all vascularized organs. Morphologically, mast cells in tissues range between 7 and 20 μm in diameter and appear as round, spindle-shaped, or spider-like cells in tissues with round or oval nuclei. They have thin 1–2 μm processes (microplicae) emanating from their plasma membranes (Figure 5.3). Ultrastructurally, mast cells have multiple electron-dense granules (Figure 5.3). In mice infected with helminthic worms, mast cells appear in the mucosal epithelium of the intestine with strikingly altered, stellate granule morphology (Figure 5.3). Mast cells are readily identified in tissues using cationic dyes that bind their unique secretory granules. These dyes (toluidine blue, methylene blue) impart a blue-to-purple change in color known as "metachromasia" (Figure 5.1). Of all hematopoietic cells, only basophils share this staining feature. The metachromatic staining of mast cell granules reflects their content of highly sulfated heparin and chondroitin sulfate glyosaminoglycans bound to a protein core, termed "serglycin" due to the presence of alternating serine and glycine residues in its amino acid structure. The sulfated glycosaminoglycans of the mast cell granule

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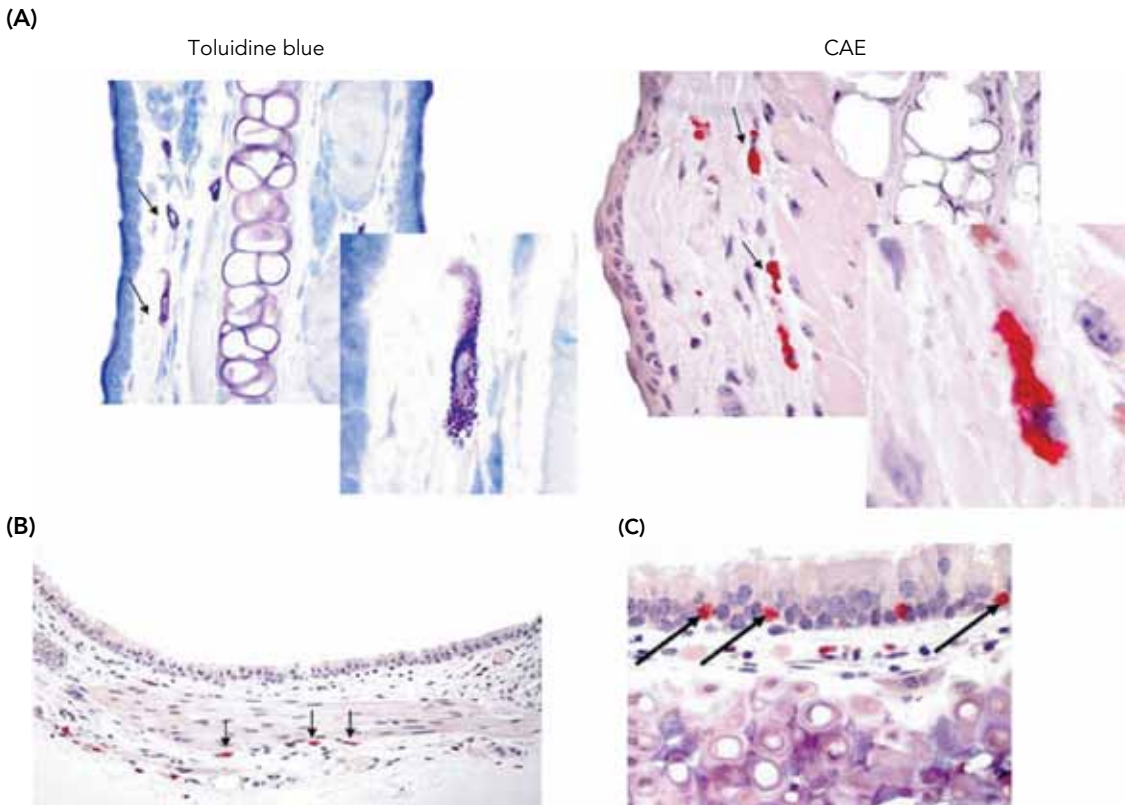


Figure 5.1. Staining features of mast cells. (A) Sections of a mouse ear showing toluidine blue (left) and chloroacetate esterase (CAE) (right) stains. Mast cells are shown using arrows. Higher magnification images (insets) are shown of typical spindle-shaped mast cells in the connective tissue of the ear. (B) CAE stains of a cross-section of a naive mouse trachea showing mast cells (arrow) in proximity to smooth muscle bundles in the submucosa. (C) Trachea of an antigen sensitized and challenged mouse showing CAE-stained intraepithelial mucosal mast cells (arrows). (Courtesy of Wei Xing, M.D., Ph.D., Harvard Medical School.)

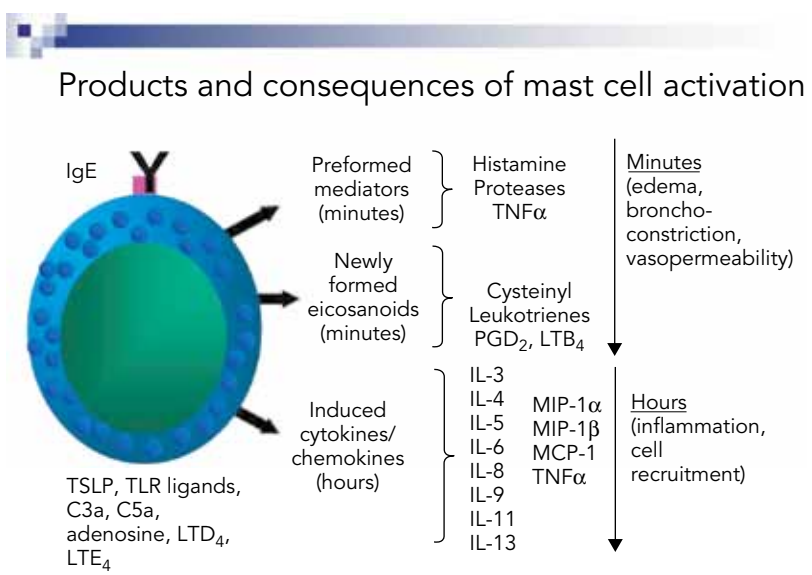


Figure 5.2. Mast cell activation products and bioactive effects. Cross-linkage of FcεRI with multivalent antigen induces exocytosis and eicosanoid formation within minutes, with attendant effects on the vascular and airway smooth muscle. Later, cytokines and chemokines are generated de novo and secreted. Several additional receptor-dependent mechanisms can induce mast cell activation independently of IgE.

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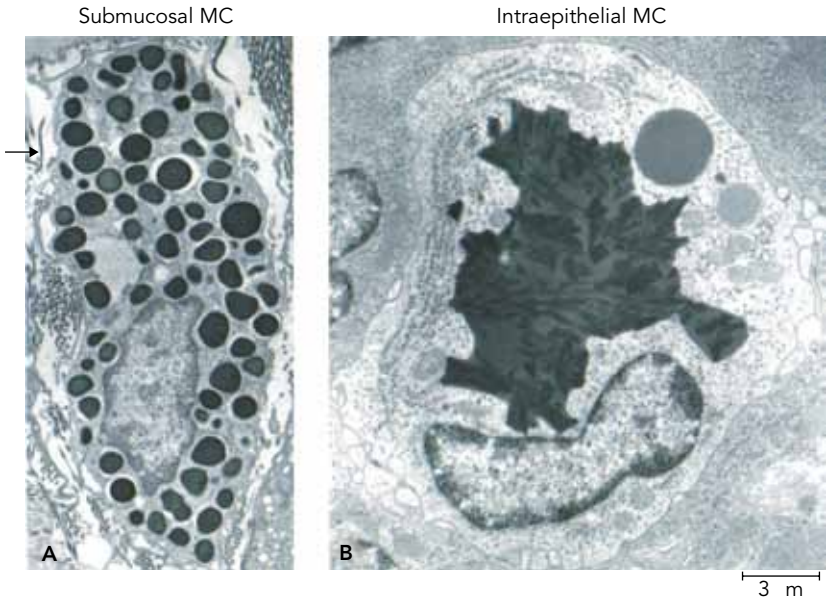


Figure 5.3. Ultrastructure of submucosa (left) and intraepithelial mast cells (right) from the jejunum of a mouse infected with *Trichinella spiralis*. Note microplicae (arrow) and dense cytoplasmic granules in the submucosal mast cell, and the large, stellate granules of the intraepithelial mast cell. (Photomicrograph courtesy of Daniel Friend, M.D., Harvard Medical School.)

core are essential for the normal storage of proteases (tryptases, chymases) and amines (histamine and serotonin). Accordingly, since proteases account for a major portion of the mast cell's granule by weight, mast cells can also be identified by immunostains or biochemical reagents that detect their proteases, such as chloroacetate esterase activity (an indicator of chymotryptic proteases) (Figure 5.1) or antibodies against tryptase or chymase in the human.

Although all known mast cells stain metachromatically with toluidine blue dye, they vary in their protease content, which is the best-established index of mast cell heterogeneity. Subsets of human tissue mast cells that contain tryptases, chymase, cathepsin G, and carboxypeptidase A (CPA) are found in perivascular connective tissues, skin, and muscularis of the uterus and intestine. In contrast, mast cells containing tryptases but lacking chymase, cathepsin G, and CPA are the dominant subtype found in lung, as well as the mucosal surfaces of the gut and nose. Although the mouse genome encodes an array of mast cell–restricted inducible chymases that do not exist in the human, it is noteworthy that two of these proteases (mouse mast cell protease [mMCP]-1 and mMCP-2) are selectively expressed by mast cells arising in the mucosal epithelium of the gut during helminth-induced intestinal inflammation. Humans with deficient T-cell function selectively lack tryptase-positive mast cells in the gut mucosa, whereas mice without functional T cells cannot develop mMCP-1- and mMCP-2-positive mast cells in the intestinal mucosa. Thus mast cell development in mucosal epithelial surfaces is T-cell-dependent, whereas mast cell development in other locations is independent of T cells. Moreover, the profile of protease expression is a marker of the subsets (see the following section). Rodent and human studies indicate that regional differences in protease expression are accompanied by heterogeneity in other effector properties, such as proteoglycan content, cytokine generation, and eicosanoid production. It seems plausible that mast cells can alter their phenotypic properties in response to factors derived from the adaptive immune system so as to fine tune their capabilities in accord with the context. It is important to note that numerous additional members of the tryptase family have been identified in recent years that are also present in mast cell granules. On the basis of precedent in rodents, it seems very likely that human mast cells will prove to be even more heterogeneous than currently thought, again reflecting "fine tuning" of their functions by the microenvironment.

MAST CELL GROWTH AND DEVELOPMENT

Role of the Kit Receptor Tyrosine Kinase and Stem Cell Factor

While mast cells originate from the bone marrow, the fact that they are found only in tissues implies that their terminal development occurs in their ultimate tissue destinations. The molecular basis of mast cell development has been studied primarily in rodents.

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The receptor protein-tyrosine kinase Kit (CD117) is encoded by the proto-oncogene c-*kit*. Two mast cell–deficient strains of spontaneously emerging laboratory mice proved to have loss-of-function mutations at the W locus at which c-*kit* resides (termed W/W^v and W^{sh}/W^{sh} mice, respectively). Another strain of mast cell–deficient mouse (Sl/Sl^d) has a loss-of-function mutation at the Steel (Sl) locus that encodes the membrane-bound form of the tissue-associated growth factor, stem cell factor (SCF). Since SCF is the ligand for Kit, the molecular defects of these three mouse strains define the fundamental requirements for mast cell development. Although no mast cell deficiency is reported in humans, a gain-of-function point mutation in Kit is associated with systemic mastocytosis, a disease of increased mast cell burden. This confirms the importance of the SCF-Kit axis in mast cell development across species. SCF is constitutively expressed as a membrane-bound protein by fibroblasts, endothelial cells, and other cell types that juxtapose to mast cells in peripheral tissues. In vitro studies using mouse and human mast cells confirm that soluble SCF protects mast cells from apoptosis, induces proliferation, can cause migration, and can induce some degree of activation and secretion independent of FcεRI and IgE. The mutant mice lacking normal Kit or SCF function are often used for diseases models to implicate functional roles for mast cells.

The identity of the Kit-positive cell population giving rise to mast cells in the tissue became evident over the past three decades as a result of several independent experimental observations. Schrader and colleagues reported that the intestinal mucosa of mice contained cells with lymphocyte-like morphology that were Thy-1 negative and capable of mast cell differentiation in vitro when treated with a splenocyte-conditioned medium. Rodewald and colleagues identified a population of cells isolated from fetal mouse blood that expressed high levels of c-Kit and low levels of Thy-1, which gave rise to pure colonies of FcεRI-positive mast cells when cultured with a combination of recombinant SCF and IL-3. These cells, termed "promastocytes," lacked developmental potential for other hematopoietic lineages. Intraperitoneal injection of purified promastocytes reconstituted the population of peritoneal mast cells in genetically mast cell–deficient W/W^v mice. More recently, fully committed mast cell progenitors, marked by their expression of CD34, FcεRI, and β_7 integrin, were identified in the intestine and in the allergen-challenged lungs of adult mice.

Human mast cell progenitors in the bone marrow and circulation have not been definitively identified but are contained within the CD34+/Kit+ cell subpopulation and are clearly distinct from the basophil lineage. Cells isolated from human blood that are CD34+/Kit+ and also positive for CD13 (aminopeptidase N), give rise to three colony types: mast cell colonies, macrophage colonies, and colonies containing mixtures of both cell types. Thus fully committed human mast cell progenitors are likely a subpopulation of the CD34+Kit+CD13+ population of cells in peripheral blood and in tissues.

Comparatively little is known about the mechanisms that control the trafficking of mast cell progenitors to tissue, their movement within tissue, and their incremental recruitment with inflammation. In mice, the β_7 integrin is instrumental to the homing of mast cell–committed progenitors to the intestine, and is also involved in the de novo recruitment of these cells to the lung during experimentally induced allergic pulmonary inflammation. Subsequent studies have implicated potential roles for the chemokine receptor, CXCR2, and for the inflammatory leukotriene LTB_4 in the recruitment of mast cell progenitors to sites of inflammation. A range of chemokines are active on human and mouse mast cells in vitro, but to date, no mast cell–selective chemokine has been identified.

Accessory Mast Cell Growth Factors

Allergic mucosal inflammation is characteristic of asthma, rhinitis, and other human diseases associated with atopy. Each of these diseases involves increases in the numbers of mast cells in the involved mucosa surface. The likely mechanisms responsible for these increases have been identified largely through studies of helminth infection in rodents. As is the case for human allergic disease, antihelminthic intestinal immune responses typically induce eosinophilia, IgE production, and Th2 cytokine generation, along with striking increases in the numbers of mast cells arising in the intestinal epithelial surface. This mucosal mast cell hyperplasia is essential for the elimination of some helminths. Not surprisingly, mast cell hyperplasia does not occur with normal kinetics or magnitude in mice lacking SCF and Kit. However, mast cell hyperplasia is also severely blunted or absent in T-cell–deficient mice, as well as mice with targeted deletions in the genes encoding IL-3 or IL-4. Antibody neutralization of either IL-3 or IL-4 also decreases the magnitude of helminth-induced mast cell hyperplasia. In contrast to the mucosal surface, the losses of IL-3 or IL-4 do not change the number of mast cells in the intestinal submucosa or in other connective tissue locations. Furthermore, mice with tissue-specific overexpression of an IL-9 transgene display a hyperplasia of mast cells in the mucosal compartments of both the intestine and lung. Thus in vivo studies implicate at least three T-cell–derived cytokines in mast cell hyperplasia that are generated during Th2 immunity in mice.

In vitro studies indicate that IL-3, IL-4, and IL-9 can control the rate of SCF-driven mast cell proliferation

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(comitogenesis), rate of apoptosis, or expression of certain key genes. IL-3 can substitute for SCF for inducing mast cell growth in vitro from mouse bone marrow (although it lacks this activity in cultures of human bone marrow or cord blood). Although human IL-3 fails to promote mast cell growth by itself, the IL-3 receptor is constitutively expressed by human mast cell progenitors, is inducible on mature human mast cells, and mediates an SCF-driven comitogenic effect on human mast cells and their progenitors in vitro. In the absence of SCF, IL-3 also sustains the survival of human mast cells derived from cord blood or isolated from surgically resected intestinal tissue. IL-9 induces the expression of the mucosal proteases mMCP-1 and mMCP-2 by mouse mast cells in vitro, and upregulates the expression of several mRNA transcripts involved in the control of cell survival and cell cycle progression in human mast cells. IL-10 shares the feature of inducing mMCP-1 and mMCP-2 expression by mouse mast cells with IL-9, and also strongly stimulates histamine synthesis and granule development.

Of the T-cell–derived cytokines, IL-4 may be especially pivotal regulating functional responses of mast cells in mucosal inflammation. First, IL-4 is important for Th2 cell development and the production of IgE. Second, IL-4 has several direct effects on the development and/or function of mast cells. In vitro, IL-4 amplifies SCF-driven proliferation of both mouse and human mast cells in vitro, and strongly upregulates the expression of leukotriene C_4 synthase (LTC_4S), the terminal enzyme needed for the production of LTC_4, the parent of the cys-LTs, in the mast cells of both species. Surprisingly, the induction of LTC_4S is necessary for a normal proliferative response to IL-4, which is markedly blunted in human or mouse mast cells in which cys-LT generation is blocked by either pharmacologic or genetic means. Extracellular cys-LTs can induce mast cell proliferation in vitro by acting at the type 1 receptor for cys-LTs ($CysLT_1R$) and inducing transactivation of Kit. These findings may explain a profound deficiency in intraepithelial mast cells identified in LTC_4S knockout mice subjected to a model of allergen-induced pulmonary inflammation. Collectively, these studies confirm the dynamic nature of mast cell phenotype and its dependency on tissue-specific determinants such as local cytokines, and implicate a major potential role for cys-LTs as signaling molecules in amplifying mast cell development with mucosal inflammation.

Transcription Factors and Signaling Molecules

The microphthalmia transcription factor (MITF) is a basic helix–loop–helix leucine zipper-type protein that is central to mast cell and melanocyte development. Mice with a loss-of-function mutation in the *mitf* gene have reduced the number of tissue mast cells, albeit not to the same extent as mice bearing mutations in the *c-kit* or *scf* genes. MITF controls c-*kit* transcription in mast cells, and regulates the expression of several proteases. MITF is also important in controlling the expression of hematopoietic prostaglandin D_2 synthase (H-PGDS), the terminal enzyme required for the synthesis of prostaglandin D_2 (PGD_2), the most abundant cyclooxygenase product of mast cells. The guanine nucleotide exchange factor RasGRP4, which is highly mast cell restricted, is also required for the normal expression of H-PGDS, suggesting that RasGRP4 and MITF might cooperate in a coordinated transcriptional system. To date, the parallel roles of MITF and RasGRP4 have not been defined in human mast cells.

The GATA family of zinc finger transcription factors are also involved in mast cell development. GATA-1 is involved in the differentiation and survival rates of mast cell precursors, and promotes expression of the FcεRI α- and β-chains. GATA-2 is important for the differentiation of mast cells from yolk sac precursors. The GATA consensus sequence is found in the promoter region of genes expressed by mast cells, including CPA, several proteases, and FcεRIα. PU.1, an Ets family transcription factor, may also play a role in the regulation of mast cell development, possibly in cooperation with GATA-2.

MAST CELL ACTIVATING RECEPTORS

FcεRI

FcεRI-dependent mast cell activation is a primary mechanism by which allergens induce effector responses in allergic disease. All tissue mast cells express the variant of FcεRI composed of an Fc-binding α subunit, a β subunit, and two γ subunits. Occupied by monomeric allergen-specific Ig, FcεRI is cross-linked by multivalent allergens, resulting in signal transduction events that result in immediate granule fusion and exocytosis, arachidonic acid metabolism, and induction of cytokine and chemokine gene transcription. The importance of these processes to the pathophysiology of allergic responses is undisputed. Early and late-phase allergic responses to allergen challenge in the nose and lung of atopic individuals are blocked by the administration of cromolyn, a drug that inhibits mast cell activation, and can also be blocked by the administration of a therapeutic anti-IgE antibody. IgE-driven early and late-phase reactions induced experimentally in mice are also mast cell dependent.

While the role of FcεRI in allergic disease is well established, its role in the normal function of the immune system is less clear. Mice genetically deficient in FcεRI develop normally and have normal T- and B-cell functions, and can eliminate *Schistosoma*

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mansoni from the intestine with normal kinetics after experimental infection. Nonetheless, they do develop increased hepatic granulomas and hepatic fibrosis. Deficiencies in mast cells or IgE also impair the ability of mice to eliminate *Haemaphysalis longicornis* ticks, suggesting a role for IgE-mediated hypersensitivity for resistance against this parasite. The elimination of adult *T. spiralis* worms from the intestine is delayed in an IgE-null strain of mice, and the number of viable larvae encysting in the skeletal muscles doubled in the IgE-null mice compared to infected wild-type controls. Thus, IgE may serve an amplifying function in the containment and/or elimination of certain multicellular parasites. However, there is no infectious disease recognized to date that absolutely requires IgE or FcεRI for protective immunity.

FcεRI expression on the surface of mast cells and basophils is regulated at several levels. FcεRI is stabilized at the cell surface on encounter with IgE. Peritoneal mast cells from IgE-null mice show strikingly lower levels of surface FcεRI than do wild-type littermates, which increase with the intravenous administration of IgE. Circulating basophils from atopic individuals show levels of surface FcεRI that correlate with circulating levels of IgE. Moreover, depletion of IgE in humans using a therapeutic antibody sharply decreases FcεRI surface expression by circulating basophils and mast cells in both the lungs and the skin. Surface expression of FcεRI on mast cells ex vivo are increased, in a dose-dependent manner, by the addition of IgE. Incubation of in vitro bone marrow–derived mast cells with IgE enhanced not only their FcεRI expression but also signal transduction and mediator release in response to FcεRI cross-linking. Certain clones of IgE can also induce cytokine secretion and sustain the survival of mouse mast cells independently of antigen in vitro. It is not clear whether this mechanism operates in vivo or in humans.

Cytokines can also regulate FcεRI expression and signaling. Human mast cells derived in vitro respond to the addition of exogenous IL-4 with dose- and time-dependent enhancement of their surface FcεRI expression. Mast cells from human intestine have similar responses to IL-4. In both instances, the effect of IL-4 is to drive transcription of FcεRIα, providing more subunits for surface expression. Not surprisingly, this mechanism is synergistic with the stabilization of surface receptors by the addition of exogenous IgE. Both mechanisms amplify mediator release in response to cross-linkage of FcεRI.

Fcγ Receptors

In rodents, anaphylaxis can occur through IgG-dependent activation of the low-affinity receptor for IgG, FcγRIII. This receptor shares β and γ subunits with the FcεRI, but has not been identified on human mast cells. Human mast cells exposed ex vivo to IFNγ inducibly express high-affinity FcγRI receptors, and generate the same mediator profile in response to cross-linkage of this receptor as they do via FcεRI. The importance of mast cell activation by these IgG-dependent pathways in human health and disease is unknown.

Pattern Recognition Receptors

In recent years, several studies have linked the activities of mast cells to the innate immune response, reflecting their capacity for activation in response to pathogen-derived molecules and endogenously generated mediators. On the basis of studies of *c-kit* mutant mice, mast cells are required for protective neutrophilia occurring in response to experimentally induced Gram-negative septic peritonitis and pneumonia. The activation of mouse mast cells in response to virulent strains of *Klebsiella pneumoniae* was linked to CD48, a glycosylphosphatidylinositol-anchored protein. Mouse mast cells also express a number of Toll-like receptors (TLRs). In Gram-negative septic peritonitis, expression of TLR4 by mast cells is required to induce normal protective neutrophilia for microbial clearance. Skin mast cells are activated by bacterial peptidoglycan through TLR2 to mediate the influx of granulocytes to the site of intracutanous challenge with staphylococcal peptidoglycan.

G-Protein–Coupled Receptors

The complement fragments C3a and C5a ("anaphylotoxins") are established agonists for mast cell activation that can be generated as by-products of both adaptive and innate immune responses. Each induces signaling through a specific GPCR. Human mast cells stimulated by exogenous cys-LTs secrete cytokines and chemokines but do not release histamine. LTD_4, the initial extracellular metabolite of LTC_4, triggers mast cell activation through the type 1 receptor for cys-LTs ($CysLT_1R$). LTE_4, the terminal product of the cys-LTs, also activates human mast cells ex vivo to generate PGD_2 and chemokines, but does so through a pathway involving an uncharacterized GPCR that induces a secondary signal through peroxisome proliferator-activated receptor (PPAR)γ, a lipid-activated transcription factor. Mast cell activation through the adenosine A3 receptor can also induce mediator release that potentiates airway reactivity in mice. It is likely that these GPCR-dependent mechanisms for mast cell activation can elicit or modify mediator release in nonallergic diseases and host defense responses that involve contributions from mast cells.

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Effector Cytokine Receptors

Recently, tissue-derived effector cytokines have been identified as potentially major inducers of mast cell activation. IL-33, which signals through T1/ST2, and thymic stromal lymphopoietin (TSLP), which signals through a heterodimeric receptor sharing a subunit with the IL-7 receptor, are both generated by injured or inflamed epithelium. Mast cells express the receptors for both cytokines, and both can potently induce mast cells to generate cytokines and chemokines. Thus, IL-33 and TSLP provide potential mechanisms whereby mucosal injury can induce a direct response from mast cells without a requirement for antibody or FcεRI.

INHIBITORY RECEPTORS

Immunoglobulin-like and Fc Receptors

Mast cells express several receptors that inhibit their activation, potentially contributing to homeostasis in mast cell–dependent inflammatory responses. Mouse mast cells possess at least four inhibitory receptors that contain an immunoreceptor tyrosine-liked inhibitory motif (ITIM). These include the closely related low-affinity Fc receptors for IgG, FcγRIIb1 and FcγRIIb2, leukocyte immunoglobulin-like receptor B4 (LILRB4, previously called GP49B1), and CD300a. Each of these receptors inhibits exocytosis and eicosanoid generation when colligated with FcεRI. Mice with targeted disruption of FcγRIIβ or LILRB4 show exaggerated experimentally induced IgE-dependent anaphylactic responses in vivo. RNA transcripts for LILRB4 have been detected in human mast cells, but their function has not been confirmed. Unlike FcγRII, the natural counterligands for LILRB4 and CD300a are not known.

G-Protein–Coupled Receptors

Mast cell activation can be inhibited by GPCRs that couple to G_s proteins and adenylyl cyclase (AC). Stimulation of mast cells through the β_2 adrenergic receptor, the A2B receptor for adenosine, and the EP_2 receptor for PGE_2 all block IgE-dependent mast cell activation responses. Inhalation of PGE_2 by subjects with asthma prevents both early and late asthmatic responses. A recent study revealed that the A2B receptor for adenosine plays a major role in regulating endogenous levels of cyclic AMP in mouse mast cells, and A2B knockout mice display exaggerated IgE-dependent anaphylactic responses. Since GPCRs are good targets for pharmacologic compounds, these G-coupled receptors all represent promising therapeutic targets, and agonists of the β_2 adrenergic receptor are stables of asthma treatment.

MAST CELL PATHOBIOLOGY

Anaphylaxis

IgE-dependent activation of mast cells (and possibly of basophils) is thought to underlie the pathophysiology of all systemic allergic reactions. The panel of mediators derived from mast cells include those that induce urticaria (histamine), vascular leak (histamine, LTD_4, PGD_2), hypotension (histamine), bronchoconstriction (histamine, LTC_4, PGD_2), flushing (PGD_2), and intestinal peristalsis (LTC_4), all of which are associated signs or symptoms of anaphylaxis. The serum levels of mature β-tryptase, a protease stored in abundance in the granules of mast cells, are elevated during the first 2–4 hours after many anaphylactic events. Because of its relatively long half-life when compared with histamine, β-tryptase is a more useful marker of mast cell activation in vivo than is histamine. Interestingly, "pseudoallergic" or "anaphylacticoid" systemic reactions to cyclooxygenase inhibitors and radiocontrast media, which are not thought to be mediated by IgE, can also be associated with elevated levels of β-tryptase, indicating the likely involvement of mast cells in these syndromes after their activation by idiosyncratic mechanisms. Surprisingly, β-tryptase levels are often not elevated in anaphylaxis caused by food ingestion. Although this may indicate a principal role for basophils, the lack of basophil-specific markers of activation precludes a definitive explanation at this time.

Asthma

Most evidence from human studies supports a role for mast cells and their mediators as effectors in asthma. Bronchial biopsies from patients with asthma contain increased numbers of mast cells relative to biopsies obtained from nonasthmatic control subjects, irrespective of whether the asthmatic patients are atopic or nonatopic. The increased numbers of mast cells reflect distributions in the mucosal epithelium and in the bronchial smooth muscle layer, the latter being a region in which mast cells are rarely found in nonasthmatic individuals or patients with eosinophilic bronchitis (a syndrome of bronchial mucosal inflammation without AHR or airflow obstruction). In contrast, substantial numbers of mast cells are found at this tissue site in asthmatic patients. The increase in the numbers of mast cells in the mucosal epithelial surface of biopsy specimens from asthmatic patients likely reflects the T-cell–driven pathway that amplifies mast cell development. The numbers of mast cells in bronchial biopsy specimens from asthmatic patients predict therapeutic failure during weaning from glucocorticoids.

Mast cell degranulation is prominent in postmortem bronchial tissues from patients who die from

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asthma and is also observed in bronchial biopsy specimens from living asthmatics. The levels of tryptase and other mast cell mediators often are markedly increased in the bronchoalveolar lavage (BAL) fluid of asthmatic patients relative to their levels in specimens obtained from nonasthmatic controls, even without allergen provocation. Thus, both mast cell hyperplasia and ongoing mast cell activation are characteristic features of asthma. Therapeutic anti-IgE is an efficacious treatment for atopic asthma. However, it is unlikely that FcεRI-dependent mechanisms account for the entirety of mast cell contributions to asthma, particularly since mast cell numbers (and indices of local mast cell activation) are increased in both atopic and nonatopic individuals.

Mastocytosis

Mastocytosis refers to a group of disorders that are characterized by increased mast cell burden in the skin and/or internal organs. When mastocytosis is limited to the skin (cutaneous mastocytosis), it may be isolated to one lesion (mastocytoma), or more commonly is manifested by multiple brown-to-tan, flat lesions (urticaria pigmentosa) that swell with gentle pressure (Darier's sign). The lesions of urticaria pigmentosa are usually present in cases of systemic mastocytosis as well, defined by mast cell infiltration of internal organs (lymph nodes, spleen, bone marrow). As many as 80% of patients with systemic mastocytosis bear a somatic mutation of the *c-kit* gene resulting in a single amino acid substitution (D816V) that renders the Kit receptor constitutively active. There are several subtypes of systemic mastocytosis (indolent, aggressive, and others) that vary in clinical presentation and prognosis, all of which are associated with this single mutation, raising the likelihood of additional genetic or environmental factors that modify the clinical course. Unfortunately, the D816V variant of Kit resists inhibition by currently available tyrosine kinase inhibitors. Most of the signs and symptoms of systemic mastocytosis (flushing, pruritis, headache, diarrhea and abdominal pain, and hypotension) are attributable to nonspecific mast cell mediator release. Total pro-β-tryptase levels (but not mature β-tryptase levels) are generally elevated in systemic, but not cutaneous, mastocytosis, as a reflection of total body mast cell burden.

Other Allergic Conditions

All atopic diseases (i.e., associated with allergen-specific IgE) involve mast cell activation. These include allergic rhinitis, rhinosinusitis, allergic conjunctivitis, oral allergy syndrome, and atopic dermatitis. The comparative success of H1 histamine receptor blockade for the treatment of these conditions provides strong support for the pathogenetic contributions of mast cells. In addition, diseases in which allergen-specific IgE is not demonstrable, such as chronic urticaria, urticaria, or anaphylaxis provoked by physical stimuli (such as exercise, cold exposure, or heat exposure) also involve contributions from mast cells. In most of these circumstances, the mechanism of activation remains obscure but may involve autoantibodies or neural circuits.

Atherosclerosis

Degranulated mast cells characteristically present in atherosclerotic plaques and sites of plaque rupture. The serum of patients presenting with acute myocardial infarction show elevations of both histamine and mature β-tryptase, suggesting mast cell activation occurs concomitantly with ischemia. Mast cell granule constituents (heparin, proteases) can contribute to foam cell formation and cholesterol accumulation through effects on lipoproteins, and plaque instability through effects on the extracellular matrix and inhibition of smooth muscle proliferation. Mast cell chymase also facilitates angiotensin-II formation and endothelin metabolism. Mast cell–derived eicosanoids can serve as vasoconstrictors and can promote neutrophil recruitment. Thus, there are several potential mechanisms by which mast cells could contribute to the pathogenesis of acute vascular events.

Roles Implicated by Animal Models

The increasing use of *c-kit* mutant mice in experimental models has led to a steadily increasing number of diseases involving mast cells. As noted earlier, mast cells are essential for the initiation of protective innate immunity to bacteria in certain experimental models. This function in part reflects their ability to store and rapidly generate and release TNFα. More recently, a critical role for the tryptase mMCP-6 in septic peritonitis was inferred from studies of mice lacking this protease. Other studies have linked the protective neutrophilia to the production of LTB_4, a powerful neutrophil chemotactic factor. Models of contact dermatitis, inflammatory arthritis, aortic aneurisms, bullous pemphigoid, and multiple sclerosis all implicate potential roles for mast cells as effectors. Although evidence for mast cell contributions to these diseases in humans is inferential at present, it seems possible that mast cells may prove to be therapeutic targets in at least some of these disorders, provided the responsible mediators and activation mechanisms become evident. Since animal models also support prominent roles for mast cells and their mediators in protective immunity to helminths, bacteria, and a range of other pathogens, the challenge

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for therapeutic development is to maintain these benefits without sacrificing efficacy.

SUGGESTED READINGS

Bischoff, S. C., Sellge, G., Lorentz, A., et al. 1999. IL-4 enhances proliferation and mediator release in mature human mast cells. *Proc Natl Acad Sci USA* **96**(14):8080–8085.

Enerback, L. 1966. Mast cells in rat gastrointestinal mucosa. I. Effects of fixation. *Acta Pathol Microbiol Scand* **66**(3):289–302.

Godfraind, C., Louahed, J., Faulkner, H., et al. 1998. Intraepithelial infiltration by mast cells with both connective tissue-type and mucosal-type characteristics in gut, trachea, and kidneys of IL-9 transgenic mice. *J Immunol* **160**(8):3989–3996.

Hsieh, F. H., Lam, B. K., Penrose, J. F., et al. 2001. T helper cell type 2 cytokines coordinately regulate immunoglobulin E-dependent cysteinyl leukotriene production by human cord blood-derived mast cells: profound induction of leukotriene C(4) synthase expression by interleukin 4. *J Exp Med* **193**(1):123–133.

Jiang, Y., Feng, C., Backskai, B., and Boyce, J.A. 2007. CysLT2 receptors interact with CysLT1 receptors and down-modulate cysteinyl leukotriene-dependent mitogenic responses of mast cells. *Blood* **100**:3263–3270.

Jiang, Y., Kanaoka, Y., Feng, C. L., Nocka, K., Rao, S., and Boyce, J.A. 2006. Cutting edge: interleukin 4-dependent mast cell proliferation requires autocrine/intracrine cysteinyl leukotriene-dependent signaling. *J Immunol* **177**:2755–2759.

Kambe, N., Hiramatsu, H., Shimonaka, M., et al. 2004. Development of both human connective tissue-type and mucosal-type mast cells in mice from hematopoietic stem cells with identical distribution pattern to human body. *Blood* **103**(3):860–867.

Kirshenbaum, A. S., Goff, J. P., Semere, T., et al. 1999. Demonstration that human mast cells arise from a progenitor cell population that is CD34(+), c-kit(+), and expresses aminopeptidase N (CD13). *Blood* **94**(7):2333–2342.

Kitamura, Y., and Go, S. 1979. Decreased production of mast cells in S1/S1d anemic mice. *Blood* **53**(3):492–497.

Kitamura, Y., Go, S., and Hatanaka, K. 1978. Decrease of mast cells in W/Wv mice and their increase by bone marrow transplantation. *Blood* **52**(2):447–452.

Li, L., Yang, Y., and Stevens, R. L. 2003. RasGRP4 regulates the expression of prostaglandin D2 in human and rat mast cell lines. *J Biol Chem* **278**(7):4725–4729.

Madden, K. B., Urban, J. F., Jr., Ziltener, H. J., et al. 1991. Antibodies to IL-3 and IL-4 suppress helminth-induced intestinal mastocytosis. *J Immunol* **147**(4):1387–1391.

Morii, E., and Oboki, K. 2004. MITF is necessary for generation of prostaglandin D2 in mouse mast cells. *J Biol Chem* **279**(47):48923–48929.

Paruchuri, S., Jiang, Y., Feng, C., Francis, S. A., Plutzky, J., and Boyce, J. A. 2008. Leukotriene E4 activates peroxisome proliferator activated receptor gamma and induces prostaglandin D2 generation by human mast cells. *J Biol Chem* **283**:16477–16487.

Reynolds, D. S., Stevens, R. L., Lane, W. S., et al. 1990. Different mouse mast cell populations express various combinations of at least six distinct mast cell serine proteases. *Proc Natl Acad Sci USA* **87**(8):3230–3234.

Rodewald, H. R., Dessing, M., Dvorak, A. M., et al. 1996. Identification of a committed precursor for the mast cell lineage. *Science* **271**(5250):818–822.

Ruitenberg, E. J., and Elgersma, A. 1976. Absence of intestinal mast cell response in congenitally athymic mice during *Trichinella spiralis* infection. *Nature* **264**(5583):258–260.

Schrader, J. W., Scollay, R., and Battye, F. 1983. Intramucosal lymphocytes of the gut: Lyt-2 and thy-1 phenotype of the granulated cells and evidence for the presence of both T cells and mast cell precursors. *J Immunol* **130**(2):558–564.

Schwartz, L. B., Metcalfe, D. D., Miller, J. S., et al. 1987. Tryptase levels as an indicator of mast-cell activation in systemic anaphylaxis and mastocytosis. *N Engl J Med* **316**(26):1622–1626.

Toru, H., Ra, C., Nonoyama, S., et al. 1996. Induction of the high-affinity IgE receptor (Fc epsilon RI) on human mast cells by IL-4. *Int Immunol* **8**(9):1367–1373.

Vyas, H., and Krishnaswamy, G. 2006. Paul Ehrlich's 'Mastzellen' – from aniline dyes to DNA chip arrays: a historical review of developments in mast cell research. *Methods Mol Biol* **315**:3–11.

Weidner, N., and Austen, K. F. 1991. Ultrastructural and immunohistochemical characterization of normal mast cells at multiple body sites. *J Invest Dermatol* **96**(3 Suppl):26S–31S.

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6 Basophils

Jonathan Arm and David Sloane

INTRODUCTION

Named after their affinity for alkaline stains and possessed of numerous potent inflammatory substances, basophils are the rarest of the granulocytes, typically constituting 0.5%–1.5% of peripheral blood leukocytes. Studied both for their suspected effector roles in parasitic and allergic diseases and for their similarity to mast cells, these peripheral blood cells are increasingly hypothesized to have important immunoregulatory functions. This chapter focuses primarily on the human basophil, but makes reference to the basophils of other species to point out phenotypic differences from human basophils, to describe phenomena that have been elucidated best in nonhuman basophils, and to describe murine and other mammalian models of disease in which the role of the basophil has been explored.

IDENTIFICATION

Historical Discovery

Paul Ehrlich, the German Nobel prize winner who made foundational contributions to immunology, is credited with the first description of the human basophil in 1879 in a paper in which he also proposed the name *Mastzellen* for connective tissue cells that also took up aniline dyes in their numerous cytoplasmic granules. Because of their infrequency among peripheral blood leukocytes, basophils were given scant attention until their functional similarity to mast cells in terms of IgE-mediated histamine release was appreciated more than eight decades later.

Development

Like other leukocytes, basophils derive from totipotent hematopoietic stem cells, and are believed to arise from the common myeloid precursor by means of the common granulocyte precursor. However, the penultimate and ultimate steps in basophil development are presently unclear. There are two models of basophil ontogeny, both of which are supported by experimental data. The first posits that basophils and mast cells share a common parent cell distinct from that which produces eosinophils. This hypothesis is most strongly supported by a murine model in which a bipotent cell could differentiate into either mast cells or basophils in the mouse spleen. The discovery of a cell with ultrastructural features of mast cells and basophils in humans with chronic myelogenous leukemia agrees with this model. However, a genetic analysis using the D816V mutation in the c-kit receptor in patients with the disease systemic mastocytosis as a trackable marker did not suggest a close lineage relationship between mast cells and basophils, as basophils were found to carry the mutation in a minority of patients (5 of 33) and only when other lineages (monocytes, neutrophils) also harbored D816V. If a common basophil mast cell precursor cell exists, it would be expected to carry the D816V mutation more frequently in mastocytosis and would occur at least occasionally in the absence of other lineage involvement.

The second, competing hypothesis is that there is a basophil–eosinophil precursor distinct from a cell that gives rise to mast cells. This hypothesis is supported by the finding that peripheral blood cells from atopic donors can, when cultured in methylcellulose and with conditioned media, produce colonies from single cells that are either pure basophils or mixtures of basophils and eosinophils, suggesting the presence of a circulating basophil–eosinophil precursor. Quantification of such precursors demonstrates that atopic patients have fluctuating numbers of these bipotent progenitors, with fewer observed during steroid treatment and more detected after allergen challenge. Similar cells have been obtained from the bone

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marrow and peripheral blood of patients with myeloid leukemias, and hybrid cells with histochemical and ultrastructural features of both basophils and eosinophils can be derived from normal human cord blood precursors.

Drawing firm conclusions from these studies is limited by the potentially confounding effects of pathological conditions in human donors (allergic diseases, mastocytosis, malignancy), medications donors take to treat these diseases, and the lack of details and variation in culture conditions (cytokines, growth factors, and conditioned media) among studies. The possibility that there may be two or more basophil subpopulations, one more closely related to mast cells, another more akin to eosinophils, has not been explored, but would resolve the apparently conflicting data and would suggest that basophils resemble dendritic cells in their developmental heterogeneity. An alternative reconciling hypothesis is that basophils and/or eosinophils may, in some contexts, (de)differentiate into hybrid cells, but definitive studies documenting this are presently lacking.

The current dogma posits that basophils exit the bone marrow as fully differentiated cells, as they are identified in peripheral blood relatively easily, and once isolated are functional for mediator release. However, basophils in inflamed tissues have distinct biochemical properties and may survive significantly longer than those in the circulation, suggesting that blood basophils may be partially functional cells that only fully mature after diapedesis or that they have a more plastic phenotype than currently appreciated.

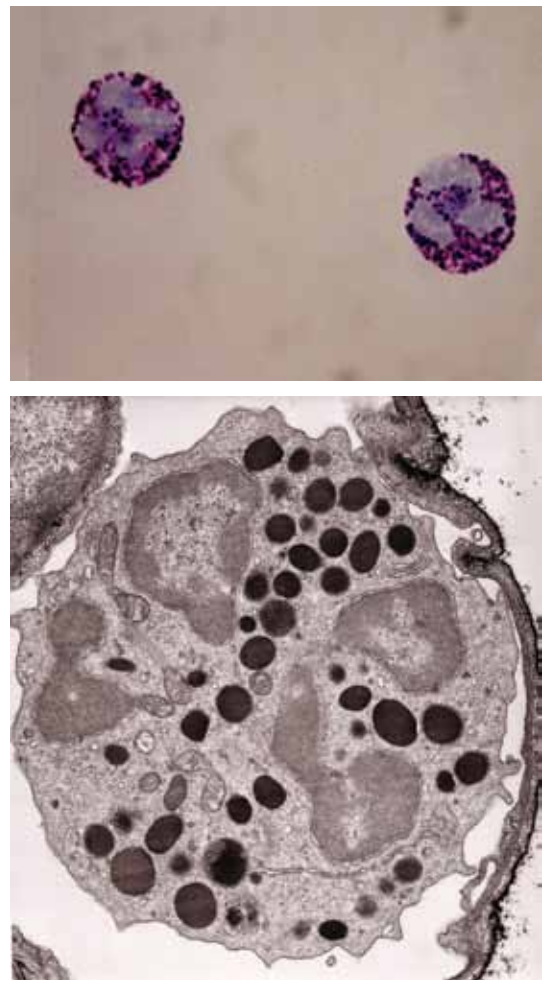


Figure 6.1. (A) Light micrographs of human peripheral blood basophils stained with Wright's Giemsa. Note the numerous dark purple, metachromatic granules, and the bilobed or trilobed (close up) nucleus. (B) Electron micrograph of a human basophil in a capillary.

Phenotypic Characteristics

Appearance

As at the time of their discovery, basophils are still identified routinely by their microscopic appearance. In addition, their identity is assessed by analysis of their pattern of expression cell surface proteins.

Light Microscopy Image

Basophils from peripheral blood avidly take up alkaline stains, such as alcian blue and trypan blue. Importantly, the granules of basophils (and mast cells) acquire a purple, not blue (as seen in other granulated cells such as monocytes and neutrophils), color upon interaction with these dyes; hence, the staining is termed "metachromatic" (Figure 6.1A). Basophils have numerous, moderately sized, metachromatic cytoplasmic granules and a bean-shaped or bilobed nucleus. Immunohistochemically, peripheral blood basophils from normal donors generally have a very low content of tryptase and do not contain chymase, classical markers of the mast cell.

Electron Microscopy Image

Viewed by electron microscopy, human basophils have a diameter of 5–7 microns, numerous amorphous round granules, and "stubby" cell surface processes (Figure 6.1B). Compared to basophils, tissue mast cells are larger; have more numerous, smaller, and more electron dense granules that have either a lattice or scroll substructure; and possess distinctly longer and thinner plasma membrane processes. Basophils undergo dramatic ultrastructural changes with activation, but the pattern and degree of alteration vary depending on the particular stimulus used. Thus, for example, "anaphylactic degranulation," with granule-to-granule fusion or granule-to-plasma membrane fusion was seen best with stimuli that engage the high-affinity Fc receptor for IgE (FcεRI), while recombinant histamine releasing factor was most potent in inducing uropod formation, and the more gradual process of piecemeal degranulation

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was observed after stimulation by monocyte chemotactic protein-1.

Cell Surface Markers

Basophils are often identified by their pattern of plasma membrane proteins. Like mast cells, basophils have high cell surface binding of IgE by means of their expression of FcεRI. CD123, the α chain of the IL-3 receptor, is also easily detected on these cells, as are CD11b and CD13. Often, the positive expression of these two proteins is used in conjunction with the absence of expression of a panel of lineage markers (such as CD2, CD14, CD16, CD19, and MHC class II) to identify basophils on flow cytometry (Figure 6.2).

Over a decade ago, the murine monoclonal antibody 2D7 was found to bind to the secretory granules of basophils (but not those of eosinophils, neutrophils, or mast cells). Likewise, the antibody BB1+ has been reported to be basophil specific. Antibody J175–7D4 stains basophils based on its binding to promajor basic protein, which is not expressed by mature eosinophils.

Cell surface protein expression is highly dynamic. With basophil activation through cross-linking of FcεRI, there is increased expression of a number of basophil cell surface markers, including CD13, CD63, CD107a, CD164, and CD203C. This observation is the basis of various *in vitro* basophil activation assays that have been used to identify circulating allergen/antigen-specific IgE. Expression of specific cell surface receptors through which basophils may be activated is described later.

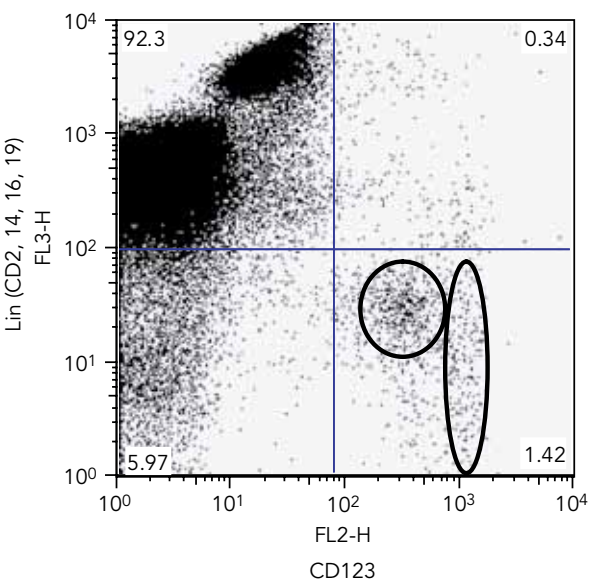


Figure 6.2. Flow cytometric identification of peripheral blood basophils. Basophils (right lower quadrant, within the bold circle) are typically positive for CD123 (the IL-3 receptor α chain, x-axis) and negative for the lineage markers IL-2, -14, -16, and -19 (y-axis). The cells within the ellipse are plasmacytoid dendritic cells, which may be distinguished from basophils by their expression of other cell surface markers such as blood dendritic cell antigen-2 (BDCA-2, CD303), and BDCA-4 (neuropilin-1, CD304), not shown here.

Products of Basophils

Histamine

Histamine, the best known mediator released by basophils, is present preformed in cytoplasmic granules, and is rapidly released in response to stimuli that induce degranulation. While the platelets of some species (e.g., rabbits) also contain histamine, basophils seem to be the exclusive source of this mediator in human peripheral blood. The histamine content of basophils is typically on the order of 1.0 pg per cell. When quantifying the release of histamine from basophils, results are often reported as a percent of histamine release, with the total histamine content being determined by complete lysis of the cells using hypotonic and/or acidic media. Thus, for example, extensive cross-linking of FcεRI on a population of basosphils with allergen, anti-IgE, or anti-FcεRI may result in 50%–90% histamine release, with exocytosis beginning within 2–5 minutes of activation and completed by 30–45 minutes.

Lipid Mediators

Unlike histamine, the major lipid mediators of basophils, leukotriene C4 (LTC_4), and platelet activating factor (PAF) are newly synthesized upon cell activation. Although not preformed, the kinetics of their release after basophil activation with physiological and nonphysiological stimuli rivals that of histamine, initially being detected 1–5 minutes after activation and approaching maximal release by 15 minutes. When purified basophils are activated, the released LTC_4 (<5–80 ng/million basophils) remains stable in solution. However, when basophils are selectively stimulated in the presence of other leukocytes, such as the mononuclear cells (monocytes, T cells, B cells, NK cells, and dendritic cells) with which they copurify during density gradient centrifugation, there is further metabolism of LTC_4 to the most stable of the cysteinyl leukotrienes, LTE_4. Unlike mast cells, basophils lack the enzyme hematopoietic prostaglandin (PG)D_2 synthase, and are thus incapable of generating PGD_2.

Cytokines and Chemokines

The discovery that basophils release significant amounts of important immunoregulatory proteins is arguably the seminal finding that rescued the cell from being relegated to the status as a "mere" innate immune effector cell or surrogate mast cell. It also

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justified the exploration of what roles basophils might play in the initiation, maintenance, and resolution of inflammation, especially that of allergic diseases.

Chief among the cytokines produced by basophils is interkeukin-4 (IL-4), which is released by basophils in response to stimulation both through FcεRI (by means of specific allergen, anti-IgE, or anti-FcεRI – see the discussion of this receptor later) and via IgE-independent stimuli such as the complement fragment C5a. Basophils possess mRNA encoding IL-4 at baseline, but cell activation augments this by more than threefold. The mass of IL-4 released varies among donors, ranging from less than 20 to more than 600 pg/million basophils. The kinetics of secretion are slower than that of histamine and the lipid mediators, with a half time to maximal release of approximately 1.5 hours and maximal release generally seen by 4–8 hours. The inhibition of IL-4 release by basophils in the presence of cyclohexamide indicates that new protein synthesis is required.

Basophils also release IL-13 (<20–2,000 pg/million cells) in response to IgE cross-linking and to non-IgE-mediated stimulation. As with IL-4, new protein synthesis is required, but maximal release may not be achieved until 16–24 hours, depending on the agonist. One study of normal and asthmatic subjects revealed that during the first 6 hours after specific allergen challenge *in vitro*, basophils outnumbered T cells by fourfold as sources of IL-4 and by twofold as sources of IL-13.

Basophils release macrophage inflammatory protein-1α (MIP-1α) in amounts and according to kinetics similar to those of IL-13 secretion, and are also sources of the cytokine GM-CSF chemokines produced by basophils in response to IgE cross-linking include IL-8, MIP-5, and eotaxin.

Others

Although best known as the products of eosinophils, major basic protein and Charcot–Leyden crystals containing lysophospholipase are produced by basophils. Likewise, basophils contain small amounts of mast cell tryptase, and release this mediator in response to anti-IgE. Recently, the cytotoxic serine protease granzyme B has been detected from activated basophils, and basophils were able to directly kill target cells at least in part by means of this mediator, in a caspase-dependent mechanism and to a degree rivaling natural killer (NK) cells.

ISOLATION TECHNIQUES

Positive Selection

Basophils comprise 0.5%–1.5% of circulating leukocytes. They can be enriched modestly by density gradient centrifugation over Ficoll or Percoll, during which they co-sediment with the mononuclear cells with rather efficient removal of contaminating polymorphonuclear neutrophil (PMN) and eosinophils. This is sufficient for many purposes, in which selective basophil activation can be effected by stimulation through FcεRI and monitored by measuring histamine release. Employing multistep density gradient centrifugation, peripheral blood mononuclear cells can be enriched for basophils from their baseline frequency 10–20 fold. Thereafter, basophils can be purified further by countercurrent elutriation. Relying on the rarity of circulating mast cell precursors, the absence of FcεRI expression by resting eosinophils, and the relative paucity of FcεRI expression by plasmacytoid dendritic cells, early techniques positively selected basophils by means of magnetic microbeads coated with anti-FcεRI. Once bound, the basophils are separated from the non-basophils by application to a column placed in a magnetic field that restrains the microbeads while all unbound cells flow through the column.

Negative Selection

Because positive selection techniques that depend on ligation of cell surface receptors such as FcεRI risk activating the cells, they have largely been replaced by the inverse method – the binding and removal of all non-basophils in a sample to purify unbound basophils by negative selection. Such an approach depends on a panel of cell surface markers strongly and consistently expressed by eosinophils, neutrophils, monocytes, NK cells, T cells, B cells, and various subtypes of dendritic cells. To succeed in purifying basophils, an antibody against at least one marker on each non-basophil must be included. In this case, magnetic beads bind to non-basophils and the cell mixture is again subjected to column filtration. The unlabeled basophils flow through the magnetic field and are collected while other cells are trapped. Basophils of >95% purity and >50% recovery can be achieved using negative selection techniques.

STIMULI FOR SECRETION

FcεRI and ITAM-Linked Receptors

FcεRI has been the most extensively studied cell surface receptor on the basophil. As in mast cells, this receptor is composed of an α chain whose extracellular domain binds to the CH3 region of the IgE heavy chain, a β chain, and two copies of a γ chain shared with numerous other receptors such as those for IgG. Although other cells, including eosinophils and plasmacytoid dendritic cells, express the α and γ chains, and thus bind IgE and react to its cross-linking, they lack the β chain. The intracellular domains of the β and γ chains contain immunoreceptor tyrosine-based

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activating motifs (ITAMs). Cross-linking of IgE results in the phosphorylation of critical tyrosine residues in these motifs, providing docking sites for proteins such as the tyrosine kinase Syk and the induction of a complex intracellular signaling network that culminates in mediator release. In particular, it was first demonstrated in basophils (and later confirmed in mast cells) that the expression of FcεRI is dynamic and directly proportional to the ambient concentration of IgE. This has been exploited clinically with the development of therapy for allergic diseases such as asthma and food allergy with monoclonal antibodies directed against FcεRI. These antibodies bind to circulating IgE and thereby prevent its interaction with FcεRI. Not only does this lead to reduced binding of specific IgE to FcεRI, but also reduced expression of FcεRI, which together lead to blunted IgE-dependent responses. The presence of FcεRI is often detected by means of flow cytometry using a monoclonal antibody directed against the extracellular portion of the α chain, such as 22E7 (which binds to the α chain whether occupied by IgE or not) and 15A5 (which only binds unoccupied α chain), or by binding of labeled IgE. Because of the dynamic expression of FcεRI, basophils may have anywhere from a few thousand to hundreds of thousands of these receptors on their surface. The cross-linking threshold for mediator release from a given basophil may likewise range broadly from a few hundred to tens of thousands of FcεRI receptors.

Basophils also consistently express the Ig superfamily member leukocyte immunoglobulin-like receptor A2 (LILRA2, also known in alternative nomenclature systems as LIR7, ILT1, and CD85h). The ligands for this receptor are as yet unidentified, but cross-linking by means of a monoclonal antibody results in mediator release from all three basophil "compartments" – preformed histamine from granules, newly synthesized LTC_4, and IL-4. As with FcεRI, Syk tyrosine kinase is involved in signal transduction from LILRA2, as predicted for a receptor that also employs the common γ chain.

GPCR

Formyl Peptide Receptors

The tripeptide formylated methionine-leucine-phenylalanine (fMLP) is a bacterial product capable of inducing histamine release and LTC_4 generation by basophils. However, as a direct agonist, it is unable to elicit significant release of the cytokines IL-4 and IL-13 from these cells. Basophils consistently express at least three formyl peptide receptors: FPR, FPRL1, and FPRL2.

C5a

After FcεRI, the most potent activating basophil receptor is that of the complement fragment C5a. Agonization of C5aR leads not only to mediator release from all three basophil compartments when basophils are primed or stimulated in the presence of IL-3, but to chemotaxis as well. Basophils also express C3aR, the receptor for a second anaphylotoxin, but this molecule is a less potent basophil secretogogue when compared to C5a.

Chemokines

The impressive (and likely as yet incompletely determined) variety of chemokine receptors expressed on the surface of basophils allows them to respond to a broad gamut of factors that influence cell trafficking (see Table 6.1). Many ligands for basophil surface receptors have both chemotactic or chemokinetic and variable prosecretory activities, making impossible a sharp

TABLE 6.1. Basophil GPCR receptors and their counterligands

Receptor	Ligands
Platelet activating factor receptor	PAF
C5aR	C5a
FPR, FRPL1, FPRL2	fMLP, urokinase plasminogen activator (uPA)
CCR1	MIP-1α, MCP-3
CCR2	MCP-1 (CCL2), MCP-3
CCR3	Eotaxin (CCL11), RANTES, MCP-3
CXCR1	IL-8
CXCR2	Nap-2
CXCR4	Stromal cell–derived factor 1 (SFD1, CXCL12)
Oxoeicosanoid receptor (OXE)	5-oxo-ETE

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distinction among these components of cell activation. Thus, for example, C5a and fMLP elicit both mediator release from and chemotaxis of basophils, while IL-3 augments basophil chemokinesis but not chemotaxis. Basophils express the stem cell factor (SCF) receptor c-kit, allowing SCF, which is not directly chemotactic, to augment basophil movement in response to C5a, MIP-1α, MCP-1, and eotaxin. Some chemotactic factors promote only selected mediator release, as exemplified by the abilities of MIP-1α, MCP-1, and eotaxin, which all direct basophil movement, to elicit LTC_4 generation but not IL-4 or IL-13 production.

One study of the transendothelial migration of basophils using human vascular endothelial cells (HUVEC) documented a tenebrous array of molecules that initiate or influence basophil movement, often with complex additive or synergistic effects and distinct (sometimes partially overlapping) time courses of maximal receptor expression and ligand potencies.

Basophils express $\alpha_4\beta_1$ (CD29/CD49d) and β_2 integrins (specifically CD11a/CD18 and CD11b/CD18), both of which contribute to diapedesis by allowing basophils to bind VCAM-1 and ICAM-1 and adhere to endothelial cells. As noted earlier, basophils produce a number of chemotactic factors, many of which may bind to their receptors contributing to an autocrine and/or a positive feedback system in which basophils promote their own recruitment. The results of numerous studies suggest a convoluted system of receptors and ligands that orchestrate the movement, modulate the activation, and promote the longevity of basophils in inflamed tissues. Although labyrinthine in its details, this network resembles that regulating eosinophils, a fact that has been emphasized by those who assert a close onotologic relationship between these two granulocytes.

CRTH2

Among the most prominent biochemical differences between mast cells and basophils is the ability of the former and the inability of the latter to produce the lipid mediator PGD_2. However, basophils express the CRTH2 receptor (chemotactic receptor homologous molecule expressed on TH2 cells), which is the second receptor for PGD_2, DP2. Thus, basophils can respond to this chemotactic factor, suggesting that activation of mast cells (e.g., through IgE by allergen or parasite antigen) may lead to the recruitment of basophils to inflamed tissues. Agonization of CRTH2/DP2 also leads to Ca^{2+} influx, augmentation of degranulation induced by anti-IgE and by fMLP, and increased expression of CD11b.

Histamine and Leukotriene Receptors

Interestingly, basophils possess cell surface receptors for the inflammatory mediators they produce at least one for histamine (H_2) and two for LTC_4 (CysLTR1 and CysLTR2). While this suggests autocrine actions of histamine and cysteinyl leukotrienes, the implications of this have not been fully elucidated. In addition, expression of BLT1 allows LTB4 to induce degranulation of basophils.

Cytokines

IL-3 is the best studied cytokine influencing basophils, and the α chain of the IL-3 receptor (CD123) is so strongly and consistently expressed on the cell surface that it is often used as a marker to help identify basophils. Exposure of basophils to IL-3 has profound effects on both cell viability and mediator release in response to activation. IL-3 at a concentration of 10 ng/mL retards basophil apoptosis and augments the release of histamine, LTC4, and IL-4 in response to IgE-dependent and -independent stimuli by "priming" (e.g., a 15-minute exposure before activation). At this and higher concentrations, IL-3 can elicit some mediator release on its own. *In vitro* experiments suggest that IL-3 is necessary (and possibly sufficient) to drive precursor cells toward the basophil phenotype.

Basophils also express receptors for IL-2 (CD25), IL-4, IL-5, and IL-8. The influence of these cytokines as well as of GM-CSF and SCF on basophil development and the longevity and function of mature basophils are less well established than those of IL-3. IL-5, and GM-CSF are able to prime basophils for LTC_4 synthesis and the release of IL-4 and IL-13 in response to C5a, but with less potency than IL-3. Unlike IL-3, these cytokines do not elicit mediator release directly. Nerve growth factor (NGF) augments mediator release by basophils in response to C5a.

Toll-Like Receptors

Basophils have been shown to express both mRNA and protein for Toll-like receptors (TLRs), TLR2 and TLR4. In importance, extensive experience demonstrates little reproducible effects of the TLR4 ligand lipopolysaccharide (LPS) on basophils, possibly because these cells do not express CD14, the LPS co-receptor present on LPS responsive cells such as macrophages. In contrast, the TLR2 ligand peptidoglycan directly induces IL-4 and IL-13 secretion from basophils. This activation appears to be cytokine selective, as it occurs in the absence of any histamine or LTC_4 release, though peptidoglycan augments their release in response to anti-IgE.

Histamine Releasing Factors

Originally named for their ability to induce degranulation, these enigmatic mediators also induce IL-4 release from basophils and augment cytokine release

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in response to IgE-mediated stimulation of these cells. Early reports suggested the existence of at least two histamine releasing factors (HRFs), one that depends on cell surface IgE and at least one that is IgE-independent. Some HRF activity was subsequently attributed to the action of chemokines. HRFs are produced by a variety of cells, such as endothelial cells, platelets, monocytes, and B lymphocytes. Much about HRFs remains to be explored. The receptors for HRF are currently unknown, and basophils from some donors do not respond to HRF. One particular HRF, human recombinant HFR (HrHRF) has been purified, sequenced, and studied in detail. This 23 kDa protein induces histamine release from the basophils of a subpopulation of atopic individuals. Responsiveness to HrHRF depends on IgE present on the basophils and in the serum of these donors, termed IgE+. This IgE+ can confer HrHRF responsiveness to any basophil, as opposed to the IgE from HrHRF unresponsive donors, termed IgE–, which cannot. While HrHRF depends on IgE+, it appears to have its own receptor, as it also primes unresponsive basophils for histamine release in response to anti-IgE and fMLP. In addition to inducing histamine release from the basophils of responders, it elicits IL-4 secretion from these cells (also in an IgE+ dependent manner) and primes even nonresponder basophils for IL-4 and IL-13 secretion in response to anti-IgE. The signal transduction mechanisms of HRF-mediated activation involves the tyrosine kinase Syk, but other critical details, such as the identity of the HrHRF receptor, are lacking.

Inhibitory Receptors

Appreciation of the importance of inhibitory receptors on the functions of basophils (as with those of other inflammatory cells) has increased dramatically during the past decade. Basophils consistently express the leukocyte immunoglobulin-like receptor B3 (LILRB3), a member of the inhibitory subgroup of LILRs. Although the related inhibitory receptors LILRB1 and 2, which are expressed on the surface of basophils from a minority of donors, are known to bind nonpolymorphic regions of MHC class I molecules, the ligands for LILRB3 are currently undetermined. Co-ligation of LILRB3 to an activating receptor such as FcεRI or LILRA2 results in the attenuation of mediator release from all three basophil compartments. LILRB3 possesses a relatively long cytoplasmic tail that contains immunoreceptor tyrosine-based inhibitory motifs (ITIMs). When phosphorylated, the tyrosine residues in these regions allow the docking and activation of phosphatases that antagonize the kinases involved in the propagation of activating signal transduction networks. It is tempting to hypothesize that evolutionarily related receptors such as LILRB3 and LILRA2 (whose genes are found along with those encoding the other LILRs in the leukocyte receptor complex on chromosome 19q13.42) are paired immunoregulatory switches, differentially expressed on various leukocytes and potentially binding related ligands with distinct affinities and kinetics to exert fine control over the functions of basophils and other inflammatory cells.

Basophils also express the inhibitory IgG receptor FcγRIIB (CD32), which possesses a single ITIM. Costimulation of basophils through both FcεRI and FcγRIIB delays and limits Ca^{2+} influx and attenuates degranulation and IL-4 release. These effects appear to be mediated by phosphorylation of the ITIM tyrosine residues, providing a docking site for the SH2 domain-containing phosphatase SHP-1, which then dephosphorylates key activating signal transduction components such as Syk. These observations form the basis of two novel approaches to antagonizing basophil and mast cell contributions to allergic inflammation. The first is a bifunctional Fcγ–Fcε fusion protein composed of part of the Fc portion of IgG that binds to FcγRIIB linked to a part of the Fc portion of IgE that binds to FcεRI. This protein effectively co-ligates FcγRIIB to FcεRI, constraining all IgE-mediated activation of basophils and mast cells. The second is a fusion protein composed of the same part of the Fc portion of IgG that binds to FcγRIIB linked to the major cat allergen Fel d1. By co-ligating FcγRIIB to Fel d1 specific IgE, this protein is an effective allergen-specific antagonist of basophil and mast cell activation appropriate for immunotherapy.

CD200R is a further example of an inhibitory receptor expressed by basophils. Intriguingly, a number of pathogenic viruses, including the human herpes viruses HHV-6, -7, and -8 express homologues of the ligand CD200, and are capable of downregulating activation of basophils.

FUNCTIONAL IMPLICATIONS OF BASOPHIL ACTIVATION

Tissue Effects of Mediator Release

The widespread activation of circulating basophils, with the release in minutes of preformed granule contents and the rapid *de novo* synthesis of lipid mediators, is predicted to result in systemic vasodilation and thus a decreased blood pressure. Indeed, since histamine, LTC_4, and PAF possess such vasoactive properties, this is a model of anaphylaxis especially attractive to explaining the rapid systemic allergic reactions to specific allergens administered intravenously (e.g., those to IV antibiotics) and to nonspecific secretogogues such as radiocontrast media and hyper- and

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hypotonic solutions. However, unlike mature, tissue-bound mast cells, circulating basophils must be recruited to tissues to participate in more typical allergic or antiparasitic inflammatory reactions, and this process may take hours to days. In the skin, the manifestations of basophil activation due to the release of these mediators include edema, erythema (wheal and flare reactions), and pruritus. In the gut, mucosal changes such as edema and the transmucosal loss of fluid, accompanied by increased peristalsis result in nausea, vomiting, diarrhea, and cramping abdominal pain (colic) as seen in food allergies. In the upper respiratory tract, similar changes in the nasal mucosa appear as congestion and rhinorrhea, while bronchoconstriction and wheezing typify the effects of LTC_4 and other mediators in the lower respiratory tract. The relative contribution of mast cells and basophils to IgE-dependent allergic responses has been the subject of debate. However, the observation that histamine but not PGD_2 is released into the nasal secretions during late-phase allergic reactions are argued for involvement of the basophil.

Recruitment and Modulation of Other Immune Cells

Their expression of numerous cytokines and chemokines suggests that over a longer time course (hours to days), basophils may recruit other immune cells to inflammatory lesions, and kinetic studies support this by demonstrating that basophils arrive in inflammatory tissues concomitant with or before other cells such as eosinophils and lymphocytes. Once there, basophils may modulate the function of these cells by means of extracellular mediators and by cell surface receptors recognizing their counterligands. The details of such cell–cell interactions await complete elucidation, but likely vary among different disease models.

The most intriguing advance in the appreciation of the potential immunoregulatory role of the basophil is the observation that activation of this innate immune cell can induce ε isotype switching and IgE production by human B cells when the two cell types are cocultured *in vitro*. Originally observed in response to nonphysiological stimulation of peripheral blood basophils (with PMA and ionomycin), this effect on B cells is also seen with activation of KU812 cells (a basophil-derived cell line) and with FcεRI-mediated activation of human cord blood–derived basophils. In all these systems, B-cell isotype switching is accompanied by and dependent on the production of IL-4 and IL-13 by the basophil (or KU812 cell) and expression of the costimulatory molecule CD40L (CD154). This phenomenon has traditionally been interpreted as representing the ability of basophils to *reinforce* previous Th2 skewing: once IgE production is established by Th2 cells, the IgE sensitizes basophils, and IgE-dependent stimulation of these basophils leads to their elaborating additional pro-IgE production signals, further encouraging B cells to make IgE. Importantly, however, recent data from *in vivo* murine models suggests that basophils may be an early, innate influence on B cells that *initiates* IgE production. This is not a direct effect of basophils on B cells, but rather is mediated by basophils inducing Th2 skewing, and it is the Th2 cells that directly induce B cells to undergo ε isotype switching. Thus, the basophil may be an answer to the mystery of the origins of Th2 skewing as well as a player in a potential positive feedback loop of IgE production for those antigens that can activate basophils for IL-4 production and CD40L expression by an IgE-independent mechanism.

EVIDENCE FOR ROLES IN DISEASES

Asthma and Rhinitis

Determining the precise contribution of the basophil to human diseases has been challenging. Basophils are considered to participate in allergic and antiparasitic inflammatory reactions, based on their biochemical and morphological resemblances to mast cells and eosinophils.

Their role in asthma was suggested by early studies that documented increased numbers of peripheral blood basophils and histamine concentrations in symptomatic asthmatics. Numerous basophils are present in sputum samples from patients with untreated asthma exacerbations, while 95% of the IgE-positive cells in bronchoalveolar lavage (BAL) fluid from asthmatic subjects have the morphologic and flow cytometric characteristics of basophils. Allergen inhalation by subjects with atopic asthma induces increases in sputum basophils at both early (7 hours after inhalation) and late phase (24 hours after inhalation) time points. The magnitude of this increase correlated with airway hyperresponsiveness (AHR) in response to methacholine in some studies, but this finding is controversial as other studies do not support or confirm this result.

Using the basophil-specific antibody 2D7, significantly more basophils were detected in postmortem lung section of patients with fatal asthma than in control patients who had asthma but died of other causes. The basophils were observed in both large and small airways and were present in the epithelium, submucosa, and alveolar wall. A subsequent study using 2D7 to identify basophils in biopsy specimens after allergen-induced late asthmatic responses showed that basophils accounted for a significant percentage of cells positive for IL-4 mRNA and the majority (72%)

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of cells positive for IL-4 protein. Employing the BB1 antibody that recognizes basophil granules and defining basophils and tryptase negative, increased numbers of basophils were seen at baseline in the lungs of atopic asthmatics compared to atopic subjects without asthma and control patients with nonatopic asthma. Basophil numbers increased further after antigen inhalation, but these cells were outnumbered tenfold by eosinophils and mast cells, suggesting that the basophil may play a predominantly regulatory role rather than being an effector cell.

Using flow cytometry to detect intracellular cytokine production in peripheral blood mononuclear cells from asthmatic and nonasthmatic, nonatopic subjects, one study determined that, as expected, specific antigen elicited IL-4 and IL-13 secretion only from the basophils of the asthmatics. Of note, under such circumstances, basophils outnumbered T cells as IL-4 producing cells by fourfold. Two hours after antigen stimulation basophils were the exclusive cellular source of IL-13, suggesting that in the peripheral blood basophils are the dominant cellular source of Th2 cytokines in the first 6 hours after stimulation with allergen.

Like asthma, allergic rhinitis is an allergen-driven mucosal inflammatory condition with an immediate and a late phase (6–8 hours later). Among such biphasic responders, there is an increase in the histamine content of nasal secretions at both time points. However, elevated concentrations of PGD_2 are observed only in the immediate phase. As, in contrast to mast cells, basophils lack the ability to produce PGD_2, these findings suggest that the immediate phase is dominated by the activation of mast cells and that mediator release in the late phase is derived from newly recruited basophils. Data in support of this model were obtained in a study of patients with nasal allergy to grass pollen. At baseline, biopsies of the nasal mucosa demonstrated no basophils (as assessed by staining with the BB1 antibody). Within 1 hour after local allergen challenge, biopsies demonstrated a significant influx of basophils into the epithelium and lamina propria, and basophils remained detectable in the lamina propria at 24 hours and 1 week after challenge. Basophil numbers increase in the circulation of patients with ragweed allergic patients during ragweed season, suggesting that an antigen induced signal reaches the bone marrow and augments basophil production and/or attenuates basophil apoptosis.

Thus, the role of basophils in asthma and rhinitis is suggested by their recruitment to the airways in acute asthma and following antigen challenge; by the pattern of mediator production in the late phase allergic response; and by the increased numbers in the circulation of basophils producing Th2 cytokines after antigen challenge.

Immunologic Skin Disease

Cutaneous allergic inflammation takes a number of forms, one of the most common of which is urticaria (hives) with or without angioedema (swelling). Analogous to studies of the airways, the involvement of basophils in antigen-induced late phase skin reactions (LPR), a model of cutaneous allergy, has been assessed by staining skin biopsies with the BB1 antibody. After local antigen challenge, basophils were detected in the dermis (especially around blood vessels) at 6 hours, peaked at 24 hours, and started to resolve at 48 hours. Maximal basophil numbers reached 40% of peak eosinophil numbers, and approximately one-third of the basophils had an appearance consistent with partial degranulation. A similar study employing the 2D7 antibody also demonstrated a dramatic influx of basophils into the dermis around blood vessels 6 and 24 hours after allergen injection. Importantly, these 2D7-positive dermal basophils were negative on stating for metachromatic granules with toluidine blue, suggesting that degranulation had occurred and indicating that light microscopy might underestimate the presence of basophils.

A comparison of biopsies of allergen-induced cutaneous LPR with those from patients with chronic idiopathic urticaria (CIU) and those from normal controls demonstrated increased numbers of BB1 positive basophils in the LPR and CIU tissues compared with nonatopic controls. Approximately 35%–50% of patients with CIU have IgG autoantibodies directed against IgE or FcεRI, but there were no statistically significant differences in the numbers of dermal basophils between those with and without such antibodies. In situ hybridization demonstrated mRNA for IL-4 in both LPR and CIU biopsies, but it was unclear what fraction of IL-4 mRNA positive cells were basophils (as opposed to Th2 lymphocytes). A number of studies have used peripheral blood basophils in assays for histamine release to detect anti-IgE or anti-FcεRI autoantibodies in the serum of patients with CIU, and found that basophil histamine is dependent on or augmented by the complement fragment C5a.

Basophils have been detected infiltrating the lesions of bullous pemphigoid, erythema multiforme, incontinentia pigmenti, and rejection of first set skin allographs. Thus, basophils may be involved in the pathobiology of many immunologic skin diseases.

Anaphylaxis

Anaphylaxis is a severe, rapidly progressive, life threatening systemic allergic reaction typically triggered by allergens such as foods (e.g., peanut, shellfish), drugs (e.g., penicillin, aspirin), insect venom, and

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latex. Typical routes of allergen entry include ingestion (foods and some drugs), injection (intradermal for venom, intravenous for some drugs), and inhalation (latex). Anaphylaxis is a clinical diagnosis, dependent on the appearance of common symptoms and signs in specific organ systems (e.g., pruritus, urticaria, and angioedema; asthma; and abdominal pain, nausea, vomiting, and diarrhea) in the context of allergen exposure. With the exception of food-induced anaphylaxis, an elevation in the serum concentration of mast cell tryptase is so frequently seen that its absence casts some doubt upon the diagnosis, though it does not categorically rule out anaphylaxis. This in conjunction with decades of histological and biochemical data have established that mast cells are central to most forms of anaphylaxis.

Whether basophils are activated in one or more subtypes of anaphylaxis, depending perhaps on the triggering allergen and the route of exposure is the subject of investigation. The absence of elevations in serum tryptase in food induced anaphylaxis has led some to hypothesize that basophils may play a dominant role in this subtype of the disease. Individuals who develop anaphylaxis to the venom of stinging insects can be effectively treated with incremental subcutaneous injection of increasing doses of the responsible antigen over several hours or, more commonly, several weeks. This treatment is called immunotherapy. One recent study of peripheral blood basophils from patients with venom-induced anaphylaxis and those from patients with only large local sting reactions indicated that CD63 expression was higher at baseline (i.e., prior to sting challenge) among patients with systemic reactions (anaphylaxis) who were receiving venom immunotherapy compared to patients with large local reactions who were receiving immunotherapy. Basophils increased their expression of CD69 and CD203C in response to *in vivo* sting challenge. The authors suggested that basophils may thus be useful biomarkers of venom-induced anaphylaxis, but whether basophil activation is either necessary or sufficient for human anaphylaxis is unclear.

Parasitic Infections

Parasitic infections typically elicit elevated serum concentrations of IgE and eosinophilia. Evidence for basophils participating in immune responses against parasites is largely extrapolated from animal models (see later). While basophilia (increased numbers of circulating basophils) is commonly observed in non-human models, it is very rarely observed in human infections. One study documented that basophilia occurred in only 4 out of 668 patients with confirmed parasitic infection. However, data from other studies suggest that human basophils are functional in these diseases. Basophils from patients with hookworm infection (*Necator americanus*) degranulate in response to hookworm antigen, and the proteases secreted by this parasite have been shown to induce cytokine release from purified human basophils. This latter finding suggests that hookworm proteases may be innate (i.e., IgE-independent) activators of basophils important in early antiparasitic responses that later include IgE production (see earlier section for a discussion of basophil induction of B-cell ε isotype switching). Likewise, peripheral blood basophils from patients harboring active infection with the filarial worm *Brugia malayi* produced significant amounts of IL-4 in response to *B. malayi* antigen. Importantly, this IL-4 response was quantitatively on a par with that of parasite-specific T cells and was triggered by lower concentrations of antigen. Together, these studies suggest that at least for some infections, parasite antigens may elicit IL-4 release from basophils in an IgE-independent manner and that basophils may therefore be influential in establishing Th2 immune antiparasitic immune responses.

Other Human Diseases

Using the BB1 antibody, basophils have been identified in the gastric mucosa of patients with moderate to severe gastritis due to *Helicobacter pylori* infection. Hp(2–20), a peptide derived from the gut pathogen, is a potent chemoattractant for basophils, likely through the FPR-like 1 and 2 receptors (Table 6.1). While Hp(2–20) could induce Ca^{2+} influx into basophils, this stimulus did not elicit histamine release or cytokine (IL-4 or IL-13) secretion from these cells *in vitro*. The nature, mechanisms, and effects of basophil activation, if it occurs in the context of *H. pylori* infection, remain to be fully elucidated.

Peripheral blood basophils from normal individuals, unlike their mast cells, express a poorly characterized 45 kDa cell surface protein that binds the IgM mAb Bsp-1. As noted earlier, these cells have trace amounts or no detectable mast cell tryptase, chymase, or carboxypeptidase in their granules, and little cell surface expression of *c-kit*. However, in the peripheral blood of patients with asthma, atopy (allergic rhinitis, atopic dermatitis, urticaria), and drug allergy there are significant numbers of metachromatic, Bsp-1 positive cells with segmented nuclei (i.e., with the morphology of basophils) that possess tryptase, chymase, and/or carboxypeptidase in their granules. In addition, these cells expressed detectable amounts of *c-kit* on their surface. These data suggest that basophils in allergic disease may have a plastic phenotype and/or that basophils and mast cells are developmentally related, though this is controversial (see the discussion of basophil ontogeny earlier).

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Guinea Pig and Mouse Models of Disease

Decades ago, studies of basophils from other mammalian species were frequently performed in guinea pig models and occasionally in rats and mice. Recently, the murine model has come to dominance, but advances have been limited by two difficulties. The first is in definitively identifying mouse basophils, as murine basophils appear to differ morphologically and in terms of cell surface receptor expression from their human counterparts. The second is the lack of a genetic basophil "knockout" model corresponding to mast cell deficient mice strains.

Guinea pig and murine models of parasitic infections, such as ectoparasites (ticks) in the former and *Nippostrongylus brasiliensis* in the latter, have been used to explore both the contribution of basophils to host defense and the ontogeny of basophils, as both species (unlike humans) typically develop basophilia in response to parasite invasion. These studies established basophils as important sources of cytokines such as IL-4 and IL-13.

Recently, a murine model of response to an allergen (the protease papain) indicated a critical immunoregulatory role for basophils. After immunization with papain, basophils entered draining lymph nodes and initiated Th2 differentiation by releasing IL-4 and the cytokine thymic stromal lymphopoietin (TSLP), leading to papain-specific IgE production. Depletion of basophils using the MAR-1 antibody revealed the dependence of Th2 development on basophils (by means of IL-4 and TSLP) but not on mast cells. Other recent studies have investigated the roles of murine basophils in a model of IgG– (as opposed to IgE–) mediated anaphylaxis and in antibody memory responses.

PHARMACOLOGICAL REGULATION OF BASOPHIL RESPONSES

Several drugs that are commonly used to treat asthma and allergic diseases have been shown to have an effect on basophil numbers and function. The relevance of these findings to the therapeutic efficacy of these medications depends on the putative role of basophils in these conditions. As outlined earlier, this remains somewhat controversial, and is the subject of ongoing investigation.

Steroids

Administration of intravenous hydrocortisone and inhaled beclomethasone was shown three decades ago to decrease the number of peripheral blood basophils concomitant with their therapeutic effects in asthmatics. Indeed, steroids appear to augment basophil apoptosis (programmed cell death) just as they do in eosinophils.

LT Receptor Antagonist

Basophils both produce and respond to cyLTs. Antagonism of the cysLT1 receptor by montelukast in patients with asthma had no effect on basophil degranulation in response to anti-IgE, but did result in modest inhibition of LTC_4 release from these cells, indicating differential regulation of these distinct mediator compartments and a possible cysLT feedback mechanism.

Anti-IgE

As noted earlier, the expression of FcεRI by basophils is directly proportional to the ambient concentration of free IgE. Humanized monoclonal antibodies (e.g., omalizumab) that bind to the C_H3 domain of the IgE heavy chain compete with the binding site of the FcεRI α chain, thus preventing IgE from adhering to the surface of basophils by means of the high-affinity receptor. This leads to a dramatic and nearly complete decline in FcεRI expression on the surface of basophils. This greatly impairs FcεRI-mediated activation of basophils, including histamine release, LT secretion, and cytokine production. Over the course of weeks after cessation of treatment, free IgE concentrations, FcεRI expression, and basophil mediator responses return to baseline.

DIAGNOSTIC TESTS USING BASOPHILS

A number of tests using basophils have been developed to assist in the diagnosis of allergic diseases by means of an *in vitro* assay that is ideally easily performed, minimally invasive (requiring a peripheral blood sample as opposed to skin testing or allergen challenge), rapid, reproducible, and widely applicable. Generally, these tests involve exposing basophils, either in a peripheral blood sample or purified to varying degrees, to a suspected allergen, such as environmental substances (e.g., plant pollens, animal dander, dust mite antigens), drugs (e.g., NSAIDs, antibiotics, chemotherapeutic agents), foods, and Hymenoptera venom. Because basophils are believed to be the exclusive source of histamine in human peripheral blood, measuring histamine release 30–60 minutes after exposing the patient's blood cells to the antigen determines whether basophil activation occurred. Other tests rely on changes in the expression of certain basophil cell surface markers, such as CD63, CD69, and CD203C as assayed by flow cytometry. Some assays combine measurements of mediator release with flow cytometry.

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Some potential pitfalls with these tests include their inability to determine whether basophil activation is IgE dependent or independent; the assumption that an absence of basophil activation correlates with nonresponsiveness of mast cells to a given antigen; and the possibility that an allergen might elicit the release of one or more important but less cell-specific mediators (e.g., LTC_4, PAF) but not histamine, resulting in a false negative test. It is important also to note that the basophils of some individuals do not release histamine or leukotrienes in response to FcεRI cross-linking. This so-called nonreleasing phenotype has been attributed to a delayed and blunted Ca^{2+} influx in response to FcεRI cross-linking, possibly due to impaired expression and/or activity of the signaling molecule, Lyn kinase.

CONCLUSIONS

Human basophils are the rarest of peripheral blood granulocytes. Their expression of the high-affinity Fc receptor for IgE, the range of mediators they produce, and their presence in tissues and circulation following allergen challenge suggest that they are involved in asthma, allergies, and selected infections. While their precise roles in health and disease are still incompletely understood, recent data suggest that they may be important immunoregulators, especially in allergic diseases and parasitic infections. Their role as immunoregulatory cells, particularly a role in "priming" Th2-dependent allergic reactions, as opposed to effector cells of allergic disease would be consistent with their presence in low numbers in the resting state. This is the subject of intense investigation, the conclusions of which will depend in part on the relevance of animal models to human disease and our ability to directly interrogate basophil function in humans. There is still much to explore in terms of the basic mechanisms and intercellular interactions of basophils.

SUGGESTED READINGS

Devouassoux, G., Foster, B., Scott, L.M., Metcalf, D.D., and Prussin, C. 1999. Frequency and characterization of antigen-specific IL-4 and IL-12-producing basophils and T cells in peripheral blood of healthy and asthmatic subjects. *J Allergy Clin Immunol* **104**:811–819.

Falcone, F.H., Haas, H., and Gibbs, B.F. 2000. The human basophil: a new appreciation of its role in immune responses. *Blood* **96**(13):4028–4038.

Gauchat, J.-F., Henchoz, S., Mazzei, G., et al. 1993. Induction of human IgE synthesis in B cells by mast cells and basophils. *Nature* **365**:340–343.

Lee, J.J., and McGarry, M.P. 2007. When is a mouse basophil not a basophil? *Blood* **109**(3):859–861.

MacGlashan, D.W., Jr., Bochner, B.S., Adelman, D.C., et al. 1997. Down-regulation of FcεRI expression on human basophils during in vivo treatment of atopic patients with anti-IgE antibody. *J Immunol* **158**:1438–1445.

Marone, G., Triggiani, M., and de Paulis, A. 2005. Mast cells and basophils: friends as well as foes in bronchial asthma? *Trends Immunol* **26**(1):25–31.

Ochensberger, B., Tassera, L., Bifrare, D., Rihs, S., and Dahinden, C.A. 1999. Regulation of cytokine expression and leukotriene formation in human basophils by growth factors, chemokines, and chemotactic agonists. *Eur J Immunol* **29**:11–22.

Sokol, C.L., Barton, G.M., Farr, A.G., and Medzhitov, R. 2008. A mechanism for the initiation of allergen-induced T helper type 2 responses. *Nat Immunol* **9**(3):310–318.

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7 Eosinophils

Sophie Fillon, Steven J. Ackerman, and Glenn T. Furuta

INTRODUCTION

Eosinophilic granulocytes, commonly referred to as eosinophils, or less commonly as acidophils, were first identified by Paul Ehrlich in 1879. He named these bilobed nucleated cells as eosinophils because of their intense staining with the acidic dye eosin. Ehrlich completed his original studies that defined the eosinophil's morphological features during a time when he was committed to a clinical practice as well as basic research, an early example of a physician scientist. Combining his clinical expertise with his laboratory discoveries, he proposed a variety of functions for eosinophils that included phagocytosis, granule secretion, and chemotaxis.

Eosinophils are normally present as a minority of the peripheral blood leukocyte pool and primarily reside within tissues containing mucosal surfaces such as the uterus and gastrointestinal tract. While their exact function is not certain, eosinophils are thought to possess both beneficial and deleterious properties as participants in both innate and adaptive host immune responses. As an arm of the innate defense system, eosinophils are strongly proposed to be responsible in part for combating parasitic infections, particularly with helminths. When recruited in excess, eosinophils are currently thought to participate in mediating the pathogenesis of allergic diseases such as asthma, atopic dermatitis, gastrointestinal diseases, and hypereosinophilic syndromes.

EOSINOPHIL MORPHOLOGY

Eosinophils are approximately 12–17 μm in diameter, and represent 1%–6% of the total blood leukocyte population (Figure 7.1). The human eosinophil's nucleus is characteristically bilobed, and usually circular or ovoid [1]. The cytoplasm of the eosinophil is filled with a number of secretory organelles that include primary and secondary (crystalloid/specific) granules, small granules, and secretory vesicles. Primary and secondary granules together number approximately 200 per cell and contain an electron-dense matrix containing the eosinophil's characteristic cationic proteins including eosinophil peroxidase, the eosinophil-derived neurotoxin, and the eosinophil cationic protein (the eosinophil's two ribonucleases RNase2 and RNase3, respectively) (Figure 7.2). In addition, the secondary granules contain an electron-dense crystalloid core composed of two related cationic proteins, major basic protein-1, and major basic protein-2. The small, amorphous granules contain arylsulfatase and acid phosphatase [2]. The Charcot–Leyden crystal (CLC) protein is present within the primary granule population (which represent less than 5% of total granules) and in the agranular spaces beneath the eosinophil's plasma membrane. When released by eosinophils that undergo cytolytic degranulation in tissues such as the lung and gastrointestinal (GI) tract, the CLC protein may form the characteristic hexagonal bipyramidal structures known as CLCs, a hallmark of eosinophil involvement in host allergic and other immune responses. These crystals can be observed in the stool or sputum of patients with nonspecific eosinophilic inflammation associated with inflammatory bowel diseases, allergic intestinal disease and asthma, respectively (Figure 7.3). The CLC are slender, hexagonal bipyramidal crystals that stain purplish-red in tissue sections stained with hematoxylin and eosin (H&E). Eosinophils also contain lipid bodies, nonmembrane-bound lipid-rich organelles often confused with granules and present in many types of leukocytes and other cells [3]. Eosinophils contain greater numbers of lipid bodies than do neutrophils, and the numbers of lipid bodies increase during eosinophil activation in vitro and engagement in inflammatory reactions in vivo [4]. The functions of lipid bodies include incorporation of fatty acids such as arachidonate. They likely serve as intracellular depots for its storage and metabolism,

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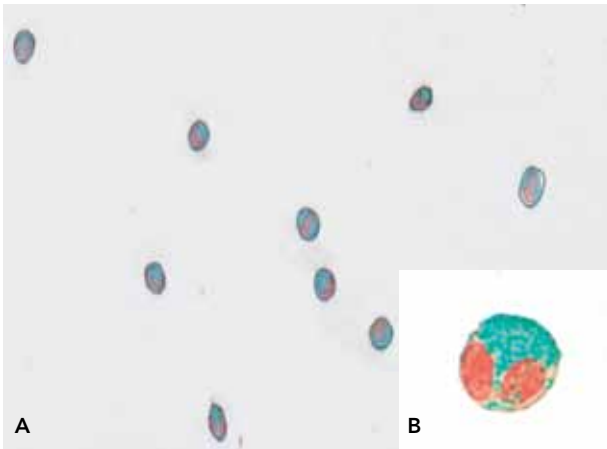


Figure 7.1. Eosinophils in peripheral blood smear. Fast green staining of eosinophils in human peripheral blood with a neutral red counterstain. Cytoplasmic granules appear brilliant turquoise green and the bilobed nucleus is stained red (Magnification A: 20×, B: 100×).

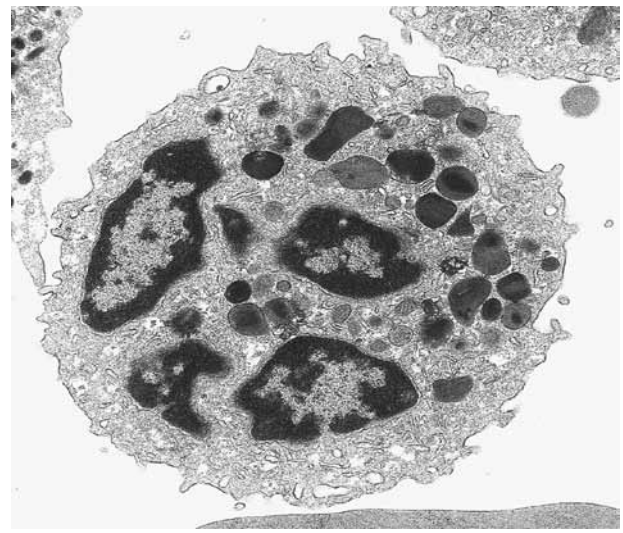


Figure 7.2. Electron microscopic view of the human eosinophil. Electron micrograph of a human eosinophil with granules showing electron dense cores (2500×, Courtesy of Dr. Ann Dvorak, Beth Israel Deaconess Medical Center, Boston, MA).

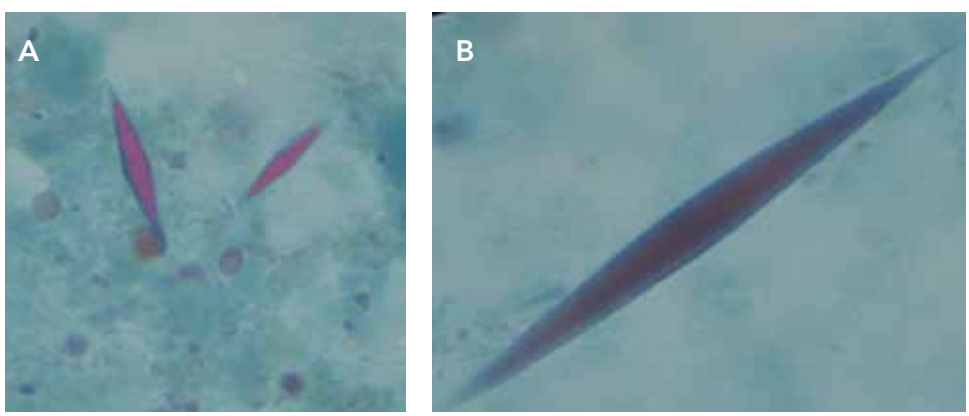


Figure 7.3. Charcot–Leyden crystals from bronchial lavage fluid. Photomicrographs of birefringent Charcot–Leyden crystals, a hallmark of degranulated eosinophils (A: 40×, B: 100×).

as suggested by the localization of many of the eosinophil's eicosanoid-forming enzymes including 5-lipoxygenase, leukotriene C4 synthase, and cyclooxygenase in the lipid body [5]. These four organelles, along with vesiculotubular structures and small vesicles involved in transport and secretion in the activated cell, serve as the major subcellular sites for the eosinophils' armamentarium of preformed cytotoxic and inflammatory mediators.

EOSINOPHIL MEDIATORS

Granule Proteins

Studies of the biochemistry, biologic activities, and, in particular, the localization in tissues of the distinctive enzymatic and nonenzymatic cationic protein constituents of the eosinophil's secondary (specific) granule provided the first convincing evidence for the role of the eosinophil in the pathogenesis of inflammation and tissue damage in eosinophil-associated diseases [6]. In addition, identification of eosinophil-specific granule constituents in tissues by a variety of sensitive and specific immunochemical methods provided additional evidence for the participation of the eosinophil in diseases not normally associated with blood or tissue eosinophilia, for example, certain skin diseases [7]. Studies in the past 15–20 years of the biochemistry, functions, and localization in tissues of the unique enzymatic and nonenzymatic cationic proteins present in eosinophil granules provide compelling evidence supporting a pathologic proinflammatory and effector

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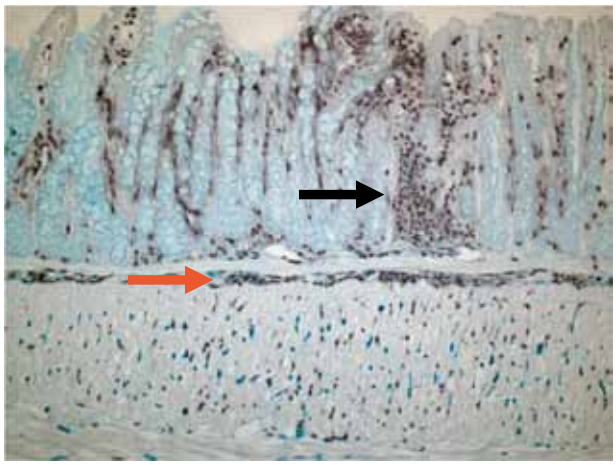


Figure 7.4. Gastrointestinal eosinophilic inflammation. Immunoperoxidase detection of major basic protein-1 in a section of inflamed murine small intestine. Black arrow shows eosinophilic infiltration in epithelial surface whereas orange arrow shows inflammation in the muscularis (20×).

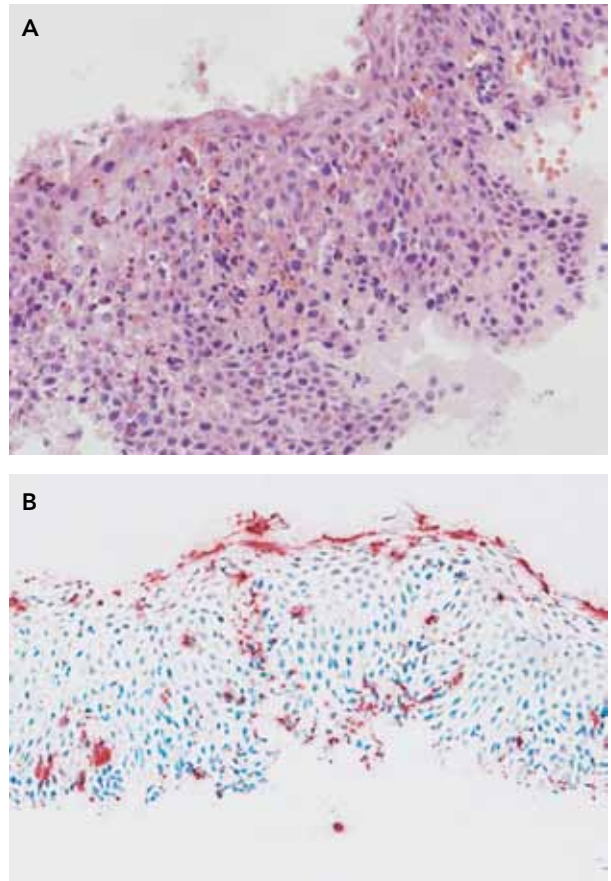


Figure 7.5. Eosinophilic esophagitis inflammation. (A) H&E of the esophageal epithelium from a patient with eosinophilic esophagitis (20×). (B) Eosinophil peroxidase staining of the esophageal epithelium from a patient with eosinophilic esophagitis (20×). (Courtesy of Cheryl Protheroe and James Lee, Mayo Clinic Scottsdale, AZ.)

role for the eosinophil in directly inducing tissue damage. The distinctive cationic granule protein constituents of the eosinophil include the two major basic proteins (MBP-1, MBP-2) (Figure 7.4), the eosinophil peroxidase (EPO) (Figure 7.5), and the ribonucleases eosinophil cationic protein (ECP), and eosinophil-derived neurotoxin (EDN). Eosinophils also have the capacity to express a variety of toxic oxidative intermediates [8], eicosanoid and other lipid mediators of inflammation, and most recently, inflammatory and hematopoietic cytokines integral to the normal and pathologic roles of this granulocyte in health and disease. Studies of the mechanisms by which the eosinophil granule cationic proteins are secreted or released into tissues have demonstrated that they can be selectively secreted, depending on the activation stimulus, by a number of different secretory pathways ranging from classical granule fusion and exocytosis (e.g., in killing parasitic helminths), piecemeal degranulation (PMD; vesicular transport from the granules in the absence of classical exocytosis), and cytolytic degranulation (release of intact membrane-bound granules directly into the tissue upon eosinophil apoptosis/cell death) [9].

Cytokines

Eosinophils are a rich source of both Th1 and Th2 cytokines that may participate in immune regulation, leukocyte chemotaxis, and cell growth (Table 7.1) [10]. While other cell types such as lymphocytes can produce larger quantities of a variety of cytokines localized within inflammatory sites, eosinophils may act in a different role. For instance eosinophils, like mast cells, have the capacity to synthesize cytokines, and then store them in their cytoplasmic granules. Thus, rather than a lag period between stimulation and release of cytokine, eosinophils are able to provide immediate and focused cytokine release at sites of inflammation [11].

Traditionally, eosinophils are associated with the generation of Th2 immune responses and are increased in immune environments rich with Th2 cytokines such as IL-4, IL-5, and IL-13 as seen in asthma. IL-5 mediates eosinophil expansion and differentiation from progenitors in the bone marrow, recruitment, priming, and activation, and prolongs tissue survival in response to allergic stimuli [12]. Activated eosinophils also express granulocyte-macrophage colony-stimulating factor (GM-CSF) which functions in an autocrine fashion to prevent apoptosis and prolong their survival in tissues such as the lung in asthma. Interestingly, basal levels of eosinophil development can occur in the absence of IL-5, and antigen-induced tissue eosinophilia can occur independent of IL-5 as

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TABLE 7.1. Eosinophils release interleukins, chemokines, and growth factors

Interleukins
IL-1, IL-2, IL-3, IL-4, IL-5, IL-6, IL-9, IL-10, IL-11, IL-12, IL-13, IL-16, IL-17
Chemokines
Epithelial cell–derived neutrophil activating peptide (CXCL5), eotaxin (CCL11), growth-related oncogene (CXCL1), interleukin-8 (CXCL8), IFN-γ-inducible protein (CXCL10), IFN-inducible T-cell alpha chemoattractant (CXCL11), macrophage inflammatory protein 1 (MIP-1α), monocyte chemoattractant protein 1 (CCL3), monokine induced by IFN-γ (CXCL9), MCP-3 (CCL7), MCP-4 (CCL13), RANTES (CCL5)
Growth factors
Heparin-binding epidermal growth factor-like binding protein (HB-EGF-LBP), nerve growth factor (NGF), transforming growth factor (TGF-α), transforming growth factor-beta (TGF-β1)
Others
Interferon (IFN-γ), tumor necrosis factor (TNF-α), granulocyte-macrophage colony-stimulating factor (GM-CSF)

suggested by the presence of tissue eosinophils in asthmatic patients treated with anti-IL-5 [13]; these observations and others suggest a redundancy in eosinophil growth factors capable of driving eosinophil terminal differentiation and prolonging eosinophil survival in tissues, factors that include IL-3 and GM-CSF. In contrast, numerous studies have shown the importance of IL-5 during allergic responses, particularly for the amplification of blood and tissue eosinophilia. Employing experimental models of asthma, investigators have demonstrated that IL-5 neutralization can block various aspects of the asthmatic response [14,15]. Animal models have also shown that inhibition of IL-5 suppresses pulmonary eosinophilia in response to antigen inhalation [16]. Moreover, eosinophil trafficking to the allergic lung is significantly reduced in IL-5–deficient (knockout) mice and in animals treated with anti-IL-5 antibodies [17–19]. Finally, overexpression of IL-5 in transgenic mice is sufficient for the development of eosinophilia [20–22]. IL-4 activates human vascular endothelial and respiratory epithelial cells to produce eosinophil chemoattractant cytokines [23]. IL-4 and IL-13 produced by Th2 cells recruits and activates IgE-producing B cells, and enhances IgE-mediated responses [24]. IL-13 appears to have its own distinct role in allergic inflammation, acting as a key regulator of allergen-induced airway inflammation, and goblet cell metaplasia [25]. Both molecules induce the expression of eotaxins by a STAT6-dependent pathway [26]. IL-13 alone is also an important eosinophil chemoattractant [12].

Alternatively, eosinophils themselves can produce Th1 cytokines such as IL-2, IL-6, IL-12, TNF-α, and IFN-γ [27–29]. Th1 cells are recruited by CXCL9 (monokine induced by IFN-γ) and CXCL10 (IFN-γ-inducible protein-10) [30]. IL-6 has been shown to inhibit Th1 and increase Th2 differentiation of eosinophils through induction of naive T-cell IL-4 production [31]. In some Th1 models of inflammation, eosinophil-derived IFN-γ is increased [10]. In addition TNF-α is essential for IFN-γ induced secretions of Th1 chemokines [28]. Eosinophils secrete chemokines such as the eotaxins, RANTES, and macrophage inflammatory protein (MIP)-1α. Eotaxins (1, 2, and 3), secreted primarily by activated epithelial cells and T cells, are highly selective eosinophil chemoattractants that bind their cognate CCR3 receptor on the eosinophil's surface. Eotaxin secretion by eosinophils strongly suggests their ability to initiate and perpetuate an allergic inflammatory response. RANTES can stimulate eosinophil and neutrophil influx while MIP-1α is primarily involved in neutrophil trafficking [32].

Eosinophils also may participate in other biological processes. IL-4-deficient mice have reduced transplant-associated eosinophilia. Impaired Th2 inflammation at the site of incompatible allograft transplants was also observed in these mice suggesting that eosinophils may modulate immune responses leading to transplant rejection [33]. Moreover, IL-4 and IL-13 are strongly associated with tumor cell death in several types of cancer [34]. In addition, the use of eosinophil-specific antibodies has demonstrated that almost all human or mouse cancers are associated with an important eosinophil infiltrate at some point in tumor growth [35,36]. Finally, eosinophils are associated with tissue remodeling (e.g., epithelial cell hyperplasia, angiogenesis) and fibrosis (fibroblast activation, transdifferentiation into myofibroblasts increased deposition of extracellular matrix) in many eosinophil-associated diseases, and are a rich source of TGF-β [37].

Lipid Mediators

Eosinophils express cysteinyl leukotriene receptors (CysLT1R and CysLT2R), prostaglandin (PG)D2 type 2 receptor, and platelet activating factor (PAF) receptor [38–40]. Eosinophil recruitment is induced by leukotrienes (LTB4, LTD4, LTE4), PAF, and 5-oxo-6,8,11,14-eicosatetraenoic acid, suggesting that these lipid mediators may play a role in eosinophil transmigration [41–43]. Eosinophils have the capacity to secrete lipid mediators including PAF and leukotrienes such as LTC_4. The release of these lipids increase leukocyte trafficking, endothelial adhesion, smooth muscle contraction, vascular permeability and mucus secretion and are considered one of the proinflammatory activities of the eosinophil [44].

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EOSINOPHIL DEVELOPMENT

Life Cycle

Eosinophils are derived from hematopoietic stem cells and myeloid-committed progenitors that arise within the bone marrow. The human eosinophil progenitor is derived from the common myeloid progenitor pool, expresses a number of unique cell surface markers including IL-5Rα, CD34, CD38, IL-3Rα, is CD45RA negative, and is distinct from the human granulocyte-macrophage progenitor (GMP) population that gives rise to neutrophils and macrophages [45]. Two groups of transcription factors, the C/EBP family and GATA-1, appear to synergistically participate in eosinophil development [46]. The importance of GATA-1 and the C/EBP factors (C/EBPα and C/EBPε) is emphasized by the fact that murine models with targeted deletions (knockouts) of these three factors are all eosinophil-deficient [47]. However, GATA-1 appears to be the most important transcription factor for eosinophil lineage specification, since transgenic deletion of the high-affinity GATA-binding site in the HS-2 region of the GATA-1 promoter itself leads to a selective loss of eosinophils in the mouse [48]. Both GATA-1-specific expression and activity in eosinophils appears to be mediated by this high-affinity palindromic GATA site. This double GATA site is present in the GATA-1 promoter and in the regulatory regions of many eosinophil specific genes (e.g., eotaxin receptor CCR3, CLC/Gal-10, MBP-1, and IL-5Rα) [48–50].

Eosinophil differentiation and expansion from IL-5Rα+ common myeloid progenitor cells occurs in response to three important cytokines, IL-3, IL-5, and GM-CSF [51–53]. These three cytokines provide permissive proliferative and differentiation signals through transcription factors that include GATA-1, PU.1, C/EBPα, and C/EBPε. IL-5 is most specific to the eosinophil lineage of these cytokines and is responsible for selective differentiation of eosinophils [54] and stimulates eosinophil release from the bone marrow into the peripheral circulation [55]. At these sites, eosinophils are primed by Type 2 cytokines released from Th2 helper cells. IL-5, GM-CSF, and IL-3 are important as well for eosinophil terminal differentiation, activation, and are antiapoptotic, promoting long-term eosinophil survival in vitro and in tissues.

Eosinophils produce and store secondary granule proteins before their exit from the bone marrow. PU.1 plays a critical role in eosinophil granule protein production as evidenced by the fact that PU.1 null mice show attenuated MBP and EDN expression and addition of PU.1 induces MBP expression. A synergistic role for PU.1 in amplifying GATA-1-mediated gene transcription has been demonstrated in the eosinophil lineage [49]. Direct PU.1 protein–protein interaction alters GATA-1 conformation and/or binding to high-affinity double GATA sites present in many eosinophil-specific gene promoters, increasing its activity and synergizing to strongly upregulate transcription of eosinophil lineage-specific genes such as MBP in developing eosinophil progenitors [49].

Eosinophil Migration

After maturation, eosinophils migrate through the vascular space and traffic to mucosal surfaces including the gastrointestinal tract (Figures 7.4 and 7.5) in response to a variety of different chemoattractants produced by cells present in the mucosa, particularly epithelial cells. These chemoattractants or chemokines (chemotactic cytokines) including CCL11 (eotaxin-1), CCL24 (eotaxin-2), CCL26 (eotaxin-3), and CCL5 (RANTES) participate in the recruitment of eosinophils to mucosal surfaces and sites of allergic inflammation. Arachidonic acid metabolites, especially certain leukotrienes such as (LTB_4) are synthesized in leukocytes from arachidonic acid by 5-lipoxygenase. Leukotrienes are thought to play a role in allergic and asthmatic reactions by stimulating bronchoconstriction and increasing vascular permeability [56]. Eotaxin production can lead to tissue eosinophilia. For instance, experimental induction of cutaneous and pulmonary late-phase responses in humans leads to synthesis of eotaxins by resident cells as well as infiltrative cells such as eosinophils [26].

Transmigration of eosinophils through the vascular endothelium into mucosal surfaces involves rolling, tethering, adhesion, and transendothelial migration [57,58]. Eosinophils express the transmembrane glycoproteins, selectins, that regulate rolling on endothelia [59]. Adhesion and transmigration of eosinophils across the vascular endothelia to tissue is a tightly regulated process. Of the cytokines implicated in modulation of leukocytes recruitment, only IL-5 and the eotaxins have been shown to selectively regulate eosinophil trafficking [60]. Eosinophil chemoattractants alter the expression of vascular and leukocyte adhesion molecules and receptors such as selectins, integrins, and integrin receptors. Mucosal addressin cell adhesion molecule (MAdCAM)-1, vascular cell adhesion molecule (VCAM)-1, and intercellular adhesion molecule (ICAM)-1 present on endothelial cells can be differentially expressed during transmigration [61]. VLA-4 of the p-integrin family is expressed on the surface of eosinophils [62] where it can interact selectively with the activated endothelial ligand, VCAM-1 [63]. Endothelial cell products and adhesion molecules (β1-, β2-, and β7-intergrins) [62] also participate in eosinophil trafficking.

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TABLE 7.2. Eosinophils' chemoattractants
Platelet activating factor (PAF), complement factor C5a, leukotriene B_4 (LTB_4), LTC_4, *N*-formyl-methyionyl-leucyl-phenylalanine (fMLP), 5-oxo-15-HETE, 5,15-diHETE
Cytokines
IL-2, IL-3, IL-5, IL-16, GM-CSF, Ecalectin
Chemokines
Eotaxins 1–3, MCP-2, MCP-3, MCP-4, RANTES, MIP-1α, IL-8, MDC

Eosinophils express the chemokine receptors CCR3 and CCR1 [64–66]. Ligation of these receptors by macrophage inflammatory protein (MIP)-1α, RANTES, macrophage chemotactic protein (MCP)-2, MCP-3, MCP-4, eotaxin-1, -2, and -3, and mucosa-associated epithelial chemokine (MEC) leads to increased expression of these receptors. Eosinophils also express chemokine receptor such as CXCR3, CXCR4, CCR5, CCR6, and CCR8 after IL-5 activation [67–69]. RANTES as well as the acidic mammalian chitinase participates in eosinophil chemoattraction [26,70] (Table 7.2). Trafficking of eosinophils into inflammatory sites involves several cytokines such as IL-4, IL-5, and IL-13. These cytokines, particularly IL-5, modulate leukocyte recruitment. IL-4 and IL-13 act as potent inducers of the eotaxins [71,72].

ACTIVATION OF EOSINOPHIL DEGRANULATION AND MEDIATOR RELEASE

For eosinophils to participate in the pathogenesis of local tissue inflammation, damage, remodeling, and fibrosis, more must occur besides their selective recruitment and accumulation. In fact, the state of activation of recruited eosinophils is now considered to be a critical aspect of their participation in disease pathophysiology. For example, a number of strains of IL-5 transgenic mice have massively increased numbers of eosinophils in their blood, spleen, and other tissues and organs, but without a second signal, these mice show little evidence of eosinophil-mediated pathology and are relatively healthy. One major pathway by which eosinophils are activated is through the cross-linking of surface immunoglobulin receptors, especially those involving IgA, and to a lesser extent IgG [73]. Although there is general agreement that murine eosinophils do not express the high-affinity FcεRI IgE receptor, its expression remains controversial for human eosinophils, which express very low levels, if any, FcεRI on their surface, and if present, clearly lack the β chain; so significant, direct activation of eosinophils via IgE remains highly unlikely [74–76]. Activation of eosinophils by a variety of cytokine and lipid stimuli leads to eosinophil granule protein release (e.g., ECP, EDN, EPO, MBP), superoxide generation, and the synthesis of LTC_4 [73,77], although secretion of the granule cationic proteins is difficult to induce with traditional eosinophil-activating stimuli. Eosinophils also elaborate the lipid mediator PAF and a wide range of cytokines and chemokines, not the least of which include IL-1β and TGF-β, two key players in the eosinophil-mediated tissue remodeling and fibrosis seen in many eosinophil associated diseases [78–81]. Although the quantities of cytokines and chemokines released by the eosinophil compared to other leukocytes vary widely, GM-CSF is among the cytokines produced in greatest quantities by eosinophils [73], and as noted earlier, functions in part in an autocrine fashion to prevent apoptosis and prolong eosinophil survival once they are recruited into sites of tissue inflammation. Eosinophil priming for increased secretory and oxidative responses, for example, by IL-5 and the eotaxins, and subsequent activation in vitro by a number of physiologically relevant agonists that include IL-5, IFN-γ, sIgA, and others, has been shown to induce secretion of the granule cationic proteins (EPO, MBP1, EDN, ECP) and eosinophil-expressed cytokines (e.g., RANTES and IL-4) in a unique process termed PMD [82,83] that involves the differential mobilization followed by vesicular transport of these proteins to the cell surface and into the extracellular space [84–86]. This is in marked contrast to more classical secondary granule fusion and exocytosis, events rarely observed for eosinophils at sites of inflammation in tissues [87] but frequently observed in eosinophil-mediated killing of the larval stages of parasites (helminths), for example, schistosomula of *Schistosoma mansoni*. Once secreted, the eosinophil granule cationic proteins are reported to have multiple proinflammatory activities that have been well-defined both in vitro and in vivo, including membrane, cell, and tissue-damaging cytotoxicity [88,89], the ability to selectively activate other inflammatory cells, such as mast cells and basophils to release inflammatory mediators (e.g., histamine) [90], and oxidative burst in neutrophils, potent blocking activity for inhibitory M2 muscarinic receptors in the airways in allergic asthma models [91], as well as the ability to augment TGF-β primed fibroblast elaboration of the inflammatory and profibrotic IL-6 family of cytokines including IL-6 and IL-11 [92], to name just a few. Thus, eosinophils (1) come fully armed with preformed mediators of inflammation, tissue damage, remodeling, and fibrogenesis that are secreted at sites of eosinophilic inflammation in tissues in many eosinophil-associated allergic diseases such as asthma and hypereosinophilic syndromes such as eosinophilic esophagitis and (2) have the capacity to generate newly formed protein (cytokines, chemokines) and lipid mediators (LTC_4, PAF) of inflammation when primed

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TABLE 7.3. Eosinophils' activators

IL-3, IL-5, GM-CSF, IFN, PAF, LTB4, eotaxins, sIgA

and further activated in the process of their recruitment from the bone marrow into the tissue in response to allergic (e.g., mast cell, basophil) and other inflammatory and/or Th2 T-cell stimuli [93] (Table 7.3).

SIGNAL TRANSDUCTION

The process in which eosinophils convert external stimuli into biochemical reactions, similar to many aspects of eosinophil biology, is understudied. A number of factors are known to rapidly activate eosinophils ranging from growth factors, extracellular matrix proteins, cytokines, and neural factors. For instance, fibronectin can modify cell signaling and subsequent migratory capacity of eosinophils [94]. This is of particular interest since fibronectin levels from bronchoalveolar lavage fluids of asthmatic patients increase after antigen challenge, and correlates with eosinophilic inflammation [95,96]. The increase in airway fibronectin may contribute to eosinophil persistence by modulating the survival, recruitment, and retention of lung eosinophils [96,97].

While a number of factors lead to cell surface activation of signal transduction pathways in eosinophils, a new body of evidence suggests that eosinophil granules may also undergo activation with release of preformed proteins and cytokines. Granule surfaces contain IFN-γ receptors and G-protein–coupled receptors for eotaxin providing a means for rapid release [98].

Other studies have investigated the role of eosinophils on activating other resident mucosal cells such as nerves. For instance, the adhesion of eosinophils to cholinergic nerves led to a rapid and sustained activation of the nuclear transcription factors such as nuclear factor (NF)-κB and activator protein (AP)-1. Eosinophil binding to neuronal ICAM-1 contributes to a rapid activation of ERK1/2 whereas eosinophil adhesion to VCAM-1 results in AP-1 activation, mediated in part by rapid activation of the p38 mitogen-activated protein kinase signaling pathway [99].

SUMMARY

Eosinophils are multifunctional leukocytes that likely participate in the initiation and propagation of inflammatory responses. Much is yet to be learned about eosinophil biology. To date, their function is not certain but their presence at mucosal surfaces, such as the gastrointestinal tract, in the healthy host suggests a protective role. Their increasing recognition in increased numbers in disease states supports a role in the pathogenesis of allergic, inflammatory, and neoplastic diseases. As more knowledge accumulates regarding the mechanisms of mediator production and release, chemotaxis, and life cycle, we will begin to understand the role of this enigmatic cell in health and disease.

KEY POINTS

Eosinophils can synthesize a number of biologically active mediators and contain a number of preformed proteins both of which can be released upon appropriate stimulation.

A variety of mediators including eotaxins, PAF, LTB_4, fMLP are potent eosinophil chemoattractants.

While eosinophils occupy 1%–6% of the peripheral white blood cell count, they are normally present in larger numbers within mucosal surfaces and increase during disease states.

ACKNOWLEDGMENTS

We are grateful to Joanna Grenawalt and Joanne Masterson for their expertise in obtaining human and murine micrographs.

REFERENCES

1. Sparrevohn, S., and Wulff, H.R. 1967. The nuclear segmentation of eosinophils under normal and pathological conditions. *Acta Haematol* **37:**120.
2. Afshar, K., Vucinic, V., and Sharma, O.P. 2007. Eosinophil cell: pray tell us what you do! *Curr Opin Pulmon Med* **13:**414.
3. Weller, P.F., Bozza, P.T., Yu, W., et al. 1999. Cytoplasmic lipid bodies in eosinophils: central roles in eicosanoid generation. *Int Arch Allergy Immunol* **118:**450.
4. Weller, P.F., Monahan-Earley, R.A., Dvorak, H.F., et al. 1991. Cytoplasmic lipid bodies of human eosinophils. Subcellular isolation and analysis of arachidonate incorporation. *Am J Pathol* **138:**141.
5. Bozza, P.T., Yu, W., Penrose, J.F., et al. 1997. Eosinophil lipid bodies: specific, inducible intracellular sites for enhanced eicosanoid formation. *J Exp Med* **186:**909.
6. Gleich, G.J., Adolphson, C.R., and Leiferman, K.M. 1993. The biology of the eosinophilic leukocyte. *Annu Rev Med* **44:**85.
7. Leiferman, K.M. 1991. A current perspective on the role of eosinophils in dermatologic diseases. *J Am Acad Dermatol* **24:**1101.
8. Weiss, S.J., Test, S.T., Eckmann, C.M., et al. 1986. Brominating oxidants generated by human eosinophils. *Science* **234:**200.
9. Moqbel, R., and Lacy, P. 2000. New concepts in effector functions of eosinophil cytokines. *Clin Exp Allergy* **30:**1667.
10. Spencer, L.A., Szela, C.T., Perez, S.A., et al. 2009. Human eosinophils constitutively express multiple Th1, Th2, and immunoregulatory cytokines that are secreted rapidly and differentially. *J Leukoc Biol* **85:**117.
11. Lacy, P., and Moqbel, R. 2000. Eosinophil cytokines. *Chem Immunol* **76:**134.
12. Rosenberg, H.F., Phipps, S., and Foster, P.S. 2007. Eosinophil trafficking in allergy and asthma. *J Allergy Clin Immunol* **119:**1303.

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13. Flood-Page, P.T., Menzies-Gow, A.N., Kay, A.B., et al. 2003.Eosinophil's role remains uncertain as anti-interleukin-5 only partially depletes numbers in asthmatic airway. *Am J Respir Crit Care Med* **167:**199.
14. Foster, P.S., Hogan, S.P., Ramsay, A.J., et al. 1996. Interleukin 5 deficiency abolishes eosinophilia, airways hyperreactivity, and lung damage in a mouse asthma model. *J Exp Med* **183:**195.
15. Hamelmann, E., and Gelfand, E.W. 2001. IL-5-induced airway eosinophilia – the key to asthma? *Immunol Rev* **179:**182.
16. Boyce, J.A., and Austen, K.F. 2005. No audible wheezing: nuggets and conundrums from mouse asthma models. *J Exp Med* **201:**1869.
17. Foster, P.S., Mould, A.W., Yang, M., et al. 2001. Elemental signals regulating eosinophil accumulation in the lung. *Immunol Rev* **179:**173.
18. Hogan, S.P., Koskinen, A., Matthaei, K.I., et al. 1998. Interleukin-5-producing CD4+ T cells play a pivotal role in aeroallergen-induced eosinophilia, bronchial hyperreactivity, and lung damage in mice. *Am J Respir Crit Care Med* **157:**210.
19. Mattes, J., Yang, M., Mahalingam, S., et al. 2002. Intrinsic defect in T cell production of interleukin (IL)-13 in the absence of both IL-5 and eotaxin precludes the development of eosinophilia and airways hyperreactivity in experimental asthma. *J Exp Med* **195:**1433.
20. Lee, J.J., McGarry, M.P., Farmer, S.C., et al. 1997. Interleukin-5 expression in the lung epithelium of transgenic mice leads to pulmonary changes pathognomonic of asthma. *J Exp Med* **185:**2143.
21. Lee, N.A., McGarry, M.P., Larson, K.A., et al. 1997. Expression of IL-5 in thymocytes/T cells leads to the development of a massive eosinophilia, extramedullary eosinophilopoiesis, and unique histopathologies. *J Immunol* **158:**1332.
22. Mishra, A., Hogan, S.P., Brandt, E.B., et al. 2002. Enterocyte expression of the eotaxin and interleukin-5 transgenes induces compartmentalized dysregulation of eosinophil trafficking. *J Biol Chem* **277:**4406.
23. Terada, N., Hamano, N., Nomura, T., et al. 2000. Interleukin-13 and tumour necrosis factor-alpha synergistically induce eotaxin production in human nasal fibroblasts. *Clin Exp Allergy* **30:**348.
24. LaPorte, S.L., Juo, Z.S., Vaclavikova, J., et al. 2008. Molecular and structural basis of cytokine receptor pleiotropy in the interleukin-4/13 system. *Cell* **132:**259.
25. Wills-Karp, M. 2004. Interleukin-13 in asthma pathogenesis. *Immunol Rev* **202:**175.
26. Zimmermann, N., Hershey, G.K., Foster, P.S., et al. 2003. Chemokines in asthma: cooperative interaction between chemokines and IL-13. *J Allergy Clin Immunol* **111:**227.
27. Kita, H. 1996. The eosinophil: a cytokine-producing cell? *J Allergy Clin Immunol* **97:**889.
28. Liu, L.Y., Bates, M.E., Jarjour, N.N., et al. 2007. Generation of Th1 and Th2 chemokines by human eosinophils: evidence for a critical role of TNF-alpha. *J Immunol* **179:**4840.
29. Straumann, A., and Simon, H.U. 2004. The physiological and pathophysiological roles of eosinophils in the gastrointestinal tract. *Allergy* **59:**15.
30. Bisset, L.R., and Schmid-Grendelmeier, P. 2005. Chemokines and their receptors in the pathogenesis of allergic asthma: progress and perspective. *Curr Opin Pulmon Med* **11:**35.
31. Rincon, M., Anguita, J., Nakamura, T., et al. 1997. Interleukin (IL)-6 directs the differentiation of IL-4-producing CD4+ T cells. *J Exp Med* **185:**461.
32. Ottonello, L., Montecucco, F., Bertolotto, M., et al. 2005. CCL3 (MIP-1alpha) induces in vitro migration of GM-CSF-primed human neutrophils via CCR5-dependent activation of ERK 1/2. *Cell Signal* **17:**355.
33. Surquin, M., Le Moine, A., Flamand, V., et al. 2002. Skin graft rejection elicited by beta 2-microglobulin as a minor transplantation antigen involves multiple effector pathways: role of Fas-Fas ligand interactions and Th2-dependent graft eosinophil infiltrates. *J Immunol* **169:**500.
34. Debinski, W., Obiri, N.I., Pastan, I., et al. 1995. A novel chimeric protein composed of interleukin 13 and Pseudomonas exotoxin is highly cytotoxic to human carcinoma cells expressing receptors for interleukin 13 and interleukin 4. *J Biol Chem* **270:**16775.
35. Cormier, S.A., Taranova, A.G., Bedient, C., et al. 2006. Pivotal advance: eosinophil infiltration of solid tumors is an early and persistent inflammatory host response. *J Leukoc Biol* **79:**1131.
36. Jacobsen, E.A., Taranova, A.G., Lee, N.A., et al. 2007. Eosinophils: singularly destructive effector cells or purveyors of immunoregulation? *J Allergy Clin Immunol* **119:**1313.
37. Aceves, S.S., Newbury, R.O., Dohil, R., et al. 2007. Esophageal remodeling in pediatric eosinophilic esophagitis. *J Allergy Clin Immunol* **119:**206.
38. Wang, H., Tan, X., Chang, H., et al. 1999. Platelet-activating factor receptor mRNA is localized in eosinophils and epithelial cells in rat small intestine: regulation by dexamethasone and gut flora. *Immunology* **97:**447.
39. Fujii, M., Tanaka, H., and Abe, S. 2005. Interferon-gamma up-regulates expression of cysteinyl leukotriene type 2 receptors on eosinophils in asthmatic patients. *Chest* **128:**3148.
40. Zinchuk, O., Fukushima, A., Zinchuk, V., et al. 2005. Direct action of platelet activating factor (PAF) induces eosinophil accumulation and enhances expression of PAF receptors in conjunctivitis. *Mol Vis* **11:**114.
41. Bandeira-Melo, C., Bozza, P.T., Diaz, B.L., et al. 2000. Cutting edge: lipoxin (LX) A4 and aspirin-triggered 15-epi-LXA4 block allergen-induced eosinophil trafficking. *J Immunol* **164:**2267.
42. Ohshima, N., Nagase, H., Koshino, T., et al. 2002. A functional study on CysLT(1) receptors in human eosinophils. *Int Arch Allergy Immunol* **129:**67.
43. Powell, W.S., Chung, D., and Gravel, S. 1995. 5-Oxo-6, 8,11,14-eicosatetraenoic acid is a potent stimulator of human eosinophil migration. *J Immunol* **154:**4123.
44. Rothenberg, M.E., and Hogan, S.P. 2006. The eosinophil. *Annu Rev Immunol* **24:**147.
45. Mori, Y., Iwasaki, H., Kohno, K., et al. 2008. Identification of the human eosinophil lineage-committed progenitor: revision of phenotypic definition of the human common myeloid progenitor. *J Exp Med*.206**:**183–193.
46. Bedi, R, Du, J., Sharma, A.K., et al. 2009. Human C/EBP-epsilon activator and repressor isoforms differentially reprogram myeloid lineage commitment and differentiation. *Blood* **113:**317.
47. McNagny, K., and Graf, T. 2002. Making eosinophils through subtle shifts in transcription factor expression. *J Exp Med* **195:**F43.
48. Yu, C., Cantor, A.B., Yang, H., et al. 2002. Targeted deletion of a high-affinity GATA-binding site in the GATA-1 promoter leads to selective loss of the eosinophil lineage in vivo. *J Exp Med* **195:**1387.
49. Du, J., Stankiewicz, M.J., Liu, Y., et al. 2002. Novel combinatorial interactions of GATA-1, PU.1, and

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Fundamentals of Inflammation, edited by Charles N. Serhan, et al., Cambridge University Press, 2010. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/yale-ebooks/detail.action?docID=802921.
Created from yale-ebooks on 2026-06-19 00:03:59.

C/EBPepsilon isoforms regulate transcription of the gene encoding eosinophil granule major basic protein. *J Biol Chem* **277:**43481.
50. Zimmermann, N., Daugherty, B.L., Kavanaugh, J.L., et al. 2000. Analysis of the CC chemokine receptor 3 gene reveals a complex 5' exon organization, a functional role for untranslated exon 1, and a broadly active promoter with eosinophil-selective elements. *Blood* **96:**2346.
51. Metcalf, D., Begley, C.G., Nicola, N.A., et al. 1987. Quantitative responsiveness of murine hemopoietic populations in vitro and in vivo to recombinant multi-CSF (IL-3). *Exp Hematol* **15:**288.
52. Metcalf, D., Burgess, A.W., Johnson, G.R., et al. 1986. In vitro actions on hemopoietic cells of recombinant murine GM-CSF purified after production in *Escherichia coli*: comparison with purified native GM-CSF. *J Cell Physiol* **128:**421.
53. Yamaguchi, Y., Suda, T., Suda, J., et al. 1988. Purified interleukin 5 supports the terminal differentiation and proliferation of murine eosinophilic precursors. *J Exp Med* **167:**43.
54. Sanderson, C.J. 1992. Interleukin-5, eosinophils, and disease. *Blood* **79:**3101.
55. Collins, P.D., Marleau, S., Griffiths-Johnson, D.A., et al. 1995. Cooperation between interleukin-5 and the chemokine eotaxin to induce eosinophil accumulation in vivo. *J Exp Med* **182:**1169.
56. Capra, V., and Rovati, G.E. 2004. Leukotriene modifiers in asthma management. *IDrugs* **7:**659.
57. Wardlaw, A.J., Walsh, G.M., and Symon, F.A. 1994. Mechanisms of eosinophil and basophil migration. *Allergy* **49:**797.
58. Wardlaw, A.J. 2000. The role of adhesion in eosinophil function. *Chem Immunol* **78:**93.
59. Sriramarao, P., von Andrian, U.H., Butcher, E.C., et al. 1994. L-selectin and very late antigen-4 integrin promote eosinophil rolling at physiological shear rates in vivo. *J Immunol* **153:**4238.
60. Rankin, S.M., Conroy, D.M., Williams, T.J. 2000. Eotaxin and eosinophil recruitment: implications for human disease. *Mol Med Today* **6:**20.
61. Hogan, S.P., Rothenberg, M.E., Forbes, E., et al. 2004. Chemokines in eosinophil-associated gastrointestinal disorders. *Curr Allergy Asthma Rep* **4:**74.
62. Bochner, B.S., and Schleimer, R.P. 1994. The role of adhesion molecules in human eosinophil and basophil recruitment. *J Allergy Clin Immunol* **94:**427.
63. Elices, M.J., Osborn, L., Takada, Y., et al. 1990. VCAM-1 on activated endothelium interacts with the leukocyte integrin VLA-4 at a site distinct from the VLA-4/fibronectin binding site. *Cell* **60:**577.
64. Ponath, P.D., Qin, S., Post, T.W., et al. 1996. Molecular cloning and characterization of a human eotaxin receptor expressed selectively on eosinophils. *J Exp Med* **183:**2437.
65. Phillips, R.M., Stubbs, V.E., Henson, M.R., et al. 2003. Variations in eosinophil chemokine responses: an investigation of CCR1 and CCR3 function, expression in atopy, and identification of a functional CCR1 promoter. *J Immunol* **170:**6190.
66. Elsner, J., Dulkys, Y., Gupta, S., et al. 2005. Differential pattern of CCR1 internalization in human eosinophils: prolonged internalization by CCL5 in contrast to CCL3. *Allergy* **60:**1386.
67. Sullivan, S.K., McGrath, D.A., Liao, F., et al. 1999. MIP-3alpha induces human eosinophil migration and activation of the mitogen-activated protein kinases (p42/p44 MAPK). *J Leukoc Biol* **66:**674.
68. Nagase, H., Miyamasu, M., Yamaguchi, M., et al. 2000. Glucocorticoids preferentially upregulate functional CXCR4 expression in eosinophils. *J Allergy Clin Immunol* **106:**1132.
69. Oliveira, S.H., Lira, S., Martinez, A.C., et al. 2002. Increased responsiveness of murine eosinophils to MIP-1beta (CCL4) and TCA-3 (CCL1) is mediated by their specific receptors, CCR5 and CCR8. *J Leukoc Biol* **71:**1019.
70. Zhu, Z., Zheng, T., Homer, R.J., et al. 2004. Acidic mammalian chitinase in asthmatic Th2 inflammation and IL-13 pathway activation. *Science* **304:**1678.
71. Moser, R., Fehr, J., and Bruijnzeel, P.L. 1992. IL-4 controls the selective endothelium-driven transmigration of eosinophils from allergic individuals. *J Immunol* **149:**1432.
72. Sher, A., Coffman, R.L., Hieny, S., et al. 1990. Ablation of eosinophil and IgE responses with anti-IL-5 or anti-IL-4 antibodies fails to affect immunity against Schistosoma mansoni in the mouse. *J Immunol* **145:**3911.
73. Kita, H., Adolphson, C.R., and Gleich, G.J. 2003. Biology of eosinophils. In *Allergy Principles and Practice*, Adkinson, J., Yunginger, J.W., Busse, W.W., et al. (eds.), 6th Ed., p. 305 Philadelphia: Mosby.
74. Kayaba, H., Dombrowicz, D., Woerly, G., et al. 2001. Human eosinophils and human high affinity IgE receptor transgenic mouse eosinophils express low levels of high affinity IgE receptor, but release IL-10 upon receptor activation. *J Immunol* **167:**995.
75. Kita, H., Kaneko, M., Bartemes, K.R., et al. 1999. Does IgE bind to and activate eosinophils from patients with allergy? *J. Immunol* **162:**6901.
76. Sihra, B.S., Kon, O.M., Grant, J.A., et al. 1997. Expression of high-affinity IgE receptors (FcεRI) on peripheral blood basophils, monocytes, and eosinophils in atopic and nonatopic subjects: relationship to total serum IgE concentrations. *J Allergy Clin Immunol* **99:**699.
77. Bandeira-Melo, C., and Weller, P.F. 2003. Eosinophils and cysteinyl leukotrienes. *Prostaglandins Leukot Essent Fatty Acids* **69:**135.
78. Gharaee-Kermani, M., and Phan, S.H. 1998. The role of eosinophils in pulmonary fibrosis (Review). *Int J Mol Med* **1:**43.
79. Gomes, I., Mathur, S.K., Espenshade, B.M., et al. 2005. Eosinophil-fibroblast interactions induce fibroblast IL-6 secretion and extracellular matrix gene expression: implications in fibrogenesis. *J Allergy Clin Immunol* **116:**796.
80. Levi-Schaffer, F., Garbuzenko, E., Rubin, A., et al. 1999. Human eosinophils regulate human lung- and skin-derived fibroblast properties in vitro: a role for transforming growth factor beta (TGF-beta). *Proc Natl Acad Sci USA* **96:**9660.
81. Spry, C.J. 1989. The pathogenesis of endomyocardial fibrosis: the role of the eosinophil. *Springer Semin Immunopathol* **11:**471.
82. Dvorak, A.M., Ackerman, S.J., Furitsu, T., et al. 1992. Mature eosinophils stimulated to develop in human-cord blood mononuclear cell cultures supplemented with recombinant human interleukin-5. II. Vesicular transport of specific granule matrix peroxidase, a mechanism for effecting piecemeal degranulation. *Am J Pathol* **140:**795.

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83. Melo, R.C., Perez, S.A., Spencer, L.A., et al. 2005. Intragranular vesiculotubular compartments are involved in piecemeal degranulation by activated human eosinophils. *Traffic* **6:**866.
84. Logan, M.R., Odemuyiwa, S.O., and Moqbel, R. 2003. Understanding exocytosis in immune and inflammatory cells: the molecular basis of mediator secretion. *J Allergy Clin Immunol* **111:**923.
85. Moqbel, R., and Coughlin, J.J. 2006. Differential secretion of cytokines. *Sci STKE* **2006:**pe26.
86. Moqbel, R., and Lacy, P. 1999. Exocytotic events in eosinophils and mast cells. *Clin Exp Allergy* **29:**1017.
87. Dvorak, A.M., Ackerman, S.J., and Weller, P.F. 1990. Subcellular morphology and biochemistry of eosinophils. In *Blood Cell Biochemistry: Megakaryocytes, Platelets, Macrophages and Eosinophils*. Harris, J.R. (ed.), Vol 2, p.237. London: Plenum Publishing Corporation.
88. Gleich, G.J. 2000. Mechanisms of eosinophil-associated inflammation. *J Allergy Clin Immunol* **105:**651.
89. Martin, L.B., Kita, H., Leiferman, K.M., et al. 1996. Eosinophils in allergy: role in disease, degranulation, and cytokines. *Int Arch Allergy Immunol* **109:**207.
90. Thomas, L.L., and Page, S.M. 2000. Inflammatory cell activation by eosinophil granule proteins. *Chem Immunol* **76:**99.
91. Jacoby, D.B., Costello, R.M., and Fryer, A.D. 2001. Eosinophil recruitment to the airway nerves. *J Allergy Clin Immunol* **107:**211.
92. Rochester, C.L., Ackerman, S.J., Zheng, T., et al. 1996. Eosinophil-fibroblast interactions. Granule major basic protein interacts with IL-1 and transforming growth factor-beta in the stimulation of lung fibroblast IL-6-type cytokine production. *J Immunol* **156:**4449.
93. Sedgwick, J.B., Calhoun, W.J., Vrtis, R.F., et al. 1992. Comparison of airway and blood eosinophil function after in vivo antigen challenge. *J Immunol* **149:**3710.
94. Holub, A., Byrnes, J., Anderson, S., et al. 2003. Ligand density modulates eosinophil signaling and migration. *J Leukoc Biol* **73:**657.
95. Meerschaert, J., Kelly, E.A., Mosher, D.F., et al. 1999. Segmental antigen challenge increases fibronectin in bronchoalveolar lavage fluid. *Am J Respir Crit Care Med* **159:**619.
96. Meerschaert, J., Vrtis, R.F., Shikama, Y., et al. 1999. Engagement of alpha4beta7 integrins by monoclonal antibodies or ligands enhances survival of human eosinophils in vitro. *J Immunol* **163:**6217.
97. Anwar, A.R., Moqbel, R., Walsh, G.M., et al. 1993. Adhesion to fibronectin prolongs eosinophil survival. *J Exp Med* **177:**839.
98. Neves, J.S., Perez, S.A., Spencer, L.A., et al. 2008. Eosinophil granules function extracellularly as receptor-mediated secretory organelles. *Proc Natl Acad Sci USA* **105:**18478.
99. Walsh, M.T., Curran, D.R., Kingham, P.J., et al. 2004. Effect of eosinophil adhesion on intracellular signaling in cholinergic nerve cells. *Am J Respir Cell Mol Biol* **30:**333.

SUGGESTED READINGS

Fillon, S., Robinson, Z.D., Colgan, S.P., and Furuta, G.T. 2009. Epithelial function in eosinophilic gastrointestinal diseases. *Immunol Allergy Clin North Am* **29**(1)**:**171–178.

Gleich, G.J., and Leiferman, K.M. 2009. The hypereosinophilic syndromes: current concepts and treatments. *Br J Haematol* **145**(3)**:**271–285.

Hogan, S.P., Rosenberg, H.F., Moqbel, R., et al. 2008. Eosinophils: biological properties and role in health and disease. *Clin Exp Allergy* **38**(5)**:**709–750.

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8 Macrophages

Sarah Fox and Adriano G. Rossi

INTRODUCTION

Macrophages are a major leukocyte involved in orchestrating inflammatory responses. Their name is derived from the Greek term "big eaters" (*makros*, large; and *phagein*, eat). This gives some insight into the primary function of this cell in clearance of invading pathogens, cell debris, and apoptotic cells by a process of engulfment and digestion called "phagocytosis." However, the role of the macrophage goes beyond that of what its name suggests. They are endowed with the ability to rapidly react to and secrete a plethora of biological agents and mediators that can influence the initiation, progression, and resolution of an inflammatory response and coordinate processes to establish acquired immunity against specific pathogens. This chapter is an overview of the basics of macrophage biology and function, with particular insights into the involvement of macrophages in disease pathogenesis as well as pharmacological modulation of macrophage responses as targets for treatment of disease.

IDENTIFICATION

Historical Discovery

Elie Metchnikoff first used the term "macrophage" to describe large mononuclear phagocytic cells he observed in tissues over 100 years ago. Macrophages are now recognized as the major phagocytic cell type with diverse characteristics and localities around the body where they are important for both innate and acquired immune responses as well as maintenance of tissue homeostasis and regulation of various processes subsidiary to the immune defense such as hematopoiesis.

Hematopoietic Lineage

In man, macrophages are derived from bone marrow hematopoietic cells and constitute the terminally differentiated cell type from the mononuclear phagocytic cell lineage. Monocytes are the immediate precursor cell types of macrophages and these cells have important functions in inflammatory responses on their own. It is monocytes that are liberated from bone marrow to circulate in the bloodstream where they are recruited to sites of inflammation. Tissue macrophages are present constitutively in the absence of inflammation where they are thought to contribute to the maintenance of many processes (see heterogeneity). During an inflammatory response, monocytes differentiate into macrophages within the tissue where they remain until their useful function is completed. It is thought that they either die by apoptosis *in situ* or are cleared by drainage to the lymphatic system.

Phenotypic Characteristics

Macrophages are characterized by their large size (approx 5–50 μm in diameter) and irregular shape. They generally contain a large kidney-shaped nuclear lobe (mononuclear). The cytoplasm is abundant and contains many large and small granules, which contain the chemical cocktail needed to kill and digest invading pathogens.

Cell Surface Markers, Isolation of Macrophages

Macrophages are endorsed with an impressive repertoire of surface receptors, which allow these cells to have functionally diverse actions. Many of these receptors are not exclusive to macrophage cell populations

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and are shared by other cell types such as granulocytes and lymphocytes. There are macrophage-specific cell surface markers, which are used to identify and isolate these cells from unpurified cell populations. The F4/80 antibody was developed as one of the first macrophage-specific markers and remains one of the most common markers used for this purpose. The F4/80 antibody recognizes human CD97, a glycoprotein 7-transmembrane domain protein receptor resembling the G-protein-coupled peptide hormone receptor family. It is expressed highly in tissue macrophage populations, dendritic cells (DCs), Langerhans cells, and microglia. The receptor is also expressed in small to moderate levels in blood monocytes and eosinophillic granulocytes. No clear function of CD97 is apparent although the high degree of posttranslational modification of this protein could be linked to diverse functional characteristics of macrophages including cell–cell recognition and adhesion. Fluorescent markers conjugated to antibodies against the F4/80 antigen can be used to distinguish macrophages from other cell types. In this context, a cell sorter can distinguish those cells with higher fluorescent profiles, and due to the relatively specific binding capacity of the antibody for the macrophage F4/80 antigen, a purified macrophage population can then be singled out for analysis.

Cell Surface Receptors; Pathogen Recognition Receptors

The ability to discriminate between self and foreign particles by cells involves the recognition motifs expressed on prokaryotic cells. These motifs are highly conserved within the pathogenic organisms and are subsequently not subjected to the high rates of mutation occurring in other aspects of prokaryotic biology. Pathogen recognition receptors (PRRs) are receptors located both on the cell surface and in the intracellular environment that can recognize an array of pathogen-associated molecular patterns (PAMPs). On the cell surface, the Toll-like receptors (TLRs) form the major class of PRRs found on the surface of macrophages. There are 10 known TLRs in humans and 13 in mice, and most of these TLRs are located on the cell membrane although some are found in intracellular membranes such as those found on endosomes or lysosomes. These receptors endow the cell with the ability to recognize a range of PAMPs including lipid and carbohydrate residues from Gram-positive bacteria (TLR1, TLR2, and TLR6), LPS from Gram-negative bacteria (TLR4), fungal pathogens (TLR2), bacterial proteins such as flagellin and profilin (TLR5 and TLR11 respectively), nucleic acid and its derivatives (TLR3, TLR7, TLR8, and TLR9), double-stranded RNA (TLR3), synthetic antiviral compounds and guanine nucleotide analogues (murine TLR7 and human TLR8), and DNA containing unmethylated CpG-motifs abundant in bacterial DNA (TLR9). The TLRs can also form heterodimers with other subtypes of TLRs or form complexes with accessory molecules (LPS-binding protein and CD14 in the case of TLR4). TLR-independent recognition of pathogens is important particularly for intracellular recognition of pathogens and can occur via the nucleotide-binding oligomerization domain (NOD)-like receptors (NLRs) or retinoic acid–inducible gene I (RIG-I)-like helicases (RLHs). Macrophages use these receptors to form robust responses to various types of pathogenic infection, which stimulate an array of second messenger pathways to mount diverse functional immune responses.

Heterogeneity

All members of the mononuclear phagocytic system (MPS) cell lineage share homology in morphological and ultrastructural features, expression of specific enzymes and surface receptors, and the ability to engulf various particles. However, there are many different morphologically and functionally diverse subsets of macrophages that have adapted to perform tasks ideally suited to the surrounding environment.

Bone Marrow Macrophages

Cells of the monocytes-macrophage line are recruited into the bloodstream where they circulate and migrate into tissues where they differentiate into specific tissue macrophages or, depending on the inflammatory status of the tissue environment, they will form activated inflammatory macrophages. However, mature macrophages are also found within the bone marrow where they play an important role in support and regulation of hematopoiesis. Release of granulocyte-macrophage colony-stimulating factor (GM-CSF) can stimulate maturation of granulocyte macrophage and eosinophil colony formation in vitro.

Alveolar Macrophages

Alveolar macrophages are a distinct population of cells found within the lung where they provide protection against pathogens and particles inhaled from the air. They are the only type of macrophage that are exposed to environmental air and, as such, have developed characteristics appropriate for the task of clearing the abundance of potentially harmful bodies entering the body through this route. They characteristically contain a large number of membrane-bound cytoplasmic inclusions, indicative of their ability to rapidly release a barrage of proteolytic enzymes and reactive oxygen species to kill and clear foreign bodies. Damage to these cells is an important factor associated with increased host susceptibility to airborne infection and environmental toxins.

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Splenic Macrophages

Resident macrophages in the spleen are thought to function to trap and process foreign antigens, coordinate with T and B cells in the lymphoid areas, as well as the engulfment and digestion of red blood cells in the red pulp. The specific phenotype of macrophages in the spleen varies between different locations and likely reflects the degree of functional specialization of these cells.

Kupffer Cells

The resident macrophages in the liver called Kupffer cells are responsible for clearance of particulate and soluble substances from the portal circulation primarily by phagocytosis. They are responsible for the detoxification of endotoxin from the portal circulation and can present antigens to initiate T-cell lymphocyte responses. Ingestion of microorganisms by Kupffer cells triggers the release of a deluge of inflammatory mediators that rapidly recruits monocytes to the liver to aid in clearance of invading pathogens. Thus, these cells are considered as an important front in the defense against infection but also contribute to acute liver damage sustained during liver toxicity (e.g., alcohol consumption) and cancer.

Microglial Cells

Microglial cells are the first and main line of defense against pathogens of the central nervous system (CNS). They differentiate from myeloid progenitor cells normally found within the bone marrow and differ from macrophages in their responses to various stimuli and their turnover rate. Microglial cells do not need to be replaced as often as other macrophages and have adapted a range of quiescent and activated states to maintain their function over longer periods. They are termed nonprofessional phagocytes due to their need to be activated before functions such as phagocytosis and antigen presentation are performed. They can rapidly respond to CNS challenges due to the ability of these cells to use potassium channels and changes in potassium levels as sensors of injury or infection.

Langerhans Cells

Langerhans cells are a type of DCs found predominantly and in large numbers in the epidermis where they function to recognize and capture antigens. DCs are the main cells responsible for antigen processing and presentation. They are derived from activated monocyte precursors and found in areas likely to come into contact with antigens such as the skin, nose, lungs, stomach, and intestines. They differ from tissue macrophages in their morphology and primary function that in the case of the latter is phagocytosis. During an immature stage, they extend large cytoplasmic projections called dendrites (not the same as neuronal dendrites but from which they get their name) that increase their surface area and increase contact area with surrounding antigens. The surface of DCs are endowed with PRRs such as the TLRs that recognize and uptake various bacterial and viral antigens for processing via the major histocompatibility complex (MHC) molecule. Upregulation of the surface chemotactic receptor (CCR7) encourages the transport of the DCs to lymph nodes where phenotypic changes occur to facilitate the presentation of the processed antigens on their surface with naive T cells, thus orchestrating the adaptive immune response.

Mesangial Cells

Mesangial cells are specialized phagocytic cells located in the glomerular lobules of the kidney. They are thought to contribute to the support of these kidney structures and to aid in filtration and regulation of blood flow to the glomerular capillaries through contractile responses, similar to smooth muscle cells. In fact, these cells are rare examples of phagocytic cells derived from smooth muscle and not monocytes. Mesangial cells have many other characteristics that differ from that of the smooth muscle, they can respond to and secrete a number of vasoactive and immunomodulatory substances as well as antigen presentation making these cells important in the local response to injury in the kidney.

Other

Macrophages are present in many other areas of the body particularly where you would expect exposure to or processing of foreign pathogens such as the gut and lymphoid tissues. Interestingly, macrophages can fuse to form multinucleated cells often termed giant cells. The giant cells tend to form two main phenotypes (Figure 8.1): the Langhans cells (Figure 8.1A) where the nuclei are arranged in a horseshoe or ring shape around the outer periphery of the cell and the so-called foreign body giant cell (Figure 8.1B) where the nuclei are more disorganized but often form a group of nuclei in the center of the cell. The precise functions of multinucleated giant cells, however, are unknown although they appear to be found in granulomatous disease and the origin of multinucleated osteoclasts.

MACROPHAGE ACTIVATING STIMULI

The surface of the macrophage is equipped with many receptors that stimulate macrophage function such as adhesion, chemotaxis, secretion, and phagocytosis. Activating stimuli are often found in the inflammatory milieu and are commonly produced by macrophages themselves, other inflammatory cell types, or the invading organisms. Thus, the inflammatory environment

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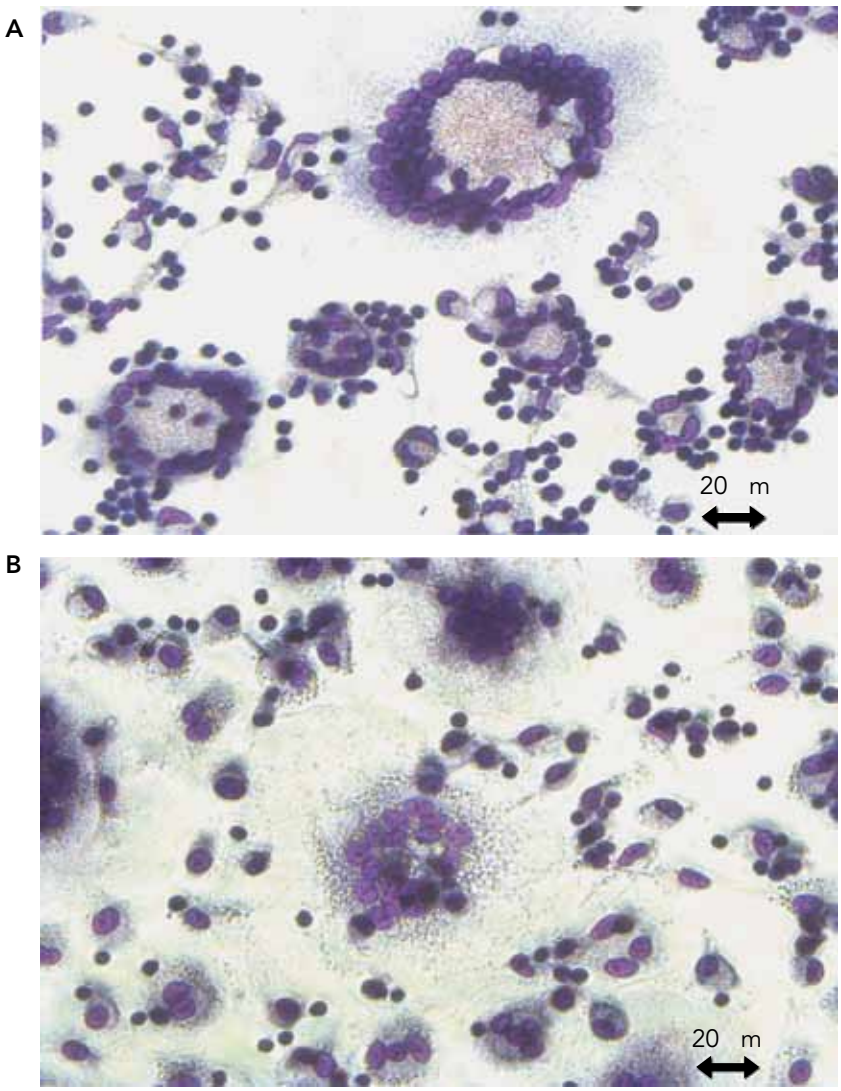


Figure 8.1. Multinucleated giant cells of human macrophages differentiated from blood-derived monocytes. (A) Represents typical images of Langhans cells and (B) depicts examples of foreign body cells. (Images derived from work by Martin F. Lister, Charlotte M. Hamilton, and Adriano G. Rossi.)

is instrumental in determining the type and intensity of the inflammatory response by the macrophage. It is imperative for macrophages to be able to distinguish between cells that are "self" and those that are "non-self" during the inflammatory process. Surface receptors expression by both macrophages and surrounding cells are a way in which macrophages are able to do this.

Fc and Complement

Immunoglobulin (Ig) antibody complexes and complement proteins are found in abundance in the serum. They bind to the outside of pathogenic material to facilitate the recognition and clearance of these particles by phagocytes. This process is called opsonization and, thus, Ig antibodies and complement are important particle opsonins. The Fc portion of the Ig complex is prominent upon opsonization of a particle and it is this portion that is readily recognized by Fc receptors (FcRs) on the surface of the macrophage. The FcRs involved in the internalization of Ig complexes are the FcγRI, FcγRIIA, and the FcγRIII. Particles opsonized by complement proteins are recognized and facilitate phagocytosis through the stimulation of the complement receptors (CRs), also located on the surface of the macrophage. Several CRs exist on macrophages, namely CR1, CR3, and CR4, which recognize several complement proteins. For example, the CR1 binds C3b, C4b, and

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C3bi. The CRs can also function to facilitate different aspects of the phagocytic process such as particle binding (CR1) and internalization (CR3 and CR4). The Fc and complement pathways of particle uptake are different in that the Fc-mediated process is constitutively active whereas the complement system can require the presence of additional stimuli, such as tumor necrosis factor alpha (TNF-α) for phagocytosis to occur. Furthermore, the initial macrophage surface changes differ between these two receptor-mediated phagocytic response with Fc receptors producing membrane projections to envelop the particle, whereas the complement system does not and the particle appears to sink into the cell. Differences also arise in the inflammatory response of macrophages following phagocytosis mediated by either receptor; the FcR process is coupled with the release of proinflammatory mediators such as reactive oxygen species (ROS) and arachidonic acid metabolites whereas CR-stimulated phagocytosis does not induce the release of either of these classes of molecules and is considered less proinflammatory.

Cytokines

During inflammation, cytokine release from inflammatory cells is greatly increased. Many different cytokines exist that orchestrate a number of important immunomodulatory roles. Macrophages can be either classically activated or alternatively activated depending on the cytokine profile of the surrounding inflammatory environment. Classically activated macrophages are driven by cytokine activators interferon gamma (IFN-γ), TNF-α, and also by microbial products such as LPS. Classically activated macrophages are highly proinflammatory and microbicidal cells that produce exaggerated levels of ROS and cytokines responsible for immune defense. In contrast, alternatively activated macrophages are a consequence of predominant IL-4 and IL-13 production from Th2 cells dominant during parasitic infection. They induce macrophage production of specific chemokines, macrophage-derived chemokine (MDC) and thymus and activation-regulated chemokine (TARC), and upregulate the expression of receptors responsible for endocytosis and antigen presentation and enzymes responsible for arginase metabolism and granuloma formation. The alternatively activated macrophage does not produce a major respiratory burst and ROS products are considered less proinflammatory but nonetheless are critical for clearing of parasitic infections. Other cytokines are also important for macrophage deactivation such as IL-10 and transforming growth factor-beta (TGF-β), which promotes the resolution of the macrophage inflammatory response.

Chemokines

Chemokines are a specific group of peptide inflammatory mediators of the cytokine family that act as chemoattractants to induce the recruitment of inflammatory cells to the site of injury or infection. There are various subclasses of chemokines that are described as CC (e.g., monocyte chemoattractant protein-1; MCP-1), CXC (e.g., IL-8), and CX3C (fractalkine). They are classed depending on the molecular structure with CC chemokines containing two cysteine residues together and CXC chemokines containing two cysteine residues separated by another amino acid. A number of CC receptors and CXC receptors have been identified on the macrophage that can bind chemokines including MIP-1α, MIP-1β, RANTES, MCP-1, and IL-8.

Lipoproteins

Macrophages express receptors that allow them to recognize, uptake, and digest cholesterol-containing lipoproteins such as low-density lipoproteins (LDLs). Macrophages are also involved in oxidation of lipoproteins by macrophage produced ROS, a process linked with atherosclerotic plaque progression. The uptake of LDLs occurs through macrophage recognition receptors for apoplipoprotein E and apoplioprotein B-100 expressed on the LDL molecule. Scavenger receptors have also been implicated in the uptake of modified lipoproteins.

Integrins

Integrin molecules are expressed on the surface of many cell types including neutrophils, endothelial cells, and macrophages. They are important for cell adhesive interactions and are commonly upregulated on the surface of cells during inflammation. Cell adhesion during inflammation is particularly important during macrophage migration and cell adhesion during phagocytosis. Integrins are recognized by macrophage receptors to facilitate adhesion and to initiate macrophage behaviors that allow cell movement including actin rearrangement.

Oligosaccharides

Oligosaccharide molecules terminating in fucose, mannose, galactose, and sialic acid are recognized by specific macrophage receptors. The discrimination of self and nonself particles is partly due to expression of cell specific oligosaccharide molecules. Bacterial and fungal membranes characteristically contain terminal monosaccharide residues that are recognized by the macrophage mannose receptor to facilitate phagocytic clearance of these particles.

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TABLE 8.1. Major macrophage secretory products

Complement proteins
- C1
- C2
- C3
- C3b
- C3b inactivator
- C4

Enzymes
- Lysozyme
- Plasminogen activator
- Collagenase
- Elastase
- Angiotensin converting enzyme
- Cysteine proteinase
- Proteases
- Lipases
- Deoxyribonucleases
- Phosphatases
- Glycosidases
- Sulfatases
- Arginase

Proteins
- Alpha-proteinase inhibitor
- Alpha-macroglobulin
- Plasminogen activator inhibitor
- IL-1 inhibitor
- Neutrophil migration inhibitor
- Inhibitor of fibroblast growth
- Lipomodulin
- Glycoproteins
- Fibronectin
- Transferrin
- Transcobalamin II
- Apolipoprotein A and E

Cytokines
- Interleukin (IL)-1α and β
- IL-6
- MIP-2
- IFN-α and γ
- TGF-β
- TNF-α
- IL-8
- Colony stimulating factors (G-CSF, M-CSF, and GM-CSF)

Reactive oxygen species
- Superoxide anions
- Hydrogen peroxide
- Hydroxyl radical
- Nitric oxide

Antioxidants
- Glutathione

Coagulation factors
- Factor V, VII, IX, and X
- 1,25-dihydroxyvitamin D3
- Thromboplastin
- Prothrombin
- Prothrombinase
- Tissue factor
- Plasminogen activator

Bioactive lipids
- Prostaglandin E, $F_2\alpha$
- Cyclooxygenase
- Thromboxane A_2
- Leukotriene B_4, C_4, D_4, E
- Platelet-activating factor

Others

The above description of macrophage activating agents is by no means an exhaustive list as there are many other equally important mediators that have been demonstrated to activate or modulate macrophage responsiveness. For example, there are many lipid mediators including prostaglandins, leukotrienes, lipoxins, and ether lipids (e.g., platelet-activating factor) which can influence macrophage function. Other agents known to affect macrophage function include adenosine, ATP, neuropeptides, nitric oxide, and so on, which are covered in other chapters.

MACROPHAGE SECRETORY PRODUCTS

Much has been discovered about macrophages over the past few decades that go beyond the phagocytic function from which they were originally described. These cells release many agents that influence and orchestrate many processes including enhancement and resolution of the inflammatory response, microbial killing, antigen presentation, and wound healing to name a few (Table 8.1). The specific profile of agents released is dynamic and can switch from proinflammatory to antiinflammatory phenotypes, further demonstrating the multi-faceted roles of the macrophage in inflammation.

Enzymes

Various enzymes are released from macrophages both constitutively and in response to Fc or complement receptor engagement, cytokines, or bacterial products. They are generally released to aid digestion of cell debris and pathogens by breaking down components of cell membranes, connective tissue, and nuclear material (DNA). Lysozyme is secreted constitutively and in large amounts by macrophages where it functions to digest bacterial cell walls. Enzymes that are released upon macrophage stimulation include proteases, lipases, deoxyribonucleases, phosphatases,

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glycosidases, collagenase, elastase, and sulfatases. Enzyme secretion by macrophages also serves to activate or inhibit various pathways including the complement pathway, renin-angiotensin system, and the kinin-kallikrein system.

Cytokines

Cytokines, many of which can themselves influence macrophage behavior (see above) are described as soluble peptide factors that can influence cell growth, function, and differentiation of many target cell types. Cytokine secretion by macrophages plays a pivotal role in orchestration of many stages of the immune response. Most cytokines are released during an inflammatory response when concentrations of substances such as TNF-α and IL-6 can be significantly increased in the inflammatory environment. Although TNF-α was one of the first cytokines discovered that is secreted by macrophages, these cells release many different cytokines although many share a common target and often work in synergy to achieve their function. Furthermore, the inflammatory milieu can influence the function of secreted cytokines. The cell priming phenomena is an example of how the inflammatory environment can affect macrophage responses. Stimuli that do not necessarily evoke an inflammatory response on their own, when presented with another inflammatory stimulus can cause a reaction that is massively increased. Such processes help explain how macrophages can adapt their responses in a manner appropriate to the inflammatory insult. The most well known macrophage priming agent is IFN-γ.

Complement Components

Macrophages can secrete many components of both the classical and alternative complement systems. These include C1, C2, C3, C4, C5, factor B, factor D, properdin, C3b inactivator, and α-IH. In addition, proteases secreted by macrophages can generate active fragments of the complement system such as C3a, C3b, C5a, and Bb. The complement system is involved in many immunoregulatory functions including lysing of foreign cells by forming C5–9 complex or by acting as opsonins to promote recognition and uptake of particles by phagocytosis. Complement factors can also act directly to promote or suppress the inflammatory response by binding directly to complement receptors on the macrophage surface. Factors C3a and C5a are well-known macrophage chemoattractants whereas factor Bb, a product of the alternative complement pathway, suppresses macrophage migration.

Coagulation Factors

Macrophages secrete coagulation factors VII, IX, X, and V. The coagulation cascade is a critical pathway stimulated in response to injury or trauma where it promotes blood clotting and through recruitment and activation of platelets and wound healing through deposition of fibrin.

Reactive Oxygen Intermediates

Small molecular weight molecules such as O_2^- and NO are released by activated macrophages. The rapid increase in macrophage oxygen consumption and reactive oxygen intermediate (ROI) production occurs in response to various inflammatory stimuli. This process is called the respiratory burst. These highly reactive molecules are the product of macrophage NADPH and iNOS enzyme activity. They are directly micobicidal but can also react with surrounding molecules to form other cytotoxic species including H_2O_2, $ONOO^-$, and OH^- that also contribute to defense against infection. The importance of these substances during infection is shown through the use of mouse models deficient in macrophage-derived ROS, which show markedly increased susceptibility to infection by various organisms.

Arachidonic Acid Intermediates

Macrophages are known to be important sources of products of the arachidonic acid pathway such as prostacyclin, thromboxane, PGE_2, and LTB_4. Macrophages contain the necessary enzyme components involved in synthesis, control, and metabolism of these products such as PLA_2 that forms arachidonic acid by cleaving phospholipid precursors and the cyclooxygenase and lipoxygenase enzymes that control the metabolism of arachidonic acid into active prostacyclin, prostaglandins, thromboxanes, and leukotrienes. The release of metabolites of arachidonic acid has multiple functions during inflammation including effects on macrophage functions. Lipoxin A4 production increases the phagocytosis of neutrophils by macrophages and is thus considered important in the resolution phase of the inflammatory response. Macrophages are a significant source of prostaglandin E that acts to control body temperature through its action on the temperature regulatory organ, the hypothalamus. The elevation of body temperature is considered an important defense mechanism that develops to create an environment less suitable for pathogens to thrive in while simultaneously facilitating immune functions.

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FUNCTIONAL RESPONSES OF MACROPHAGES

Phagocytosis of Cell Debris and Invading Organisms

As already mentioned, the process of engulfing and digesting particles (phagocytosis) is the characteristic most associated with, although not exclusive to, macrophages. Most cells have some phagocytic capacity, such as some epithelial cell types, but these cells are not equipped with the complex repertoire of receptors specific for recognition of a large particle range and modulation of phagocytic rate. The process of phagocytosis is complex and involves a large number of recognition receptors and signaling events to initiate this process. The fundamental steps surrounding phagocytosis are as follows: recognition of particles requiring to be phagocytosed, rearrangement of macrophage cytoskeleton to allow it to extend "arms" (pseudopodia) and surround the particle and, lastly, to digest the particle. Different particles can stimulate an array of different receptors and signaling cascades and these processes are capable of cross talk and synergy adding further complexity to the regulation of phagocytosis. Thus, phagocytosis is a very dynamic process that allows for a diverse reactionary repertoire appropriate to the inflammatory insult. For example, the ingestion of Gram-negative microbial particles involves receptors for LPS that trigger unique intracellular signaling pathways to induce phagocytosis but which also modulate transcriptional activity of the macrophage. In particular, expression of the inflammatory mediator TNF-α is upregulated and released into the inflammatory environment, where its presence can recruit and activate other inflammatory cells, thus allowing for an intense inflammatory response to clear the infection. In contrast to this, the ingestion of innate senescent cells, such as apoptotic neutrophils, occurs in a noninflammatory manner using distinct recognition mechanisms that utilize surface molecules on the surface of apoptotic cells (e.g., phosphatidylserine) and specific receptors of the surface of the phagocytic macrophage. Interestingly, when macrophages phagocytose apoptotic cells (a process termed efferocytosis) their phenotype is changed in that the macrophage starts to reduce the amount of proinflammatory mediator being synthesized and liberated. Indeed, there is now strong evidence that these macrophages start to express mediators that can have anti-inflammatory potential (e.g., TGF-β and IL-10) thereby aiding the resolution of the inflammatory response. Phagocytosis in general is a highly regulatable process that can be modulated by endogenous mediators and by pharmacological intervention. The clearance by macrophages of apoptotic cells can be modulated by complement components, CD44 ligation, certain prostaglandins, lipoxins, and glucocorticoids, to name a few. Enhancement of macrophage clearance of apoptotic cells as an anti-inflammatory strategy is a realistic target for the pharmaceutical industry as the process is likely to be important in the resolution of inflammation. Indeed, convincing evidence indicates that failed clearance of apoptotic cells contributes to the pathogenesis of diseases such as systemic lupus erythematosus.

Chemotaxis

Chemotaxis is described as the movement of cells in response to a chemical stimulus (chemokine). These stimuli are often released by foreign pathogens or inflammatory cells (e.g., neutrophils) allowing cells such as macrophages to accurately migrate to the site of infection or injury. Chemotaxis is initiated by the stimulation of receptors on the macrophage surface, which in turn facilitate the rearrangement of cytoskeletal structures, resulting in movement of the cell in the direction of the stimulus. Rearrangement of cytoskeletal structures also allows the migration of cells through blood vessel cell layers into tissues in a process called transmigration. The most common chemotactic stimuli are bacterial products including formylated peptides and LPS although other inflammatory cells release factors such as IL-8, MCP-1, RANTES, and GM-CSF which can directly stimulate or enhance the chemotactic response of macrophages.

Antigen Processing and Presentation

In addition to the crucial role macrophages play in innate immune defenses, these cells are also important in orchestrating acquired immune responses. Antigen processing and presentation by macrophages to T- and B-cell lymphocytes allows for heightened recognition and clearance of pathogen-infected cells and antibody production against the pathogen, which allows for rapid clearance of the pathogen upon subsequent exposure. DCs are the major cell type responsible for antigen processing and presentation although macrophages are also known to perform such duties and are classed as professional antigen presenting cells (APCs) along with antibody expressing B cells. These cells process pathogenic proteins in the cytoplasm following phagocytosis of virus or bacteria initiated through stimulation of PRRs on the surface of the APCs. The processing of antigens is dependent on the MHC molecules of which there are various classes (e.g., MHC I and MHC II). These complexes are brought into contact with foreign proteins within the endoplasmic reticulum (MHC I) where they are processed into denatured or peptide fragments (antigens) by catabolism. The MHC associated with the antigen is then transported to the APC cell surface where it is recognized by

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CD4+ and CD8+ lymphocytes that can initiate a rapid cytotoxic response by initiating apoptosis of APCs that are infected with viral or bacterial pathogens. APCs also promote recruitment and clonal proliferation of T cells endowed with the specific recognition motifs for the MHC:antigen complex. Memory B cells expressing the antibody complex that recognizes the specific pathogenic antigen remain in circulation following resolution of the infection where they can rapidly activate T-cell–mediated responses and clearance of the pathogen should the host be infected again. This method of immunity is exploited in vaccine development where processed or fragments of bacterial or viral antigens are given to a host along with an adjuvant to enhance the response by activating APCs. Antibody expressing memory B cells persist and protect the host from infection without ever being exposed to the full brunt of the disease producing pathogen.

Secretion

Macrophages actively produce and secrete a large number of factors that have important immunomodulatory effects. Macrophages generate and secrete (a) cytokines and chemokines usually via transcriptional activation (e.g., NF-kB and AP-1) and protein synthesis dependent mechanisms, (b) de novo generation of production of lipids and reactive oxygen species, and (c) preformed enzymes (see earlier). These factors can directly cause microbial killing, enhance or dampen an immune reaction, and facilitate the wound healing process.

Tumor Cell Control

Tumor-associated macrophages (TAMs) have recently been shown to be important in tumor biology as they secrete factors that can actively suppress cell proliferation or cause lysis of tumor cells. There is a role for cell–cell contact in macrophage-mediated tumor control, which can specifically recognize and lyse neoplastic cells in a nonphagocytic process. Release of cytostatic and cytotoxic agents includes prostaglandins, IL-1, TNF-α, ROS, and proteases.

IMPLICATIONS OF MACROPHAGE DYSREGULATION

The role of the macrophage in innate immunity has evolved to protect the host against injury and infection; however, macrophage function must be very tightly regulated to ensure minimal toxicity to host tissue and exacerbation of the inflammatory response. Dysfunctions in macrophage biology have been implicated in many disease pathologies including arthritis and atherosclerosis. Once the macrophage performs its duties, it must be decommissioned by processes such as apoptosis *in situ* or by drainage by the lymphatics to the lymph nodes. Macrophages can respond to their environment through the vast array of receptors on their surface, which influences the behavior and toxicity of the macrophage. However, the complex biology of macrophages leaves plenty of scope for things going awry. Dysfunction in maturity, migration, and phagocytic responses of macrophages ultimately result in a compromised ability to clear infection. Also, the inability to control macrophage responses can result in host tissue damage due to increased release of toxic substances.

EVIDENCE FOR ROLES IN DISEASES

Arthritis

Patients with arthritis have elevated levels of various macrophage-derived products that are associated with increased cartilage and bone degradation leading to compromised joint function. The chronic inflammatory response in the joints of patients with arthritis is intimately associated with prolonged and continued cytokine production by an abundant inflammatory cell infiltrate, which includes large numbers of macrophages. TNF-α, IL-1, and IL-6 have been identified as targets for anticytokine therapies for arthritis.

Atherosclerosis

The uptake and oxidation of cholesterol containing lipoproteins is associated with the production of foam cells; modified lipid-laden macrophages. These cells contribute to the progression and instability of atherosclerotic plaques and can promote inflammatory environments that lead to plaque rupture and thrombus formation, the leading cause of myocardial infarction and stroke.

IBD

Irritable bowel disease (IBD) includes Crohn's disease and ulcerative colitis, which are chronic inflammatory diseases of the gastrointestinal tract. These diseases are caused by inappropriate or hypersensitive inflammatory reactions to normal gut flora, a reduced capacity to resolve gut inflammation, and increased inflammatory cell burden. Increased levels of inflammatory cell cytokines are found in patients with IBD, including macrophage-produced cytokines such as TNF-α and IL-6, which can contribute to increased tissue damage (gut ulceration) and enhanced inflammatory cell recruitment, further exacerbating the inflammatory response. Patients with IBD have increased risk of developing malignancies within the gut.

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Lung Fibrosis

Lung fibrosis has a poor prognosis once developed. Idiopathic pulmonary fibrosis, asbestosis, silicosis, and interstitial lung disease all have hallmark scarring and thickening of the lung tissue which leads to impaired gas exchange in the alveolar spaces of the lungs. Fibroblast proliferation, collagen deposition, and interstitial tissue remodeling lead to lung fibrosis, which is usually the result of a chronic inflammatory reaction. Macrophages are the driving force initiating and maintaining the fibrotic response by producing profibrotic cytokines such as TGF-β and proinflammatory cytokines such as IL-6, IL-1β, and TNF-α.

Tumor Biology

Macrophages produce various factors important in the regulation of cell growth, proliferation, and survival. Macrophage products can initiate neoplastic cell death and are important in preventing cancers. In contrast, chronic inflammatory reactions can drive macrophage responses into "repair" mode where they switch functions and can promote tumor growth and progression. Tumor-associated macrophages (TAMs) (see earlier) are found at high density in some tumors where they support tumor growth through the production of angiogenic factors, promotion of tissue remodeling and repair mechanisms, and release of growth factors. The major growth factor released by macrophages is vascular endothelial growth factor (VEGF) that promotes angiogenesis and can also attract monocytes to the tumor site. Many cancers are associated with chronic inflammatory responses such as lung cancer in asbestosis patients and stomach cancer associated with *Helicobacter pylori* infection indicating a detrimental role for macrophages in cancer biology. In addition, the inflammatory microenvironment is rich is mutagenic factors such as ROS that can stimulate neighboring cells into a neoplastic phenotype.

Infection

Many pathogens have evolved defense mechanisms to avoid eradication by the immune response when invading the body. In some instances, pathogens use macrophages and their products to facilitate infection. The human immunodeficiency virus (HIV) enters the macrophage through receptors on its surface where it can then replicate. HIV also relies on macrophages to sustain reservoirs of virus that are resistant to antiretroviral therapies, thereby forming a continuing cycle of infection and disease progression. Increases in levels of M-CSF production by macrophages increases macrophage numbers and longevity associated with the maintenance of tissue reservoirs of HIV. M-CSF also promotes virus replication, underpinning the importance of macrophages in the pathogenesis and persistence of HIV. Some species of *Leishmania* parasites can target macrophages by expressing phosphatidylserine (PS) residues to encourage phagocytosis (via mechanisms evolved to ingest apoptotic cells) of the organism where they then establish themselves as intracellular parasites. In addition, the PS-mediated Phagocytosis stimulates the macrophage into an anti-inflammatory mode by increasing release of IL-10 and TGF-β and by decreasing NO production, which aid in the survival of extracellular parasites and increase the susceptibility of the host to infection.

PHARMACOLOGICAL REGULATION OF MACROPHAGE RESPONSES

Macrophages play a pivotal role in immune defense and resolution of inflammation. Thus, pharmacological manipulation of macrophage functions such as enhancement of phagocytosis have become of increasing interest in conditions of chronic inflammation. In addition, we have already touched upon how the macrophage can contribute to disease pathogenesis, highlighting a possible route by which drugs can dampen the effects of macrophages in these conditions. Proinflammatory cytokines released by macrophages, such as TNF-α, can be significantly elevated in various diseases causing tissue damage and exacerbation and perpetuation of the inflammatory response. Although many anti-inflammatory and immunomodulatory drugs can influence macrophage responsiveness two major examples of such agents are highlighted in the subsequent text.

Steroids

Pharmacological intervention in inflammatory conditions is spearheaded by the use of steroidal anti-inflammatory drugs. Glucocorticoids (GCs) are commonly used to treat inflammation such as rheumatoid arthritis and asthma but have significant drawbacks due to side effects. The primary mechanism of action of GCs is the modulation of transcription at the DNA level by activation or repression of GC response elements or by inhibition of transcription factors such as NF-κB and AP-1. In addition, the effects of GC treatment on macrophage phagocytic responses have provided further insight into the anti-inflammatory effects of these drugs. Macrophages treated with GCs such as dexamethasone were shown to markedly enhance the phagocytic capacity of macrophages for apoptotic cells. This function of GCs has obvious advantages when considering the promotion of resolution of inflammation, particularly those that

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are associated with large numbers of infiltrating neutrophils that require clearance following apoptosis to avoid release of histotoxic contents and inflammatory exacerbation.

Anti-TNF Therapy

A major inflammatory mediator associated with numerous diseases including RA and Crohn's disease is TNF-α. This has sparked research targeting this pathway as a novel treatment for chronic inflammatory conditions. Monoclonal antibodies directed against TNF and soluble TNF receptors have produced promising results in this arena. They work by inhibiting the effector functions of TNF-α by competitively binding the molecule in the extracellular environment, thus reducing the binding of TNF-α to its receptors. The rationale for targeting this cytokine came from the discovery that inhibiting TNF-α could also downregulate the production of other proinflammatory mediators including IL-1, IL-6, and GM-CSF. A potential drawback to the use of anti-TNF-α antibodies is the increased susceptibility to infection, a side effect found in TNF knockout mouse models.

CONCLUSIONS

Knowledge about macrophage biology has been greatly expanded over the past few decades. These cells are now considered integral cells in the innate immune system, involved in orchestrating many immune functions including phagocytic clearance of foreign materials or cell debris, secretion of a plethora of immunomodulatory molecules that can heighten or resolve the immune response and promote wound healing. Furthermore, these cells are also now known to have a role in initiation of the adaptive immune response. In chronic inflammatory conditions, the contribution of macrophages to the pathology of many diseases is now widely recognized and has highlighted many areas through which pharmacological manipulation of macrophage responses can be used to treat these diseases, which include anticytokine therapies and promotion of resolution of inflammation by stimulating the phagocytic capacity of macrophages. There is a continued research effort focused around these cells including using these cells as vectors for gene or drug therapy that home to areas of inflammation or injury. The complex signaling and functional responses of these cells leaves much to be understood, especially surrounding the changing inflammatory profiles of macrophages throughout an inflammatory response; the so-called alternatively or classically activated macrophages. Many new agents are being researched that can directly or indirectly enhance the anti-inflammatory potential of these cells by promoting anti-inflammatory cytokine production or by driving apoptosis of neutrophils to enhance phagocytic clearance using inhibitors of cyclin-dependent kinases (CDKs). Such therapies hold significant promise in the treatment of a number of conditions where current therapeutic strategies are limited, such as idiopathic lung fibrosis.

SUGGESTED READINGS

Aderem, A., and Underhill, D.M. 1999. Mechanisms of phagocytosis in macrophages. *Annu Rev Immunol* **17:**593–623.

Burke, B., and Lewis, C.E., eds. 2002. *The Macrophage*, 2nd ed. Oxford: Oxford University Press, ISBN 0192631977.

Fujiwara, N., and Kobayashi, K. 2005. Macrophages in inflammation. *Curr Drug Targets Inflamm Allergy* **4**(3)**:**281–286.

Gilroy, D.W., Lawrence, T., Perretti, M., and Rossi, A.G. 2004. Inflammatory resolution: new opportunities for drug discovery. *Nat Rev Drug Discov* **3**(5)**:**401–416.

Gordon, S. 2007. The macrophage: past, present and future. *Eur J Immunol* **37**(Suppl 1)**:**S9–S17.

Gordon, S., and Taylor, P.R. 2005. Monocyte and macrophage heterogeneity. *Nat Rev Immunol* **5**(12)**:**953–964.

Gregory, C.D., and Devitt, A. 2004. The macrophage and the apoptotic cell: an innate immune interaction viewed simplistically? *Immunology* **113**(1)**:**1–14.

Hallett, J.M., Leitch, A.E., Riley, N.A., Duffin, R., Haslett, C., and Rossi, A.G. 2008. Novel pharmacological strategies for driving inflammatory cell apoptosis and enhancing the resolution of inflammation. *Trends Pharmacol Sci* **29**(5)**:**250–257.

Hart, S.P., Dransfield, I., and Rossi, A.G. 2008. Phagocytosis of apoptotic cells. *Methods* **44**(3)**:**280–285.

McColl, A., Michlewska, S., Dransfield, I., and Rossi, A.G. 2007. Effects of glucocorticoids on apoptosis and clearance of apoptotic cells. *Sci World J* **7:**1165–1181.

Plüddemann, A., Mukhopadhyay, S., and Gordon, S. 2006. The interaction of macrophage receptors with bacterial ligands. *Expert Rev Mol Med* **8**(28)**:**1–25.

Serhan, C.N., Brain, S.D., Buckley, C.D., et al. 2007. Resolution of inflammation: state of the art, definitions and terms. *FASEB J* **21**(2)**:**325–332.

Sica, A., Rubino, L., Mancino, A., et al. 2007. Targeting tumour-associated macrophages. *Expert Opin Ther Targets* **11**(9)**:**1219–1229.

Szekanecz, Z., and Koch, A.E. 2007. Macrophages and their products in rheumatoid arthritis. *Curr Opin Rheumatol* **19**(3)**:**289–295.

Taylor, P.R., Martinez-Pomares, L., Stacey, M., Lin, H.H., Brown, G.D., and Gordon, S. 2005. Macrophage receptors and immune recognition. *Annu Rev Immunol* **23:**901–944.

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9 Lymphocytes

Tracy Hussell, Mary Cavanagh, Erika Wissinger, and Emily G. Findlay

Lymphocytes are a type of white blood cell or leukocyte and are critical for the defense of the body against invaders such as infectious pathogens and foreign materials. There are approximately 10^{12} lymphocytes in the average human body, though this fluctuates considerably during illness. Indeed, fluctuation from a normal range is used as an indicator of disease. For example, immune deficiency leads to reduced lymphocyte numbers in the blood whereas infection or allergy leads to an increase. Several different types of lymphocyte exist, and though they display highly specialized and diverse functions, they all derive from a common hematopoietic stem cell in the bone marrow (Figure 9.1). Mature lymphocytes are found throughout the body: in the blood and lymphatic system; concentrated in immune specialized regions (lymph nodes, thymus, spleen, and Peyer's patches); and scattered in most other tissues.

Lymphocytes can be divided on the basis of size and granularity. Natural killer (NK) cells are large and highly granular, and there are two main types of small lymphocyte: B lymphocytes (or B cells) and T lymphocytes (or T cells). As we will see in this chapter, these are further subdivided based on their expression of different surface molecules, their secretion of different products important for immunity, and their ultimate effector function. B cells (so-called because they were originally identified in birds in the Bursa of Fabricius) are not only produced in the bone marrow but also mature there. However, the precursors of T cells leave the bone marrow and mature in the thymus (which explains the notation "T"). In general, B cells secrete proteins called antibodies that recognize and bind foreign particles and toxins found free within the body interior or protruding from an infected cell. T cells assist other immune cells such as B cells (via secreted proteins called cytokines) or attack body cells when they have been infected by viruses or have become cancerous. T and B cells differ from most other white blood cells in that they express a variable receptor on their surface that specifically detects foreign molecules. T cells express the T-cell receptor (TCR) and B cells express the B-cell receptor (BCR) which is identical to their secreted antibody molecules. The method of producing these receptors leads to enormous diversity. As a result, B cells and T cells are highly specific for pathogenic molecules but cells of each specificity are relatively rare. B and T lymphocytes are therefore referred to as components of acquired or adaptive immunity. Unlike "innate" immune cells, which are more numerous but have less specific receptors, they "acquire" their specificity with time and are adaptable once differentiating. The following chapter details the function of these lymphocyte subsets, what signals cause them to become activated and factors that assist their development into further subtypes. This adaptability is extremely important as it determines their ultimate function and enables the human body to eliminate a hugely diverse range of potentially dangerous organisms and foreign particles.

ANTIGEN-PRESENTING CELLS

Though part of the "adaptive" immune system, lymphocytes must interact with the "innate" immune system and one of these interactions is antigen presentation – the key event in pathogen recognition by adaptive immune cells. To recognize pathogenic molecules, B cells use the BCR which binds pathogens directly. In contrast, T cells rely on other, more innate, antigen-presenting cells (APCs) to break down pathogenic proteins into peptides, which are then presented to the TCR by the major histocompatibility complex (MHC). Innate APCs tend to be large, phagocytic cells that patrol the body and constantly ingest particles, dead cells, and other debris from tissues. Once ingested, these particles, typically proteins, are broken down within the cell and transferred onto carrier molecules

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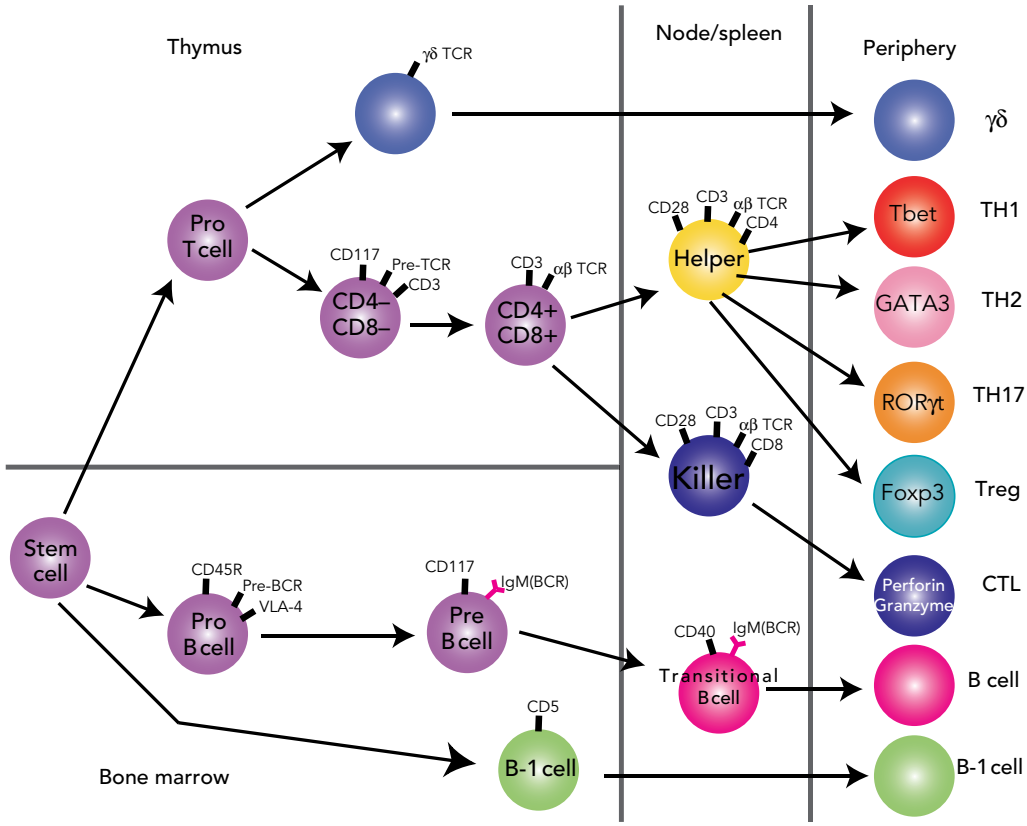


Figure 9.1. T-cell and B-cell development. T and B cells derive from a common stem cell in the bone marrow. The precursors of T cells leave the bone marrow and mature in the thymus. Here, they express different molecules which aid the maturation process. Some of these are shown in the figure. Pro B cells express VLA-4 which binds VCAM-1 on bone marrow stromal cells. This interaction induces the expression of CD117 (also known as Kit) on the B cell, which is the receptor for stem cell growth factor (SCF). In the bone marrow, BCR rearrangement occurs. This process is mirrored in the thymus where pro T cells begin by expressing Kit and rearrange their TCR to produce a highly diverse population. In the thymus, γδ T cells diverge from the αβ T-cell lineage. Once the TCR and BCR have been selected, T and B cells migrate from the thymus and bone marrow to the periphery. Now the cells have reached maturity but are naive. They then circulate in the body until they encounter specific antigen, become activated and differentiate further. The different subsets of mature T cells are shown on the right along with internal transcription factors and other internal molecules used to identify them.

known as the major histocompatibility complexes or MHC. The ways in which APCs process and present antigen are illustrated in Figure 9.2.

The two main types or classes of MHC are known as MHC class I and MHC class II. MHC class I molecules are found on the surface of all nucleated cells and present peptides derived from the cell itself and from intracellular pathogens such as viruses. MHC I molecules continuously cycle from the endoplasmic reticulum (where they pick up antigen) to the cell surface and back. If the cell is uninfected, the antigens presented on the cell surface will not be recognized by cytotoxic T cells. However, if the presenting cell is infected or cancerous, the T cell will target it for death as described later.

MHC class II molecules are generally found on the surface of professional APCs and typically present antigen from extracellular pathogens. The pathogens are ingested and broken down in endosomes and, in contrast to MHC class I, MHC class II molecules are loaded with antigen within these endosomes after passing through the endoplasmic reticulum. Once on the cell surface, MHC class II molecules present antigen to CD4+ T helper cells, which become activated and provide help to other immune cells as detailed subsequently.

For many years, it was thought that these systems of presentation were mutually exclusive but it is now known that MHC class II–destined peptides can access the MHC class I pathway by a process called cross-presentation. This process is thought to provide a way of presenting antigen from viruses that have evolved mechanisms to subvert the classical MHC class I presentation pathway.

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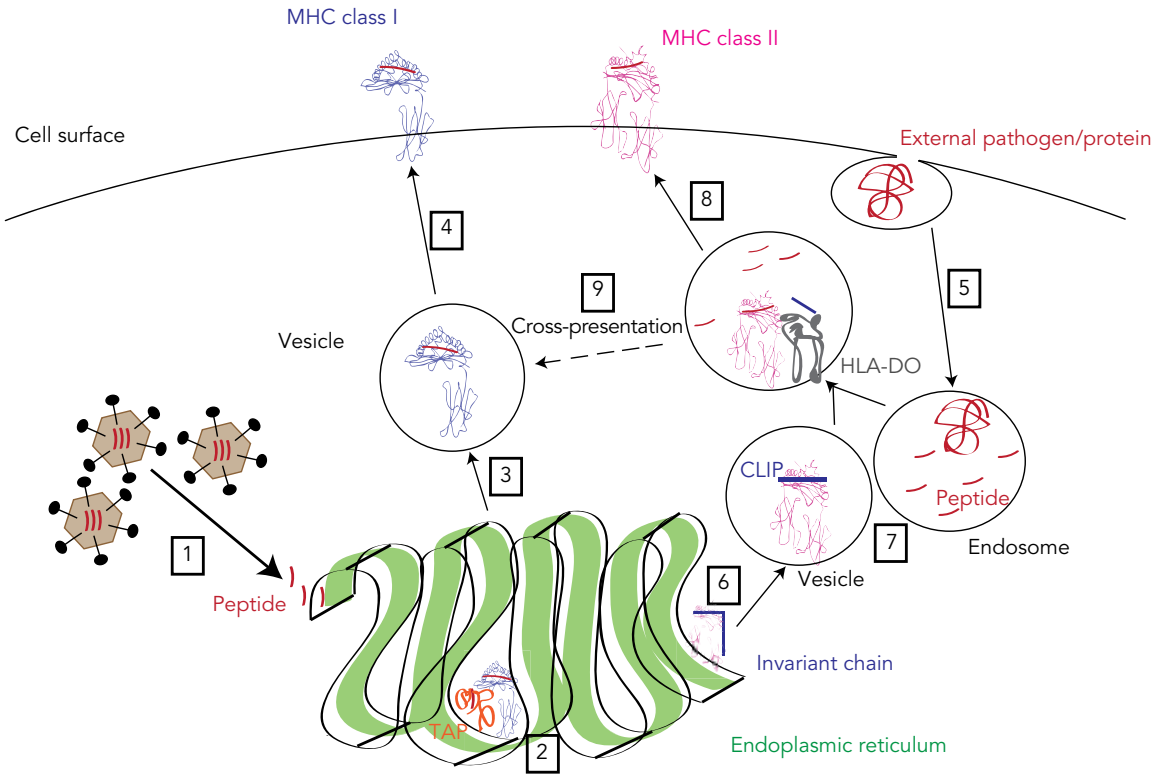


Figure 9.2. MHC processing and presentation. MHC class I molecules typically present peptides derived from intracellular sources whereas MHC class II molecules present those from extracellular sources.

During viral replication, some proteins are released from the proteosome (1) and are taken into the endoplasmic reticulum (2). Here, they encounter the membrane-associated transporter protein TAP which loads the peptide onto the MHC class I binding domain. The peptide-loaded MHC I molecule then leaves the ER in a vesicle (3) and is transported to the cell surface where it displays the peptide to passing CD8+ T cells (4).

Innate APCs can ingest proteins from the extracellular space (5). These are internalized and degraded in endosomes. Though MHC class II molecules are processed in the ER (6), they leave this compartment in vesicles before being loaded with peptide. The peptide-binding site of MHC II is protected by the CLIP fragment of the invariant chain which is only completely removed once the MHC-containing endosome has fused with the antigen-containing endosome (7). CLIP is removed by a related MHC class II protein (HLA-DM or HLA-DO), allowing the foreign peptide to bind, and the peptide:MHC complex is then transported to the cell surface where it can present antigen to CD4+ T helper cells (8).

On rare occasions, externally derived peptides can be presented by MHC class I in a process known as cross-presentation (9).

T CELLS

T-Cell Development

T cells play multiple roles in immunity and disease depending on their ultimate phenotype after differentiation. Unlike the BCR (antibody), the TCR does not see foreign particles or pathogens directly but requires them to be digested and presented by MHC molecules. The outcome of this recognition then depends on the T cell in question and will be discussed later. The TCR is a heterodimer of two chains but also requires a collection of other membrane-proximal chains for signaling (see Figure 9.3); these additional chains are known collectively as CD3. All T cells, regardless of their end phenotype, express CD3. However, broadly speaking there are two different types of CD3-expressing T cells. Helper T lymphocytes also express CD4 and cytotoxic T lymphocytes (CTLs) express CD8. As their names suggest, CD4+ T cells secrete products to help other cells of the immune system and CD8+ CTLs cause death of the cell with which they interact. As we will see later, CD4+ and CD8+ cells can be further subdivided based on the products they secrete or additional receptors they express.

Both main T-cell subsets mature in the thymus where rearrangement of the genes encoding the two variable chains (α and β) of the TCR takes place. As this rearrangement is a random process the thymus is also a sorting factory and ensures, through a process of selection, that T cells do not exit the thymus capable of recognizing our own proteins. If they did, then autoimmunity would occur. Progenitor T cells (ProT) from the bone marrow move into the outer cortex of the thymus and proliferate. At this stage, they do not express CD4 or CD8 but express Kit (the receptor for

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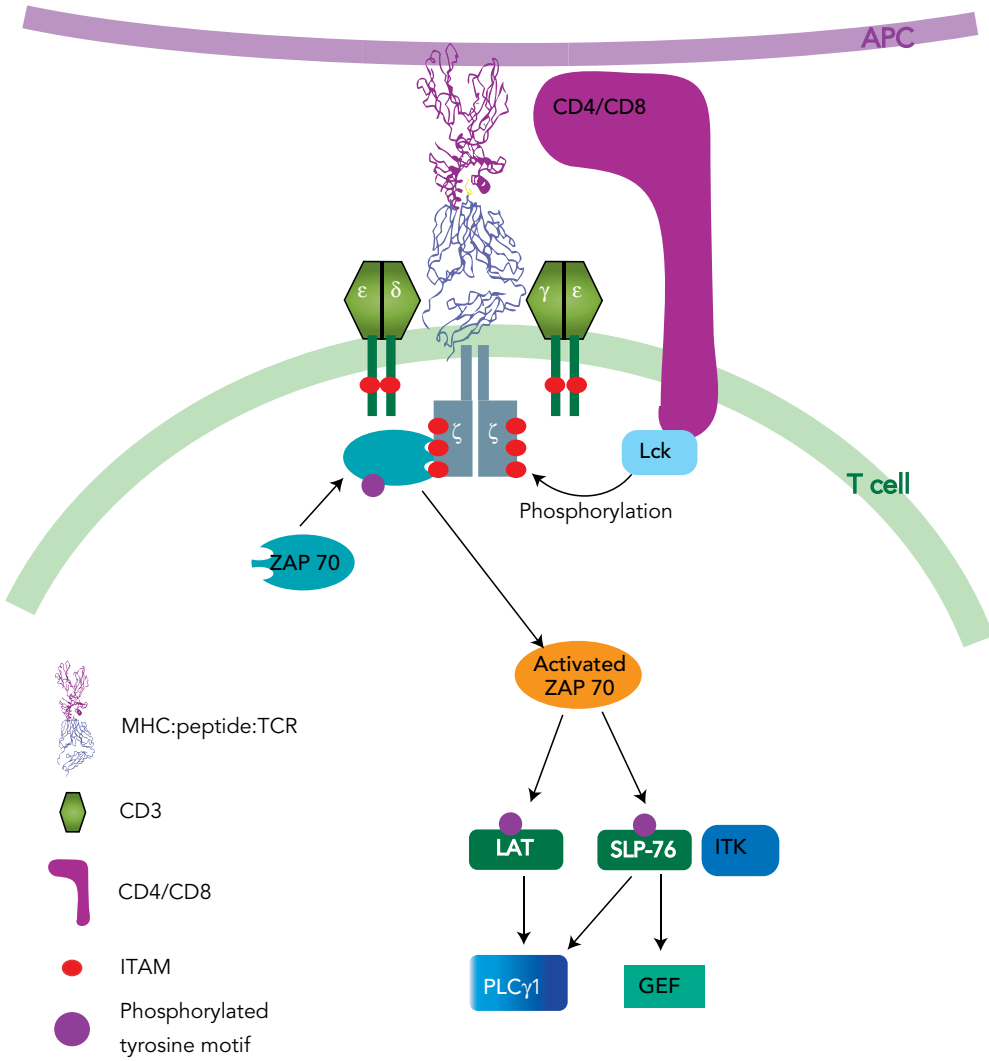


Figure 9.3. TCR signaling. A T cell is activated when its TCR binds to antigenic peptide presented by an antigen-presenting cell (APC). This peptide is bound to MHC, and a co-receptor (CD4 or CD8) stabilizes the interaction by binding to nonpolymorphic regions of the MHC molecule. Protein tyrosine kinases, such as Lck, quickly act to phosphorylate the immunoreceptor tyrosine–based activation motifs (ITAMs) located on the intracellular portions of CD3 and the ζ chains of the TCR. The ζ chain–associated protein ZAP 70 then binds and phosphorylates the membrane-bound adapter proteins, linker of activated T cells (LAT), and SH2 domain-containing leukocyte protein of 76 kDa (SLP-76). LAT, SLP-76 and another kinase, ITK, then activate phospholipase C (PLC)γ1 and recruit guanine nucleotide exchange factors (GEFs) to continue the signaling cascade.

stem cell growth factor [SCF]), CD44 and CD25 (part of the receptor for the T cell growth factor interleukin [IL]-2). Following a number of rounds of division, CD25 and CD44 are lost from the surface and the TCR gene rearrangement takes place.

The β chain genes of the TCR are first rearranged through the actions of the recombination-activating genes RAG-1 and -2, and combine with a pre-α chain to generate what is known as a pre-TCR. This combination determines whether the β chain rearrangement is successful. Successful signaling through the pre-TCR allows the T cell to progress to the next stage. Cells that pass the test move on to a CD4+CD8+ (double positive) stage, which is marked by rapid proliferation and rearrangement of the α chain. With a functional α and β chain, the TCR of the T cell is ready to be checked for function via processes called positive and negative selection. Remember, T cells require antigen to be presented to them via MHC molecules. Cells that only weakly recognize self-antigen in MHC molecules die by neglect (they do not receive sufficient stimulation to survive); those which recognize it too strongly are deemed dangerous and are induced to die by a process known as apoptosis. Only those with moderate affinity

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for MHC plus antigen survive (only approximately 5% of potential T cells). Once past this selection stage RAG-1 and -2 are downregulated and the cells become single positive for either CD4 or CD8.

The process by which cells become helper (CD4+) or cytotoxic (CD8+) has not been fully determined. Two different types of MHC molecules present antigen to CD4 or CD8 T cells (MHC class II and class I, respectively) and it is possible that the CD4 or CD8 phenotype is determined by which MHC the functional TCR encounters first; interaction with MHC II may induce CD4+ T cells whereas MHC I may promote CD8+ T cells. However, the final distinction may arise as a result of the strength of action of TCR signaling components. For example, MHC II binding the TCR enhances signaling to Lck compared to MHC I, and artificially attenuating TCR signaling allows MHC II–specific CD4+ T cells to become CD8+ over the course of the long differentiation process. Longer interactions and stronger TCR signals therefore favor CD4+ (helper) T-cell development over CD8+ (cytotoxic) T cells. Other signaling pathways, particularly involving the Notch proteins, are also important for lineage development (for an in-depth discussion of this, see Yang 2006).

T-Cell Receptor Signaling

The TCR is composed of several subunits that colocalize within the cell membrane, drawn together by virtue of their positive and negative charges. Eight subunits are required to form a complete TCR. The α and β chains form the foreign peptide:MHC recognition site but lack a signaling domain, which is provided by two intracellular ζ chains. This central αβζζ structure is flanked by two other complexes each composed of a heterodimer of either one γ and one ε or one δ and one ε chain (collectively known as CD3). The intracellular domains of the γ, δ, ε, and ζ chains provide the substrate for the first step in TCR signal transduction. Contained within them are immunoreceptor tyrosine-based activation motifs (ITAMs) which contain tyrosine residues that, once phosphorylated by the Src family kinases (such as Lck), recruit the 70 kDa ζ chain–associated protein ZAP-70. On binding to one or more ITAMs, ZAP-70 is activated setting off a series of phosphorylation events that culminate in the transcription of a number of genes. These pathways are summarized in Figures 9.3 and 9.4.

The Requirement for Co-Receptors

As described earlier, T cells differentiate during development into CD4+ and CD8+ subsets. These subsets are defined by their expression of the CD4 or CD8 co-receptor on the cell surface. CD4 and CD8 colocalize with the TCR and recognize invariant

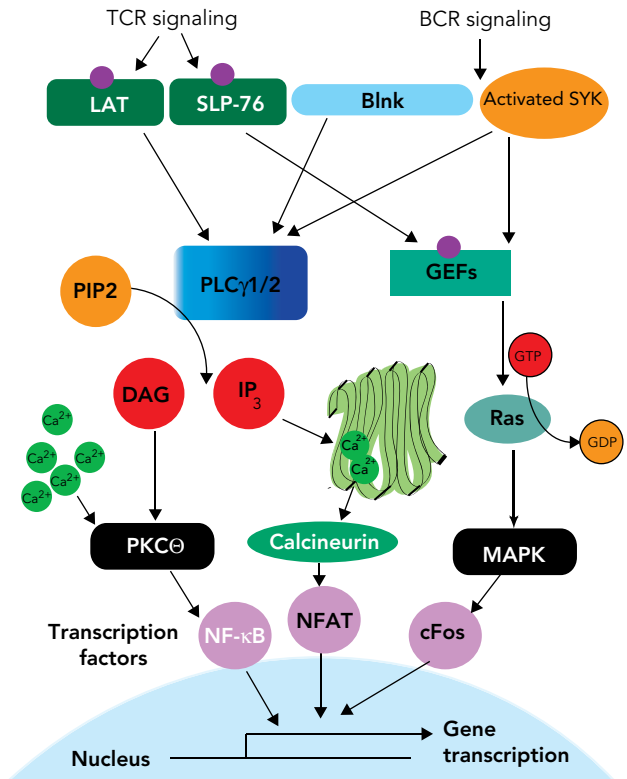


Figure 9.4. Downstream signaling following TCR or BCR activation. Signaling through SLP-76, downstream of the TCR, and through Syk, downstream of the BCR, results in the activation of guanine nucleotide exchange factors (GEFs) which promote the activation of Ras-GTPases via the dissociation of GDP allowing Ras to bind fresh cytosolic GTP. This process acts as a molecular switch and leads to the activation, via other factors, of MAP kinase and the eventual activation and nuclear translocation of the transcription factor cFos.

LAT signaling in the T cell results in the activation of the enzyme phospholipase C (PLC)γ1. Syk in the B cell interacts with the B cell linker Blnk and Tec kinases to activate phospholipase C (PLC)γ2. These enzymes convert inactive phosphatidylinositol bisphosphate (PIP2) into diacylglycerol (DAG) and inositol triphosphate (IP_3). IP_3 promotes the release of calcium ions from the endoplasmic reticulum and thus the activation of the phosphatase calcineurin. Calcineurin dephosphorylates the transcription factor nuclear factor of activated T cells (NFAT) which then translocates to the nucleus and activates the transcription of genes involved in the synthesis of interleukin (IL)-2. DAG also acts with calcium ions to activate protein kinase C (PKC)θ at the plasma membrane. This kinase also activates a transcription factor, this time NF-κB, which translocates to the nucleus to initiate the transcription of a wide variety of genes involved in T-cell activation, proliferation, and survival.

motifs on the MHC molecule. CD4 recognizes MHC class II molecules, which are mainly expressed on professionalAPCs; CD8 recognizes MHC class I molecules, which are expressed on the surface of all nucleated cells. Both receptors provide intracellular docking sites for the kinase Lck which, as described, initiates the phosphorylation cascade following TCR engagement. However, CD4 and CD8 vary in their

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downstream effects on the T cell. CD4+ T cells receive signals encouraging the production of cytokines and chemokines and the expression of molecules such as CD40L, which interacts with CD40 on B cells to encourage B-cell maturation and antibody class switching. Because CD4+ cells act, for the most part, in this indirect manner, they are known as T helper (T_h) cells. In contrast, CD8+ T cells receive signals encouraging the release of cytotoxic compounds such as perforin and granzyme. Their direct, lytic activity means that CD8+ cells are known as CTLs.

Costimulatory Signaling

We have discussed the αβ TCR that recognizes foreign particles bound to MHC, the associated intracellular chains that make up the CD3 complex and the co-receptors CD4 and CD8 that bind to MHC molecules and stabilize the interaction. Collectively this "signal 1" is still not enough for full T-cell activation. In fact, T cells that only receive signal 1 lose the ability to respond to antigen and acquire a quiescent, nonproliferative phenotype. This state of "anergy" is central to the development of peripheral tolerance and ensures that harmless antigens, encountered in the absence of other "danger" signals, do not trigger a potentially damaging autoimmune or allergic response. Full activation requires "signal 2," which is delivered by the cell that presents antigen to the T cell via MHC molecules: the APC. The additional molecules required to provide signal 2 are called costimulatory molecules. Evolutionarily, it is beneficial to require this second signal as it means that only "professional" APCs (macrophages, dendritic cells, and B cells) can fully activate a T cell. Without the need for costimulation, T-cell activation could occur widely leading to bystander tissue damage. The constitutively T-cell–expressed molecule CD28 provides the first, and strongest, of these costimulatory signals. CD28 on T cells binds to B7 molecules (CD80 and CD86) on professional APCs and results in T-cell survival and proliferation. CD28's association with lipid rafts leads to the close association of these structures with the TCR and an amplification of the protein tyrosine phosphorylation associated with TCR signaling. Downstream signals also increase the amount of IL-2 produced by the T-cell. IL-2 is the main cytokine involved in T-cell proliferation, increasing the expression of survival factors and pushing T cells to enter the cell cycle.

Late Costimulation

Though CD28 provides strong survival signals, further T cell–APC interactions are required for full T-cell effector function and to prevent T-cell death. Several other costimulatory molecules are induced on the surface of T cells following their initial recognition of peptide: MHC complexes. These induced costimulatory molecules can be divided into three families based on their structure: the Ig-like superfamily, the integrins, and the TNF/TNF receptor superfamily.

CD28 is itself a member of the Ig-like superfamily and interacts with CD80 and CD86 (also known as B7.1 and B7.2) on the surface of APCs. This interaction results in the signaling cascades described earlier. However, if this interaction continues, the T cell will remain in a proinflammatory state and may be subject to activation-induced cell death (AICD). To counteract this, CD28 signaling results in the production of a related molecule: CTL antigen (CTLA)-4. CTLA-4 is very similar in structure to CD28 and displaces it, but instead of sending a positive signal via ITAMs, it delivers inhibitory signals to the T cell that arrest the activation cascade and "turn off" the inflammatory response. In most circumstances, however, we would require the T cell to continue its effector function. How then do we override CTLA-4? If the threat persists and stimulates T cells strongly, other "late costimulators" are induced to bypass CTLA-4. The inducible costimulator (ICOS) on T cells is one such molecule and another member of the Ig-like superfamily. Integrins, such as leukocyte function-associated antigen (LFA)-1, also provide T cells with activation signals following interaction with their ligands on APCs, but are reliant on the presence of other costimulators, in particular CD28, to exert this effect. The TNF receptor superfamily (TNFRSF) includes several costimulatory molecules, which are typically upregulated late in T-cell activation, such as 4–1BB and OX40. These bind to equally inducible ligands on APCs. Without these additional late costimulators, T-cell activation would not persist and certain pathogens might escape elimination. Furthermore, these late costimulators are vital for the development of memory T-cell populations. Without them, T-cell activation and progression is prevented too early to seed the memory T-cell pool. Development of memory is vital to thwart a similar infection on re-encounter. Memory cells are more abundant and more easily activated than naive cells so can clear a second infection with the same pathogen more quickly and efficiently. A basic sequence of expression of costimulatory molecules on the T-cell surface and their ligands on the APC is illustrated in Figure 9.5.

Further Differentiation of T Cells

So far, we have discussed the T-cell subcategories of CD4+ helper and CD8+ cytotoxic T cells. CD4+ helper T cells (and recently CD8+ T cells) are further subcategorized based on the cytokines they secrete. Most work has focused on helper CD4+ T cells and will be discussed here. So far, the subcategories are called

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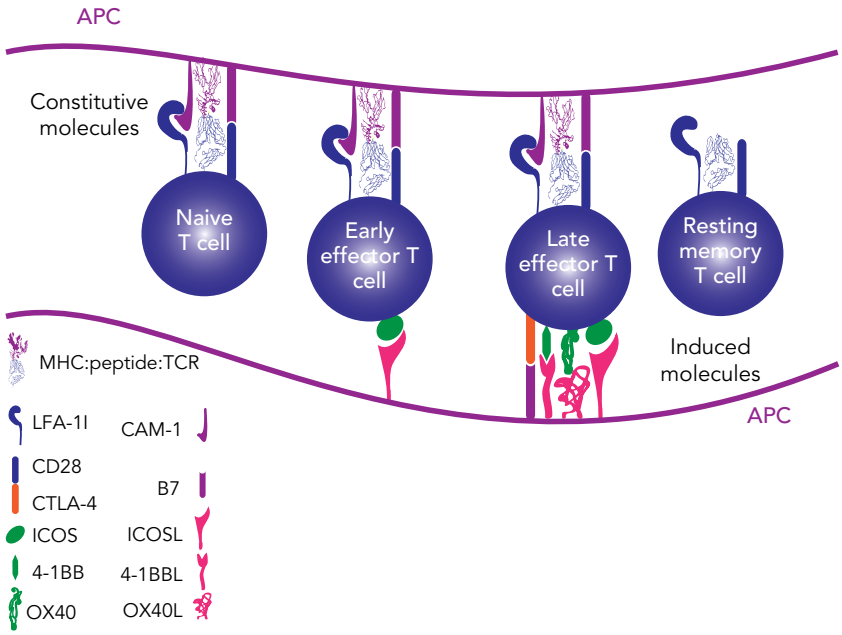


Figure 9.5. Upregulated quickly after TCR stimulation, ICOS acts to boost the signals provided by CD28. As activation continues, late costimulatory molecules begin to appear along with the coinhibitory molecule CTLA-4. These molecules result in a cessation of proinflammatory cytokine production and an increase in the expression of prosurvival factors such as members of the Bcl family of antiapoptotic genes. On receiving these signals, the T cell stops proliferating, but avoids AICD and progresses into the memory pool. Memory T cells are able to survive long term and do not continue to express costimulatory molecules. However, the kinetics of costimulatory molecule reexpression are much faster.

T_h1, T_h2, T regulatory (Treg), and T_h17 cells. T_h1 cells secrete the cytokines interferon (IFN)-γ, TNF, and IL-2; T_h2 cells secrete IL-4, IL-5, and IL-13; T_h17 cells secrete IL-17 and TNF-α; and regulatory T cells secrete IL-10 and/or TGF-β (although IL-10 is also produced by T_h2 cells). Researchers have attempted to find surface molecules that distinguish these different helper T-cell subsets for many years. However, those described so far remain equivocal and clear distinctions between the subsets, in terms of surface molecule expression, remain elusive.

The cytokine profiles of these cell types allow them to induce discrete immune responses according to the nature of the threat. T_h1 cytokines enable a cell-mediated immune response to target intracellular pathogens by activating CTLs, macrophages, and NK cells, whereas the T_h2 cytokine response induces a humoral antibody response targeting extracellular pathogens. T_h2 cell cytokines activate mast cells and basophils through IL-4 and eosinophils through IL-5. T_h subset polarization also dictates the final type of antibody produced by B lymphocytes (see the subsequent text).

It is still unclear how these different helper T-cell subtypes develop. The nature of the pathogen and its dose contribute, as do the level of costimulation, signals from the tissue microenvironment and transcription factors. At present however, there is no overriding consensus, as it seems to vary with experimental condition and/or the pathogen under investigation. Longer contact of MHC with the TCR is reported to produce a stable T_h2 response; it is also suggested that a higher degree of stimulation is required for T_h1 polarization. However, a number of parameters dictating cytokine profile outcome is standard. For example, differentiation of T_h1 responses involves activation of the *IFN-γ* gene and concomitant silencing of the IL-4 gene, while the opposite is seen on T_h2 differentiation. Type I and II interferon signaling causes the STAT1-dependent formation of the IL-12 receptor; ligation of this IL-12R leads to activation of T-bet, a transcription factor critical for the development of T_h1 responses, and STAT4-dependent potentiation of IFN-γ production. In T_h2 responses, IL-4 signaling to a naive cell activates STAT6 and the master transcription factor GATA-3. GATA-3 suppresses STAT4 and the IL-12R whilst inducing positive epigenetic changes in the T_h2 cytokine cluster of Il4, Il5, and Il13 genes (see Figure 9.6 for a summary).

For two decades, the T_h1/T_h2 polarization model was largely unaltered, until the recent description of

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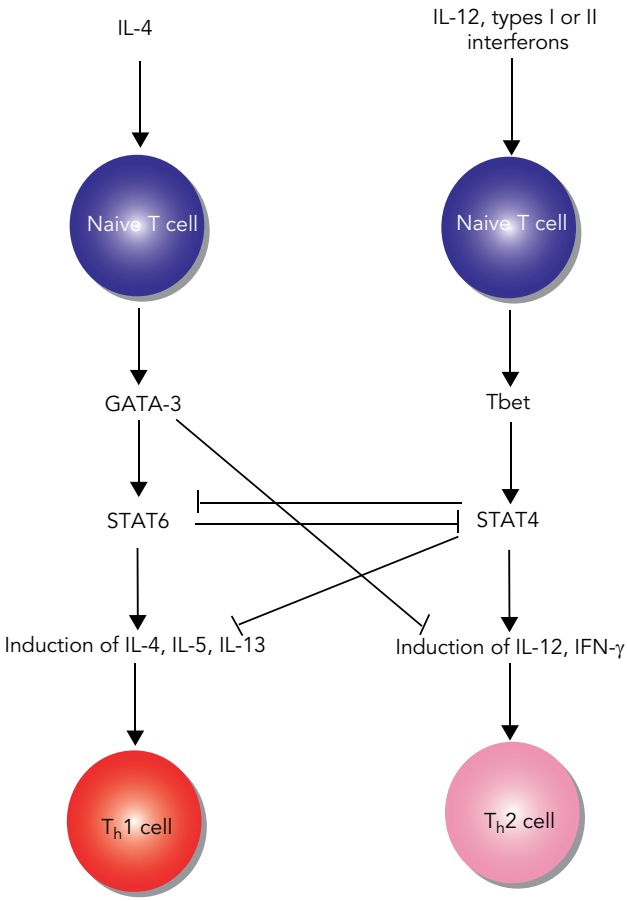


Figure 9.6. Following signals from inflammatory cytokines, naive T cells begin to polarize into T_h1 or T_h2 cells. Transcription factors, induced following cytokine signaling, direct the changes in cell phenotype. The T_h1 master transcription factor, Tbet, is induced after IL-12R ligation and STAT4 signaling. This induces further IL-12 production and expression of IFN-γ, and inhibits the activation of GATA-3: the T_h2 master regulator. Similarly, GATA-3 activation in T_h2 cells inhibits T-bet and STAT4 and allows the transcription of IL-4, IL-5, and IL-13 by inducing epigenetic changes such as induction of DNase I hypersensitive sites.

the T_h17 subset of cells. These are characterized by IL-17, IL-6, IL-22, and TNF-α production. T_h17 cells are entirely separate from the T_h1 and T_h2 lineages; cytokines from T_h1 cells antagonize T_h17 generation, and it is likely that they have evolved to deal with different pathogens to those efficiently cleared by T_h1 and T_h2 responses. IL-23 polarizes naive cells into T_h17 cells in an ICOS- and CD28-dependent manner (T_h1 cells are dependent on CD28 but not ICOS). Generation of T_h17 cells is independent of the STAT1, STAT4, and STAT6 signaling proteins but does require STAT3. In APC-free culture conditions, T_h17 cells can be generated with the addition of only TGF-β and IL-6; these cytokines induce differentiation by activation of the nuclear receptor retinoic acid receptor-related orphan receptorγ (RORγ). The specific isoform RORγt and the receptor RORα are regulated by STAT3 and are highly expressed in T_h17 cells.

Effector Function of T Helper Cell Subsets

By virtue of their secreted cytokines (IFN-γ, TNF, and IL-2), T_h1 effector cells are very efficient at promoting cell-mediated immunity; the cytokines they produce help to activate macrophages and NK cells, and promote survival of CTLs. These functions make T_h1 cells important for the clearance of intracellular pathogens such as viruses. T_h1 cytokines also induce antibody class (or isotype)-switching in B cells, leading to the production of IgG2a, as discussed later. However, an excessive T_h1 response is frequently associated with immunopathology, or immune-mediated damage to the host, and has been implicated in some autoimmune diseases and inflammatory disorders. T_h2 cells have a very different function; by producing cytokines such as IL-4, IL-5, and IL-13, they help to activate eosinophils and mast cells, and are important for the clearance of extracellular pathogens, especially parasites. T_h2 cytokines can also induce class switching in B cells, but promote the secretion of IgE, IgA, and IgG1. However, in some individuals, a disproportionate T_h2 response is mounted which is a hallmark of allergic reactions and atopic disease such as asthma. Activated T_h17 cells making IL-17, IL-21, and IL-22 are also important in host defense to infection, particularly for extracellular bacteria, as T_h17 cytokines efficiently recruit neutrophils to the site of inflammation. Once again, T_h17 cells are highly inflammatory and, like T_h1 cells, play a role in the pathogenesis of autoimmune disease. Figure 9.7 summarizes the main roles of T_h1 and T_h2 cells in health, disease, and autoimmunity, and also highlights the complications that can arise if an inappropriate response is initiated during infection.

Regulatory T cells (Treg) comprise another important T-cell subset that suppresses the activation of other cells in a contact-dependent or -independent manner. Treg interact with and suppress both T_h1 and T_h2 effector T cells. They can achieve this indirectly, by affecting APCs, or directly through T–T interactions. There are four main Treg subsets; natural Treg constitutively express CD4 and CD25 as well as the transcription factor Foxp3. Their generation is dependent on TGF-β. Tr1 cells, by contrast, are induced Treg, which do not express Foxp3 but express high levels of CTLA-4 and require IL-10 for their differentiation and suppression. In addition to these Treg subsets, anergic CD4+ and CD8+ cells are capable of suppression by competing for antigen and IL-2; they have a very high-stimulation threshold and do not therefore become activated except in situations of very high antigen exposure.

The main functions of each T_h subset and the cytokines they produce are summarized in Figure 9.8.

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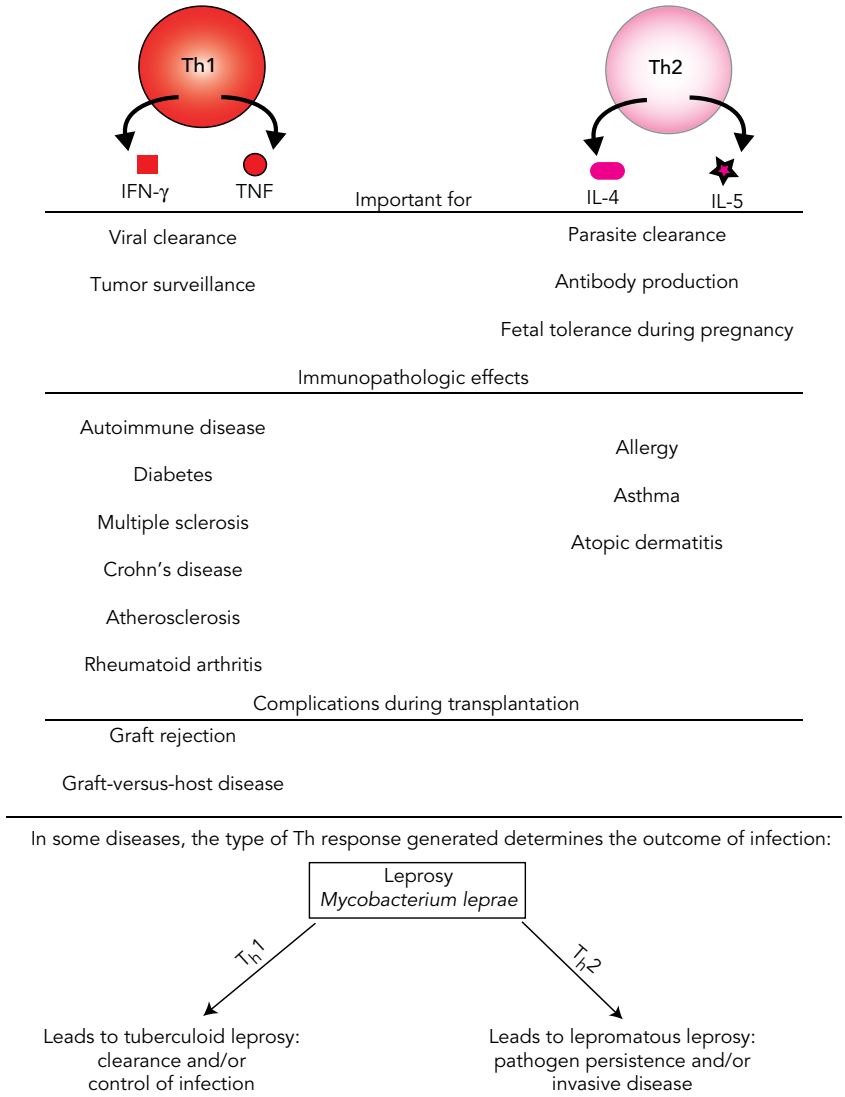


Figure 9.7. The role of T helper cell responses in pathogen clearance and immunopathology. Th1 and Th2 responses are important for pathogen clearance. However, both responses can exacerbate autoimmunity and in some diseases, the type of Th response generated can determine the outcome of infection as in the case of leprosy. The Th1/Th2 paradigm has been at the forefront of research for many years but it is now clear that Th17 cells take part in many Th1-based responses and that these cells are also involved in many autoimmune diseases. Similarly, regulatory T cells can influence T cell responses, play a role in foetal tolerance and can inhibit tumor immunity.

Intraepithelial T Cells

As with every biological system, there are a few exceptions to the rules. We cannot discuss lymphocytes without highlighting a subset of T cells that resides among the epithelial cells that line our mucosal surfaces (i.e., gut, lung, genital tract, and skin); the intraepithelial lymphocytes (IELs). There can be as many as 20 IELs per 100 epithelial cells. Within this population a high proportion (50% in humans and 90% in mice) have an alternate TCR heterodimer of γδ (do not confuse these with the γ and δ chains within the CD3 cluster). IELs are mostly CD8+ but some have CD3 but no CD4 or CD8. Their origin and development is unclear but they exist in athymic mice, albeit in reduced numbers. Unlike conventional T cells, γδ IELs do not recognize antigen presented by classical MHC molecules. Instead, it is believed that they are primed by the surrounding epithelial cells possibly via nonclassical MHC such as CD1. Despite their mucosal localization, and therefore the vast array of antigens that they must encounter, the γδ TCR is much less diverse than the αβ TCR. Rather than recognizing highly specific pathogen-derived sequences, they respond to molecules such as heat

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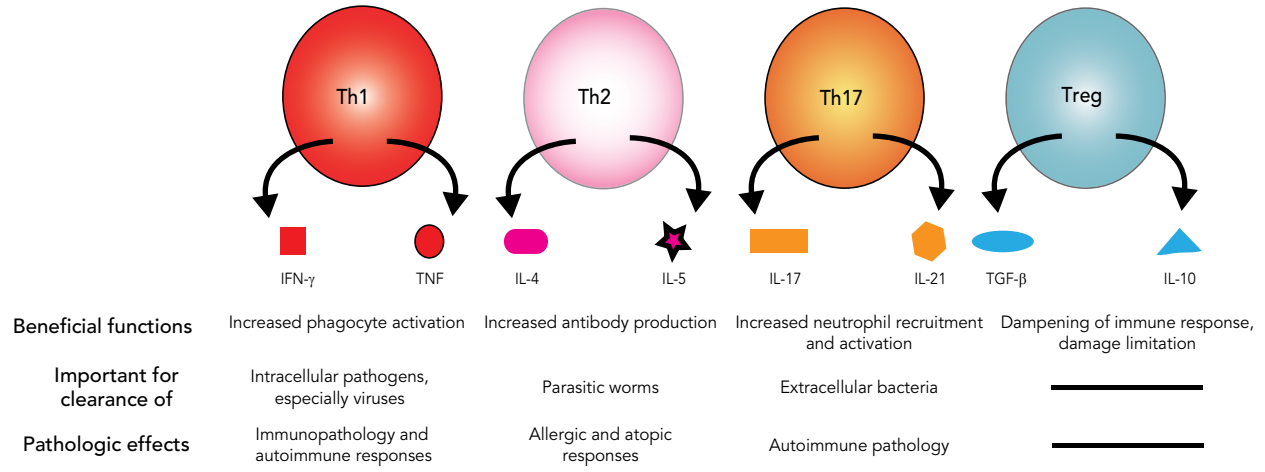


Figure 9.8. T helper cell subsets. CD4+ T helper cells can be subcategorized according to the cytokines they secrete. These cytokines are primarily responsible for the effector function of each subset and these functions are listed. If the immune response is uncontrolled or responds inappropriately to harmless antigens, for example in allergy or autoimmunity, T cells can have detrimental effects and can contribute to pathology.

shock proteins that are upregulated on the epithelial cells following infection or transformation. The suggestion therefore is that IEL behave more like innate immune cells than adaptive lymphocytes.

Effector Function of Cytotoxic (Killer) T Lymphocytes

As described previously, when an antigen-specific CD8+ T cell encounters the appropriate cognate peptide antigen bound to MHC class I on a target cell, the T cell becomes activated. For CTLs, this activation results in two major effector functions: cytokine secretion and targeted killing. Similar to effector CD4+ cells, there are subsets of effector CD8+ T cells, based on the profile of cytokines they secrete. T_c1 cells produce large quantities of IFN-γ and TNF, and T_c2 cells secrete IL-4 and IL-5. However, unlike their CD4+ counterparts, T_c2 cells are rare in vivo, and as yet no equivalent population of Treg cells has been identified for CD8+ T cells.

As well as producing cytokines, CTLs perform a vital function by killing infected cells and tumors. Following the recognition step, during which the CTL binds the antigen-bearing target cell, the TCRs, and associated co-receptors on the T cell come together, forming a large cluster called the supramolecular activation cluster (SMAC). This aggregation triggers a reorganization of the actin cytoskeleton within the CD8+ T cell, which allows it to correctly position its lytic granules to target the site of contact; a mechanism which limits nonspecific damage to surrounding cells. The highly specialized lytic granules found in CD8+ CTLs are also found in natural killer (NK) cells and contain several proteins important for triggering programmed cell death. Perforin, first, polymerizes to form pores in the target cell membrane allowing granzymes to enter. Granzymes are a family of enzymes with serine protease activity which, after entering the target cell, degrade its internal proteins. A specific member of this family, granzyme B, cleaves and activates caspase-3, which initiates the caspase cascade ultimately culminating in the degradation of cellular DNA, one of the hallmarks of cell death by apoptosis. Perforin and granzyme are both required for efficient killing of target cells, although some cytotoxic activity can be mediated by perforin alone. A further component of CTL lytic granules is granulysin, another protein, which can induce apoptosis and has antimicrobial properties. In killing a target cell, the CTL itself is unharmed and is able to kill several infected targets in rapid succession using internal stores of preformed lytic granules. The action of inducing cell death in the target cell via perforin/granzyme can be programmed in as little as 5 minutes, although the process of apoptosis may take several hours. Controlled cell death is required to contain infection because if cells die by necrosis (uncontrolled death), they can release infectious intracellular pathogens in the process. CD8+ T-cell killing can be measured using the method outlined in Figure 9.9.

Another major mechanism by which a CTL can kill an infected cell involves interactions of cell surface receptors rather than stores of toxic granules. Fas ligand (FasL), a member of the TNF family, is expressed on the surface of activated CTLs as a homotrimer. It binds to Fas, a member of the TNFR superfamily on the target cell, and in doing so, produces a conformational change in Fas allowing the recruitment of death-domain containing adaptor proteins such as FADD. This clustering of adaptor proteins ultimately results in activation of the caspase cascade, as described earlier for killing via perforin and granzyme, but is independent of these

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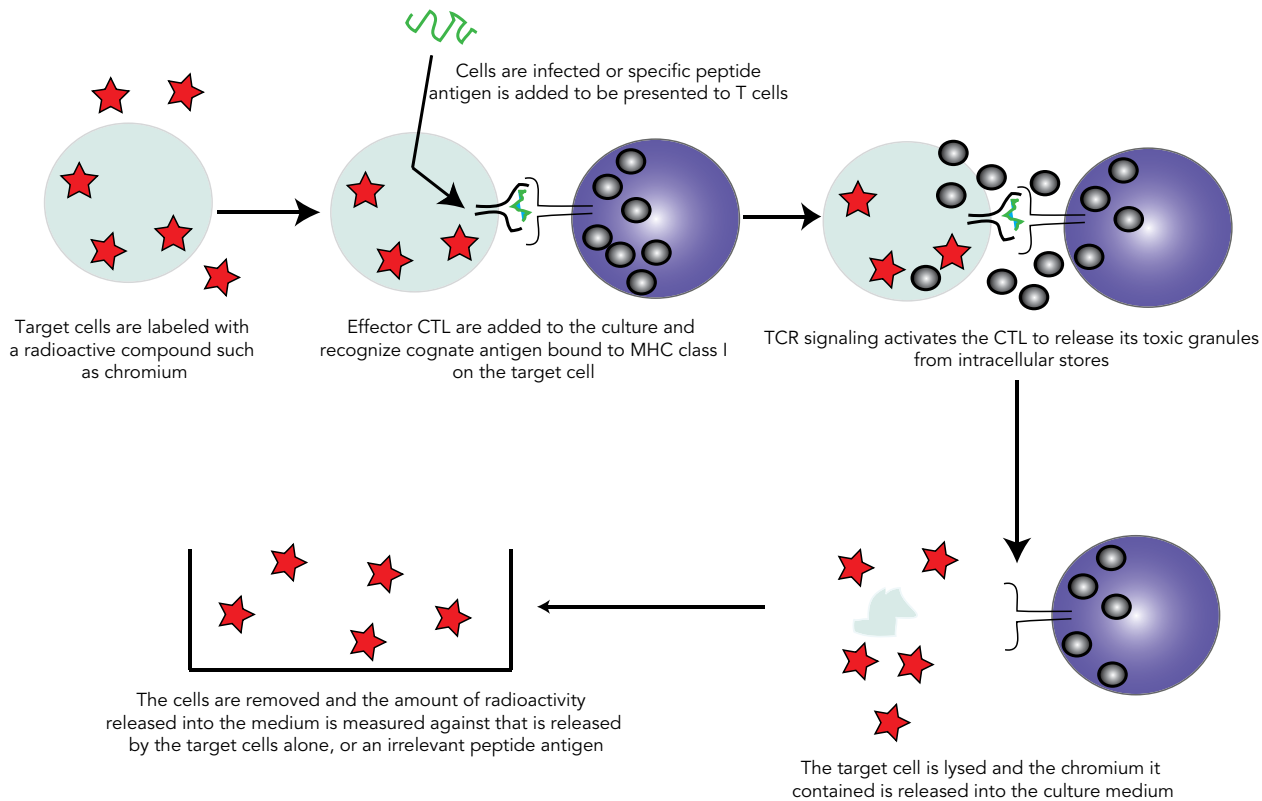


Figure 9.9. Measuring CTL activity in vitro. Activated cytotoxic T lymphocytes (CTLs) are programmed to kill cells that have been infected by viruses or that have become cancerous. However, not all CTLs are able to kill with equal efficiency and not all stimuli induce equally strong CTL responses. The assay outlined above can be used to quantify CTL killing in vitro and thus give an indication of the strength of the CTL response in vivo.

enzymes. Since activated effector T cells can express both Fas and FasL, this is one of the major mechanisms by which CTLs can kill off other CTLs (a process known as fratricide) at the end of a robust immune response to reduce cell numbers back to homeostatic levels (known as the contraction phase).

B CELLS

B-Cell Development

B cells develop in the bone marrow in adults (the liver in foeti) and are continually produced throughout life. The stromal cells present in the marrow are vital for B-cell differentiation owing to their production of adhesion molecules, which bind to and activate the progenitor cells, and to their secretion of prodifferentiation mediators such as SCF, IL-7, IL-11, and chemokines such as CXCL12. Development of a B cell in the bone marrow is marked by the expression of molecules which interact with these stromal cells, and in the periphery by the expression of immunoglobulin genes. ProB (progenitor-B) cells are distinguished from the hematopoietic stem cells from which they are derived by the upregulation of B220 (CD45R) and the integrin VLA-4. These allow the B cell to bind VCAM-1 on the stroma, which induces the expression of Kit on the B cell. Kit (CD117) is able to bind SCF, which is a proliferation factor.

The BCR is composed of two identical heavy chains (pink in Figure 9.10), of which there are five different alternatives or isotypes; and two identical light chains (green in Figure 9.10), of which there are two types (κ or λ). The two heavy and two light chains are held together by disulfide bonds. Different regions of the antibody have different functions in host defense. The two Fab portions (derived from fragment antigen binding) are shown at the top of the antibody molecule and recognize foreign antigen. Unlike the TCR, the Fab portion of the BCR can do this directly without the need for presentation. The Fc portion (derived from fragment crystalizable) binds to Fc receptors that are present on a wide variety of other immune cells including NK cells, macrophages, and dendritic cells (discussed later). This binding of soluble antibody to Fc receptors has significant consequences for the Fc receptor–bearing cells and different heavy chain isotypes have different effects. Though antibody has significant functions as a secreted molecule, B cells also retain it on their cell surface where it functions as the BCR. As with the TCR, the development of the BCR is a random process leading to a diverse and extensive

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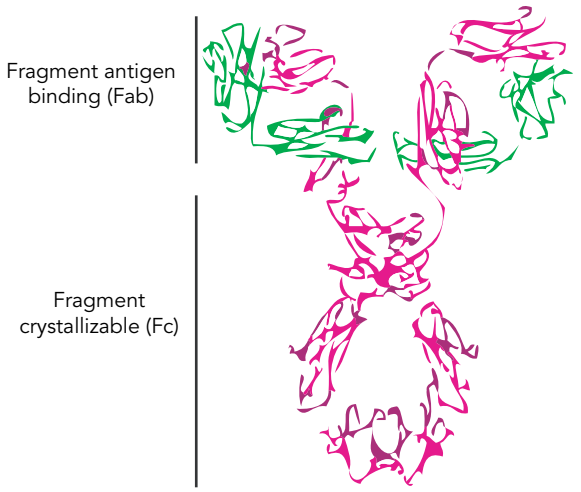


Figure 9.10. Antibody structure. The BCR is composed of an antibody molecule which itself has four chains. Two of these are the identical heavy chains and two are the identical light chains. Light and heavy chains are composed of variable and constant regions. The variable regions colocalize in the two Fab fragments at the top of the molecule and recognize antigen. The lower Fc fragment is composed of the constant regions of the heavy chains and varies between antibody isotypes allowing each to perform a different function. When tethered in the B-cell membrane as the BCR, the Fc fragment loses this function and instead associates with CD79 to transduce signals to the B cell.

repertoire capable of directly binding to a vast array of foreign proteins. Each B cell selectively expresses only one type of antibody and therefore recognizes one antigen. The specific nature of foreign protein binding by the B cell means it also functions as a highly sensitive and specific antigen-presenting cell.

During development, immunoglobulin expression depends on functional gene rearrangement, which is dependent, like TCR production, on the recombinase genes *RAG-1* and *RAG-2*. ProB cells rearrange the μ heavy chain gene to form an early BCR (preBCR). This surrogate allows testing of the ability of heavy chains to associate with light chains and other molecules important for BCR signaling such as CD79 α/β. If rearrangement of this initial heavy chain is unsuccessful or results in autoreactivity, secondary rearrangements are induced. During development in the bone marrow, signaling through the surrogate preBCR lowers the threshold for IL-7 responses, meaning cells which have productively rearranged the BCR are able to proliferate and outnumber cells which have not. All immature B cells express IgM, and strong signals received through self-antigen binding to IgM in the bone marrow deletes self-reactive B cells in the same way that self-reactive T cells are deleted in the thymus. Some of these cells can be rescued by "receptor editing," which allows the light chain recombination to continue until the self-reactive light chain has been replaced. Once the B cell has passed the test in the bone marrow, the *RAG* genes are no longer expressed and the light chain present at that time remains. These cells are now committed to the B-cell lineage and mature further in the bone marrow until the need for interaction with stromal cells is lost; the cells can then move into the center of the bone and finally out into the periphery as mature but naive B cells.

The functional maturation of the B cell, so that it is capable of recognizing antigen, begins from the earliest pro–B-cell stage and continues once the cell has left the bone marrow and moved out into the periphery. This maturation involves swapping to different heavy chain isotypes and the expression of membrane immunoglobulins. The expression of different membrane and surface immunoglobulins is the key difference between subsets of B cells, which do not diverge and mature separately as do T cells. However, the site at which B cells settle does affect their role and the molecules they express. Once they have left the bone marrow, B cells can either circulate the body in the lymphatics, or they can settle in a defined site such as the peritoneum, the pleural cavity, the lymph nodes, or the spleen. Cells in these sites are long-lived and have varied characteristics depending on the role they are required to play. Like the IEL T-cell population, a separate subset of B cells exists that expresses the CD5 molecule. CD5+ B cells (also known as B1-cells) are found in the peritoneal and pleural cavities of mice, and the tonsils and blood in humans. Like IEL they are of a separate lineage to all other B cells, are dominant early in life, and gradually decline with age. Also like IEL, they have limited receptor diversity (in this case Ig). They respond to polysaccharide, rather than protein, antigens and produce IgM very early (within 48 hours) during inflammation. A key role of CD5+ B cells is to produce "natural" antibodies; these polyreactive low-affinity antibodies bind a number of antigens including DNA and α1–3 dextran, and may thus provide a first line of defense against bacterial infection.

B-Cell Receptor Signaling

As mentioned earlier, the BCR consists of membrane-tethered immunoglobulin (mIg): a tetramer consisting of a heavy (H) chain homodimer and two κ or two λ light (L) chains. Like the TCR, the BCR requires additional molecules to facilitate signal transduction via immunoglobulin. In the case of a B cell, this is a heterodimer of Igα/Igβ (CD79 α/β) subunits, which is noncovalently bound to surface immunoglobulin. Antigen binding to the Fab portion of mIg causes rapid phosphorylation of the ITAMs in the intracellular domains of CD79 α/β by Src-family kinases, such as Lyn, leading to the activation of Syk (Figure 9.11). Syk, in turn, activates B-cell linker (Blnk) which phosphorylates and activates phospholipase Cγ2 (PLCγ2) and leads to the recruitment of guanine nucleotide exchange factors (GEFs). These then utilize the same

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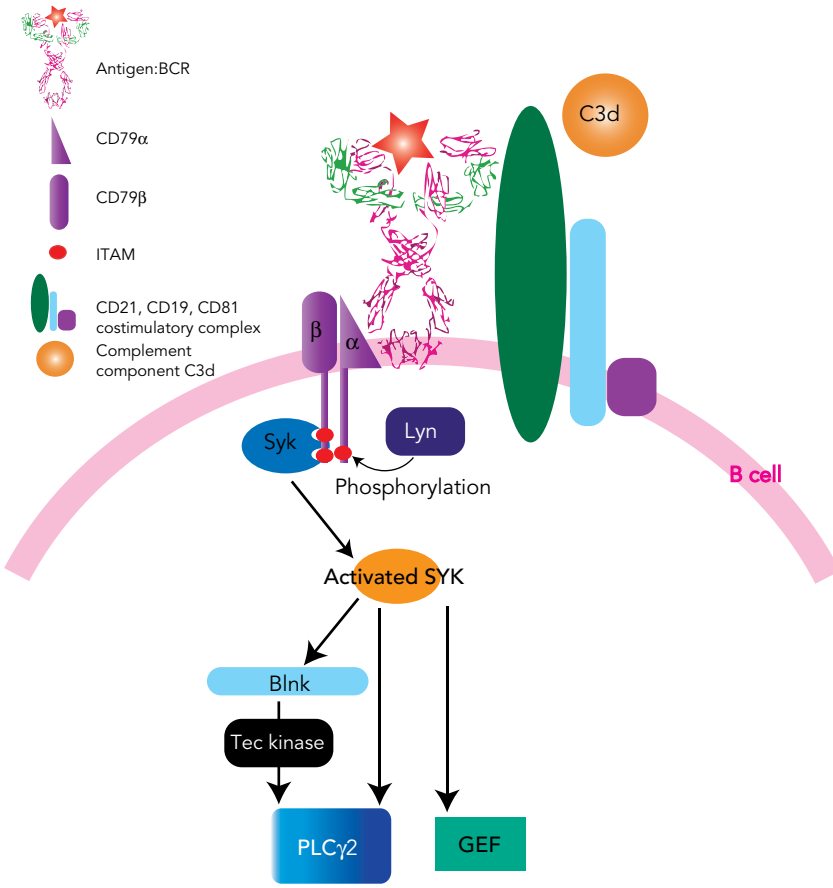


Figure 9.11. BCR signaling. Like the T cell, the B cell needs a signal through its BCR to become activated. However, unlike the T cell, the B cell does not need an APC to present processed peptides. B cells can recognize and respond to native foreign antigens. Like the TCR, the BCR lacks any significant intracellular signaling domains and must associate with accessory molecules in order to signal. In this case, signals are transmitted by CD79, the α and β chains of which associate with the BCR and also contain ITAMs. When the BCR encounters antigen, Src family kinases, such as Lyn, quickly act to phosphorylate the ITAMs located on the intracellular portions of CD79. Another kinase, Syk, is then able to bind and becomes activated going on to phosphorylate the B cell linker (Blnk) and recruiting Tec kinases and guanine nucleotide exchange factors (GEFs) to continue the signaling cascade.

signaling pathways as PLCγ1 and the GEFs involved in T-cell signaling (see Figure 9.4) to induce a variety of cellular processes, including regulation of gene expression, reorganization of the cytoskeleton, and immunoglobulin-mediated internalization of antigen complexes. Antigen internalized through the BCR is then processed in endosomal compartments and presented by MHC class II molecules to specific T cells.

The Requirement for Co-receptors

As with TCR signaling, simple stimulation through membrane immunoglobulin and its associated co-receptors does not provide the full picture. Both mIg and CD79 interact with elements of the cytoskeleton causing an accumulation of immunoglobulin–antigen complexes in a central SMAC (cSMAC). In established activation (but not required for immediate early signaling) this is encircled by a ring of integrins forming the peripheral SMAC (pSMAC).

CD19 is an essential co-receptor on the surface of the B-cell membrane and its cytoplasmic domains contain a number of binding sites for intracellular signaling and adaptor molecules. It colocalizes with CD21 and CD81 to function as an adaptor-like protein that facilitates the recruitment and activation of signaling molecules such as Vav and PI3K, to the BCR (for a review of BCR signaling, see Hasler and Zouali, 2001).

B-Cell Effector Function

All naive B cells express IgM and IgD. However, different antigens require different responses, and to fulfill these, the surface immunoglobulin isotype is changed in a process known as class switching. This involves a heavy chain gene rearrangement in the mature B cell

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resulting in a different immunoglobulin class becoming expressed.

It is one of the functions of activated T helper cells to promote immunoglobulin production and class switching in B cells. One of the most important of these interactions is binding of CD40L (CD154) on the activated T cell to CD40 on the B cell. This ligation promotes B-cell proliferation, and drives the B cell to terminally differentiate into a plasma cell that secretes large quantities of immunoglobulin or antibody. Furthermore, different T-cell subsets will cause the expression of different immunoglobulin isotypes in response to the cytokines the T-cell secrete (see Table 9.1).

The primary functions of antibodies are neutralization and opsonization of pathogens; sensitization of effector cell types, such as NK cells, phagocytes, and mast cells; and activation of complement. Opsonization is the coating of foreign particles by antibodies to facilitate their uptake via Fc receptors (see Figure 9.12). Similarly, neutralization involves the coating of pathogens or the toxic compounds they release rendering them harmless to the host (see Figure 9.13). Each isotype varies in its efficiency for each of these functions, largely determined by the anatomical location of the antibody class. For example, IgM, the only isotype of antibody that can be produced without immunoglobulin class switching, is found primarily in the blood. Its large pentameric structure makes it difficult for the IgM molecule to diffuse into tissues, but means it is very effective at activating complement proteins, which are plentiful in the blood. In contrast, IgG and IgE are always found as monomers, and diffuse easily into tissues. The method for detecting soluble levels of secreted antibody is shown in Figure 9.14.

IgG is the most abundant immunoglobulin isotype in plasma, largely as a result of its long half-life in the circulation, and is important for neutralization, opsonization, and complement fixation. IgA can be found as a monomer or dimer, and is the most abundant isotype found at mucosal epithelia, such as the gastrointestinal and respiratory tracts – sites of frequent pathogen exposure. IgA molecules are produced in very large quantities (measured in grams each day in humans) and are actively transported through epithelial cells of the mucosa (as are pentameric IgM molecules). Because of its location at the front line of immune defense, IgA is highly efficient at neutralizing invading organisms. Very little IgD is produced in the body, and it has a negligible role in humoral responses. Only small quantities of IgE are produced and yet it fulfills an important biological role. It is found at very low concentrations in the blood or tissues because once secreted it is tightly bound to the surface of mast cells, via their high-affinity Fc receptors (discussed later).

TABLE 9.1. The influence of different cytokines on the immunoglobulin expressed and secreted by B cells

	Cytokines		
Ig isotype	**IFN-γ**	**IL-4**	**TGF-β**
IgM	↓*	↓	↓
IgG1	↓	↑	
IgG2a	↑	↓	
IgG2b			↑
IgG3	↑	↓	↓
IgE	↓	↑	
IgA			↑

*↓ indicates an inhibitory effect on that immunoglobulin isotype, ↑ a stimulatory effect, and no arrow indicates no known effect.

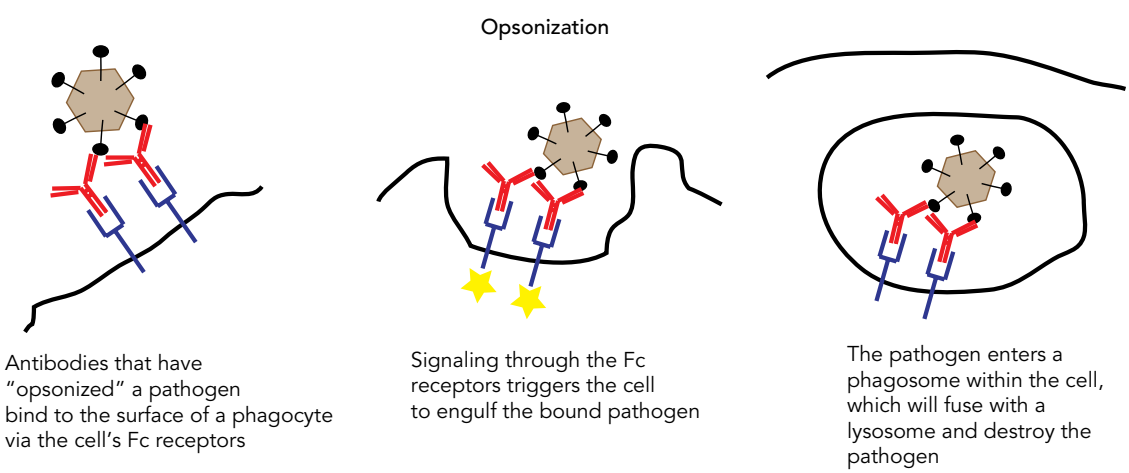
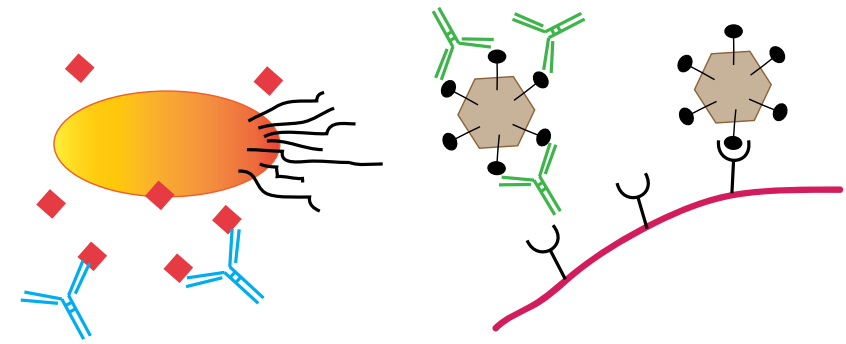


Figure 9.12. Opsonization of pathogens by antibody. Another major role of antibody is to 'opsonize' foreign particles. The variable chains of antibodies can bind specific foreign peptides on the surface of pathogens. Once the variable regions have bound, the constant regions of the antibodies are recognized by Fc receptors on phagocytes, which engulf the antibodies along with the pathogen to which they have bound. The antibody/pathogen complex is then degraded in a phagolysosome within the phagocyte.

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Antibodies can bind and neutralize harmful substances, preventing damage to host cells (e.g., secreted bacterial toxins)

Antibodies bound to pathogens can block attachment to a host cell (e.g., Ab against many viruses can prevent receptor-mediated endocytosis)

Figure 9.13. Neutralization/immune exclusion by antibody.

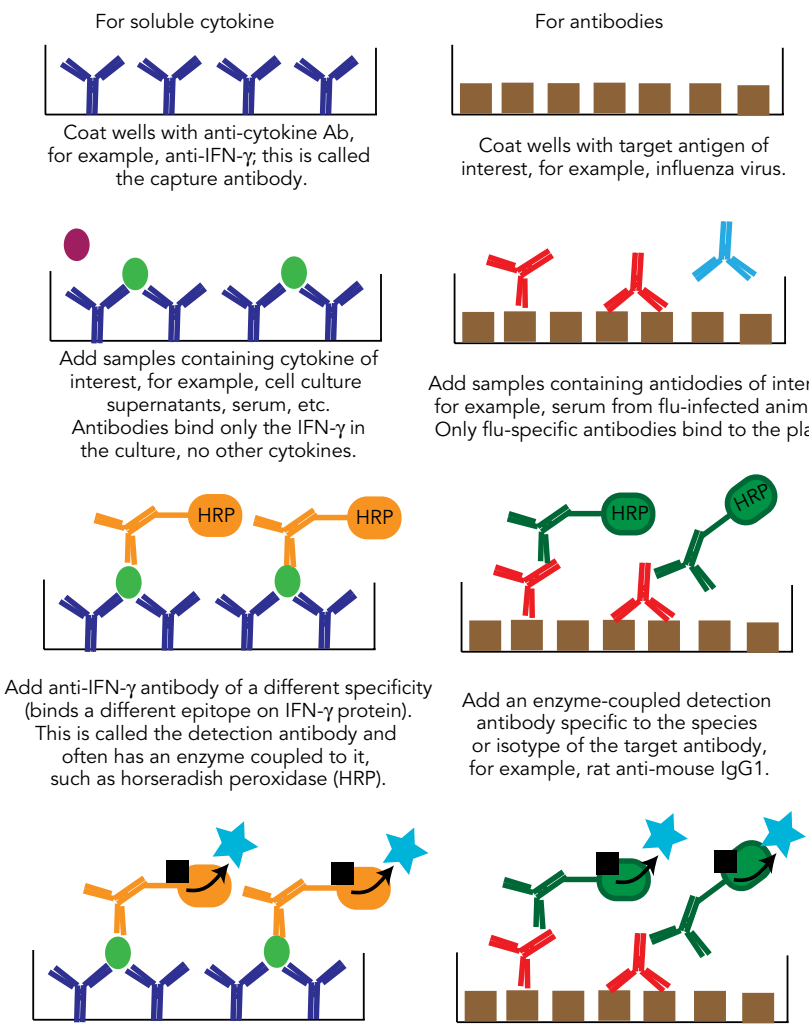


Figure 9.14. Enzyme-linked immunosorbent assay.

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TABLE 9.2. The dominant effector function (denoted in red) of secreted immunoglobulin isotypes

Ig isotype	Neutralization	Opsonization	Complement activation	NK cell activation (ADCC)	Mast cell activation
IgM					
IgD					
IgG					
IgE					
IgA					

IgE is therefore the primary activator of mast cells, which are important for T_h2-driven immune responses and feature in allergic reactions. (See Table 9.2 for a summary of antibody isotype functions.)

In addition to direct neutralization, activation of complement (leading to pore formation and lysis of the pathogen), and immune exclusion of pathogens (in which antibodies bind to and inactivate the microorganism, or block its binding to the host cell), antibodies can also exert protective effects via binding to Fc receptors (FcR) on the surface of immune cells. At least eight different FcR have been identified, including several splice variants of a single receptor. FcR are named based on the isotype of immunoglobulin they bind, so FcεR binds IgE, FcγR binds IgG, and so on. FcR vary in their distribution on cells and their binding affinity, but at least one type of FcR is expressed on DCs, macrophages, NK cells, neutrophils, eosinophils, basophils, mast cells, and B cells.

In most cases, binding of an immunoglobulin to an FcR via the Fc region results in cross-linking and signaling that activates the FcR-bearing cell. In the case of macrophages, this binding via the FcR stimulates phagocytosis and increases the antimicrobial properties of the macrophage, allowing destruction of any engulfed pathogens. FcR and complement receptors are both found on the surface of many phagocytic cells such as macrophages and neutrophils, and signaling through both of these pathways has a synergistic effect on uptake and killing by the phagocyte. This process of coating a pathogen (with antibody and/or complement) to increase its uptake by phagocytic cells is called opsonization. Cross-linking of FcR on other cell types, such as mast cells and NK cells, can trigger exocytosis of toxic mediators stored in granules within the cell. Parasitic worms are frequently too large to be engulfed by phagocytes, but when coated with IgE, become targets for activated eosinophils, which bind via their FcεR. This binding causes release of granules containing destructive enzymes and toxic proteins, helping to clear the pathogen. NK cells can be triggered to kill an infected cell by a similar mechanism. During a viral infection, viral peptides can be displayed on the surface of the infected cell and recognized by circulating immunoglobulin. The exposed Fc regions are recognized by FcRs on NK cells, triggering them to release toxic granules that result in apoptosis of the target cell, using the same mechanism as that employed by CTL discussed earlier. This NK cell activation through FcR via immunoglobulin bound to the surface of an infected cell is called antibody-dependent cell-mediated cytotoxicity (ADCC) and is shown in Figure 9.15.

NATURAL KILLER CELLS

NK cells, so called because it was initially thought that they did not require activation in order to kill cancerous cells, are a type of lymphocyte involved in innate immunity against pathogens and tumors. They arise from a common lymphocyte progenitor but are larger and more granular than T cells and B cells. NK cells account for up to 15% of peripheral blood lymphocytes and are located in several organs including the liver, peritoneal cavity, and placenta in homeostasis. However, numbers increase during disease because they are recruited and activated in response to interferons and other macrophage-derived cytokines produced during infection. By inducing apoptosis in infected cells, NK cells act early during the immune response and are able to limit viral spread until the adaptive immune response has developed. The importance of this role is apparent in patients with NK cell deficiency who are more susceptible to viral infection.

NK Cell Function

NK cells were originally defined by their ability to kill tumor cells in vitro. They sit between the innate and adaptive arms of the immune response and detect the presence of "altered self" by scrutinizing cells for the presence of MHC class I: the "missing self hypothesis." In health, all nucleated cells express MHC class I. As discussed earlier, MHC class I molecules can present viral proteins to CTLs which then recognize the cell as infected and induce apoptosis. To escape recognition by CTLs, a number of viruses and tumors downregulate MHC class I. It is thought that NK cells evolved

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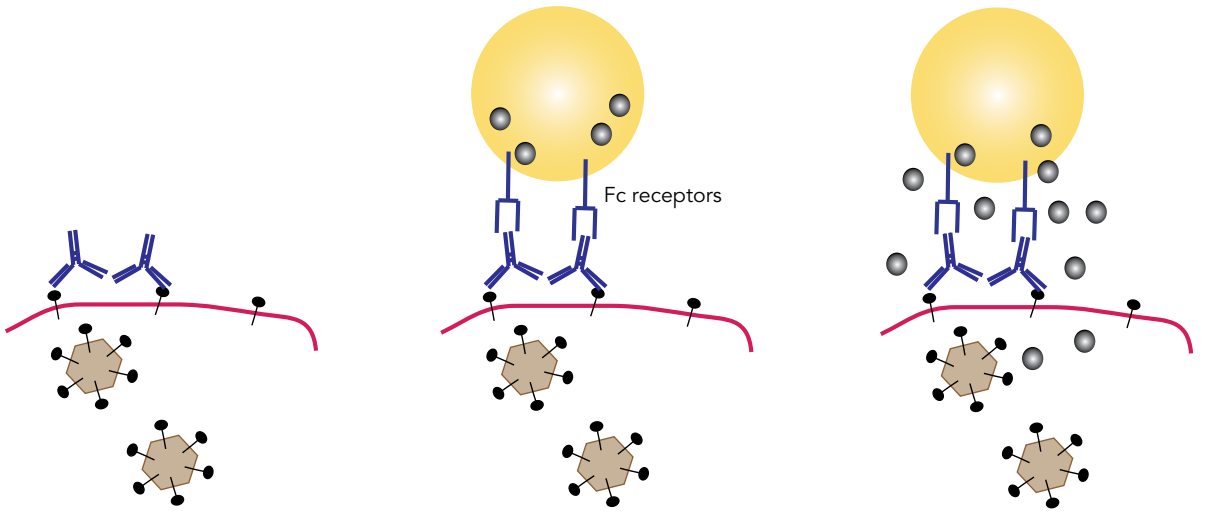


Figure 9.15. Antibody-dependent cell-mediated cytotoxicity (ADCC).

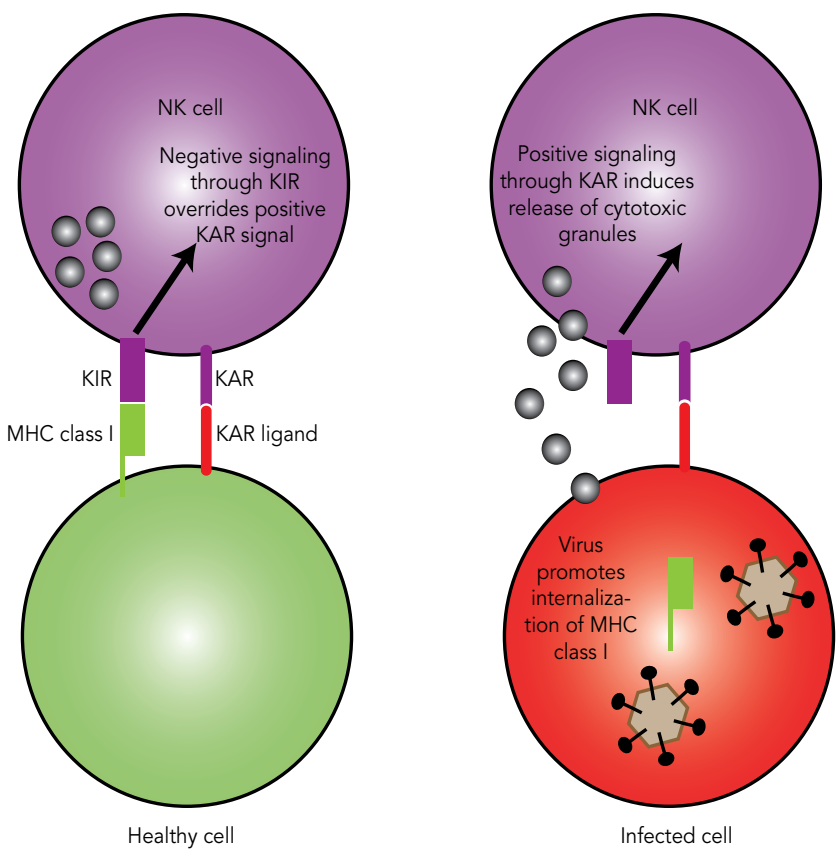


Figure 9.16. NK cell killing. NK cells have evolved to recognize MHC class I molecules. When a NK cell interacts with a healthy cell, it receives signals through both killer activating receptors (KARs) and killer inhibitory receptors (KIRs). The balance of these signals suppresses the killing response and the healthy cell is released. If, however, the cell is infected with a virus or becomes cancerous, it loses MHC class I from its surface. The NK cell no longer receives an inhibitory signal through its KIRs so the KAR signal promotes the release of cytotoxic granules from the NK cell which induces apoptosis in the infected cell.

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to counter this escape mechanism by recognizing the absence of MHC class I as a hallmark of infection and inducing apoptosis in the manner of CTLs (see Figure 9.16). In response, a number of viruses have now evolved to produce proteins that mimic MHC class I on the surface of the infected cell and thus avoid NK cell–mediated killing.

To control their cytotoxic activity, NK cells possess two types of surface receptor: killer activating receptors (KARs) and killer inhibitory receptors (KIRs). On encountering a healthy cell, KIRs recognize and bind to MHC class I molecules on the cell surface. KIRs contain immunoreceptor tyrosine-based inhibitory motifs (ITIMs) in their cytoplasmic domains that recruit intracellular tyrosine phosphatases and inhibit NK cell activation. If MHC class I is absent, then positive signals through a KAR drive the NK cell to perform its cytotoxic duties and secrete inflammatory cytokines; principally IFNγ. NK cell killing is mediated by the same factors involved in CTL killing: perforin, which polymerizes to form pores in the infected cell; granzymes, which enter the target cell and activate the caspase cascade; and the Fas–FasL interaction. In the case of KARs, the ligands inducing the killing response are still unclear though one KAR, NKG2D, has been shown to recognize a stress-induced protein with similar structure to MHC class I.

Some NK cells express the TCR. These are known as NKT cells and can recognize antigen presented by APCs. However, the TCR they express is only semivariant and recognizes a limited number of antigens on the nonpolymorphic CD1 molecule. As a result, they are also more "innate" than other T cells.

SUGGESTED READINGS

Antigen presentation

Vyas, J.M., Van der Veen, A.G., and Ploegh, H.L. 2008. The known unknowns of antigen processing and presentation. *Nat Rev Immunol* **8:**607–618.

Zinkernagel, R.M., and Doherty, P.C. 1974. Restriction of *in vitro* T cell-mediated cytotoxicity in lymphocytic choriomenigitis within a syngenic or semi-allogeneic system. *Nature* **248:**701–702.

T-cell development

Laky, K., Fleischacker, C., and Fowlkes, B.J. 2006. TCR and Notch signalling in CD4 and CD8 T cell development. *Immunol Rev* **209:**274–283.

Moroy, T., and Karsunky, H. 2000. Regulation of pre-T cell development. *Cell Mol Life Sci* **57**(6)**:**957–975.

T-cell receptor signaling

Nel, A.E. 2002. T-cell activation through the antigen receptor. Part 1: Signaling components, signaling pathways, and signal integration at the T-cell antigen receptor synapse. *J Allergy Clin Immunol* **109**(5)**:**758–770.

Nel, A.E., and Slaughter, N. 2002. T-cell activation through the antigen receptor. Part 2: Role of signaling cascades in T-cell differentiation, anergy, immune senescence, and development of immunotherapy. *J Allergy Clin Immunol* **109**(6)**:**901–915.

Costimulatory signaling and late costimulators

Alegre, M-L., Frauwirth, K.A., and Thompson, C.B. 2001. T-cell regulation by CD28 and CTLA-4. *Nat Rev Immunol* **1**(3)**:**220–228.

Croft, M. 2003. Co-stimulatory members of the TNFR family: keys to effective T-cell immunity? *Nat Rev Immunol* **3**(8)**:**609–620.

Effector function of T helper cell subsets

Reiner, S.L. 2007. Development in motion: Helper T cells at work. *Cell* **129**(1)**:**33–36.

Intraepithelial T cells

Hayday, A., Theodoridis, E., Ramsburg, E., and Shires, J. 2001. Intraepithelial lymphocytes: exploring the Third Way in immunology. *Nat Immunol* **2**(11)**:**997–1003.

Effector function of cytotoxic (killer) T lymphocytes

Anderson, M.H., Schrama, D., thor Straten, P., and Becker, J.C. 2006. Cytotoxic T cells. *J Invest Dermatol* **126**(1)**:**32–41.

B-cell development

Milne, C.D., Fleming, H.E., Zhang, Y., and Paige, C.J. 2004. Mechanisms of selection mediated by interleukin-7, the preBCR, and hemokinin-1 during B-cell development. *Immunol Rev* **197:**75–88.

B-cell receptor signaling

Hasler, P., and Zouali, M. 2001. B cell receptor signaling and autoimmunity. *FASEB J* **15**(12)**:**2085–2098.

Pierce, S.K. 2002. Lipid rafts and B-cell activation. *Nat Rev Immunol* **2**(2)**:**96–105.

The requirement for co-receptors

Rickert, R.C. 2005. Regulation of B lymphocyte activation by complement C3 and the B cell coreceptor complex. *Curr Opin Immunol* **17**(3)**:**237–243.

B-cell effector function

Casadevall, A., and Pirofski, L.A. 2006. A reappraisal of humoral immunity based on mechanisms of antibody-mediated protection against intracellular pathogens. *Adv Immunol* **91:**1–44.

Martin, F., and Kearney, J.F. 2000. B-cell subsets and the mature preimmune repertoire. Marginal zone and B1 B cells as part of a "natural immune memory." *Immunol Rev* **175:**70–79.

NK cell function

Cerwenka, A., and Lanier, L.L. 2001. Natural killer cells, viruses and cancer. *Nat Rev Immunol* **1**(1)**:**41–49.

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Nimmerjahn, F., and Ravetch, J.V. 2008. Fcγ receptors as regulators of immune responses. *Nat Rev Immunol* **8:**34–47.

Sigal, L.H. 2007. Basic science for the clinician 47: Fcgamma receptors. *J Clin Rheumatol* **13**(6)**:**355–358.

References for protein structures

MHC class I – Protein databank accession number 2vaa. Primary citation

Fremont, D.H., Matsumura, M., Stura, E.A., Peterson, P.A., and Wilson, I.A. 1992. Crystal structures of two viral peptides in complex with murine MHC class I H-2Kb. *Science* **257:**919–927.

MHC class II – Protein databank accession number 1iao. Primary citation

Scott, C.A., Peterson, P.A., Teyton, L., and Wilson, I.A. 1998. Crystal structures of two I-Ad-peptide complexes reveal that high affinity can be achieved without large anchor residues. *Immunity* **8:**319–329.

B cell receptor – Protein databank accession number 2RCJ. Primary citation

Perkins, S.J., Nealis, A.S., Sutton, B.J., and Feinstein, A. 1991. Solution structure of human and mouse immunoglobulin IgM by synchrotron X-ray scattering and molecular graphics modelling: a possible mechanism for complement activation. *J Mol Biol* **221:**1345–1366.

T cell receptor:peptide:MHC complex – Protein databank accession number 1ao7. Primary citation

Garboczi, D.N., Ghosh, P., Utz, U., Fan, Q.R., Biddison, W.E., and Wiley, D.C. 1996. Structure of the complex between human T-cell receptor, viral peptide and HLA-A2. *Nature* **384:**134–141.

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10 Fibroblasts and Stromal Cells

Andrew Filer and Christopher D. Buckley

WHAT IS A STROMAL CELL?

The architecture of organs and tissues is closely adapted to their function to provide microenvironments in which specialized functions may be carried out efficiently. The nature and character of such microenvironments are primarily defined by the stromal cells that reside within the tissues. The most abundant cell types of the stroma are fibroblasts, which are responsible for the synthesis and remodeling of extracellular matrix (ECM) components. In addition, their ability to produce and respond to growth factors and cytokines allows reciprocal interactions with adjacent epithelial and endothelial structures and with infiltrating leukocytes. As a consequence, fibroblasts play a critical role during tissue development and homeostasis and have often been described as having a "landscaping" function. In this chapter, we concentrate on fibroblasts as the prototype stromal cell. However, the stroma also consists of blood and lymphatic vessels, and a wider definition of stromal cells might include endothelial cells, specialized cells such as pericytes (blood vessel supporting cells), and even tissue resident macrophages.

Stromal Cell Identity and Microenvironments

Tissue resident macrophages in the liver (Kupffer cells) and lung (alveolar macrophages) perform very different functions compared to macrophages in the brain (glial cells) or skin (Langerhans cells), yet they are all members of the monocyte/macrophage family. Until recently, fibroblasts had been thought of as ubiquitous, generic cells with a common phenotype even within different tissues. However, we now know that fibroblasts from different organs are more like their macrophage counterparts, with unique morphology and repertoires of ECM proteins, cytokines, costimulatory molecules, and chemokines specialized to the different microenvironments in which they are found (Figure 10.1). Furthermore, when their transcriptional profiles are examined using microarray techniques, fibroblasts hold a strong memory of their position and function in the body. Early studies demonstrated that fibroblast transcriptomes (the global picture of transcribed genes measured using microarrays) could be clustered into peripheral (synovial joint or skin fibroblasts) versus lymphoid (tonsil or lymph node) groups according to their organ of origin, with the potential to shift their transcriptional profiles by treatment with inflammatory mediators such as tumor necrosis factor alpha (TNF-α), interleukin-4 (IL-4), or interferon-γ. More extensive analysis of expression profiles from primary human fibroblasts by Rinn et al. has shown large-scale differences related to three broad anatomical divisions: anterior–posterior, proximal–distal, and dermal–nondermal. Genes involved in pattern forming, cell-signaling, and matrix remodeling were found to predominantly account for these divisions. The gene expression profile of adult fibroblasts may therefore play a significant role in assigning positional identity within an organism. This in turn provides evidence for the concept of a stromal address code, analogous to the endothelial area post code. According to this hypothesis, chemokines, cytokines, and adhesion molecules produced by stromal cells control the accumulation, retention, survival, and differentiation of leukocytes in a site or organ-dependent manner, providing a plausible explanation for the well described but as yet poorly understood clinical finding that relapses in chronic inflammatory diseases are often tissue and site specific. For instance, despite some similarities at the level of leukocyte infiltrates rheumatoid arthritis (RA) affects predominantly the joints, whereas multiple sclerosis affects the central nervous system. Support for this hypothesis also comes from the cancer field, where defective or altered stromal cell

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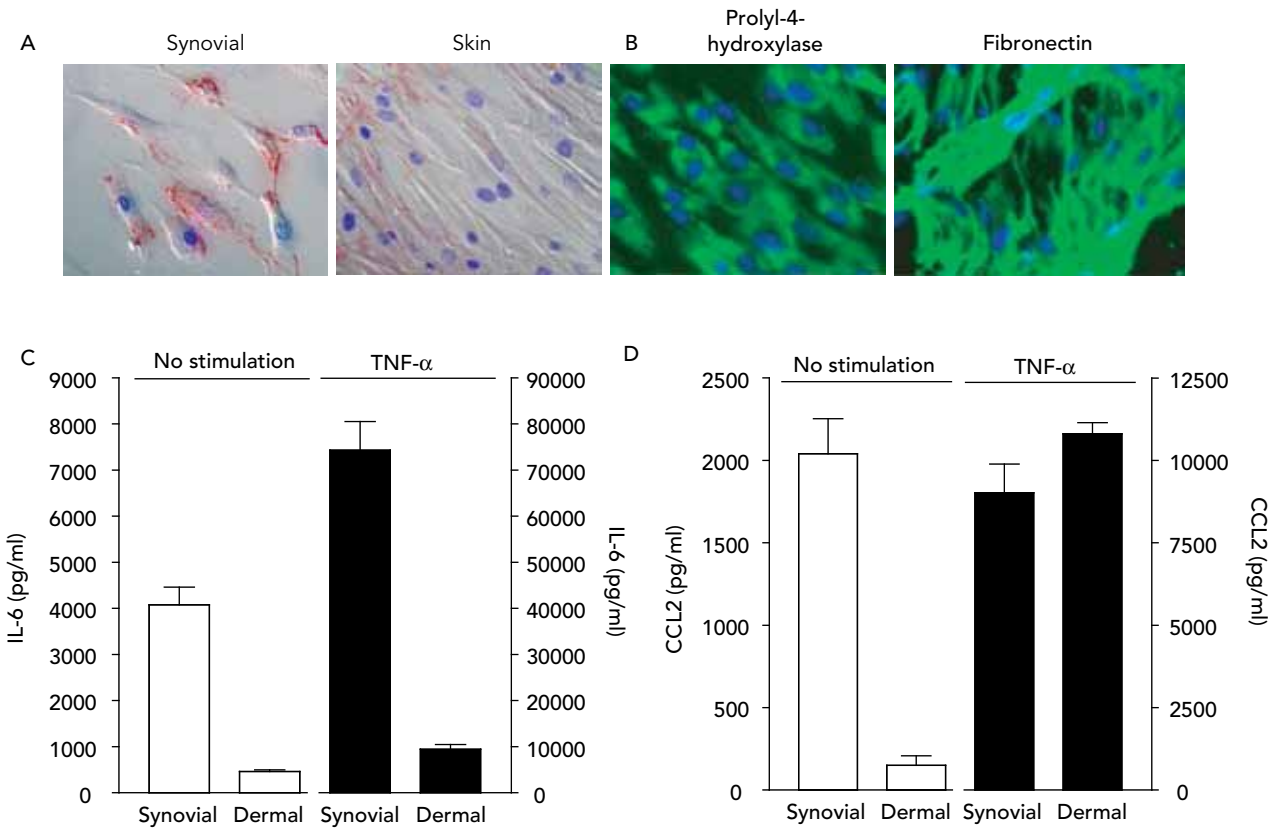


Figure 10.1. Fibroblast phenotype. (A) staining of live fibroblast cells in culture illustrating typical morphology and marked differences between synovial fibroblasts of the rheumatoid arthritis joint and skin fibroblasts. Red stain fibronectin demonstrates matrix production. (Blue stain = nuclear stain.) (B) Stromal cell status is confirmed by fluorescence microscopy of cells showing collagen synthetic enzymes (prolyl-4-hydroxylase) and matrix production (fibronectin) in skin fibroblasts. (C) Production of IL-6 measured by ELISA in unstimulated and TNF-α-stimulated fibroblasts is constitutively higher in synovial than skin cells. (D) A similar high basal CCL2 (MCP-1) chemokine production is seen in synovial fibroblasts; this is obliterated by TNF-α stimulation.

function has been strongly associated with tumorigenesis and progression of tumors to metastasis.

Stromal Cells and Inflammation

Inflammatory reactions proceed against the backdrop of specialized stromal microenvironments. The response to tissue damage involves a carefully choreographed series of interactions between diverse cellular, humoral, and connective tissue elements. For an inflammatory lesion to resolve, dead or redundant cells that were recruited and expanded during the active phases of the response must be removed. In addition, resident stromal cells, largely defined by fibroblasts, attempt to repair damaged tissue. It is becoming increasingly clear that fibroblasts are not just passive players in immune responses, but that they play active roles in determining the switches that occur governing progression from acute to chronic inflammation, and also those governing resolution or the progression to chronic, persistent inflammation. The "switch to resolution" is an important signal that permits tissue repair to take place and enables immune cells to return to draining lymphoid tissues (lymph nodes) for immunological memory to become established. However, in immune-mediated inflammatory diseases, stromal cells contribute to the inappropriate recruitment and retention of leukocytes, leading to chronic persistent inflammation. Research on the roles of stromal cells in inflammation and cancer medicine and specifically the parts played by fibroblasts in physiological and pathological inflammation will be examined later.

Stromal Cells: In Vitro and In Vivo

By virtue of their role in defining the geography of specialized tissues, fibroblasts and other stromal cells exist in living organisms within 3-dimensional environments, whereas the majority of experiments performed using fibroblasts in the laboratory are conducted within 2-dimensional environments. Furthermore, fibroblasts are frequently grown in nonphysiological stimuli such as serum, to which

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fibroblasts would not normally be exposed unless tissue damage were to occur. It has been shown that the behavior of fibroblasts is different when cultured in artificial 3-dimensional environments. It is therefore all the more remarkable that fibroblasts cultured using conventional 2-dimensional techniques retain characteristics such as positional memory and unique cytokine profiles. This suggests a degree of epigenetic imprinting may account for the persistent changes in gene expression seen in these cells.

EMBRYOLOGICAL ORIGINS OF FIBROBLASTS

As mentioned earlier, even within a single tissue there is growing evidence that fibroblasts are not a homogeneous population, but exist as subsets of cells, much like tissue macrophages and dendritic cells (DCs). It is likely that connective tissue contains a mixture of distinct fibroblast lineages with mature fibroblasts existing side by side with more immature fibroblasts that are capable of differentiating into other connective tissue cells. Stromal cells have been defined in terms of their embryological origins and lineage relationships, and are generally considered to be mesenchymal in origin. However, the embryological mesenchyme, from which fibroblasts are derived, is not in itself a germ layer (usually defined as ecto-, endo-, and mesoderm), but is variably considered to be either wholly composed of embryonic mesoderm, or a combination of the mesoderm and ecto- or endoderm layers. For instance, in the head and neck some mesenchyme is derived from neural crest cells (and hence from ectoderm). Moreover, cell populations have now been identified which appear to blur the distinction between hemopoietic and nonhemopoietic populations. In addition, other unexpected shifts in lineage have been reported, including differentiation from neural stem cells into myeloid and lymphoid hemopoietic lineage. Classification by such lineages that are no longer as restrictive as previously thought is therefore becoming increasingly awkward.

The problem of distinguishing fibroblasts of differing origin or maturity has historically been very difficult due to a lack of cell surface specific markers. Whereas the cluster of differentiation (CD) markers have revolutionized the isolation and study of leukocyte subsets, there have been relatively few, poor quality discriminatory markers allowing the identification of fibroblast subpopulations. Fibroblasts have frequently been identified by their spindle-shaped morphology (Figure 10.1), elaboration of ECM, and lack of positive markers for endothelium, epithelial, and hemopoietic cells. However, recent studies have begun to identify novel markers which demarcate distinct subpopulations of stromal cells during development and which have the potential to act as markers for different subpopulations of fibroblasts each with different roles. Such markers include smooth muscle actin, which marks out a population of secretory, activated cells termed myofibroblasts, and more recently discovered markers such as CD248 and GP38 (podoplanin) (Table 10.1). The question of the origin of fibroblasts is an important one. Both inflammation and wound healing are characterized by the formation of new

TABLE 10.1. Comparison of phenotypic expression of blood-borne mesenchymal precursors and RA synovial fibroblasts

Marker	MSC/MPC	RA synovial fibroblasts	Tissue myofibroblasts	Fibrocytes
Collagen I	+	+	+	+
Collagen III	+	+	+	+
Fibronectin	+	+	+	+
Vimentin	+	+	+	+
HLA-DR	+	+*	–	+
CD44	+	+	+	+
CD45	–	–	–	+
CD34	–	–	–	+
VCAM-1	+	+	–	–
Smooth muscle actin	+	+/–	+	–**
Bone morphogenic protein receptors	+	+	–	Not known
DAF (CD55)	+	+	–	Not known
BST-1	+	+	–	Not known

*, Rapidly lost in culture; **, fibrocytes differentiate into SMA; –, positive fibroblasts.

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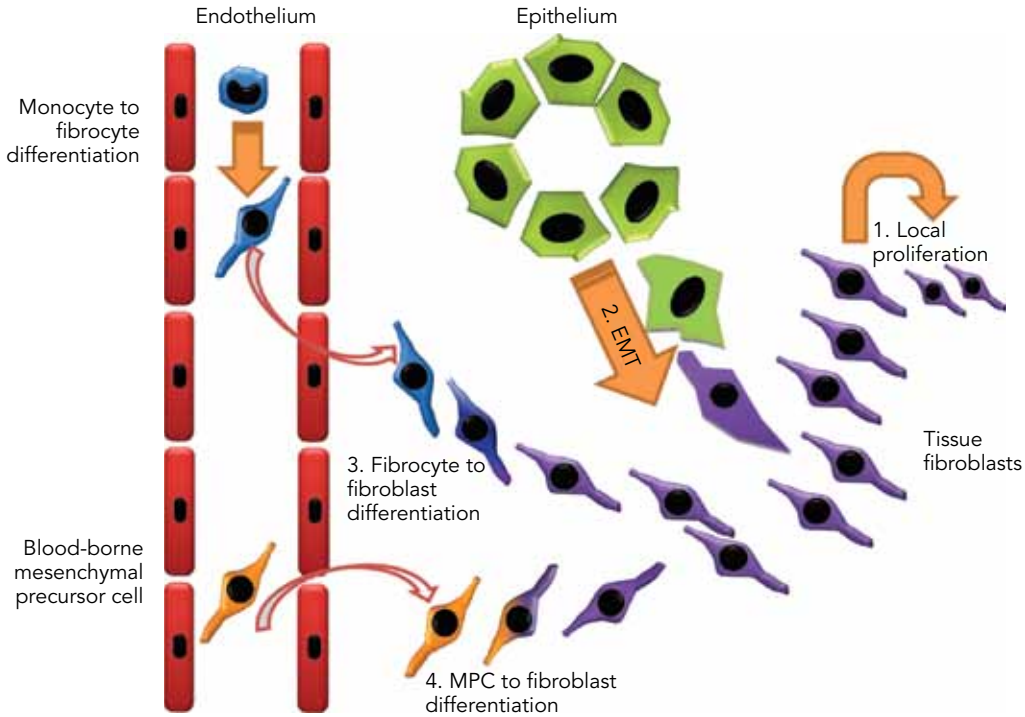


Figure 10.2. Routes of differentiation to tissue fibroblasts. In response to wounding or inflammation, increased numbers of fibroblasts are produced within tissue. (1) Fibroblasts can proliferate locally to generate new fibroblasts. (2) The transition of epithelial to stromal cell populations has been shown to occur in diseases of the lung and kidney. (3) Fibrocytes arise from the monocyte population in blood, then differentiate toward fibroblasts in tissue. (4) Blood borne mesenchymal precursor cells may be recruited to tissues and undergo local differentiation to tissue fibroblasts.

tissue. However, recent findings have suggested that the new cells that form the remodeled tissues are not, as was hitherto assumed, necessarily derived from the proliferation of cells that are resident in the adjacent noninjured tissue.

Sources of Tissue Fibroblasts

It is widely accepted that the principle origin for fibroblasts is from primary mesenchymal cells and that upon appropriate stimulation fibroblasts can proliferate locally to generate new fibroblasts. Though an increase in fibroblast numbers caused by local proliferation does occur, fibroblasts may arise from other sources (Figure 10.2). The first of these is local epithelial to mesenchymal transition (EMT). This is an essential, physiologically important developmental mechanism for diversifying cells in the formation of complex tissues. However, fibroblasts also appear to be derived by this process in adult tissue following epithelial stress such as inflammation or tissue injury. EMT disaggregates epithelial cells and reshapes epithelial cells for movement. The epithelium loses polarity as defined by the loss of adherens junctions, tight junctions, desmosomes, and cytokeratin intermediate filaments. Epithelial cells also rearrange their F-actin stress fibers and express filopodia and lamellopodia. A combination of cytokines and matrix metalloproteinases (MMPs) associated with digestion of the basement membrane is believed to be secreted and important in the process. The transition of epithelial to mesenchymal cell populations has been shown to occur in diseases of the lung and kidney where the process has been implicated in fibrotic disease (see Kalluri et al.).

An alternative explanation for the accumulation of stromal cells in chronic inflammatory conditions such as RA lies in the possibility of blood borne precursors. In the mid 1990s it was shown that vascular precursors (angioblasts) could be found circulating in the blood of normal individuals, and that they could be recruited to sites of vasculogenesis in a rabbit ischemic hind limb model. This demonstrated that circulating mesenchymal precursors exist outside the hemopoietic system. Subsequent work has confirmed the presence of circulating cells of a mesenchymal phenotype in human subjects. These cells bear a remarkable resemblance to the synovial fibroblasts found in the joints of patients with RA, which accumulate in enormous quantities in the joint lining despite little evidence of proliferation. Interestingly, Marinova-Mutafchieva et al. showed

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that an influx of such cells preceded inflammation in a mouse collagen-induced arthritis model, suggesting that there may be a role for blood-borne stromal cell precursors in the initiation of inflammatory diseases.

Another circulating precursor cell that could account for the accumulation of fibroblasts in some diseases is a cell called the fibrocyte. Fibrocytes appear to comprise 0.1%–0.5% of nonerythrocytic cells in peripheral blood, and have been shown to rapidly enter sites of tissue injury and contribute to tissue remodeling in models of inflammatory lung disease. They are adherent cells with a spindle-shaped morphology which express MHC class II as well as type I collagen and which arise from within the CD14-positive (monocyte) fraction of peripheral blood. Fibrocytes are capable of matrix elaboration, and have been proposed to differentiate along a fibroblast lineage under the influence of cytokines, particularly TGF-β. The mere fact that a cell type apparently arising from within the monocyte lineage may become a "mesenchymal" stromal cell such as a fibroblast implies a further degree of plasticity and blurring of the apparently clear line between hemopoietic and nonhemopoietic lineages.

Fibroblasts versus Mesenchymal Precursor Cells

The potential role of circulating mesenchymal cell precursors (variously termed mesenchymal stem cells [MSC], mesenchymal stromal cells or mesenchymal precursor cells [MPC]) as sources of tissue fibroblasts is highlighted by the remarkable capacity of these cells to differentiate into other members of the connective tissue family including cartilage, bone, adipocyte, and smooth muscle cells. This ability was initially demonstrated in bone marrow stromal cells, RAL synovial fibroblasts, and circulating mesenchymal cells, and was therefore suggested to define a characteristic mesenchymal phenotype, based on the hypothesis that the rheumatoid synovium could become populated by a large proportion of circulating mesenchymal precursor cells exported from the bone marrow. However, the property of trilineage differentiation ("pluripotentiality") has now been shown to be a property of many adult tissue fibroblasts, though varying somewhat between fibroblasts from different tissues, implying a hitherto unsuspected degree of plasticity in the body's stromal cell populations. The two previously separate fields of mesenchymal precursor cell biology and largely disease-centered fibroblast biology have therefore rapidly converged.

STROMAL CELLS IN IMMUNE RESPONSES

Physiological inflammation is not a stable state. Such inflammation either resolves, and the tissue reverts to normal, or develops into chronic persistent inflammation. Once established, chronic inflammation is hard to "cure" as evidenced by the fact that despite increasingly focused approaches to suppressing active inflammation within tissues, a cure for chronic immune-mediated inflammatory diseases such as RA remains elusive, since on the withdrawal of effective anti-inflammatory therapies the disease inevitably relapses.

As an inflammatory process reaches its conclusion, the resolution of inflammatory leukocyte infiltrates within a microenvironment is governed by a number of dynamic factors: first, the balance between cell recruitment and emigration; second, the balance between cell death and proliferation; and third, the coordinated release of proresolution factors such as resolvins and adenosine (which will not be discussed further in this chapter). Recent evidence suggests that tissue stromal cells are able to determine the type and duration of leukocyte infiltrates in an inflammatory response. At the resolution of such responses, stromal cells contribute to the withdrawal of survival signals and normalization of chemokine gradients, allowing infiltrating cells to undergo apoptosis or leave via draining lymphatics (Figure 10.3). Subversion of these pathways results in a switch to persistent inflammation that remains remarkably stable over time. We shall use the stromal microenvironment of the rheumatoid joint as a model to illustrate the role played in persistence of inflammation by stromal cells.

Stromal Cells as Innate Immune Sentinels

Classically, macrophages have been studied as sources of inflammatory cytokines and chemokines in response to innate immune stimuli, and portrayed as immune sentinel cells accordingly. However, when activated by substances released during tissue injury or the products of invading microorganisms, fibroblasts are capable of elaborating a broad repertoire of inflammatory mediators which fully justifies their classification as immune sentinel cells. Through expression of toll-like receptors, fibroblasts respond to bacterial products such as LPS by generating chemokines capable of recruiting inflammatory cells. Furthermore, fibroblasts are capable of bridging the innate and adaptive immune responses through expression of the molecule CD40. This molecule was initially assumed to be restricted in its expression to antigen-presenting cells such as macrophages and DCs. However, it is widely expressed by fibroblasts within discrete tissues. Engagement of CD40 by its ligand CD40L expressed on a restricted population of immune cells including activated T lymphocytes (Figure 10.4) is critical for the further induction of proinflammatory cytokines and chemokines during an immune response, as well

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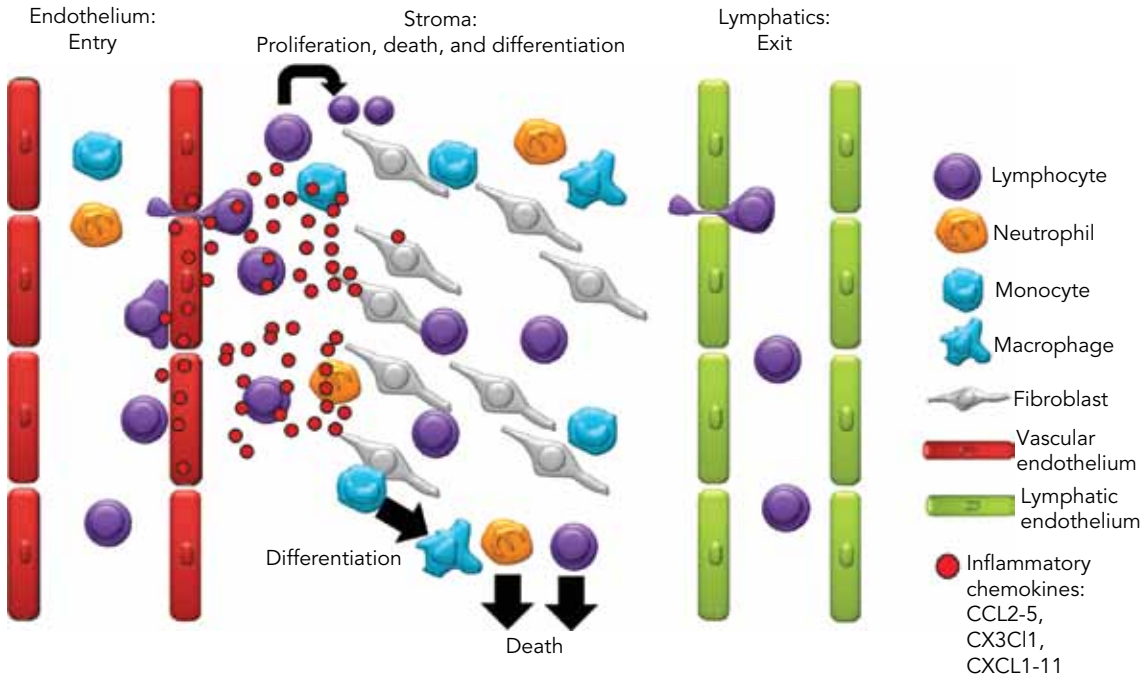


Figure 10.3. Regulation of inflammatory responses by the stroma. Evolution of an inflammatory response is characterized by recruitment of immune system cells via production of inflammatory chemokines. Recruited cells may undergo proliferation and/or differentiation. As an inflammatory process reaches its conclusion, the resolution of inflammatory leukocyte infiltrates within a microenvironment is governed by a number of dynamic factors including the balance between cell recruitment and emigration and the balance between cell death and proliferation. Recent evidence suggests that tissue stromal cells are able to determine the type and duration of leukocyte infiltrates in an inflammatory response. At the resolution of such responses, stromal cells contribute to the withdrawal of survival signals and normalization of chemokine gradients, allowing infiltrating cells to undergo apoptosis or leave via draining lymphatics.

as for antibody production by CD40 expressing B lymphocytes.

In some pathological states, activation of fibroblasts becomes persistent, leading to the hypersecretion of cytokines, chemokines, and matrix, which contribute to disease. The rheumatoid synovial fibroblast is a prime example of such persistent activation of fibroblasts. In RA, the fibroblasts take on a characteristic activated phenotype that remains stable even after culturing in vitro for many months. These cells play a direct role in tissue damage through secretion of multiple MMPs and cathepsins that degrade cartilage and bone tissues in the joint (Figure 10.4). Attachment to and direct, autonomous invasion of fibronectin-rich matrix such as cartilage has been demonstrated in the SCID mouse model of arthritis, in which cultured rheumatoid, but not osteoarthritis synovial fibroblasts invade and destroy co-implanted human cartilage. In vitro models such as the matrigel invasion assay produce corresponding results, in which the degree of invasion correlates with the degree of damage seen in the joints of patients from whose samples the fibroblast were initially grown. These models demonstrate the remarkably stable and disease-specific phenotype of cultured RA synovial fibroblasts which includes high basal and stimulated expression of signature cytokines such as IL-6, and chemokines (discussed later). RA synovial fibroblasts also express characteristic adhesion and immune modulating molecules such as VCAM-1, galectin-3, and a specific repertoire of Toll-like receptors, which initiate innate immune cellular responses. A satisfactory molecular explanation for the stable phenotype of RA synovial fibroblasts has until recently evaded the field. However, epigenetic changes including DNA methylation, histone modifications such as acetylation, and microRNA expression have now been suggested to underlie the observed persistent changes in fibroblast gene transcription and posttranscriptional repression.

Signal Transduction Processes

Both innate and adaptive immune responses triggered in synovial fibroblasts initiate characteristic signal transduction events. These result from stimulation such as Toll-like receptor engagement, or exposure to a proinflammatory cytokine network within a persistent chronic inflammatory infiltrate, as is seen in the case of RA. Many of the genes activated by such stimuli belong to the classical proinflammatory signaling

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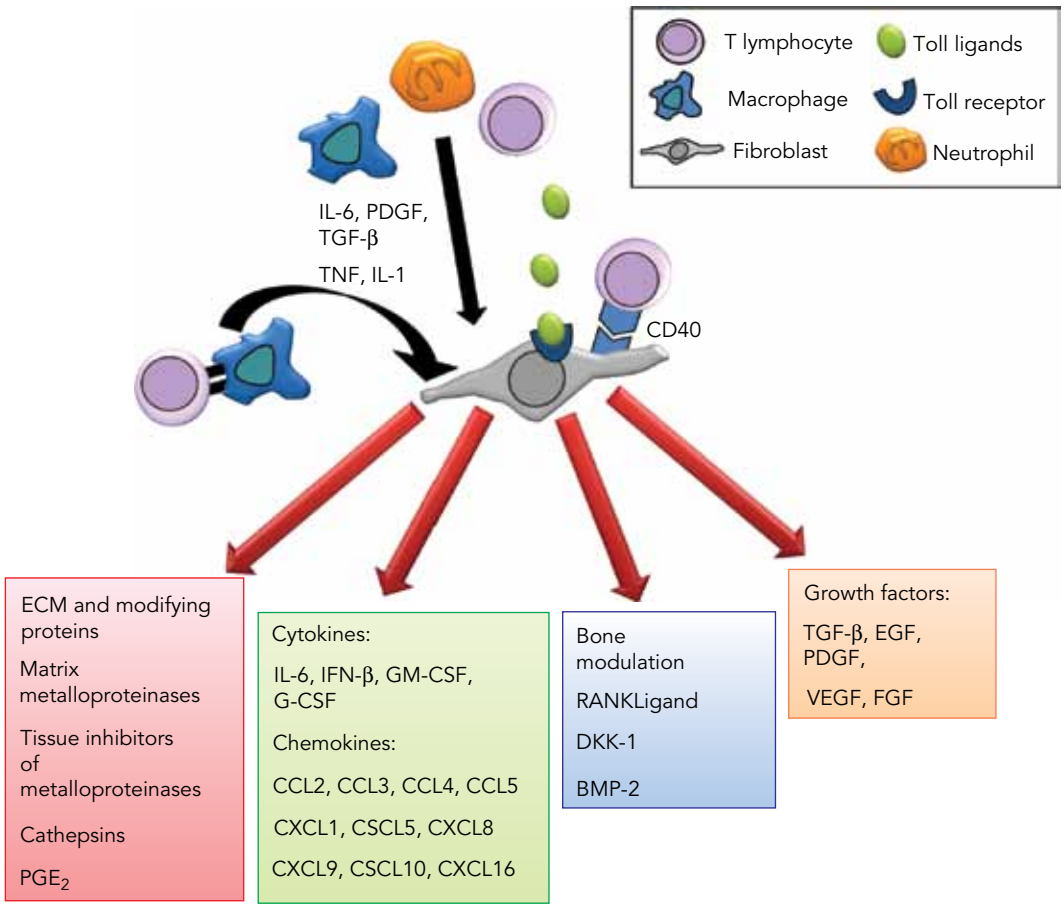


Figure 10.4. Fibroblast roles in inflammation. Fibroblasts may be subjected to various inflammatory stimuli. Bacterial cell products activate toll receptors, proinflammatory cytokines, such as TNF, IL-1, and IL-6 are most abundantly secreted by infiltrating macrophages and neutrophils. Infiltrating T lymphocytes may activate macrophages by direct contact to produce further inflammatory cytokines, and also activate fibroblasts by CD40Ligand–CD40 interactions. In response, fibroblasts break down and elaborate new extracellular matrix (ECM) by means of metalloproteinases and prostaglandin E2 (PGE_2). A wide range of cytokines including IL-6 and GM-CSF and a broad range of chemokines may be elaborated. In conditions such as rheumatoid arthritis, interaction with bone metabolism is important: RANKLigand promotes osteoclast differentiation and activation, while DKK-1 (Dickkopf) inhibits osteoblast activation. Alternatively, bone formation may be stimulated by bone morphogenic proteins (BMP) and vascular endothelial growth factor (VEGF), which also has a primary role in promoting vasculogenesis. Fibroblasts are a rich source of other growth factors which modulate stromal cell survival such as platelet-derived growth factor (PDGF) and epidermal and fibroblast growth factors (EGF, FGF).

pathways such as NF-κB and the mitogen-activated protein kinases (MAPK) with increasing recognition of the importance of the phosphatidylinositol 3-kinase (PI3K) signaling pathways, particularly of the delta and gamma subtypes.

All three subpathways within the MAPK system (JNK, ERK-p42/p44, and p38) are heavily implicated in inflammatory signaling cascades of fibroblasts in the rheumatoid synovium. Furthermore, MAPK activation frequently overlaps with proinflammatory NF-κB pathways. Activation of these pathways leads to further induction of proinflammatory cytokines and chemokines, as well as regulating the survival of stromal cells by inhibiting apoptosis. Work using knockout mice has demonstrated that JNK MAPK is crucial to the elaboration of MMPs such as the collagenases, while P38 MAPK has also been implicated in regulation of other MMPs by synovial fibroblasts. The importance of these signaling pathways has led to the suggestion that MAPK should be targets for rational drug design. PI3K is frequently implicated in production of mononuclear cell attracting chemokines such as CCL2 and CCL5 in stromal cells. For instance, PI3K and JNK are involved in IL-18 regulation of CCL2 production by synovial fibroblasts. PI3K signaling frequently lies upstream of NF-κB activation, which plays an important role in regulating the cellular response to proinflammatory

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stimuli. Furthermore, signaling due to IL-6 binding results in activation of the JAK-STAT pathways (Janus kinases [JAKs] and signal transducers and activators of transcription [STATs], particularly STAT-3). These pathways govern cell proliferation, differentiation, and apoptosis.

Fibroblasts also express chemokine receptors and respond to chemokine ligands. All functioning chemokine receptors appear to signal through pertussis toxin–sensitive heterotrimeric G proteins. Shape changes, firm adhesion, and chemotaxis, which involve restructuring of the cytoskeleton, utilize the Rho GTPase and phosphatidylinositol-3-OH kinase (PI3K) pathways. Other pathways downstream from PI3K include the modulation of gene transcription via the MAPK. Activation of pertussis toxin insensitive pathways such as the JAK-STAT pathway has been proposed to result from either hetero- or homo-oligomerization of chemokine receptors, leading to a range of functional outcomes such as cellular adhesion. A further emerging signaling pathway of note is PLC-γ (phospholipase-C-γ). This phospholipid hydrolase phospholipase plays a major role in regulation of cell proliferation, development, and cell motility. It has been implicated in the control of stromal cell apoptosis and production of MMPs, which are important both in progression of cancer mediated by stromal cells and in damage to structures within the rheumatoid joint. PLC-γ downstream effects are usually mediated through the ERK MAPKinases.

STROMAL CELL–LEUKOCYTE INTERACTIONS AND PERSISTENT INFLAMMATION

The maintenance of a persistent leukocyte infiltrate at sites of chronic inflammation reflects a distorted balance between those factors that enhance cellularity (leukocyte recruitment, proliferation, and retention) and those that decrease cellularity (cell death and emigration) (Figure 10.3). While the mechanisms responsible for the recruitment of leucocytes into and their proliferation within tissues have been well studied, those responsible for their survival, retention, and emigration within tissues have attracted much less attention.

Recruitment of Inflammatory Infiltrates into the Joint

Stromal elements such as synovial fibroblasts are subject to a proinflammatory cytokine network within the inflamed synovium. Direct contact interactions with other infiltrating cells such as T lymphocytes leads to high levels of expression of many inflammatory chemokines (Figure 10.3). Neutrophil-attracting chemokines are expressed at high levels by stimulated fibroblasts and include CXCL8 (IL-8), CXCL5 (ENA-78, epithelial-cell–derived neutrophil attractant 78), and CXCL1 (GROalpha, growth-related oncogene alpha). Monocytes and T cells are recruited by a range of chemokines found at high levels in the synovium; CXCL10 (IP-10) and CXCL9 (Mig) are highly expressed in synovial tissue and fluid. CXCL16 is also highly expressed in the RA synovium and acts as a potent chemoattractant for T cells. CCL2 (MCP-1) is found in synovial fluid and known to be produced by synovial fibroblasts; it is considered to be a pivotal chemokine for the recruitment of monocytes. CCL3 (Mip-1alpha), CCL4 (Mip-1beta), and CCL5 (RANTES) are chemotactic for monocytes and lymphocytes, and are known products of synovial fibroblasts. CCL20 (Mip-3alpha) is also overexpressed in the synovium, and has a similar chemoattractant profile via its specific receptor, CCR6. CX3CL1 (Fractalkine) is also widely expressed in the rheumatoid synovium. A number of chemokine receptors have been shown to differ between peripheral blood and synovial leucocytes, suggesting that they are enriched in the synovium either though their selective recruitment by endothelial expressed chemokines, or following upregulation by the microenvironment after their recruitment.

Stromal Support for Leukocyte Survival

Stromal cell support for the survival of leukocyte populations fulfills a physiological role in certain organs within the body. The selective recruitment and support of hemopoietic subsets is an essential physiological function of stromal cells in specific microenvironments. For instance, immature B lymphocytes are completely dependent on factors such as IL-6 produced by bone marrow stromal cells. Though the bone marrow niche plays a critical role in the early development of all hemopoietic leukocyte populations, it also acts as an active reservoir for terminally differentiated leukocyte subpopulations, including CD4 and CD8 T cells and neutrophils. The bone marrow stromal microenvironment therefore maintains not only the selective survival, differentiation, and proliferation of all lineages of immature hemopoietic cells, but, in some cases, also the survival of their mature counterparts. The stromal microenvironment plays a crucial role in the maintenance of such survival niches, which are not generic, but highly specific to certain organs and tissues, resulting in site-specific differences in the ability of different stromal cells to support the differential accumulation of leukocyte subsets.

In the case of an inflammatory response, successful resolution requires the removal of the vast majority of immune cells that were recruited and expanded during the active phase of the inflammation. A

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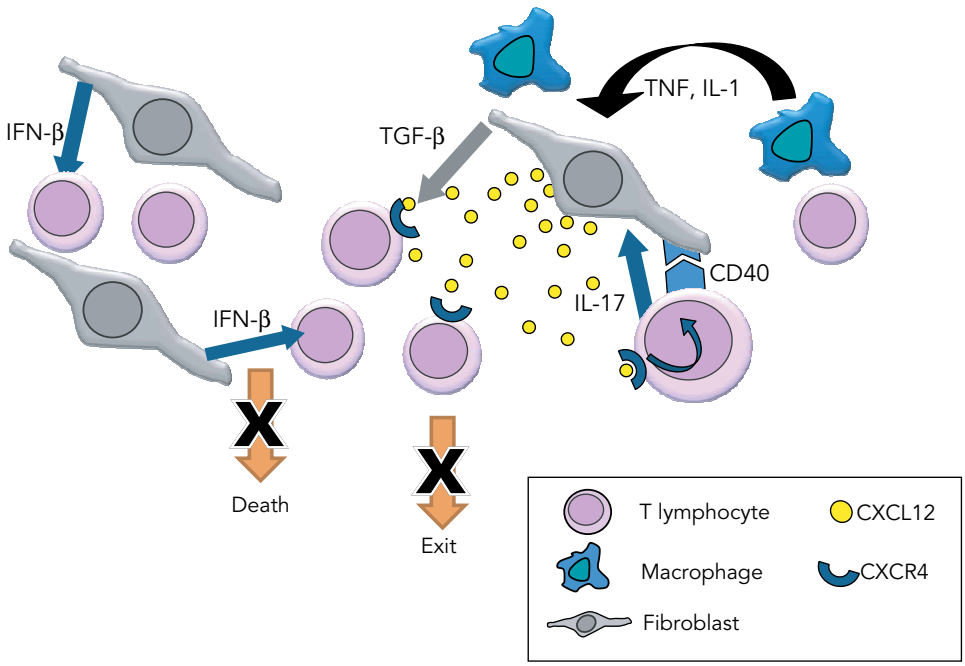


Figure 10.5. Inappropriate survival and retention of T lymphocytes mediated by synovial fibroblasts. Interferon-β, produced by synovial fibroblasts, has been identified as one of the principal factors responsible for prolonged T-cell survival in the rheumatoid joint. Similar mechanisms exist for stromal-induced survival of B cells. It is likely that these mechanisms of stromal cell–induced leukocyte survival occur in many chronic inflammatory conditions in which lymphocytes accumulate. In addition, aberrant ectopic expression of constitutive chemokines such as CXCL12 by synovial stromal cells contributes to the retention of T cells within the RA synovium. CXCL12 (SDF-1) and its receptor CXCR4 plays an important role, both in the constitutive traffic of lymphocytes and in the recruitment and retention of hemopoietic cells within the bone marrow. In the rheumatoid joint, fibroblast-derived TGF-β induces high levels of CXCR4 receptors in T cells, leading to retention of cells. CXCL12 is highly expressed on endothelial cells and fibroblasts at sites of T-cell accumulation. Positive feedback loops enhance CXCL12 production by fibroblasts via CD40–CD40 ligand interactions.

number of studies have shown that during the resolution phase of viral infections, the initial increase in T cell numbers in peripheral blood that is seen within the first few days is followed by a wave of apoptosis occurring in the activated T cells. This situation is mirrored within tissues, where apoptosis induced by the molecule Fas occurs at the peak of the inflammatory response and may be responsible for limiting the extent of the immune response. In contrast, the resolution phase appears to be principally triggered by cytokine-deprivation–induced apoptosis, during which leukocytes compete for a shrinking pool of survival factors provided by the microenvironment, leading to programmed death of those cells that are surplus to requirements.

In RA, the resolution phase of inflammation becomes disordered. Recent studies have shown that a failure of synovial T cells to undergo apoptosis or programmed cell death contributes to the persistence of the inflammatory infiltrate. The T-cell survival pathway shares all the essential hallmarks of a stromal cell, cytokine-mediated mechanism (high Bcl-X_L, low Bcl-2, and lack of cell proliferation). Type I interferons (interferons alpha and beta), produced by synovial fibroblasts and macrophages, have been identified as one of the principal factors responsible for prolonged T-cell survival in the rheumatoid joint (Figure 10.5). Interestingly while type I interferon has been shown to be beneficial in multiple sclerosis (a disease in which tissue scarring and low levels of T-cell infiltrates are observed) these results suggest that type I interferon is not likely to be a successful therapy for RA patients, a prediction which has been borne out in clinical trials. It is likely that this mechanism of stromal cell–induced leukocyte survival occurs in many chronic inflammatory conditions in which T cells accumulate. Not surprisingly, other leukocyte subpopulations have been shown to derive support from stromal cells. While fibroblast support for T-cell and B-cell survival exhibits site-specific properties,

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neutrophil survival is dependent on prior cytokine activation of fibroblasts, and shows no differences between fibroblasts taken from different anatomical sites.

Stromal Cell Retention of Leukocytes in Persistent Inflammatory Disorders

While the inhibition of T-cell death by stromal cells at sites of chronic inflammation contributes to T-cell accumulation, it is unlikely to be the only mechanism, because lymphocytes should be able to leave the inflamed tissue during the resolution of inflammation, even if their death is inhibited. A number of studies have recently reported that the synovial microenvironment contributes directly to the inappropriate retention of T cells within the joint, by an active chemokine-dependent process. The presence of high levels of inflammatory chemokines, produced by stromal cells, is a characteristic of environments such as the rheumatoid synovium. However, recent data suggest that paradoxically constitutive chemokines, which are involved in the recruitment of lymphocytes to secondary lymphoid tissues, are ectopically expressed in immune-mediated inflammatory diseases. The constitutive chemokine CXCL12 (SDF-1) and its receptor CXCR4 emerged as unexpected but crucial players in the accumulation of T lymphocytes within the rheumatoid synovial microenvironment (Figure 10.5). This chemokine receptor pair plays an important role, both in the constitutive traffic of lymphocytes and in the recruitment and retention of hemopoietic cells within the bone marrow. Unexpectedly, CD45RO+ T lymphocytes in the rheumatoid synovium were found to express CXCR4 receptors at high levels in the rheumatoid synovium. Its ligand CXCL12 was highly expressed on endothelial cells at the sites of T-cell accumulation. In addition, stromal cell–derived TGF-β is responsible for upregulation of CXCR4 receptors on T cells in the synovium. Evidence also suggests that the stability of lymphocyte infiltrates is reinforced by a positive feedback loop, whereby tissue CXCL12 promotes CD40 ligand expression on T cells, which in turn stimulates further CXCL12 production by CD40 expressing synovial fibroblasts. Furthermore, levels of CXCL12 secreted by synovial fibroblasts have recently been shown to be controlled in part by T cell derived IL-17.

There is therefore clear evidence in support of the hypothesis that aberrant ectopic expression of constitutive chemokines such as CXCL12, CCL19, and CCL21 by synovial stromal cells contributes to the retention of T cells within the RA synovium.

Other cell constituents of the rheumatoid inflammatory infiltrate may be affected by the CXCL12/CXCR4 axis. Blades and colleagues have shown increased expression of CXCL12/CXCR4 by monocyte/macrophage cells in RA compared with osteoarthritis. In addition, using implanted human synovial tissue in SCID mice, they demonstrated that monocytes are recruited into transplanted synovial tissue by CXCL12. Contact-mediated B-cell survival induced by synovial fibroblasts has also been shown to depend on CXCL12 and CD106 (VCAM-1)-dependent mechanisms which are independent of TNF-α. Overexpression of CXCL12 has also been identified as a distinct feature of rheumatoid, as opposed to osteoarthritis synovia using cDNA arrays. Data validating these findings in vivo have come from a collagen-induced arthritis model of RA in DBA/1 (interferon-γ receptor deficient) mice, where administration of the specific CXCR4 antagonist AMD3100 significantly ameliorated disease severity. In another murine collagen-induced arthritis model, the small molecule CXCR4 antagonist 4F-benzoyl-TN14003 ameliorated clinical severity and suppressed DTH (delayed type hypersensitivity) responses. The CXCL12/CXCR4 constitutive chemokine pair therefore seems to play an important role in lymphocyte retention in RA.

Constitutive Chemokines and Lymphoid Neogenesis

Rheumatoid arthritis is one of a number of inflammatory diseases in which the organization of the inflammatory infiltrate shares characteristics of lymphoid tissue. Follicular hyperplasia with germinal center formation can occur in autoimmune thyroid disease, myasthenia gravis, Sjögren's disease, and RA, and may occur during infection with *Helicobacter pylori* and *Borrelia burgdorferi*. The lymphoid infiltrates in the rheumatoid synovium can be divided into at least three distinct histological groupings, varying from diffuse lymphocyte infiltrates through organized lymphoid aggregates to clear germinal center reactions. Moreover, there is conflicting evidence that such distinct histological types correlate with other serum indicators of disease activity. This form of inflammatory lymphoid neogenesis relies upon inappropriate, but highly organized temporal and spatial expression by stromal cells of the constitutive chemokines, particularly CXCL13 and CCL21, which are required for physiological lymphoid organogenesis (Figure 10.6). The elegant choreography of lymphocyte–stromal interactions within lymph nodes is organized by expression of adhesive and chemotactic cues in overlapping and combinatorial fashions. Once they have encountered new antigen, DCs specialized in the presentation of antigen to lymphocytes undergo a process of maturation under the local influence of inflammatory cytokines and bacterial and viral products. As a result inflammatory chemokine receptors are

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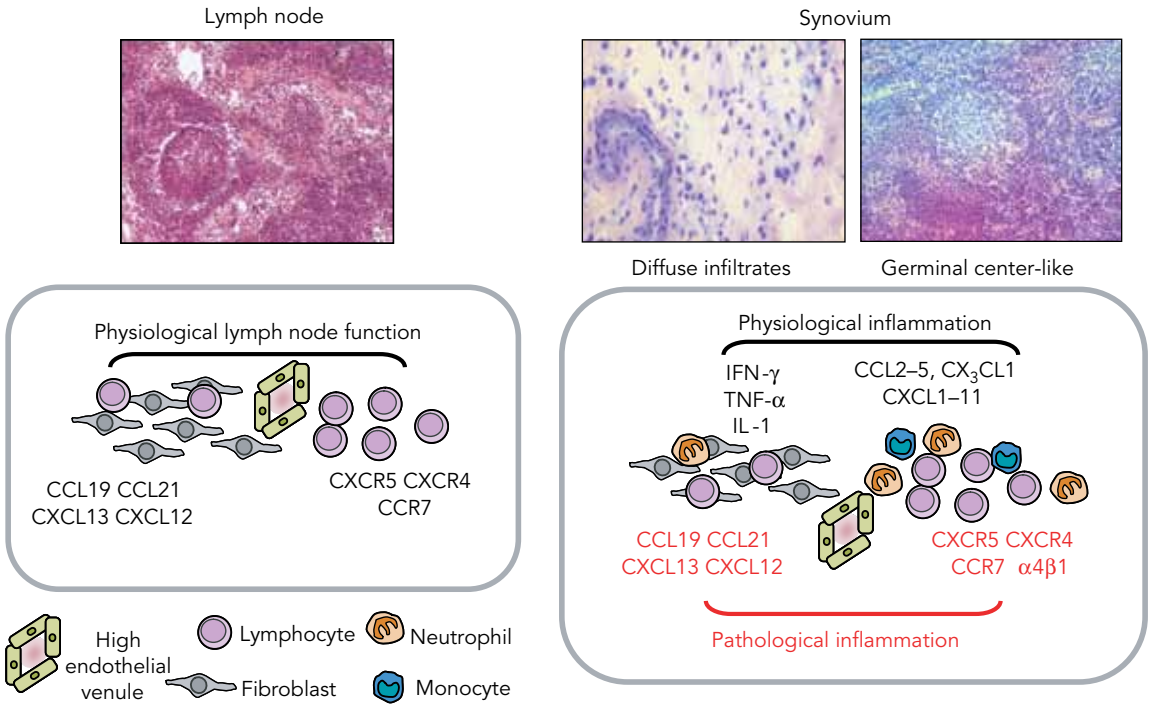


Figure 10.6. Inappropriate expression of constitutive chemokines leads to lymphoid neogenesis. Homeostatic chemokines (CXCL12, CXCL13, CCL19, CCL21) are components of the stromal code that help define niches such as the lymph node and bone marrow, governing leukocyte accumulation, differentiation, and survival. Stromal cells produce/express the appropriate cytokine/chemokine/adhesion receptor that is recognized by cognate receptors on infiltrating leukocytes. During physiological inflammation, inflammatory chemokines (CCL2–5, CX3CL1, and CXCL1–11 and inflammatory mediators such as IFN-γ, TNF-α, and IL-1 are produced by stromal cells and lead to the recruitment of inflammatory cells (lymphocytes, neutrophils, and monocytes). However, in persistent, pathological inflammation occurs in RA, stromal cells begin to aberrantly produce/express components of the physiological stromal code normally associated with lymphoid tissues, leading to lymphoid neogenesis.

downregulated, and upregulation of the constitutive receptors CCR4, CCR7, and CXCR4 occurs, causing DCs to migrate into local draining lymphatics and thereby into peripheral lymph nodes. Trafficking of B and T cells is regulated by CXCL13 (BCA-1, B cell–attracting chemokine 1), its receptor CXCR5, and CCL21 and CCL19 (EBL-1-ligand chemokine, ELC), which are both CCR7 agonists. Within the lymph node CXCR5-bearing B cells are attracted to follicular areas, while T cells and DCs are maintained within parafollicular zones by local expression of CCL21 and CCL19. Some T cells which have been successfully presented with their cognate antigen by DCs then upregulate CXCR5, allowing them to migrate toward and interact with B cells.

The genesis of lymphoid follicular structures in diseases such as diabetes and RA appears to rely upon expression of such constitutive chemokines, in association with the lymphotoxins alpha and beta (LT-α and LT-β) and TNF-α. In this context, it is important to note that transgenic animals overexpressing the TNF-α gene display increased formation of focal lymphoid aggregates and develop a chronic arthritis similar to RA. Clearly one of the many mechanisms of action of anti-TNF therapy may be the dissolution of such aggregates. In transgenic mouse models, expression of CXCL13 in the pancreatic islets was sufficient for the development of T- and B-cell clusters, but as they lacked follicular DCs, was not sufficient for true germinal center formation. CCL21 does appear to be sufficient in some cases for lymph node formation; murine pancreatic islet models have demonstrated formation of lymph node like structures in the presence of CCL21, and lymphoid infiltrates in response to CCL19 expression. Weyand and colleagues used the histological heterogeneity seen in RA to identify those factors critical to the formation of lymphoid microstructures, showing that transcription levels of CXCL13 and CCL21 were increased 10–20 times in tissues with germinal centers compared to tissues with other histological patterns. Multivariate analysis showed that LT-β and CXCL13 were necessary, but not sufficient for lymphoid neogenesis. It has also been shown that CXCR5 is overexpressed in the rheumatoid synovium, consistent with a role in

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recruitment and positioning of B and T lymphocytes within lymphoid aggregates of the RA synovium. It therefore seems likely that expression of lymphoid constitutive chemokines contributes significantly to the entry, local organization, and exit of lymphocytes in the RA synovium. It also seems that the ectopic expression of chemokines is a general characteristic of a number of chronic rheumatic conditions, since another B-cell attracting chemokine CXCL13 (BCA-1) is inappropriately expressed by stromal cells in the salivary glands of patients with Sjögren's syndrome.

MULTIPLE CELL–CELL INTERACTIONS INVOLVING FIBROBLASTS

A further pathological model in which fibroblasts become shifted into an activated phenotype is that of liver fibrosis. Models of hepatic fibrosis suggest that resident populations of hepatic macrophages are able to modulate the processes of fibrotic progression and subsequent liver recovery. Stellate cells (specialized liver fibroblasts) are known to differentiate during the process of hepatic injury and fibrosis toward a profibrotic, secretory myofibroblast phenotype under the influence of cytokines such as TGF-β1, of which hepatic macrophages are a potentially important source. High expression of tissue inhibitor of metalloproteinase-1 (TIMP-1) by stellate cells/myofibroblasts appears to maintain their survival and inhibit degradation of matrix leading to fibrosis. During recovery from fibrosis in animal models, myofibroblasts undergo apoptosis accompanied by a fall in TIMP-1 expression and degradation of scarring matrix. Expression by macrophages of TNF-related apoptosis-inducing ligand (TRAIL) is one mechanism by which stellate cell apoptosis has been proposed to occur. Using a transgenic mouse model of fibrosis that allowed the conditional depletion of macrophage lineage cells, Duffield et al. were able to demonstrate a fascinating divergence of regulation of these processes. Depletion of macrophages during the induction/progression phase of fibrosis resulted in decreased scarring and fewer myofibroblasts, implying a role for macrophages in the activation of myofibroblasts. However, when macrophages were depleted in the early phases of recovery, sustained accumulation of scarring matrix occurred (Figure 10.7), suggesting that at this stage of the inflammatory response macrophages are required to inactivate myofibroblasts.

These experiments demonstrate that modeling the behavior of stromal cells and leukocytes within microenvironments necessarily requires that we take

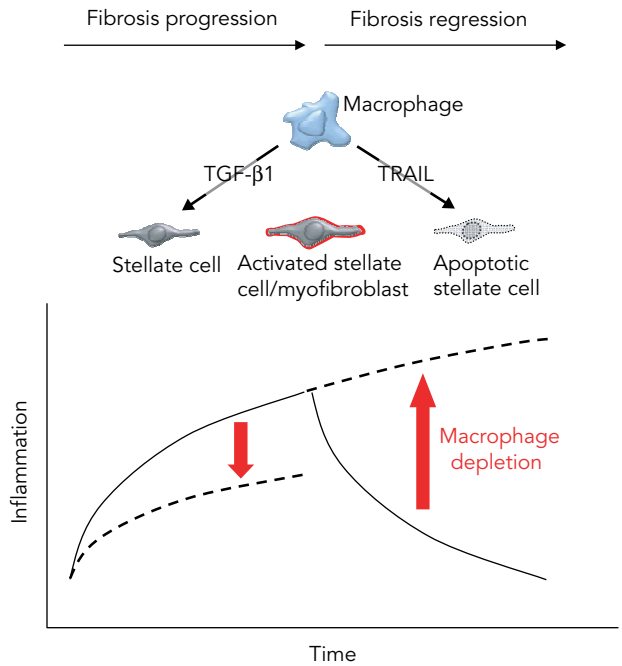


Figure 10.7. Macrophage–stromal cell interactions in the regulation of fibrotic liver disease. Hepatic fibroblasts induce divergent effects on liver stellate cell activation in models of fibrosis. During fibrosis progression, TGF-β1 is a potential macrophage-derived stimulator of stellate cell activation. Depletion of macrophages during this phase results in decreased scarring and fewer myofibroblasts. During fibrosis regression, stellate cell apoptosis may occur under the influence of TRAIL. Depletion of macrophages early during fibrosis regression results in sustained accumulation of scarring matrix. Depletion of macrophages during the healing phase leads to sustained fibrosis.

into account the interactions of both leucocytes and stromal cells. A very elegant example of this approach in vitro is the work of Smith and Lally, who developed a model of cellular recruitment to the rheumatoid synovium. Models of recruitment in which leukocytes adhere and migrate beneath cultured endothelial cells under flow conditions are well established as a means of investigating the mechanisms governing recruitment to tissues from the blood, and will be considered elsewhere. Coculturing fibroblasts from skin and RA synovial membrane with endothelial cells showed that IL-6 released from synovial (but not skin) fibroblasts was able to induce production of chemokines and adhesion molecules, resulting in greater neutrophil recruitment by synovial fibroblasts. Subsequent work interrogating the system using low-density gene arrays demonstrated that the effect of neutrophil attracting chemokines such as CXCL5 released from synovial fibroblasts was dependent on the function of the chemokine transporter molecule DARC (Duffy antigen receptor for chemokines) which was also induced by coculture (Figure 10.8).

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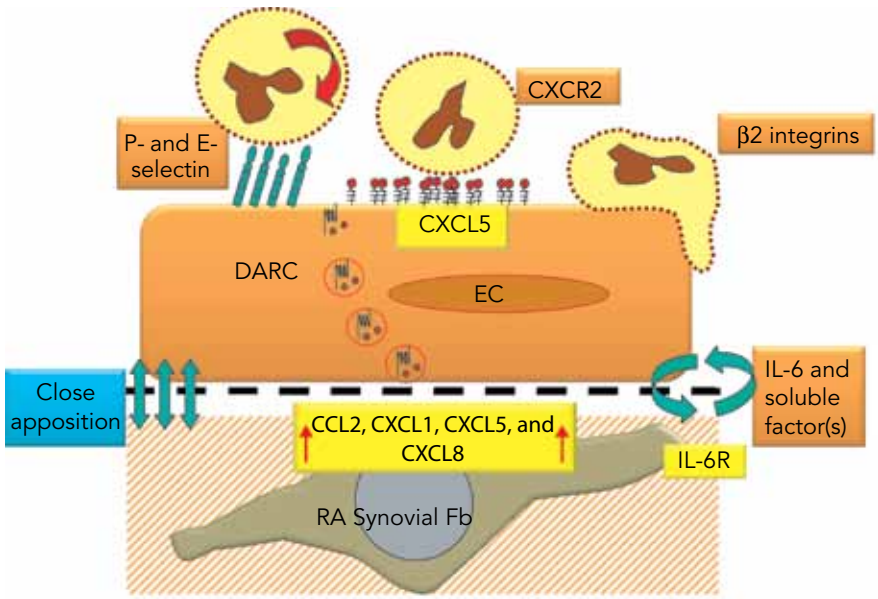


Figure 10.8. Three cell model of recruitment to the inflamed rheumatoid synovium. A multicellular coculture model of RA synovium was used to reconstruct the persistently inflamed RA microenvironment in vitro, by coculturing RA synovial fibroblasts or skin fibroblasts with human endothelial cells (EC). After a period of conditioning, fibroblasts and EC were isolated and screened using microarray analysis. Recapitulating the RA environment upregulated message for several CXC chemokines in ECs and RASFs, an effect that was absent in cocultures of ECs and skin fibroblasts. Flow adhesion assays demonstrated that only CXCL5 was functional on ECs cocultured with RASFs and recruited flowing neutrophils. EC expression of DARC was induced by coculture with RASFs, and antibody-blocking CXCL5 interactions with DARC or small interfering RNA (siRNA) targeting DARC expression abolished neutrophil recruitment. Thus, DARC edits the leukocyte recruitment code in a chemokine-specific manner in a model of human inflammatory disease, demonstrating the impact of inflammatory "cross-talk" between cells of the stromal microenvironment.

SUMMARY

The demonstration of inflammatory "cross-talk" between cells of the stromal microenvironment resulting in recruitment of inflammatory cells illustrates the significant contribution within microenvironments of cells such as fibroblasts. Populations of leukocytes recruited to sites of inflammation should not therefore be considered or studied in isolation, but should be considered in the context of their complementary stromal microenvironment, which provides survival, differentiation, and positioning cues upon which the formation and persistence of leukocyte infiltrates depends.

KEY POINTS

What Is a Stromal Cell?

- Fibroblasts are a heterogeneous population of stromal cells with specialized roles within different organs and tissues.
- Fibroblasts have a role in determining the location of tissue-specific diseases.

Embryological Origins of Fibroblasts

- Stromal cells in tissues may differentiate from a variety of precursor cells.
- Fibroblast precursor cells may have important roles in initiation of disease.

Stromal Cells in Immune Responses

- Stromal cells function as innate immune sentinels elaborating cytokines and chemokines in response to innate *stimuli* such as bacterial products and CD40 ligand.
- Persistent activation of fibroblasts may lead to pathological disease via an inflammatory stromal microenvironment.
- Classical inflammatory signal transduction pathways including NF-κB, MAPK, and PI3K are activated in fibroblasts during inflammation.

Stromal Cell–Leukocyte Interactions and Persistent Inflammation

- Fibroblasts play an active role in recruiting leukocytes to persistently inflamed sites.
- Infiltrating leukocytes are rescued from apoptosis by fibroblasts in persistent inflammation.
- Inappropriate expression of constitutive chemokines by stromal cells results in recruitment and retention of leukocytes within persistently inflamed microenvironments.

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- Lymphoid neogenesis in inflammatory disorders results from a stromal defect in chemokine expression.

Multiple Cell–Cell Interactions Involving Fibroblasts

- A shift into a profibrotic fibroblast phenotype regulated by tissue macrophages accounts for disease progression in models of liver fibrosis.
- Inflammatory cross-talk between fibroblasts and endothelial cells drives recruitment of neutrophils in three cell coculture models.

SUGGESTED READINGS

Asahara, T., Murohara, T., Sullivan, A., et al. 1997. Isolation of putative progenitor endothelial cells for angiogenesis. *Science* **275:**964–967.

Bhowmick, N.A., Neilson, E.G., and Moses, H.L. 2004. Stromal fibroblasts in cancer initiation and progression. *Nature* **432:**332–337.

Bjornson, C.R., Rietze, R.L., Reynolds, B.A., Magli, M.C., and Vescovi, A.L. 1999. Turning brain into blood: a hematopoietic fate adopted by adult neural stem cells in vivo. *Science* **283:**534–537.

Blades, M.C., Ingegnoli, F., Wheller, S.K., et al. 2002. Stromal cell-derived factor 1 (CXCL12) induces monocyte migration into human synovium transplanted onto SCID Mice. *Arthritis Rheum* **46:**824–836.

Brouty-Boye, D., Pottin-Clemenceau, C., Doucet, C., Jasmin, C. and Azzarone, B. 2000. Chemokines and CD40 expression in human fibroblasts. *Eur J Immunol* **30:**914–919.

Bucala, R., Spiegel, L.A., Chesney, J., Hogan, M., and Cerami, A. 1994. Circulating fibrocytes define a new leukocyte subpopulation that mediates tissue repair. *Mol Med* **1:** 71–81.

Buckley, C.D., Amft, N., Bradfield, P.F., et al. 2000. Persistent induction of the chemokine receptor CXCR4 by TGF-beta 1 on synovial T cells contributes to their accumulation within the rheumatoid synovium. *J Immunol* **165:** 3423–3429.

Buckley, C.D., Pilling, D., Lord, J.M., Akbar, A.N., Scheel-Toellner, D., and Salmon, M. 2001. Fibroblasts regulate the switch from acute resolving to chronic persistent inflammation. *Trends Immunol* **22:**199–204.

Burger, J.A., Zvaifler, N.J., Tsukada, N., Firestein, G.S. and Kipps, T.J. 2001. Fibroblast-like synoviocytes support B-cell pseudoemperipolesis via a stromal cell-derived factor-1- and CD106 (VCAM-1)-dependent mechanism. *J Clin Invest* **107:**305–315.

Duffield, J.S., Forbes, S.J., Constandinou, C.M., et al. 2005. Selective depletion of macrophages reveals distinct, opposing roles during liver injury and repair. *J Clin Invest* **115:**56–65.

Filer, A., Parsonage, G., Smith, E., et al. 2006. Differential survival of leukocyte subsets mediated by synovial, bone marrow, and skin fibroblasts: site-specific versus activation-dependent survival of T cells and neutrophils. *Arthritis Rheum* **54:**2096–2108.

Haniffa, M.A., Wang, X.N., Holtick, U., et al. 2007. Adult human fibroblasts are potent immunoregulatory cells and functionally equivalent to mesenchymal stem cells. *J Immunol* **179:**1595–1604.

Kalluri, R. and Neilson, E.G. 2003. Epithelial-mesenchymal transition and its implications for fibrosis. *J Clin Invest* **112:**1776–1784.

Kim, K.W., Cho, M.L., Kim, H.R., et al. 2007. Up-regulation of stromal cell-derived factor 1 (CXCL12) production in rheumatoid synovial fibroblasts through interactions with T lymphocytes: role of interleukin-17 and CD40L-CD40 interaction. *Arthritis Rheum* **56:**1076–1086.

Lally, F., Smith, E., Filer, A., et al. 2005. A novel mechanism of neutrophil recruitment in a coculture model of the rheumatoid synovium. *Arthritis Rheum* **52:**3460–3469.

Luther, S.A., Bidgol, A., Hargreaves, D.C., et al. 2002. Differing activities of homeostatic chemokines CCL19, CCL21, and CXCL12 in lymphocyte and dendritic cell recruitment and lymphoid neogenesis. *J Immunol* **169:**424–433.

Manzo, A., Paoletti, S., Carulli, M., et al. 2005. Systematic microanatomical analysis of CXCL13 and CCL21 in situ production and progressive lymphoid organization in rheumatoid synovitis. *Eur J Immunol* **35:**1347–1359.

Marinova-Mutafchieva, L., Williams, R.O., Funa, K., Maini, R.N., and Zvaifler, N.J. 2002. Inflammation is preceded by tumor necrosis factor-dependent infiltration of mesenchymal cells in experimental arthritis. *Arthritis Rheum* **46:**507–513.

Merville, P., Dechanet, J., Desmouliere, A., et al. 1996. Bcl-2+ tonsillar plasma cells are rescued from apoptosis by bone marrow fibroblasts. *J Exp Med* **183:**227–236.

Muller-Ladner, U., Kriegsmann, J., Franklin, B.N., et al. 1996. Synovial fibroblasts of patients with rheumatoid arthritis attach to and invade normal human cartilage when engrafted into SCID mice. *Am J Pathol* **149:**1607–1615.

Parsonage, G., Falciani, F., Burman, A., et al. 2003. Global gene expression profiles in fibroblasts from synovial, skin and lymphoid tissue reveals distinct cytokine and chemokine expression patterns. *Thromb Haemost* **90:**688–697.

Parsonage, G., Filer, A.D., Haworth, O., et al. 2005. A stromal address code defined by fibroblasts. *Trends Immunol* **26:**150–156.

Pierer, M., Rethage, J., Seibl, R., et al. 2004. Chemokine secretion of rheumatoid arthritis synovial fibroblasts stimulated by toll-like receptor 2 ligands. *J Immunol* **172:**1256–1265.

Postlethwaite, A.E., Shigemitsu, H., and Kanangat, S. 2004. Cellular origins of fibroblasts: possible implications for organ fibrosis in systemic sclerosis. *Curr Opin Rheumatol* **16:**733–738.

Rinn, J.L., Bondre, C., Gladstone, H.B., Brown, P.O., and Chang, H.Y. 2006. Anatomic demarcation by positional variation in fibroblast gene expression programs. *PLoS Genet* **2:**e119.

Ritchlin, C. 2000. Fibroblast biology. Effector signals released by the synovial fibroblast in arthritis. *Arthritis Res* **2:**356–360.

Salmon, M., Scheel-Toellner, D., Huissoon, A.P., et al. 1997. Inhibition of T cell apoptosis in the rheumatoid synovium. *J Clin Invest* **99:**439–446.

Sanchez-Pernaute, O., Ospelt, C., Neidhart, M., and Gay, S. 2008. Epigenetic clues to rheumatoid arthritis. *J Autoimmun* **30:**12–20.

Schmeichel, K.L. and Bissell, M.J. 2003. Modeling tissue-specific signaling and organ function in three dimensions. *J Cell Sci* **116:**2377–2388.

Schmidt, M., Sun, G., Stacey, M.A., Mori, L. and Mattoli, S. 2003. Identification of circulating fibrocytes as precursors of bronchial myofibroblasts in asthma. *J Immunol* **171:**380–389.

Smith, E., McGettrick, H.M., Stone, M.A., et al. 2008. Duffy antigen receptor for chemokines and CXCL5 are essential for the recruitment of neutrophils in a multicellular model of rheumatoid arthritis synovium. *Arthritis Rheum* **58:**1968–1973.

Smith, R.S., Smith, T.J., Blieden, T.M. and Phipps, R.P. 1997. Fibroblasts as sentinel cells. Synthesis of chemokines and regulation of inflammation. *Am J Pathol* **151:**317–322.

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Sweeney, S.E., and Firestein, G.S. 2004. Rheumatoid arthritis: regulation of synovial inflammation. *Int J Biochem Cell Biol* **36:**372–378.

Szekanecz, Z., Kim, J., and Koch, A.E. 2003. Chemokines and chemokine receptors in rheumatoid arthritis. *Semin Immunol* **15:**15–21.

Takemura, S., Braun, A., Crowson, C., et al. 2001. Lymphoid neogenesis in rheumatoid synovitis. *J Immunol* **167:**1072–1080.

Thalhamer, T., McGrath, M.A., and Harnett, M.M. 2008. MAPKs and their relevance to arthritis and inflammation. *Rheumatology (Oxford)* **47:**409–414.

Thelen, M. 2001. Dancing to the tune of chemokines. *Nat Immunol* **2:**129–134.

Thurlings, R.M., Wijbrandts, C.A., Mebius, R.E., et al. 2008. Synovial lymphoid neogenesis does not define a specific clinical rheumatoid arthritis phenotype. *Arthritis Rheum* **58:**1582–1589.

Tolboom, T.C., van der Helm-van Mil, A.H., Nelissen, R.G., Breedveld, F.C., Toes, R.E., and Huizinga, T.W. 2005. Invasiveness of fibroblast-like synoviocytes is an individual patient characteristic associated with the rate of joint destruction in patients with rheumatoid arthritis. *Arthritis Rheum* **52:**1999–2002.

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11 Neutrophil–Endothelial Cell Interactions

János G. Filep and Sean P. Colgan

INTRODUCTION

Neutrophils (polymorphonuclear leukocytes, PMN) have a clearly defined role in inflammation. In response to injury or infection, PMN migration across vascular endothelial cells is a first line of defense against infectious agents, and defects in such PMN–endothelial interactions contributes to fulminate microbial infections, mucosal ulcerations, and delayed tissue healing. The protective aspects of PMN in disease are objectively exemplified by the clinical observation that patients with primary defects in PMN function, including neutropenia and genetic PMN immunopathologies (e.g., leukocyte adhesion deficiencies, chronic granulomatous disease, Chediak–Higashi syndrome, myeloperoxidase deficiency, etc.), exhibit ongoing mucosal infections.

This chapter focuses on our current understanding of how PMN interact with vascular endothelial cells under physiologic and pathophysiologic conditions (Figure 11.1).

MOLECULAR MECHANISMS OF PMN ADHESION AND TRANSMIGRATION

PMN migration across the endothelial surface is a result of an orchestrated series of events, ultimately resulting in PMN accumulation at sites of tissue injury. The recruitment signals, the cell–cell interaction steps, and the regulatory pathways for these events have been an area of extensive exploration in the past two decades. A number of recent reviews have addressed these steps in detail [1–4]. Here, we will summarize some of the major steps and guide the reader to the primary literature for more insight into this dynamic process.

It is now appreciated that adhesion-based interactions involving specific cell adhesion epitopes are the primary means by which PMNs interact with endothelial cells. For example, PMN β2 integrins are required for PMN to migrate across both endothelial and epithelial surfaces. These integrins, like others, are heterodimeric glycoproteins that exist in four forms on the PMN. Each displays a unique α-subunit (CD11a, b, c, or d) and an identical β2-subunit (CD18). These receptors are best demonstrated in the genetic disorder leukocyte adhesion deficiency (LAD), in which patients lack normal expression of the CD18 β-subunit, and as a result, show increased susceptibility to infection due to abnormal leukocyte function [4]. These patients manifest severe mucosal disease, characterized primarily by severe bacterial infections.

At the level of the vasculature, P-selectin (induced in acute injury) and E-selectin (upregulated in inflammatory conditions) mediate the initial capture and subsequent rolling of neutrophils along the wall of postcapillary venules. The selectins bind to specialized fucosylated sialoglycoconjugates, including the tetrasaccharide sialyl Lewis X (sLex), that decorate selected surface glycoproteins. P-selectin glycoprotein ligand-1 (PSGL-1) interacts with all selectins under physiological inflammatory situations. This molecule is expressed as a dimer on most leukocytes, and binding activity of PSGL-1 is conferred by the N-terminal region. Like the integrins, the importance of functional selectins is illustrated by the human genetic disease LAD type II, observed as recurrent infections as a result of deficiency in selectin ligands as a result of a mutation in the GDP-fucose transporter gene [4].

Slow rolling on selectins allows neutrophils to sample chemokines presented at the surface of the endothelium. Chemokines and their receptors provide cell- and tissue-specific activation signals that selectively regulate recruitment. Neutrophils as best characterized to bind to interleukin-8 (CXCL8) through their surface receptors CXCR1/2. Other leukocytes utilize a different chemokine repertoire. T cells, for example, use

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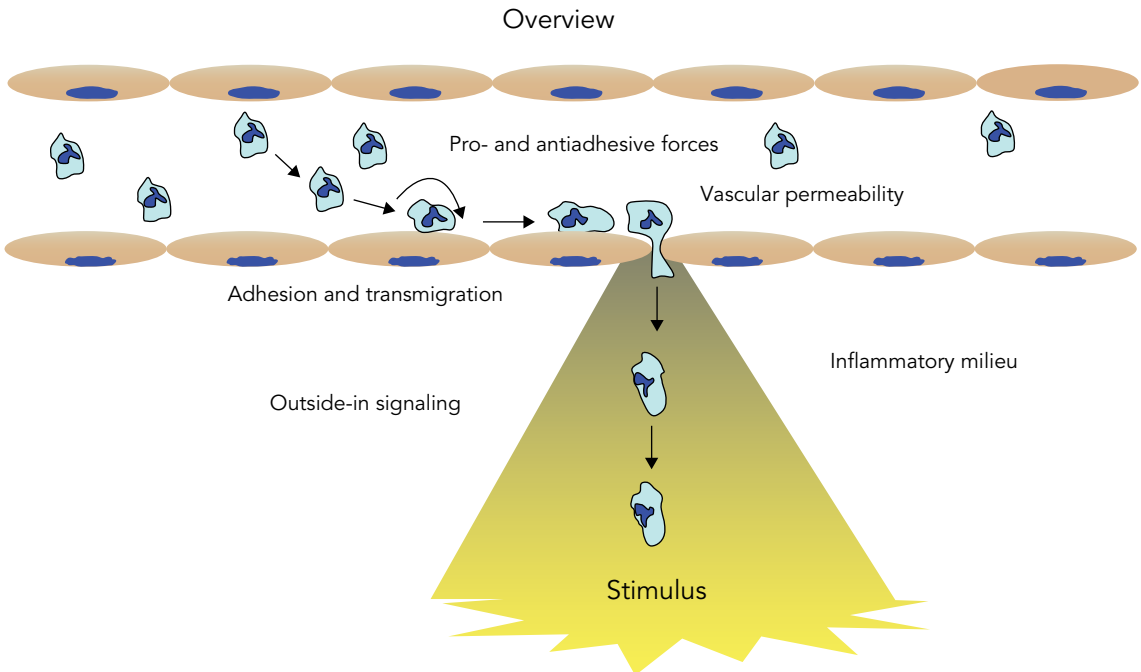


Figure 11.1. Graphical overview of this chapter, highlighting the areas of focus during neutrophil–endothelial interactions.

CCR7 for directed migration toward CCL21 and CCL19, while monocytes respond most avidly to CCL2 through its receptor CCR2, though they also bind CXCL8. The relative importance of chemokine signaling is further substantiated by the genetic disorder LAD type III, a defect wherein chemokine-triggered integrin activation on platelets and leukocytes is impaired due to defective Rap1 activation by the guanine nucleotide exchange factor CalDAG-GEFI [4].

The anatomical location of leukocyte extravasation within the vasculature has been the topic of much debate. Electron microscopy analyses and numerous in vitro studies using human umbilical vein endothelial cells have supported the opinion that leukocytes emigrate between endothelial cells (i.e., the paracellular route). The determinants that mediate paracellular transmigration have been characterized and shown to involve adhesion molecules expressed and distributed at intercellular junctions. These molecules include platelet-endothelial adhesion molecule-1 (PECAM-1), junctional adhesion molecules (JAMs), and CD99 [5]. Conversely, migration directly through individual endothelial cells (i.e., the transcellular route) has been suggested. Electron microscopy of transmigrating PMNs in vivo have indicated that transcellular migration occurs in certain physiological settings, representing the route of emigration for 5%–20% of transmigrating PMNs. These more recent studies have supported the possibility that both the transcellular and paracellular pathways likely coexist. Table 11.1 summarizes the steps and the molecules mediating leukocyte transmigration across endothelial cells.

BALANCE OF PRO- AND ANTIADHESIVE FORCES

Endothelial cells play a critical role in the regulation of leukocyte adhesion and trafficking into inflamed or injured tissues. Alterations in shear patterns and multiple cytokines and growth factors present at the sites of inflammation can potentially influence these events and therefore development and outcome of the inflammatory response. Leukocyte adhesion can be limited by anti-inflammatory counter-regulatory mechanisms that maintain the integrity and homeostasis of the vessel wall. Consistently, impaired protective mechanisms (e.g., decreased endothelial NO production or changes in laminar shear stress) would facilitate leukocyte adhesion even in the presence of low amounts of proinflammatory stimuli. The endothelium integrates the signals generated by proadhesive and antiadhesive forces.

Endothelial cell activation is predominantly mediated through the NF-κB and JNK-AP-1 signaling pathways. Unlike physiological laminar shear stress, low or turbulent blood flow patterns evoke sustained oxidative stress, resulting in activation of NF-κB and redox-sensitive gene expression. Exposure of endothelial cells to proinflammatory cytokines, including IL-1, IL-8,

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TABLE 11.1. Leukocyte adhesion cascade

Leukocytes	Endothelial cells	Major role
PSGL-1, E-selectin ligand-1	P-selectin, E-selectin	**Capture and rolling**
Glycosylated CD44		
Sialyl Lewis X		
(neutrophils, monocytes, T cells)		
L-selectin	GlyCam-1	Lymphocyte homing
Selectin signaling	Selectin signaling	Slow rolling
Chemokines → leukocyte activation		
Integrin affinity modulation		
		Migration arrest
LFA-1	ICAM-1	
VLA4	VCAM-1	
α4β2 integrin	MADCAM-1	
outside-in signaling through		
LFA-1 and Mac-1: Scr kinases, PI3K,		Adhesion strengthening and spreading
VAV1, VAV2, VAV3		
		Transendothelial cell migration
Mac-1	ICAM-1	
		Intravascular crawling (neutrophils, monocytes crawl, seeking preferred sites of transmigration)
	Junctional molecules	
LFA-1	JAM-A, JAM-1, ICAM-2	Paracellular route
Mac-1	JAM-C, ICAM-2	
PECAM-1	PECAM-1	
CD99	CD99	
Unknown	ESAM	
Membrane protrusions into endothelial cells	Vesiculo-vacuolar organelles	Transcellular route
	ICAM-1 translocation to caveolae	

The original three steps are shown in bold: (1) rolling that is mediated by selectins; (2) arrest that is mediated by integrins; and (3) transendothelial migration. The table summarizes advances that have been made in defining additional steps.

Abbreviations: ICAM-1, intercellular adhesion molecule-1; ICAM-2, intercellular adhesion molecule-2; ESAM, endothelial cell-selective adhesion molecule; JAM, junctional adhesion molecule; LFA-1, lymphocyte function-associated antigen 1; Mac-1, macrophage receptor 1; PECAM-1, platelet/endothelial cell adhesion molecule 1; PSGL-1, P-selectin glycoprotein ligand 1; PI3K, phosphoinositide 3-kinase; MADCAM1, mucosal vascular addressin adhesion molecule 1; VCAM-1, vascular cell-adhesion molecule 1; VLA4, very late antigen 4.

interferon-γ and tumor necrosis factor (TNF), vasoactive peptides, neuropeptides, minimally oxidized low-density lipoprotein, hyperglycemia and advanced glycosylated end products or smoking enhances production of reactive oxygen species (superoxide in particular), peroxynitrite formation, triggers activation of JNK, leading to transcription of proinflammatory genes (Figure 11.2). Altered hemodynamic forces and proinflammatory signals, such as oxidized lipoproteins are likely to act in concert to induce expression of a proadhesive endothelial cell phenotype as illustrated by permanent NF-κB activation and VCAM-1 expression seen in endothelial cells located in atherosclerosis-prone regions of the aorta of experimental animals.

The antiadhesion, anti-inflammatory mechanisms in the vascular wall involve antiadhesion external signals and intracellular mediators. Physiological laminar shear stress is of particular importance in protecting the endothelium against inflammatory activation. Endothelial cells are constantly exposed to hemodynamic forces generated by the pulsatile blood flow, hydrostatic pressure, cyclic strains, and wall shear

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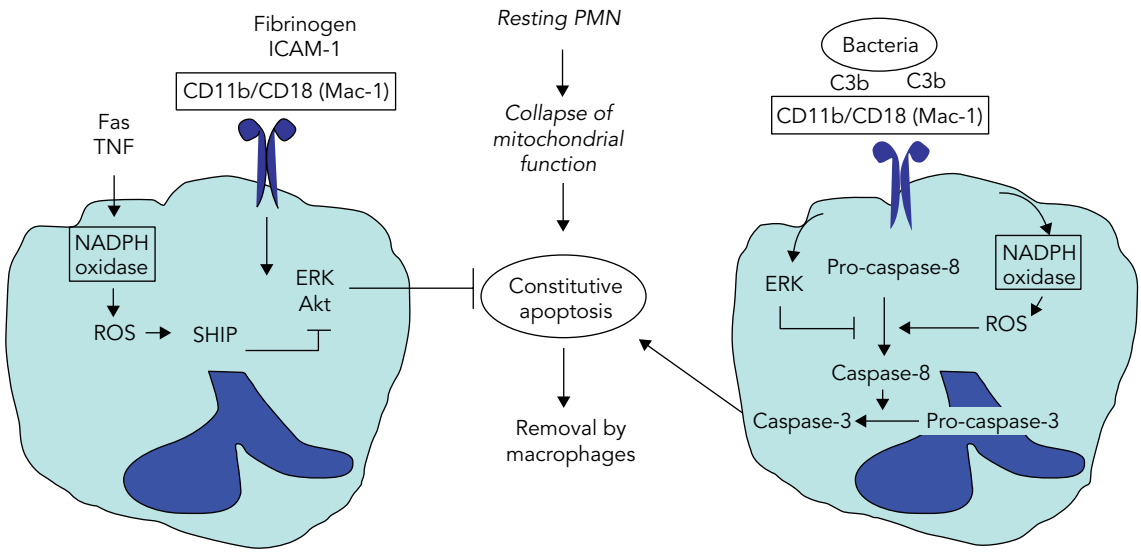


Figure 11.2. Impact of β2-integrin ligation on the fate of neutrophils. Mature neutrophils undergo constitutive apoptosis and removed from the circulation by macrophages. Neutrophil adhesion to β2-integrin Mac-1 (CD11b/CD18) ligands, fibrinogen, and ICAM-1 suppresses the cell death program through activation of the survival proteins Akt and ERK. The additional stimulation of neutrophils with death-inducing agonists, such as Fas or TNF, shifts the balance toward cell death by inducing ROS generation. ROS through lyn activates SHIP, which, in turn, inhibits Akt and induces apoptosis. Mac-1-dependent phagocytosis of complement opsonized bacteria promotes apoptosis. Phagocytosis results in the generation of ROS within the developing phagolysosomes. ROS triggers activation of caspase-8, leading to caspase-3 activation and acceleration of cell death. This overrides survival cues from Mac-1-stimulated ERK activation.

stress [6]. The antiadhesive and anti-inflammatory actions of pulsatile unidirectional flow may also prevail in conditions of activated endothelial cells. Prolonged exposure of endothelial cells to laminar flow occurs in vivo, results in downregulation of ICAM-1, VCAM-1, and E-selectin expression and inhibition of leukocyte adhesion. Molecular mechanisms of antiadhesive actions of laminar shear stress involve protection against oxidative stress and inhibition of the NF-κB and JNK-AP-1 signaling pathways (Figure 11.3). Laminar shear stress evokes expression of Cu/Zn superoxide dismutase (SOD), thereby effectively reducing superoxide. Laminar shear stress is known to be the physiological activator/inducer of NO formation by endothelial NO synthase. Besides its role in the regulation of vascular tone and permeability, endogenous NO inhibits MCP-1 and IL-6 release and cytokine-induced expression of ICAM-1 and VCAM-1 as well as leukocyte rolling and adhesion. These actions of NO are mediated by scavenging superoxide and through inhibition of NF-κB-dependent gene transcription. NO increases transcription, expression, and nuclear translocation of IκBα, resulting in acceleration of p50/p65 nuclear deactivation. Laminar shear stress specifically abrogates cytokine-induced JNK activity (Figure 11.3).

Anti-inflammatory cytokines that exert antiadhesive effects include transforming growth factor-β (TGF-β), interleukin-10 (IL-10), and IL-1 receptor antagonist (IL-1ra). TGF-β is secreted as an inactive complex with a latency-associated peptide (LAP). Cytokine activation of endothelial cells or coculture of endothelial cells with pericytes or smooth muscle cells results in the formation of active TGF-β. TGF-β downregulates cytokine-induced E-selectin, ICAM-1 and VCAM-1 expression, inhibits MCP-1 and IL-8 release, and attenuates the IL-8-dependent adhesion and transmigration of PMNs through the activated endothelial cell monolayer. The TGF-β actions are mediated through three types of cell-surface receptors coupled to their downstream effectors, known as Smad proteins. Smad proteins interact with the cAMP response element-binding protein (CREB)-binding protein (CBP), thereby blocking the association of CBP with the p65 subunit of NF-κB that is required for maximal transcriptional NF-κB activity. IL-10 is a pleiotropic cytokine produced by Th2 lymphocytes, B cells, monocytes, and macrophages. IL-10 is thought to act as a negative feedback to inhibit the production of proinflammatory cytokines, but it can also inhibit leukocyte–endothelial cell interactions in vivo. IL-10 attenuates leukocyte extravasation through decreasing endothelial expression of P-selectin, E-selectin, and ICAM-1. Conversely, IL-10-deficient mice exhibit markedly elevated expression of ICAM-1 and VCAM-1 in the vasculature and increased leukocyte adhesion in the mesenteric circulation in response to bacterial LPS. The protective effects of IL-10 are mediated

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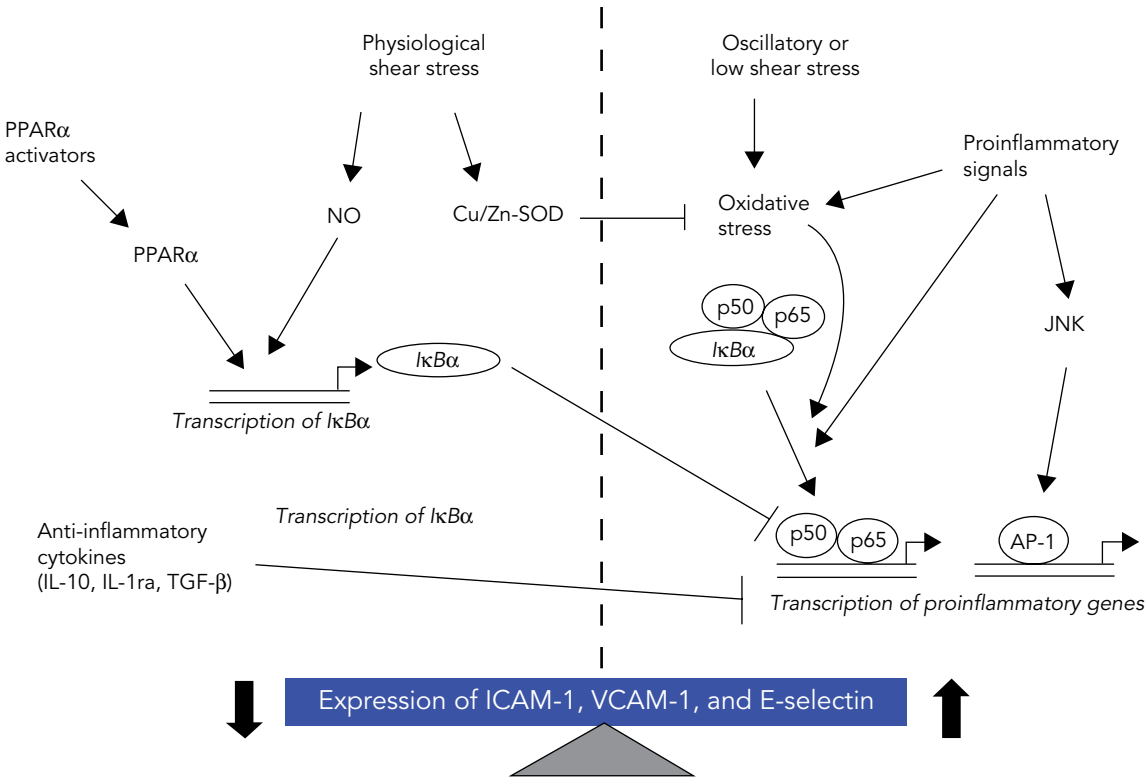


Figure 11.3. Balance of pro- and antiadhesives forces. Oscillatory or low shear stress and proinflammatory signals trigger intracellular oxidative stress and JNK activation that activate the NF-κB- and AP-1-mediated transcription of proinflammatory genes, including ICAM-1, VCAM-1, and E-selectin. Physiological laminar shear stress stimulates the expression of Cu/Zn-SOD and production of NO. Cu/Zn-SOD reduces oxidative stress. NO and activation of PPARα induces IκBα expression, resulting in inhibition of NF-κB activation and deactivation of nuclear p50/p65. Anti-inflammatory cytokines block transcription of NF-κB-driven genes. Laminar shear stress can also block the JNK pathway.

through the prevention of IκB degradation, suppression of NF-κB DNA binding activity [7], and destabilization of mRNA of proinflammatory genes with clustered AU-rich element motifs. IL-1ra is a secreted protein that binds to the IL-1 receptors without signaling. IL-1ra exhibits vascular protective effects in vivo and IL-1ra gene polymorphism is associated with coronary artery disease [8].

Peroxisome proliferators-activated receptors (PPARs) are transcription factors that regulate gene expression by forming a heterodimer with the retinoid receptor RXR that binds to specific DNA sequence elements termed PPAR-responsive elements. Activation of PPARα with its natural ligands n-3 fatty acids, or synthetic ligands, such as the lipid-lowering drugs fibrates, represses cytokine-stimulated expression of VCAM-1 and other proinflammatory mediators in human endothelial cells, resulting in reduced monocyte adhesion. PPARα activators reduce the oxidative stress and induce expression IκBα, and as a result, may inhibit NF-κB activation, in addition to direct protein–protein interactions between PPARα and NF-κB and AP-1 proteins involved in transrepression.

Endothelial cells may also acquire an autoprotective phenotype during inflammation by expressing cytoprotective genes, including the Bcl-2 family member A1, A20, as well as heme oxyganase-1 (HO-1). A1 and A20 inhibit adhesion molecule and IL-8 expression in endothelial cells by suppressing NF-κB activation. Gene transfer of HO-1 protects against hyperoxia-induced PMN infiltration and lung injury [9]. HO-1 deficiency in humans is associated with the presence of severe persistent endothelial damage [10]. Hypoxia or vascular injury induces upregulation and secretion of the glycoprotein Del-1 (developmental endothelial locus-1) that has been implicated in vascular remodeling during angiogenesis. Del-1 has recently been identified as an endogenous inhibitor of inflammatory cell recruitment. Indeed, endothelial Del-1 deficiency increased LFA-1-dependent PMN and monocyte adhesion in vitro and in mice, whereas soluble Del-1 inhibited LFA-1-dependent adhesion of PMN to immobilized ICAM-1 [11].

PMN INFLUENCES ON VASCULAR ENDOTHELIAL PERMEABILITY

Macromolecule transit across blood vessels has evolved to be tightly controlled. Relatively low macromolecular permeability of blood vessels is essential for

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maintenance of a physiologically optimal equilibrium between intravascular and extravascular compartments. Disturbances of endothelial barrier during disease states can lead to deleterious loss of fluids and plasma protein into the extravascular compartment. Such disturbances in endothelial barrier function are prominent in disorders such as shock and ischemia-reperfusion and contribute significantly to organ dysfunction. The direct relationship between PMN activation and increased endothelial permeability is not clear. While PMN accumulation and increased vascular permeability are often coincidental, PMN activation can occur with limited or no net changes in endothelial permeability. Moreover, while PMN depletion has been demonstrated to decrease organ injury in some models of ischemia and reperfusion, other models have suggested that PMN can exert protective influences in other models. Therefore, further information regarding PMN–endothelial cell interactions and their influence on endothelial permeability may provide a better understanding of the regulation of endothelial permeability.

Approximately 70 million PMN exit the vasculature per minute. These inflammatory cells move into underlying tissue by initially passing between endothelial cells that line the inner surface of blood vessels. This process, referred to as transendothelial migration (TEM), is particularly prevalent in inflamed tissues. Understanding the biochemical details of leukocyte–endothelial cell interactions is currently an area of concentrated investigation. Recent studies of genetically modified animals have suggested that specific molecules may establish "bottlenecks" to the control of the inflammatory response. For example, detailed studies have revealed that the process of leukocyte TEM entails a concerted series of events involving intimate interactions of a series of leukocyte and endothelial glycoproteins that include selectins, β2 integrins, and members of the immunoglobulin supergene family (e.g., ICAM-1). Moreover, histological studies of TEM reveal that PMN initially adhere to the endothelium, move to nearby inter-endothelial junctions via diapedesis, and insert pseudopodia into the interendothelial paracellular space. Successful TEM is accomplished by temporary PMN self-deformation with localized widening of the interendothelial junction. Following TEM, adjacent endothelial cells appear to "reseal," leaving no residual interendothelial gaps. These histological studies are consistent with the observation that leukocyte TEM may result in little or no change in endothelial permeability to macromolecules. In the absence of this tight and dynamic control of endothelial morphology and permeability, interendothelial gap formation during leukocyte TEM could lead to marked increases in endothelial permeability. However, only limited information exists regarding the biochemical events that maintain and dynamically regulate endothelial permeability in the setting of either PMN activation or TEM. A number of studies revealed that activated PMN release soluble factor(s) that support maintenance of endothelial permeability during PMN–endothelial interactions.

The predominant barrier (~90%) to movement of macromolecules across a blood vessel wall is presented by the endothelium [12]. Passage of macromolecules across a cellular monolayer may occur via either a paracellular route (i.e., between cells) or a transcellular route (i.e., through cells). In nonpathologic endothelium, macromolecules such as albumin (molecular weight ~40 kDa) appear to cross the cell monolayer by passing between adjacent endothelial cells (i.e., paracellular) although some degree of transcellular passage may also occur. Endothelial macromolecular permeability is inversely related to macromolecule size. Permeability is also dependent on the tissue of origin. For example, endothelial cells in the cerebral circulation (i.e., blood-brain barrier) demonstrate an exceptionally low permeability. Endothelial permeability may increase markedly upon exposure to a variety of inflammatory compounds (e.g., histamine, thrombin, reactive oxygen species (ROS), leukotrienes, bacterial endotoxins) or adverse conditions (e.g., hypoxia, ischemia). Mechanisms also exist to maintain or balance endothelial permeability during leukocyte TEM. For example, activated PMN release a number of soluble factors, such as ATP, AMP, adenosine, and glutamate, which promote endothelial barrier function.

Endothelial permeability is determined by cytoskeletal mechanisms that regulate lateral membrane intercellular junctions. Tight junctions, also known as zona occludens, comprise one type of intercellular junction and include the proteins zona occludens-1 (ZO-1), ZO-2, cingulin, and occludin. Tight junctions form narrow, cell-to-cell contacts with adjacent cells and comprise the predominant barrier to transit of macromolecules between adjacent endothelial cells. Disruption of cytoskeletal microfilaments (for instance, with cytochalasin B) produces reversible increases in endothelial permeability. JAMs represent another family of proteins important in the transit of PMN across the vascular interface. JAMs immunoglobulin superfamily (IgSF) proteins expressed at cell junctions in epithelial and endothelial cells as well as on the surface of leukocytes, platelets, and erythrocytes. The JAMs (JAM-A, -B, and -C) are variably expressed in endothelial and epithelial cells and mediate homophilic and heterophilic interactions within various tissues. Evidence suggests JAM proteins are important for a variety of cellular processes, ranging

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from tight junction permeability, leukocyte transmigration, platelet activation, and angiogenesis.

As alluded to earlier, activated PMN release a number of soluble factors that promote structural changes within the endothelium. One of the better understood pathways is nucleotide metabolism at the endothelial surface. Activated PMN can release nanomolar quantities of ATP, and it is now accepted that the major pathway for extracellular hydrolysis of ATP and ADP is the ecto-nucleoside triphosphate diphosphohydrolase (NTPDase), previously identified as ecto-ATPase, ecto-ATPDase or CD39 [13]. CD39 is expressed on the vascular endothelium and its role to date has been to modulate platelet purinoreceptor activity by the sequential hydrolysis of extracellular ATP or ADP to AMP. This thromboregulatory potential of CD39 has been recently demonstrated by the generation of mutant mice with disruption of the CD39 gene, and by a series of experiments where high levels of ATPDase expression are attained by adenoviral vectors in the injured vasculature [13]. Ecto-5'-nucleotidase (CD73) is a membrane bound glycoprotein that functions to hydrolyze extracellular nucleotides into bioactive nucleoside intermediates. Surface-localized CD73 converts adenine nucleotides (e.g., AMP) into adenosine, which in turn, can activate transmembrane adenosine receptors or can be internalized through dipyridamole-sensitive carriers. Activation of these pathways has been shown to enhance endothelial barrier function. Endothelial cells of many origins express constitutive CD73. The primary function attributed to endothelial CD73 has been catabolism of extracellular nucleotides, although CD73 may also mediate lymphocyte binding under some circumstances. Taken together, studies in vitro and in murine models define CD39 and CD73 as gatekeepers for the metabolic fine tuning of endothelial permeability. Such innate protective pathways share the common strategy of increasing extracellular adenosine concentrations and promoting adenosine signaling at the cell surface.

In addition to factors, which preserve barrier function during transmigration (e.g., adenosine), PMN also release factors, which increase endothelial permeability. For example, activation of PMN through β2 integrins elicits the release of soluble factor(s) that induce endothelial cytoskeletal rearrangement, gap formation, and increased permeability [14]. This PMN-derived permeabilizing factor was subsequently identified as HBP (also called azurocidin and CAP37), member of the serprocidin family of cationic peptides. HBP, but not other PMN granule proteins (e.g., elastase, cathepsin G), was shown to induce Ca^{2+}-dependent cytoskeletal changes in cultured endothelia and to trigger macromolecular leakage in vivo. Interestingly, HBP regulation of barrier may not be selective for PMN, and in fact, endothelial cells themselves are now a reported source of HPB. As such, it is possible that the endothelium may self-regulate permeability through HBP under some conditions, and that mediators found within the inflammatory milieu may also increase endothelial permeability.

OUTSIDE-IN SIGNALING EVENTS IN PMN–ENDOTHELIAL CELL INTERACTIONS

Outside-in signaling via β_2 integrins is required for PMN adhesion and transmigration across the activated endothelium as discussed earlier, as well as for other adhesion-dependent PMN function, including phagocytosis of complement-opsonized pathogens, binding to fibrinogen, immune complexes, and platelets. The biological importance of β_2 integrins is highlighted in patients with a mutation in the β_2 integrin subunit shared by the LFA-1 and Mac-1 integrins. Thus, patients with LAD1 exhibit peripheral blood neutrophilia, increased susceptibility to bacterial infections, and delayed wound healing [15]. PMN adherence to Mac-1 ligands also affects the fate of PMNs.

Mature PMNs are terminally differentiated cells that have the shortest half-life (~7 hours) among leukocytes and die rapidly via apoptosis. This constitutively expressed cell death program renders PMNs unresponsive to proinflammatory stimuli and promotes their removal from inflamed areas by scavenger macrophages with minimal damage to the surrounding tissue, thereby facilitating the resolution of inflammation. The fate of PMNs is profoundly influenced by signals from the inflammatory microenvironment. Inflammatory mediators, such as lipopolysaccharide, IL-8 and acute-phase proteins prolong PMN survival by suppressing apoptosis, whereas proapoptotic stimuli, such as TNF or Fas ligand reduce PMN longevity.

PMN transmigration across the endothelium signals a delay of PMN apoptosis through engagement of Mac-1 with its endothelial cell ligand ICAM-1. Likewise, PMN adherence to another Mac-1 ligand fibrinogen also extends their life span. The delay in apoptosis is mediated through activation of the Ser-Thr kinase Akt, and, to a lesser degree, the MAPK-ERK signaling cascade, leading to inhibition of the mitochondrial pathway of apoptosis.

Although cross-linking Mac-1 alone provides survival signals, engagement of Mac-1 in the presence of proapoptotic TNF or Fas ligand accelerates apoptosis. Phagocytosis of opsonized bacteria by PMNs also evokes programmed cell death by a Mac-1-dependent pathway. This process is referred to as phagocytosis-induced cell death or PICD [16]. Proapoptotic stimuli promote NADPH oxidase–stimulated release of ROS, which leads to activation of SHIP (Src-homology 2 [SH2]-containing inositol 5-phosphatase) that hydrolyzes products of phosphoinositide 3-kinase

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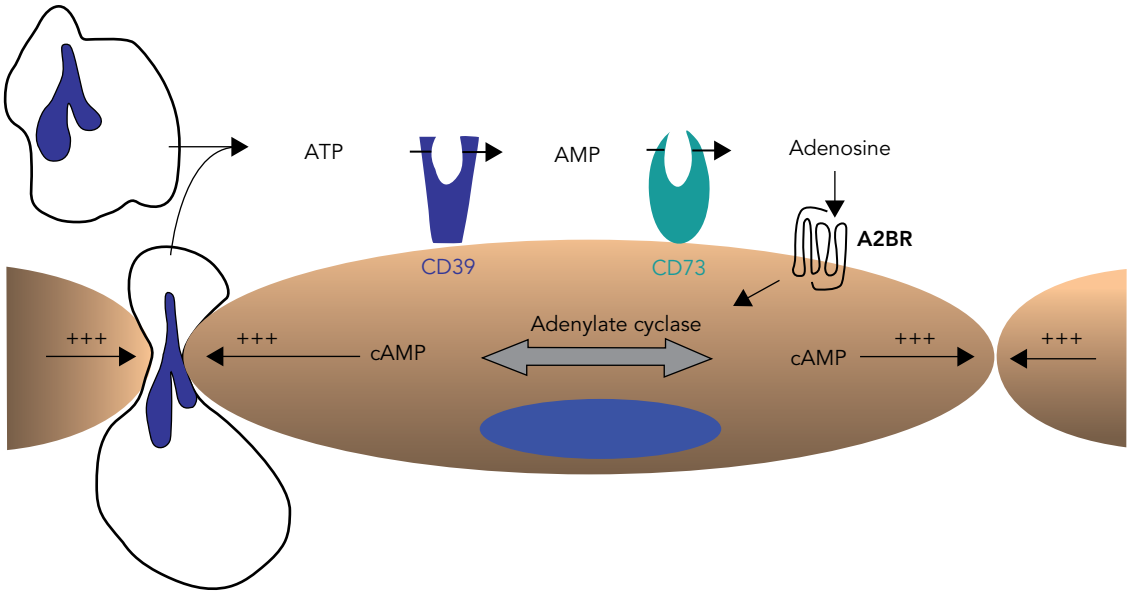


Figure 11.4. Model of coordinated nucleotide metabolism and nucleoside signaling in inflammation. In areas of ongoing inflammation, CD39 and CD73, expressed on the surface of endothelial cells, coordinate the metabolism of ATP to adenosine. Activated PMN provide a readily available extracellular source of ATP that through two enzymatic steps results in the liberation of adenosine. Adenosine generated in this fashion is available for activation of surface endothelial adenosine receptors, particularly the A2BR. Activation of A2BR leads to increases in intracellular cyclic AMP, thereby enhancing endothelial barrier function. As such, this protective mechanism may provide an innate mechanism to preserve vascular integrity and prevent intravascular fluid loss.

through the Src kinase Lyn. This signaling pathway would ultimately lead to decreased activation of the survival factor Akt. This purported mechanism of apoptosis might be particularly relevant in vivo under conditions of high PMN accumulation in tissues. Phagocytosis also results in generation of ROS within the phagolysosomes through NADPH oxidase. NADPH oxidase–derived ROS contribute to microbicidal activity and are also key intracellular trigger of PICD. ROS are required for activation of caspase-8, a signature of receptor-mediated cell death. Caspase-8, in turn, triggers activation of caspase-3, the main effector of apoptosis. Activated caspase-3 and -8 overcome Mac-1 ligation-generated survival signals, eventually favoring cell death. Additional exogenous stimuli, such as TNF or granulocyte/monocyte colony-stimulating factor, present at inflammatory sites, may interfere with phagocytosis-activated intracellular signals to shift the life-death balance of PMNs to either enhanced apoptosis or prolonged survival (Figure 11.4).

Upon leukocyte adhesion, ICAM-1 is capable of initiating outside-in signaling events in endothelial cells. ICAM-1 binding to its ligands β2-integrins or fibrinogen, leads to activation of Src tyrosine kinases, which in turn results in phosphorylation of cortactin, an actin-binding protein that modulates the F-actin cytoskeleton and endothelial cell locomotion. Following engagement, apical ICAM-1 is recruited to caveola- and F-actin–rich regions close to endothelial cell–cell borders, and translocates, with caveolin-1, to the basal plasma membrane. As caveolin-1, ICAM-1 and F-actin surround the transcellular channels around transmigrating lymphocytes, leukocyte-induced ICAM-1 recruitment to caveolae may initiate the formation of a transcellular passage for lymphocytes to cross endothelial cells [17].

Signaling pathways initiated by ICAM-1 binding to PMN LFA-1 or Mac-1 include enhanced ROS production and activation of p38 MAPK, activation of the transcription factors AP-1 and NF-kB, induction of VCAM-1 gene transcription and increased expression of VCAM-1 on the cell surface and increased production of IL-8 and RANTES. Ligation of constitutive or induced ICAM-1 increases intracellular glutathione level, thereby controlling endothelial redox status. Fibrinogen binding to ICAM-1 activates the survival signal ERK that has anti-apoptotic actions. This mechanism may be important in maintaining vascular integrity during ischemia/reperfusion injury, when fibrinogen is deposited on ECs.

INFLUENCE OF THE INFLAMMATORY MILIEU ON PMN–ENDOTHELIAL CELL INTERACTIONS

In intact tissues, endothelial cells lie anatomically positioned adjacent to and physiologically juxtaposed to number of cell types, including leukocytes, fibroblasts, and smooth muscle cells. Such a setting makes

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paracrine cross-talk pathways an important part of cell–cell communication. Locally generated mediators can bind to endothelial surface receptors, and mediate both physiologic and pathophysiologic functional responses.

A somewhat surprising observation in the past decade has been the finding that inflammatory setting can be quite oxygen deficient (hypoxic). Ongoing inflammatory responses are characterized by dramatic shifts in tissue metabolism. These changes include lactate accumulation with resultant metabolic acidosis and diminished availability of oxygen (hypoxia). Such shifts in tissue metabolism result from profound recruitment of inflammatory cell types, particularly PMN, which when activated can consume copious amounts of oxygen in the formation of oxygen radicals. At the tissue and cellular level, hypoxia induces an array of genes pivotal to survival in low oxygen states. As a global regulator of oxygen homeostasis, the αβ heterodimeric transcription factor hypoxia-inducible factor (HIF) facilitates both oxygen delivery and adaptation to oxygen deprivation. Genes induced by HIF include those necessary for cell, tissue, and whole animal adaptive responses to hypoxia. These proteins include enzymes involved in erythropoiesis, anaerobic metabolism, angiogenesis, and vasodilatation. More recently, it is appreciated that a number of inflammatory genes relevant to PMN–endothelial interactions are also HIF-regulated. For example, the CD18 chain of the β2 integrin complex is transcriptionally regulated by HIF, providing a prime example of how the inflammatory milieu might impact local PMN–endothelial responses.

With regard to paracrine signaling, an area of intense investigation is lipid mediator generation and signaling at inflammatory sites. Important in the regard, both endothelial and nonendothelial cell populations express enzymes (e.g., lipoxygenases, cyclooxygenases) capable of utilizing arachidonic acid substrates to generate bioactive lipid mediators. Such lipid mediators can signal via autocrine or paracrine pathways and depending on the tissue, microenvironment can convey a pro- or anti-inflammatory message. Of particular interest are a group of lipid mediators termed lipoxins and resolvins. Lipoxins are tetraene eicosanoids derived from membrane arachidonic acid through the combined action of 5-lipoxygenase (LO) and 12-LO or 15-LO (i.e., transcellular biosynthesis). A number of in vitro and in vivo studies revealed that lipoxins, and specifically lipoxin A4 (LXA4), function as an innate "stop signals," acting to control local inflammatory processes. At nanomolar concentrations, LXA4 has been demonstrated to inhibit PMN transmigration across confluent epithelia and endothelia. It is likely that the action(s) of LXA4 are predominantly on leukocytes and involve the activation of protein kinase C, since LXA4 inhibition required preincubation and PMN responses were sensitive to the protein kinase C inhibitor staurosporine. Additional mechanistic studies have revealed that LXA4 inhibit PMN β2 integrin (CD11/18) expression, and thus also block Mac-1 outside-in signaling. Lipoxins are rapidly (within minutes) converted to inactive compounds by myeloid cells. For this reason, stable lipoxin analogs have been synthesized and biochemically and functionally studied in detail. In vivo, both native LXA_4 and its metabolically stable have been demonstrated to block PMN trafficking, to facilitate removal of apoptotic neutrophils by macrophages, and to serve as potent anti-inflammatory and proresolution molecules in a number of vascular injury models.

More recent studies have focused on omega-3 fatty acid–derived resolvins [18]. Increasing evidence suggests that resolvin E1 (RvE1: 5*S*,12*R*,18*R*-trihydroxyeicosapentaenoic acid) contributes to resolution of inflammation via interactions with the resolvin E1 receptor (termed chemR23). RvE1 is generated at sites of inflammation through transcellular biosynthesis and has been shown to potently inhibit PMN TEM and to promote PMN clearance from the epithelial cell surface in vitro. In vivo, RvE1 has been documented to attenuate colonic mucosal inflammation in vivo and to resolve oral inflammation in a rabbit periodontitis model.

PMN–ENDOTHELIAL INTERACTION AS A THERAPEUTIC TARGET

The multistep paradigm of leukocyte recruitment implies that inhibition of any of these steps, in particular rolling and firm adhesion, should interrupt the extravasation process and therefore prevent leukocyte trafficking into inflamed or injured tissues. Indeed, the anti-inflammatory effects of glucocorticoids and many nonsteroidal anti-inflammatory drugs can be attributed, in part, to inhibition of adhesion molecules on PMN or endothelial cells [19,20]. More selective therapeutic approaches include receptor–ligand blockade, allosteric inhibitors, inhibitors of inside-out and outside-in signaling pathways, and targeting expression of immunoglobulin superfamily ligands [19,21]. Thus, inhibition of the rolling and adhesion events using monoclonal antibodies against selectins and integrins resulted in reduced leukocyte accumulation and striking protection against tissue injury in a variety of experimental models, including reperfusion injury, myocardial infarction, stroke, colitis, asthma, and rheumatoid arthritis [21]. Despite promising preclinical results, the outcome of clinical trials that followed in these indications has been disappointingly inconsistent. With the exception of some beneficial effects

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in asthma, ulcerative colitis, and psoriasis, treatment with humanized antiselectin or anti-CD18 antibodies had no overall benefit compared with placebo in patients with myocardial infarction, stroke, rheumatoid arthritis, traumatic shock, or renal transplant [21]. Considering the redundancy and overlapping functions of adhesion molecules, the negative results point toward alternative pathways that might govern leukocyte recruitment in these disorders. However, more selective targeting of adhesion molecules (e.g., targeting LFA-1 instead of CD18) or leukocyte subtypes resulted in positive clinical results. Thus, efalizumab, a humanized antibody against LFA-1 (CD11a) decreased severity index by >75% in psoriasis [22] and inhibition of α_4-integrin with natalizumab resulted in significant improvements in the clinical remission rates in patients with Crohn's disease or multiple sclerosis most likely through preventing lymphocyte infiltration [23,24]. Unfortunately, natalizumab therapy longer than 1 year had severe complications associated with lack of lymphocyte surveillance. The therapeutic potential of interfering with cell surface expression of adhesion molecules using antisense oligonucleotides, RNA silencing, or small molecule modifiers of intracellular signaling pathways remains to be investigated in the clinical setting.

Preclinical data indicate that downregulation of CD11b/CD18 expression on PMN and inhibition of PMN transmigration across endothelia and epithelia are among the central events underlying the anti-inflammatory proresolution actions of LXA4, aspirin-triggered 15-epi-LXA4, and RvE1 [18]. Thus, PMN-specific signals, which also include reduced production of proinflammatory cytokines and attenuation of oxidative and peroxidative stress, would lead to dampening of inflammation, whereas monocyte/macrophage-specific signals would promote inflammatory resolution by enhancing removal of apoptotic PMN and production of anti-inflammatory cytokines/chemokines. Accumulating evidence suggests an important role for impaired PMN apoptosis or macrophage phagocytosis in modulating the outcome of inflammation [20,25]. Consistently, induction of PMN apoptosis with cyclin-dependent kinase inhibitors [26] or overriding the potent survival signal generated by the acute-phase protein serum amyloid A with aspirin-triggered 15-epi-LXA4 [27] would enhance resolution of inflammation.

CONCLUSIONS

Our current understanding of PMN interactions with the vascular endothelium has evolved immensely in the past two decades. New technologies, better models, and the identification of human patients with genetic mutations within these pathways have provided unprecedented opportunities to define precise details of how PMN interact with endothelial cells at the molecular level. In particular, the leukocyte adhesion cascade has been augmented by several more steps, and the signaling network linking neutrophil and endothelial cell responses to activation of adhesion molecules is beginning to emerge. More detailed signaling studies on leukocytes and endothelial cells may identify new targets for therapeutic interventions. Indeed, these studies promise to provide a rational basis to develop therapies with higher selectivity for leukocyte subsets for the treatment of inflammatory diseases.

REFERENCES

1. Ley, K., Laudanna, C., Cybulsky, M.I., and Noursargh, S. 2007. Getting to the site of inflammation: the leukocyte adhesion cascade updated. *Nat Rev Immunol* **7**:678–689.
2. Yonekawa, K., and Harlan, J.M. 2005. Targeting leukocyte integrins in human diseases. *J Leukoc Biol*. **77**:129–140.
3. Smith, C.W. 2008. Adhesion molecules and receptors. *J Allergy Clin Immunol* **121**:S375–379.
4. Wagner, D.D., and Frenette, P.S. 2008. The vessel wall and its interactions. *Blood* **111**:5271–5281.
5. Weber, C., Fraemohs, L., and Dejana, E. 2007. The role of junctional adhesion molecules in vascular inflammation. *Nat Rev Immunol* **7**:467–477
6. Gimbrone, M.A., Nagel, T., and Topper, J.N. 1997. Biomechanical activation: an emerging paradigm in endothelial adhesion biology. *J Clin Invest* **99**: 1809–1813.
7. Schottelius, A.J., Mayo, M.W., Sartor, R.B., and Baldwin, A.S., Jr. 1999. Interleukin-10 signaling blocks inhibitor of κB kinase activity and nuclear factor κB DNA binding. *J Biol Chem* **274**:31868–31874.
8. Francis, S.E., Camp, N.J., Dewberry, R.M., et al. 1999. Interleukin-1 receptor antagonist gene polymorphism and coronary artery disease. *Circulation* **99**:861–866.
9. Otterbein, L.E., Kolls, J.K., Mantell, L.L., Cook, J.L., Alam, J., and Choi, A.M. 1999. Exogenous administration of heme oxygenase-1 by gene transfer provides protection against hyperoxia-induced lung injury. *J Clin Invest* **103**:1047–1054.
10. Yachie, A., Niida, Y., Wada, T., et al. 1999. Oxidative stress causes enhanced endothelial cell injury in human heme oxygensae-1 deficiency. *J Clin Invest* **103**:129–135.
11. Choi, E.Y., Chavakis, E., Czabanka, M.A., et al. 2008. Del-1, an endogenous leukocyte-endothelial cell adhesion inhibitor, limits inflammatory cell recruitment. *Science* **322**:1101–1104.
12. Colgan, S.P., Eltzschig, H.K., Eckle, T., and Thompson, L.F. 2006. Physiologic roles for ecto-5'-nucleotidase (CD73). *Purinergic Signal* **2**:351–360.
13. Robson, S.C., Sevigny, J., and Zimmermann, H. 2006. The E-NTPDase family of ectonucleotidases: structure function relationships and pathophysiological significance. *Purinergic Signal* **2**:409–430.
14. Eltzschig, H.K., Macmanus, C.F., and Colgan, S.P. 2008. Neutrophils as sources of extracellular nucleotides: functional consequences at the vascular interface. *Trends Cardiovasc Med* **18**:103–107.
15. Wehrle-Haller, B., and Imhof, B.A. 2003. Integrin-dependent pathologies. *J Pathol* **200**:481–487.

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Fundamentals of Inflammation, edited by Charles N. Serhan, et al., Cambridge University Press, 2010. ProQuest Ebook Central, http://ebookcentral.proquest.com/lib/yale-ebooks/detail.action?docID=802921.
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16. Mayadas, T.N., and Cullere, X. 2005. Neutrophil β2 integrins: moderators of life or death decisions. *Trends Immunol* **26**:388–395.
17. Millan, J., Hewlett, L., Glyn, M., Toomre, D., Clark, P., and Ridley, A.J. 2006. Lymphocyte transcellular migration occurs through recruitment of endothelial ICAM-1 to caveola- and F-actin-rich domains. *Nat Cell Biol* **8**:113–123.
18. Serhan, C.N., Chiang, N., and Van Dyke, T.E. 2008. Resolving inflammation: dual anti-inflammatory and pro-resolution lipid mediators. *Nat Rev Immunol* **8**: 349–361.
19. Ulbrich, H., Eriksson, E.E., and Lindbom, L. 2003. Leukocyte and endothelial cell adhesion molecules as targets for therapeutic interventions in inflammatory diseases. *Trends Pharmacol Sci* **24**:640–647.
20. Gilroy, D.W., Lawrence, T., Perretti, M., and Rossi, A.G. 2004. Inflammatory resolution: new opportunities fro drug discovery. *Nat Rev Drug Discov* **2**:965–975.
21. Yonekawa, K., and Harlan, J.M. 2005. Targeting leukocyte integrins in human diseases. *J Leukoc Biol* **77**:129–140.
22. Gordon, K.B., Papp, K.A., Hamilton, T.K., et al. 2003. Efalizumab for patients with moderate to severe plaque psoriasis: a randomized controlled trial. *JAMA* **290**:3073–3080.
23. Keeley, K.A., Rivey, M.P., and Allington, D.R. 2005. Natulizimab for the treatment of multiple sclerosis and Crohn's disease. *Ann Pharmacother* **39**:1833–1843.
24. Rivera-Nieres, J., Gorfu, and Ley, K. 2008. Leukocyte adhesion models of inflammatory bowel disease. *Inflamm Bowel Dis* **14**:1715–1735.
25. El Kebir, D., József, L., Pan, W., and Filep, J.G. 2008. Myeloperoxidase delays neutrophil apoptosis through CD11b/CD18 integrins and prolongs inflammation. *Circ Res* **103**:352–359.
26. Rossi, A.G., Sawatzky, D.A., Walker, A., et al. 2006. Cyclin-dependent kinase inhibitors enhance the resolution of inflammation by promoting inflammatory cell apoptosis. *Nat Med* **12**:1056–1064.
27. ElKebir, D., József, L., Pan, W., Petasis, N.A., Serhan, C.N., and Filep, J.G. 2007. Aspirin-triggered lipoxins override the apoptosis-delaying action of serum amyloid A in human neutrophils: a novel mechanism for resolution of inflammation. *J Immunol* **179**:616–622.

SUGGESTED READINGS

Butcher, E.C. 1991. Leukocyte-endothelial cell recognition: three (or more) steps to specificity and diversity. *Cell* **67**:1033–1036.

Jacobson, K.A., and Gao, Z.G. 2006. Adenosine receptors as therapeutic targets. *Nat Rev Drug Discov* **5**:247–264.

Kong, T., Eltzschig, H.K., Karhausen, J., Colgan, S.P., and Shelley, C.S. 2004. Leukocyte adhesion during hypoxia is mediated by HIF-1-dependent induction of β2 integrin gene expression. *Proc Nat Acad Sci USA* **101**:10440–10445.

Rao, R.M., Yang, L., Garcia-Cardena, G., and Luscinskas, F.W. 2007. Endothelial-dependent mechanisms of leukocyte recruitment to the vascular wall. *Circ Res* **101**:234–247

Serhan, C.N., Brain, S.D., Buckley, C.D., et al. 2007. Resolution of inflammation: state of the art, definitions and terms. *FASEB J* **21**:325–332.

Tedgui, A., and Mallat, Z. 2001. Anti-inflammatory mechanisms in the vascular wall. *Circ Res* **88**:877–887.

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